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(54) **SYSTEM FOR DISPLAYING MEDICAL MONITORING DATA**

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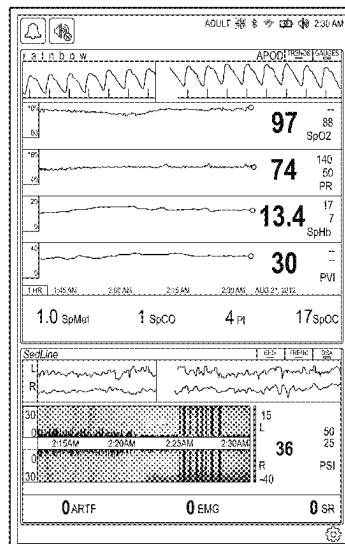
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(57) **ABSTRACT**

A first medical device can receive a physiological parameter value from a second medical device. The second physiological parameter value may be formatted according to a protocol not used by the first medical device such that the first medical device is not able to process the second physiological parameter value to produce a displayable output value. The first medical device can pass the physiological parameter data from the first medical device to a separate translation module and receive translated parameter data from the translation module at the first medical device. The translated parameter data can be processed for display by the first medical device. The first medical device can output a value from the translated parameter data for display on the first medical device or an auxiliary device.

26 Claims, 99 Drawing Sheets



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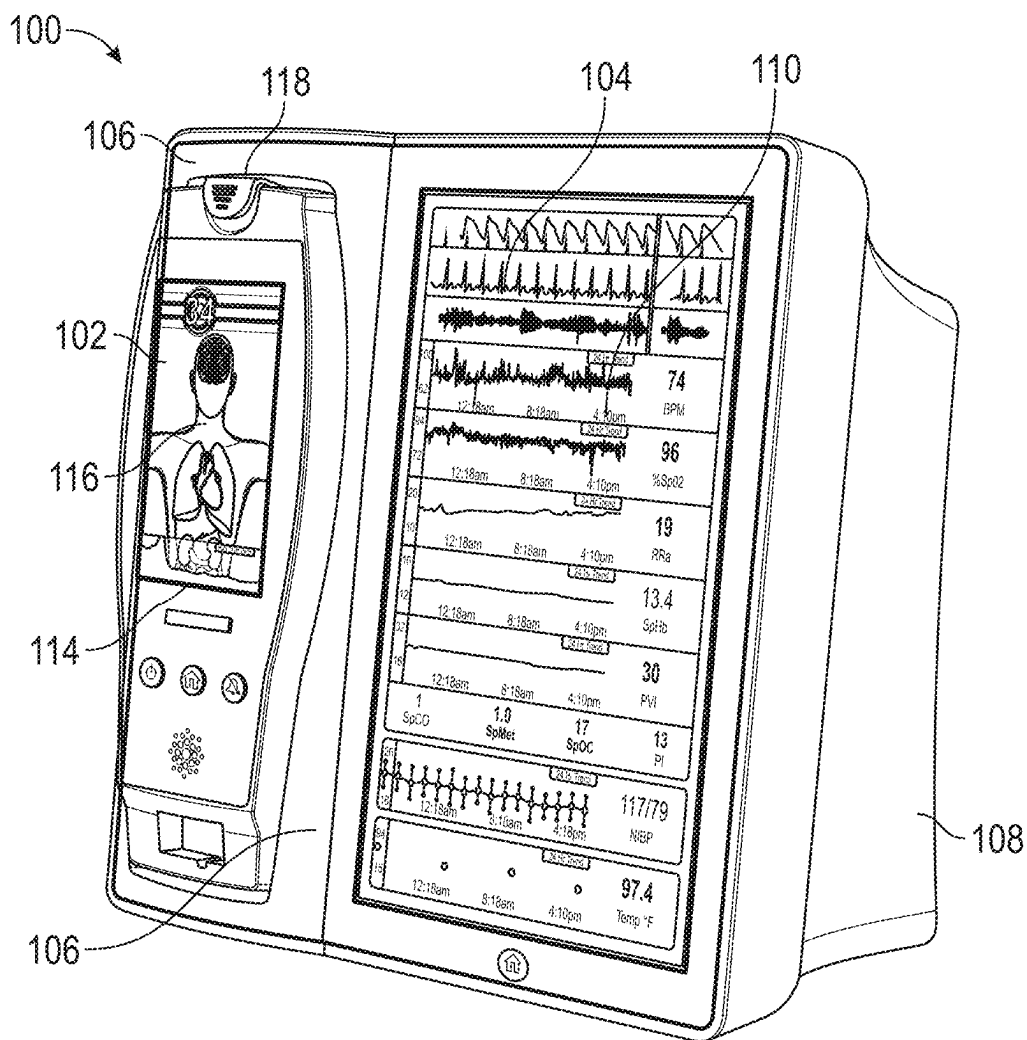


FIG. 1A

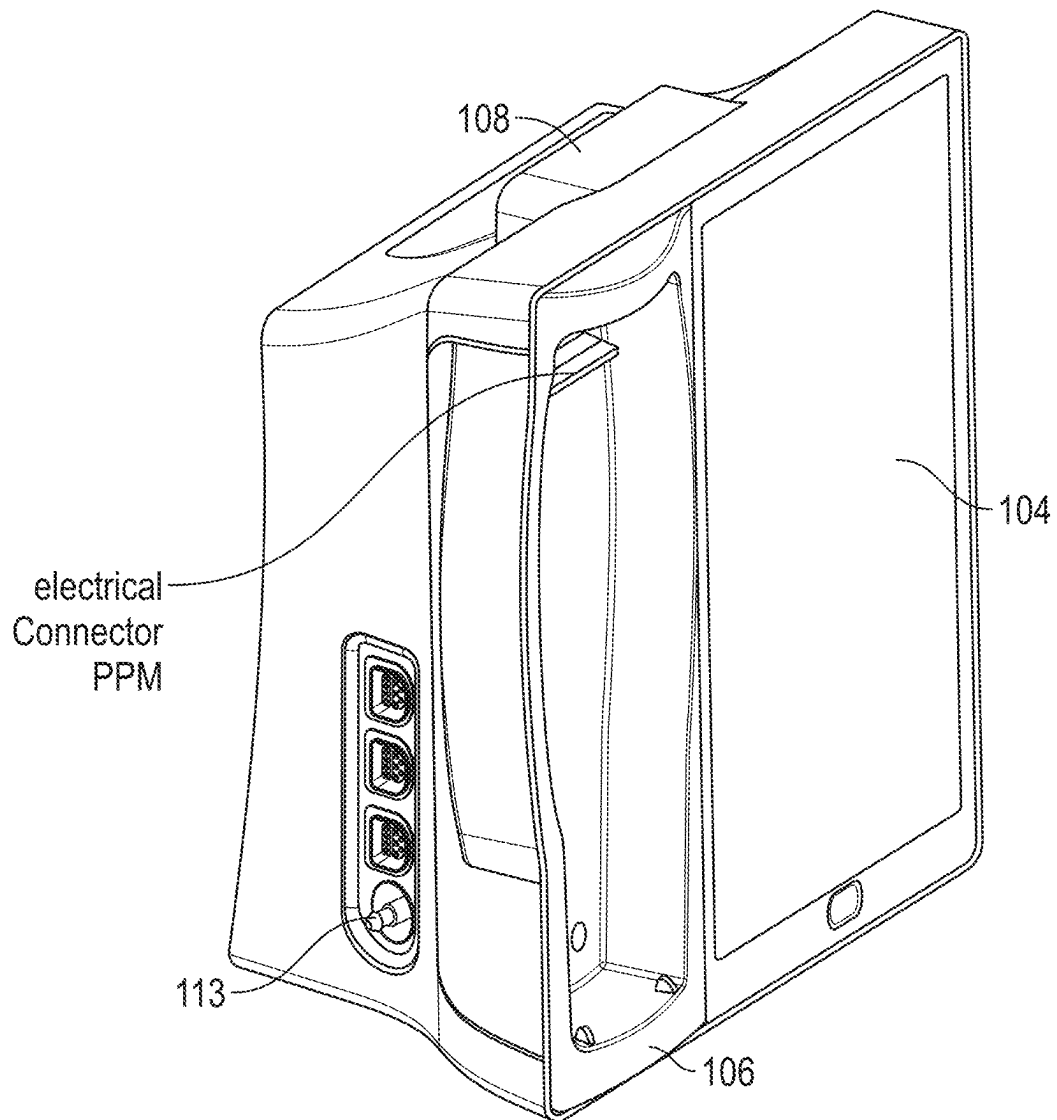


FIG. 1B

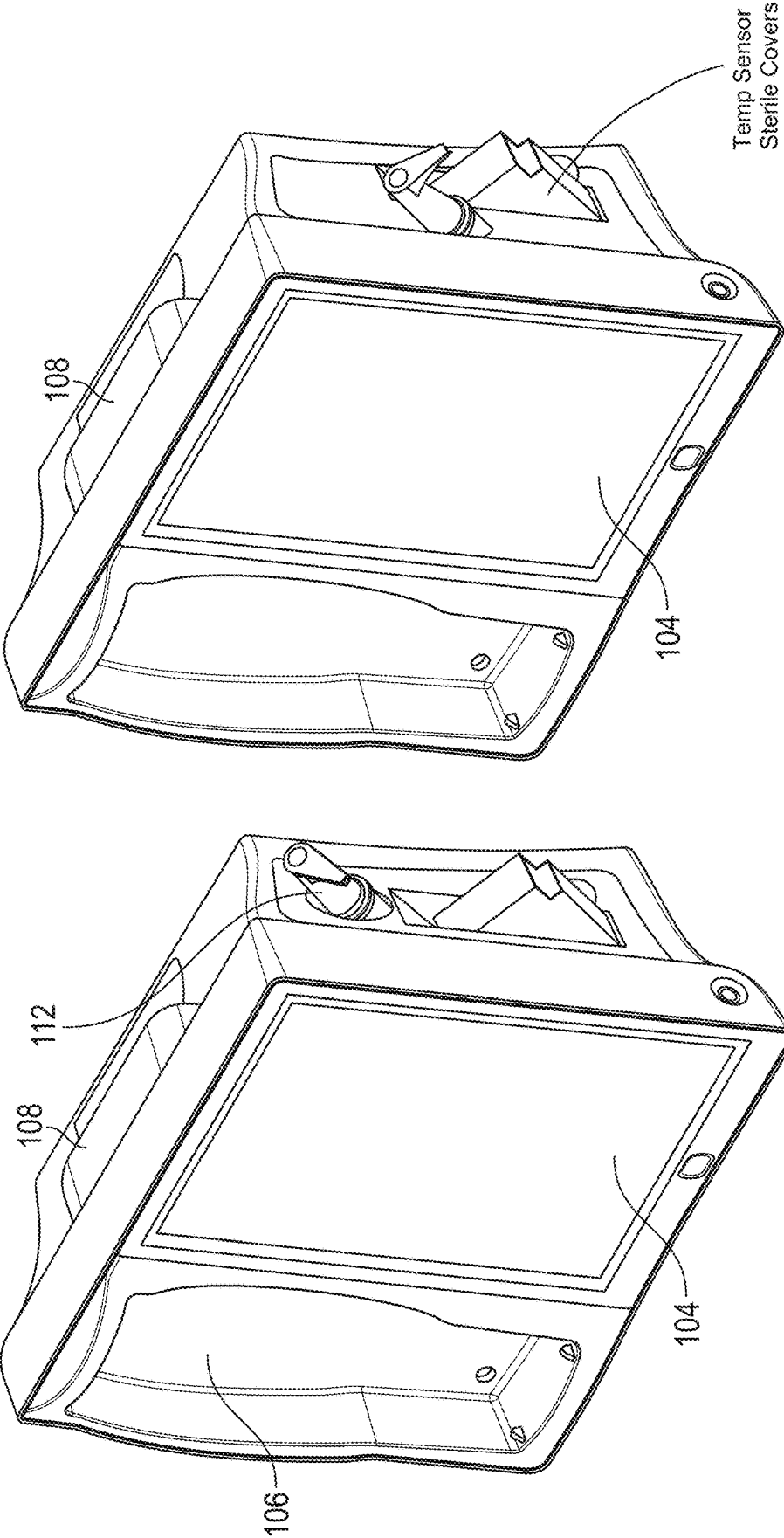


FIG. 1C

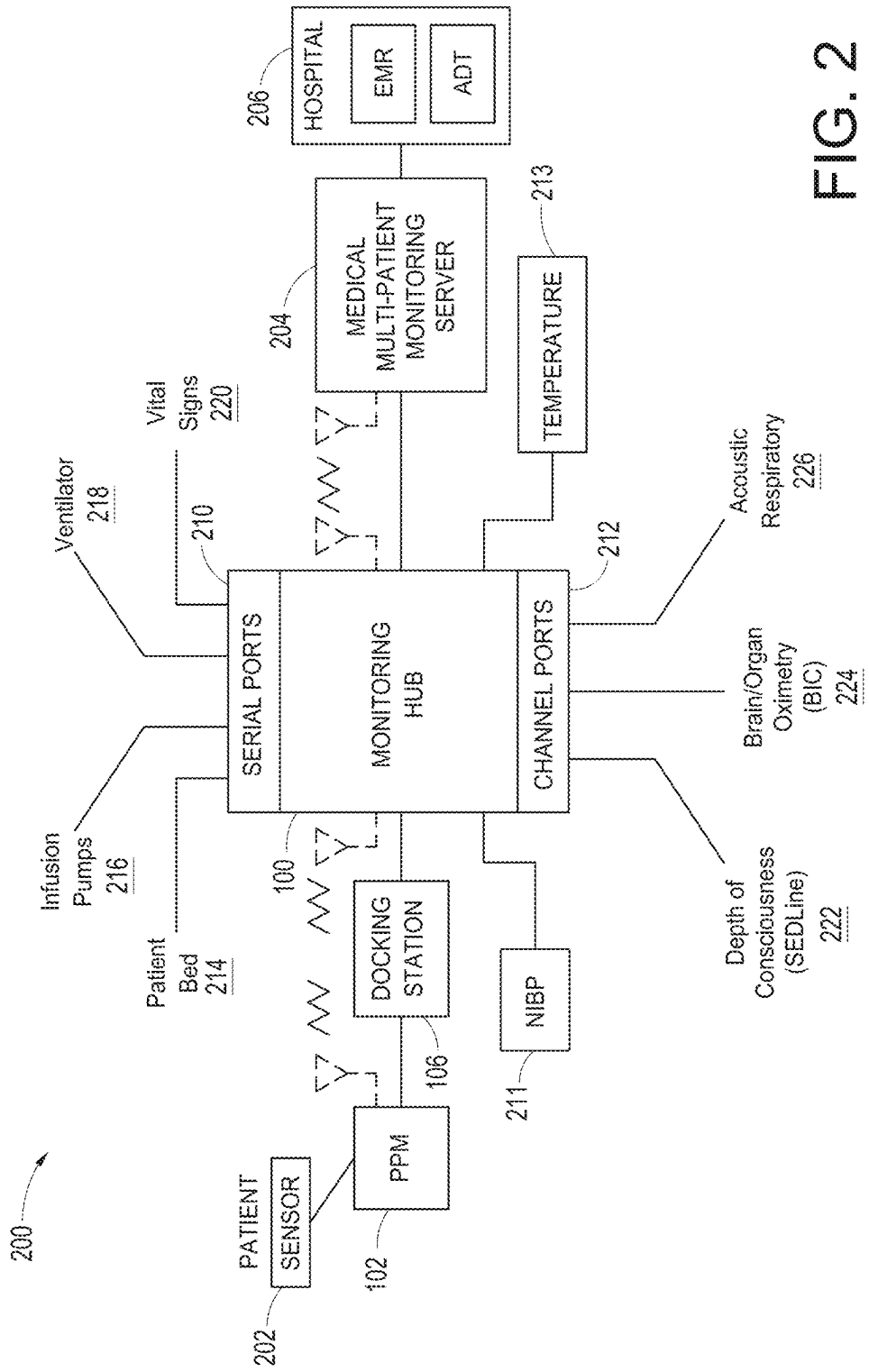


FIG. 2

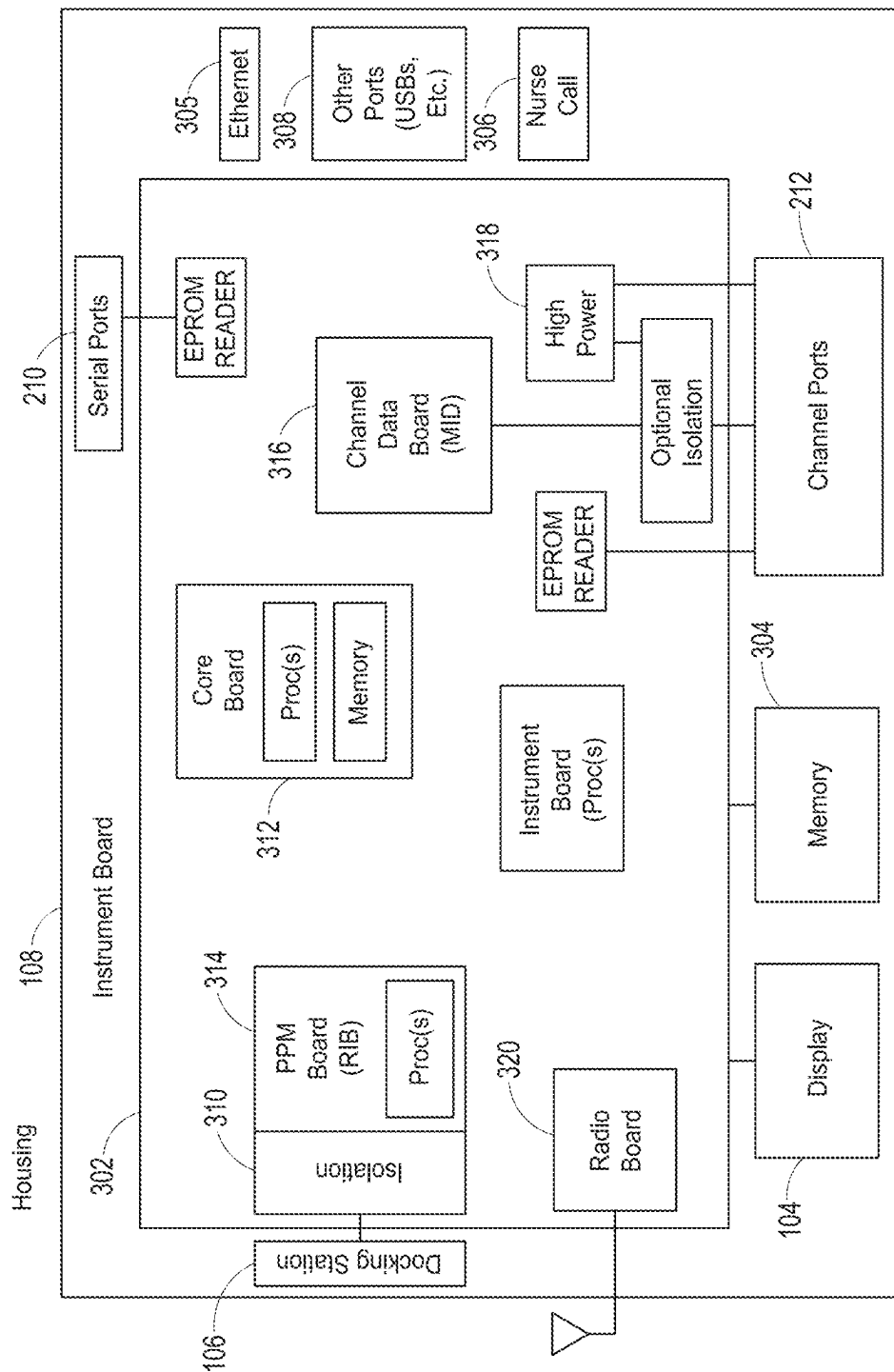


FIG. 3

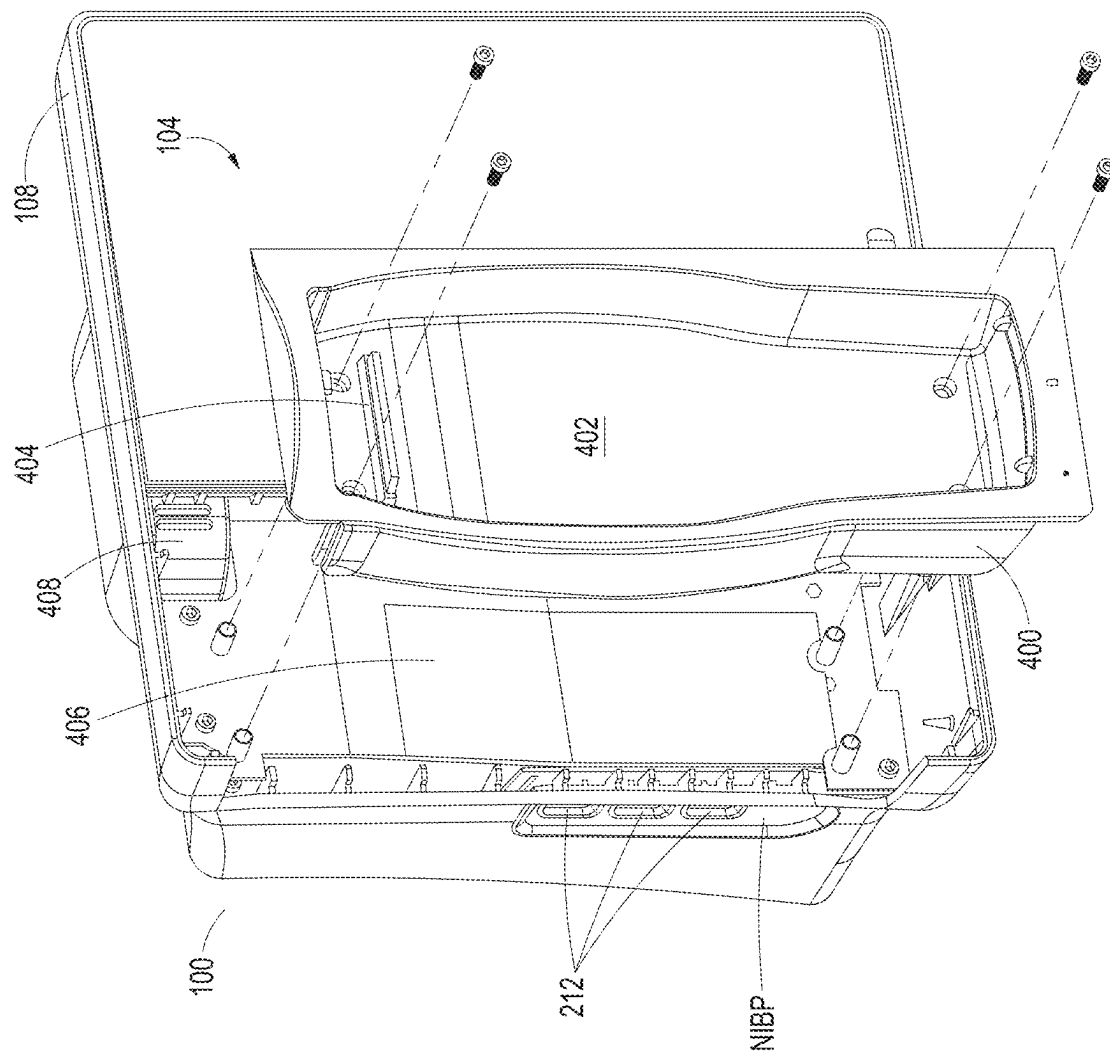


FIG. 4

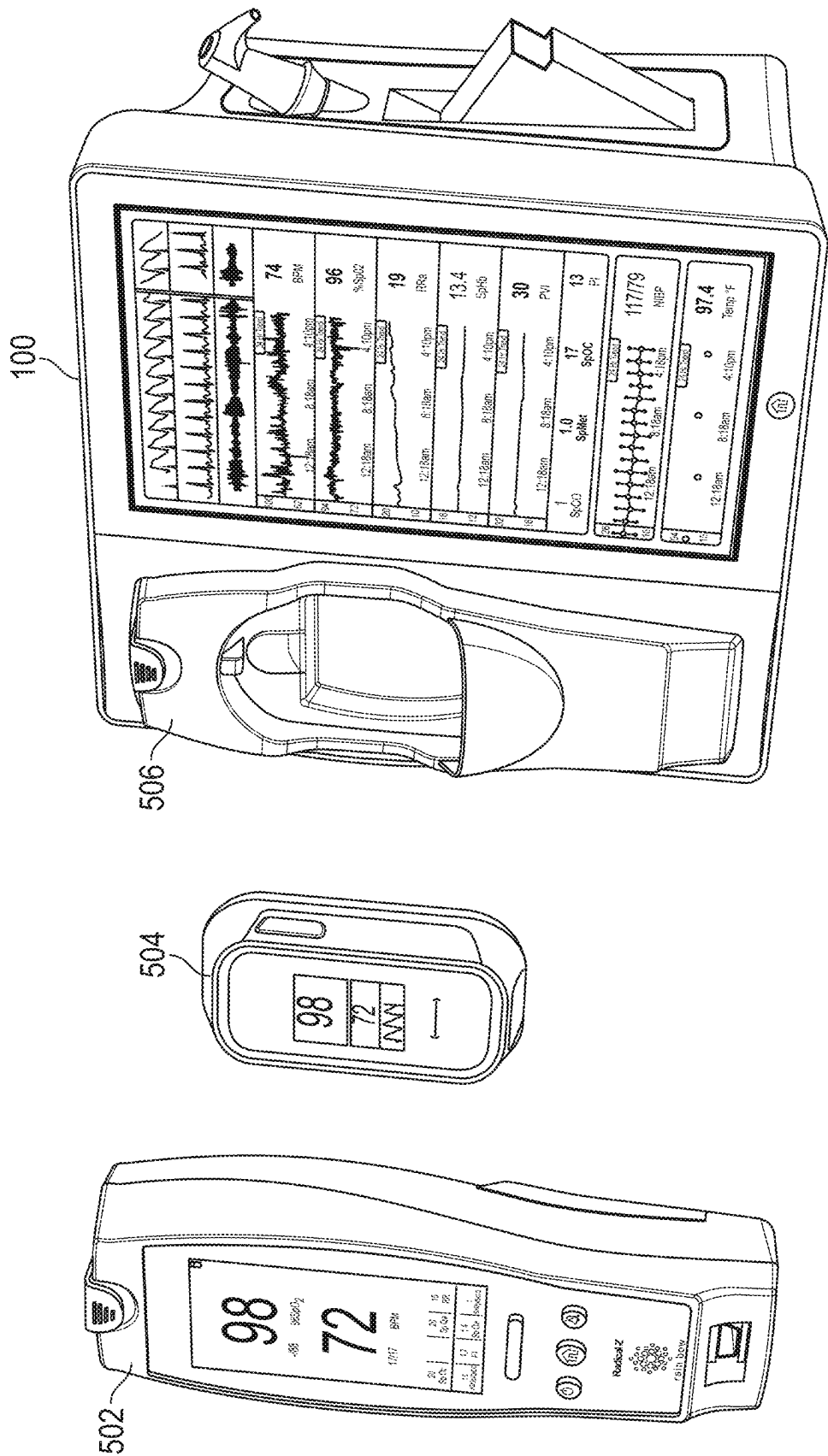


FIG. 5

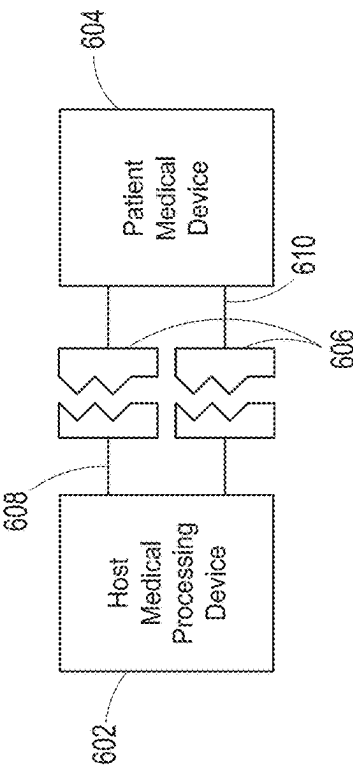


FIG. 6

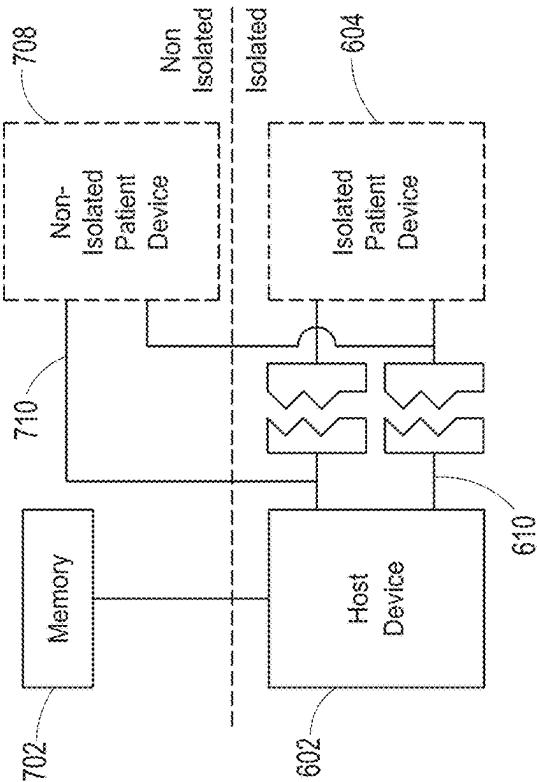


FIG. 7A

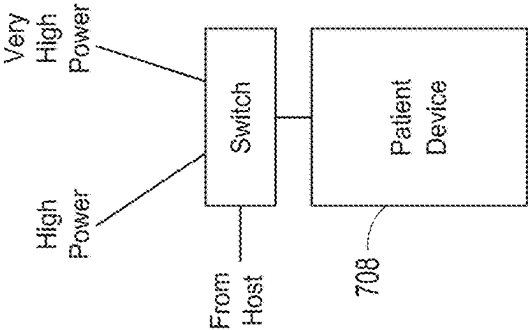


FIG. 7B

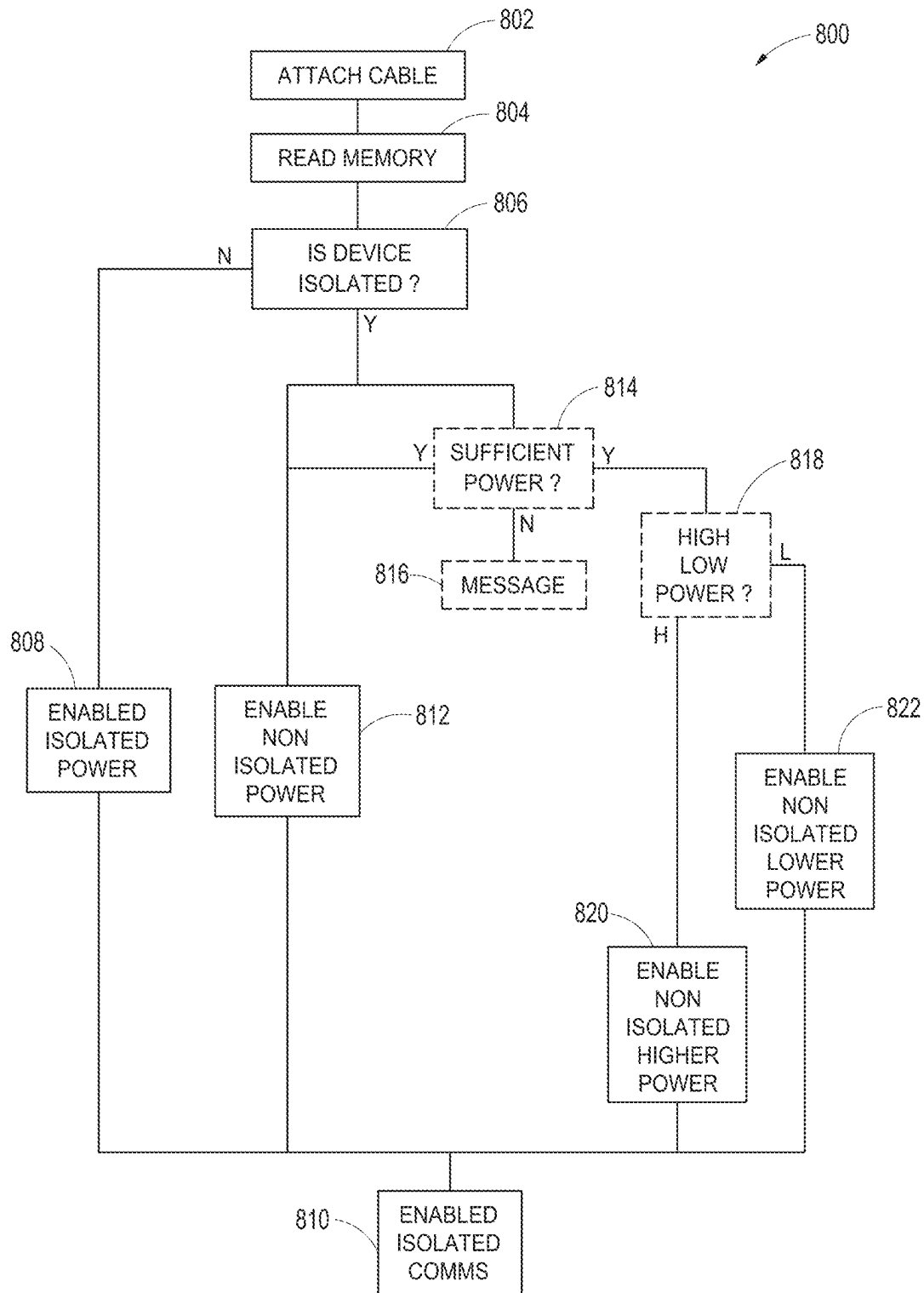


FIG. 8

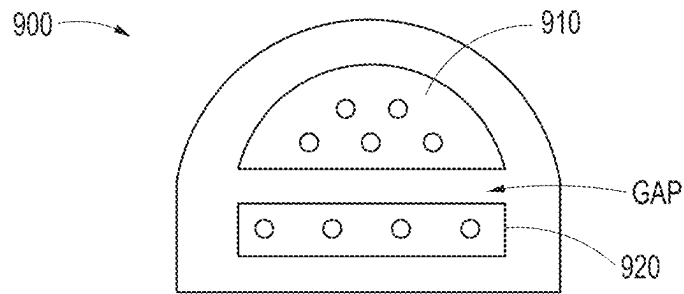


FIG. 9A

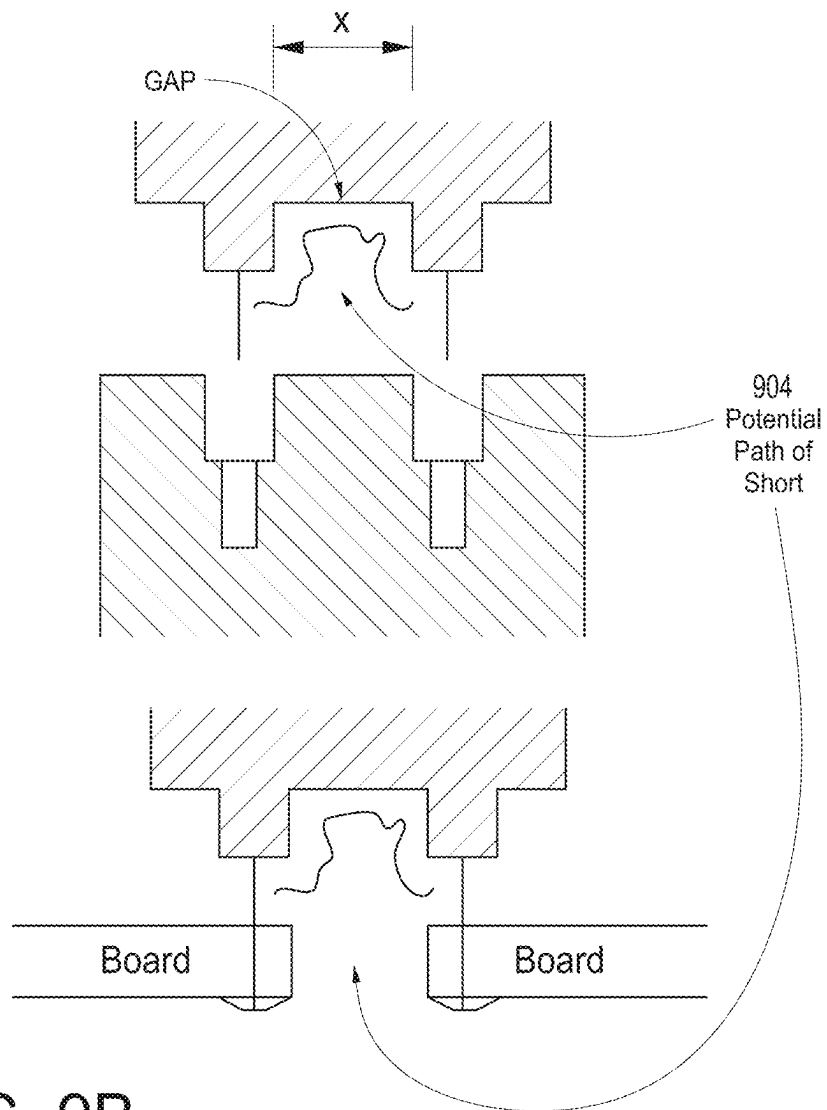


FIG. 9B

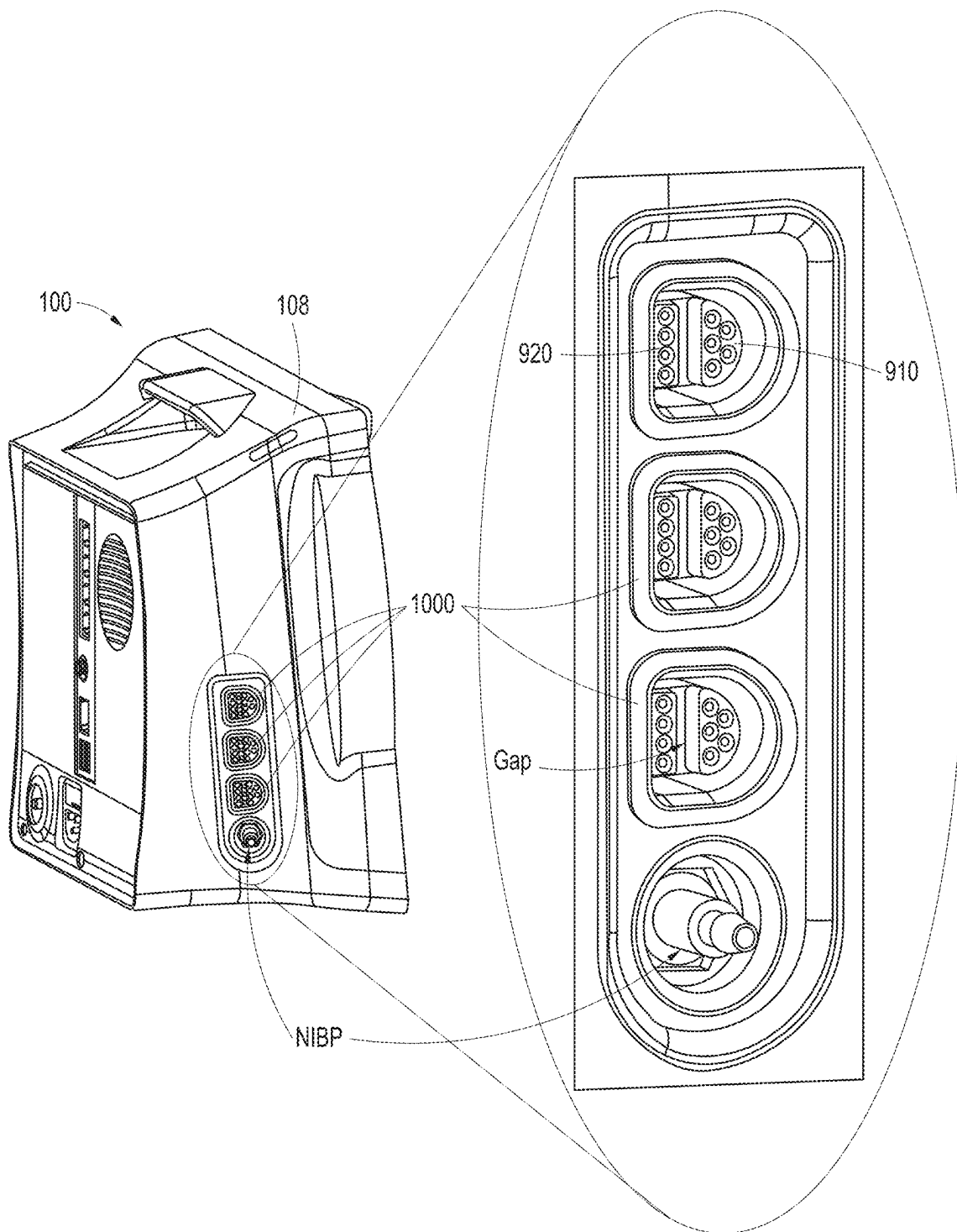


FIG. 10

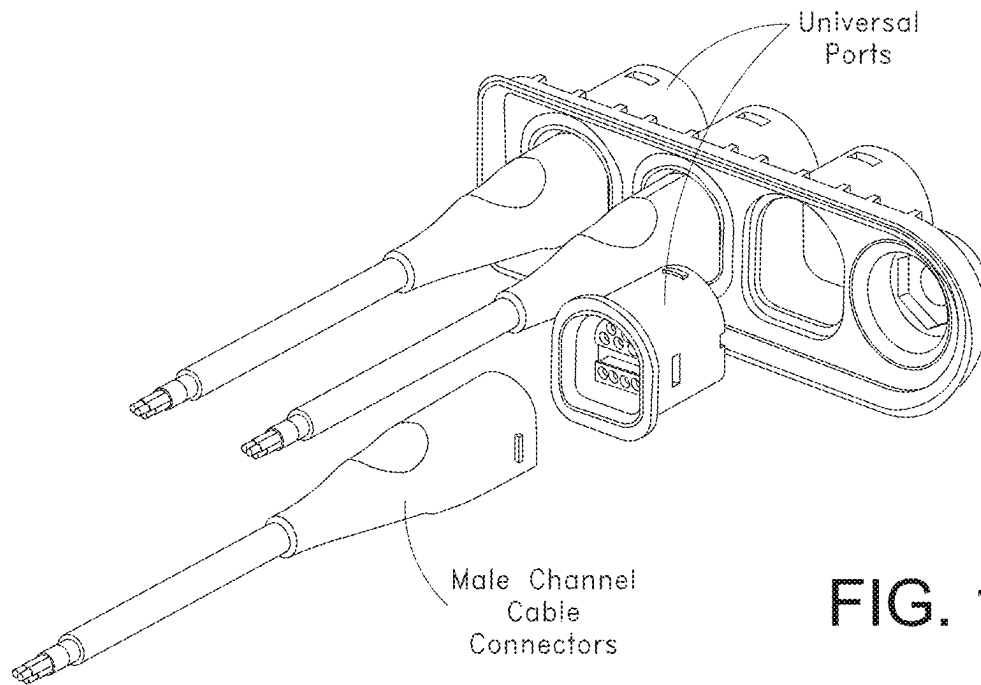


FIG. 11A

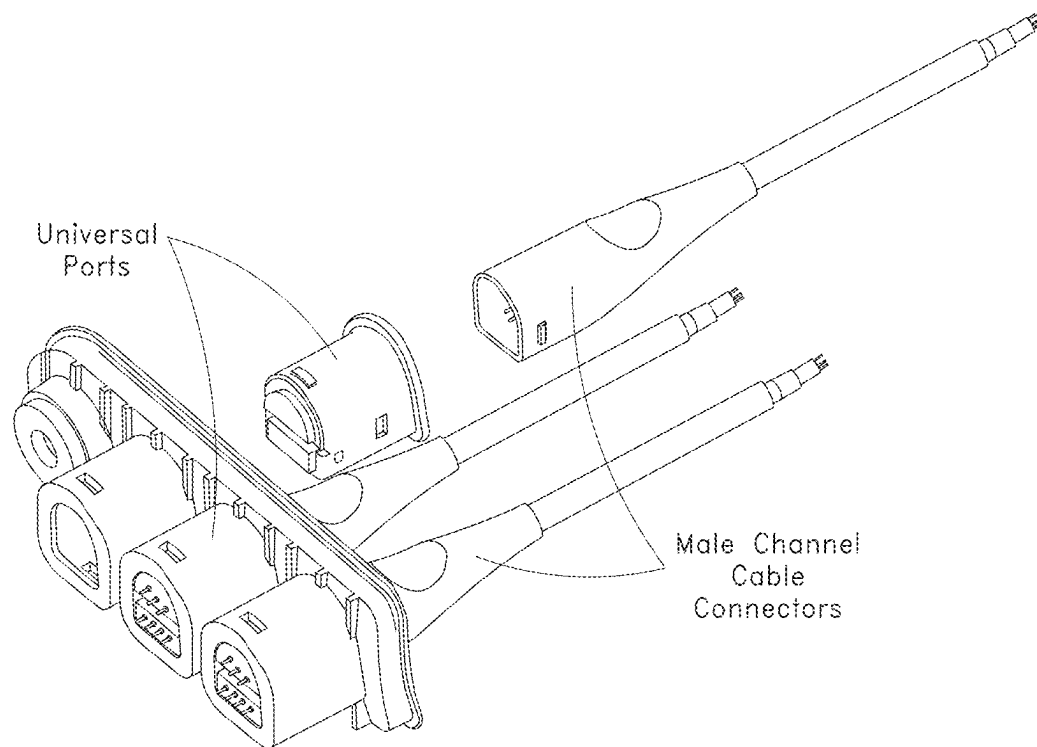


FIG. 11B

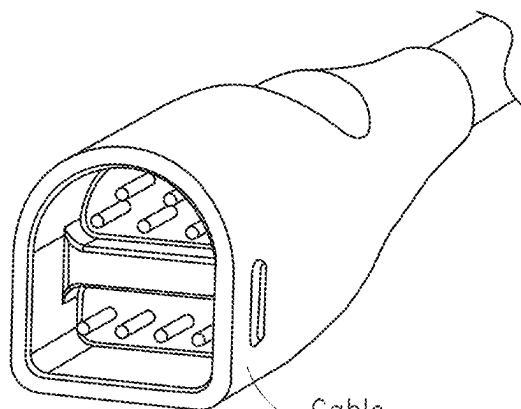


FIG. 11C

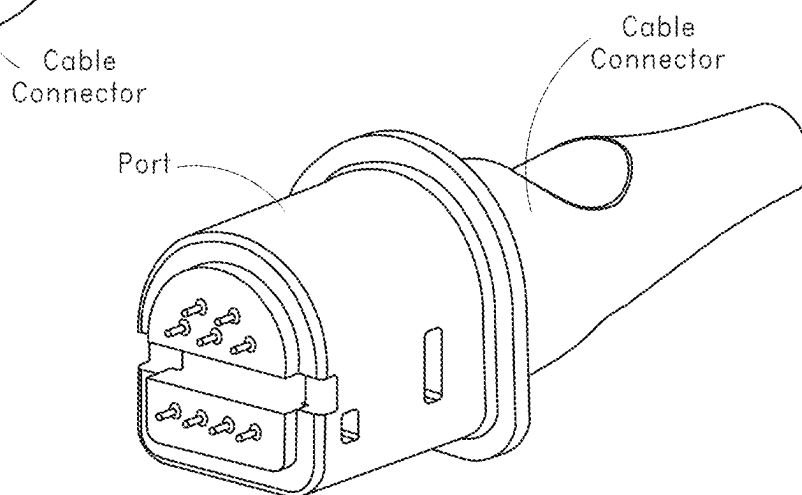


FIG. 11D

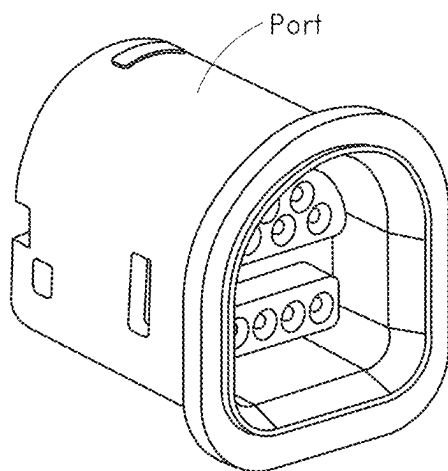


FIG. 11E

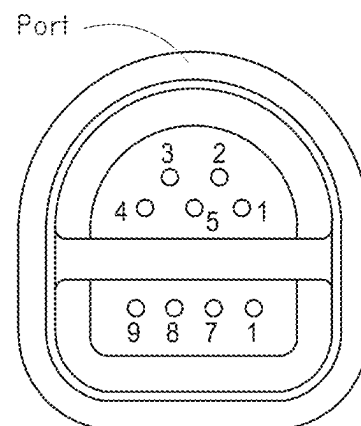
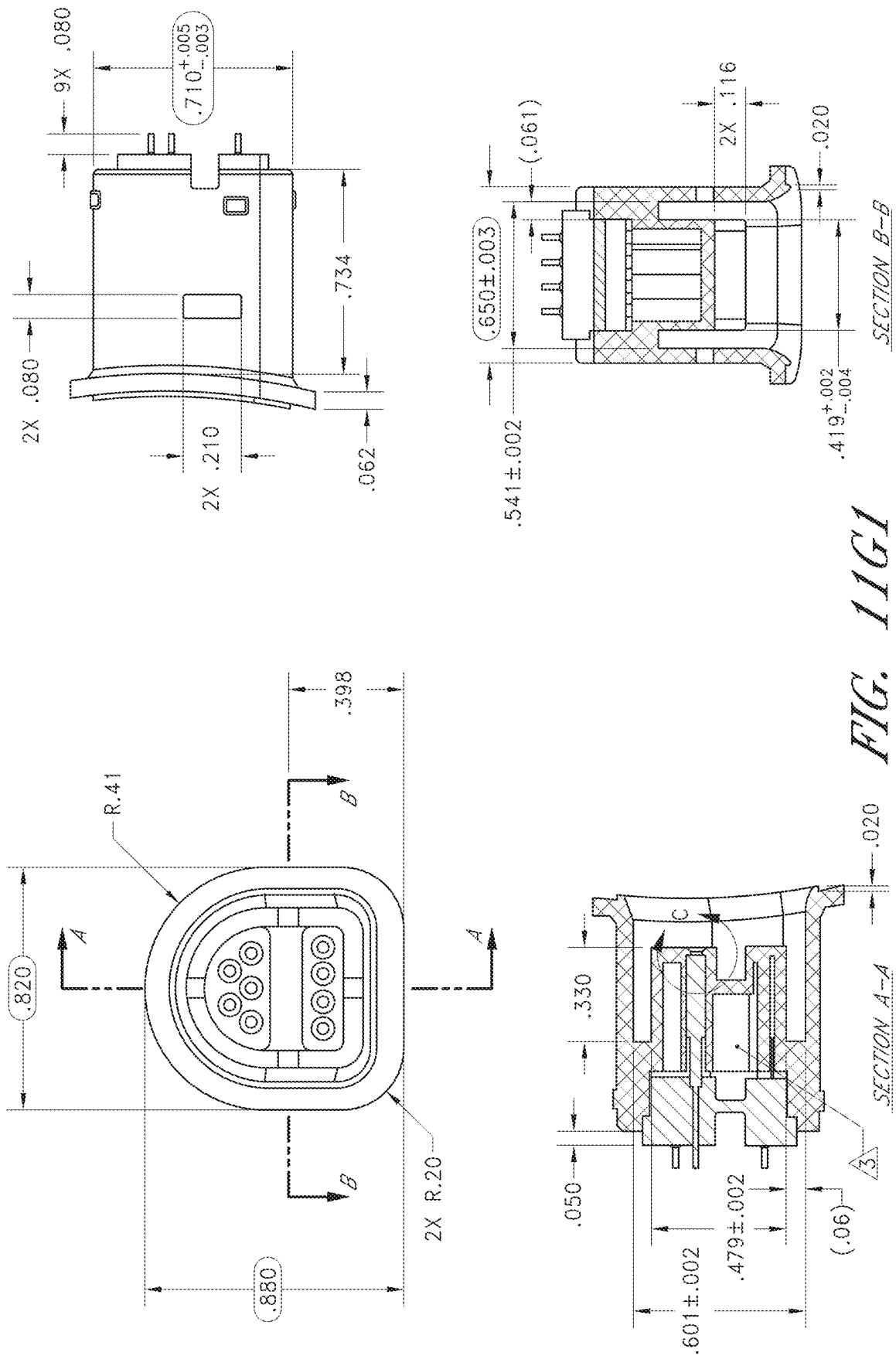
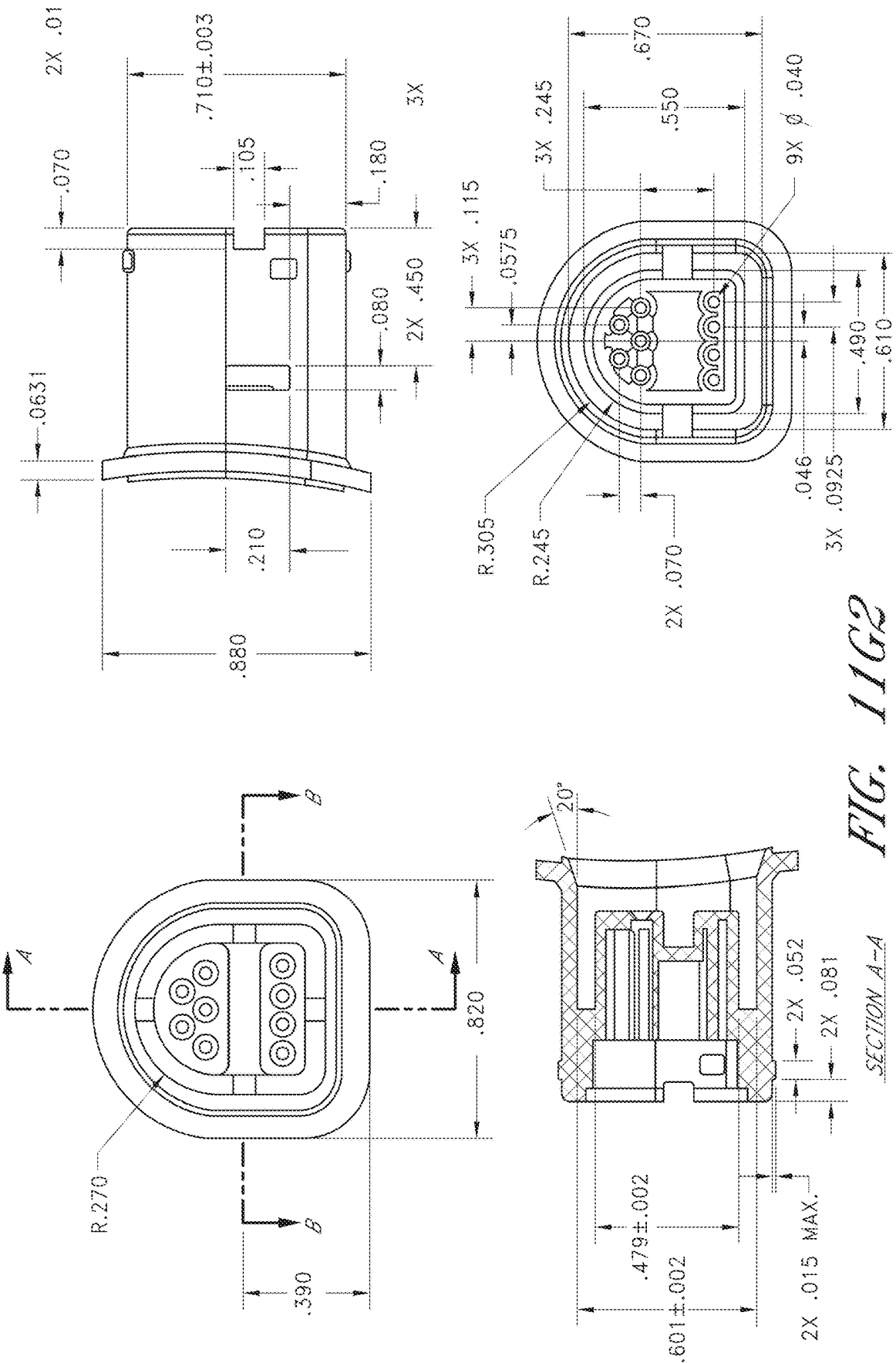


FIG. 11F





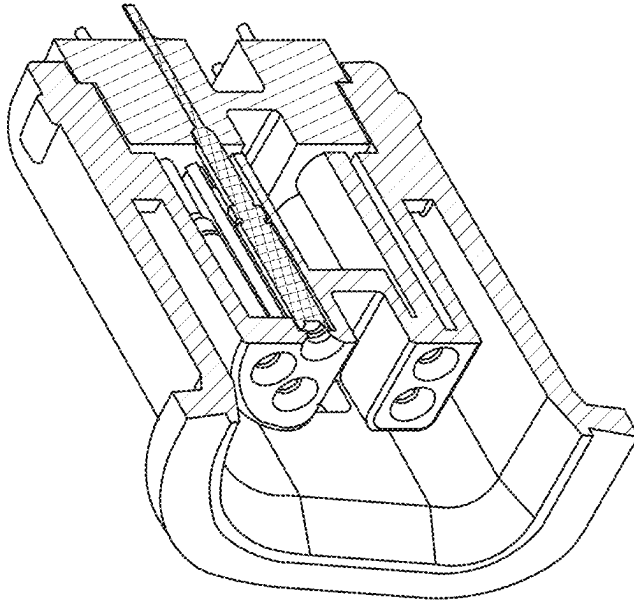
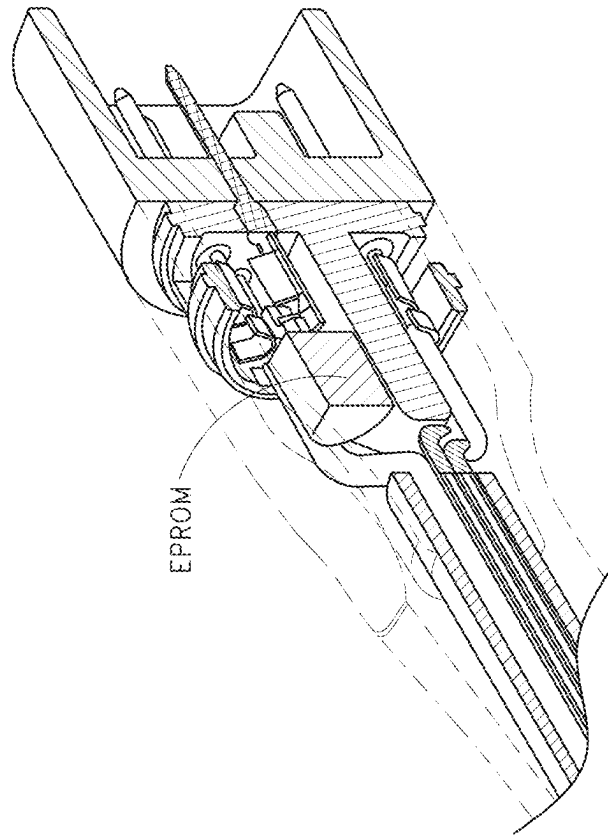


FIG. 11H



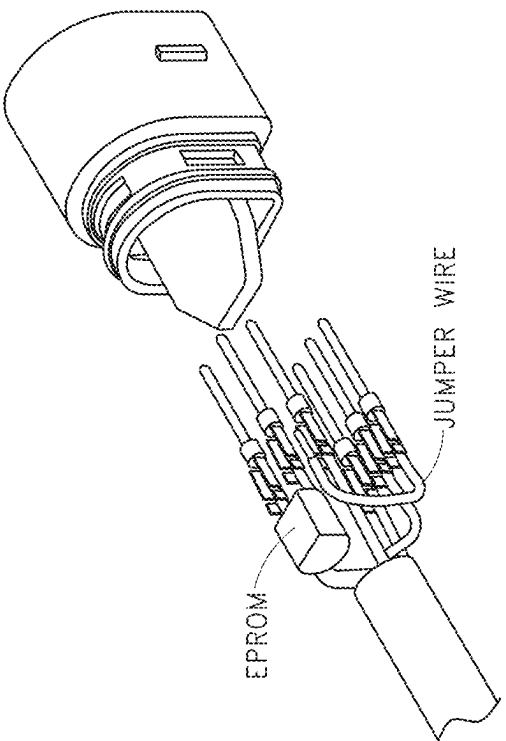


FIG. 11J

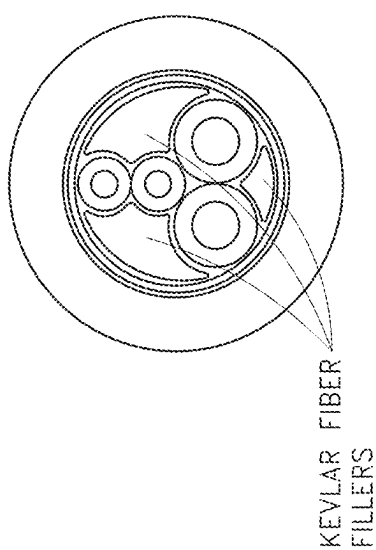


FIG. 11I

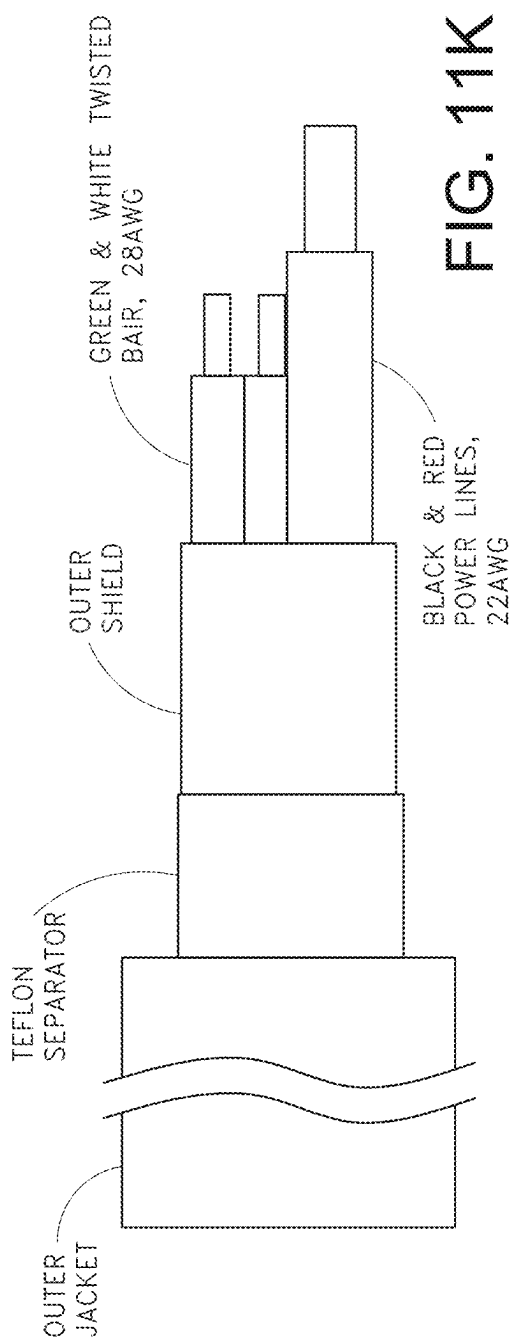


FIG. 11K

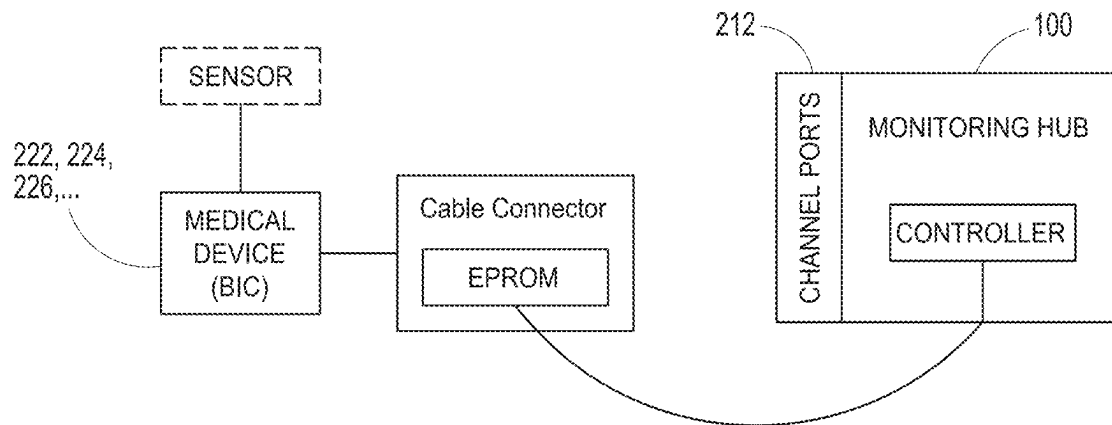


FIG. 12

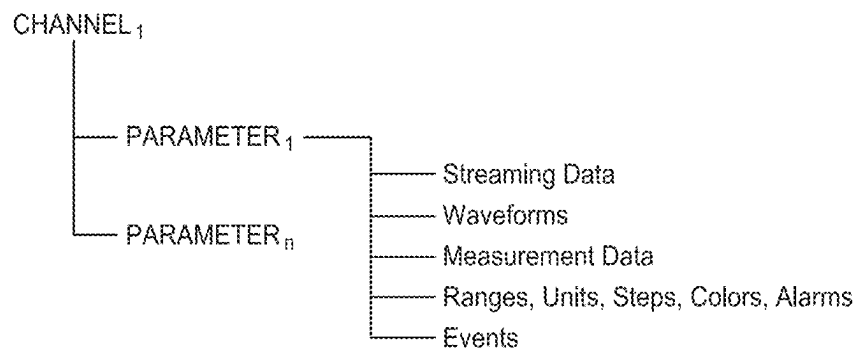


FIG. 13

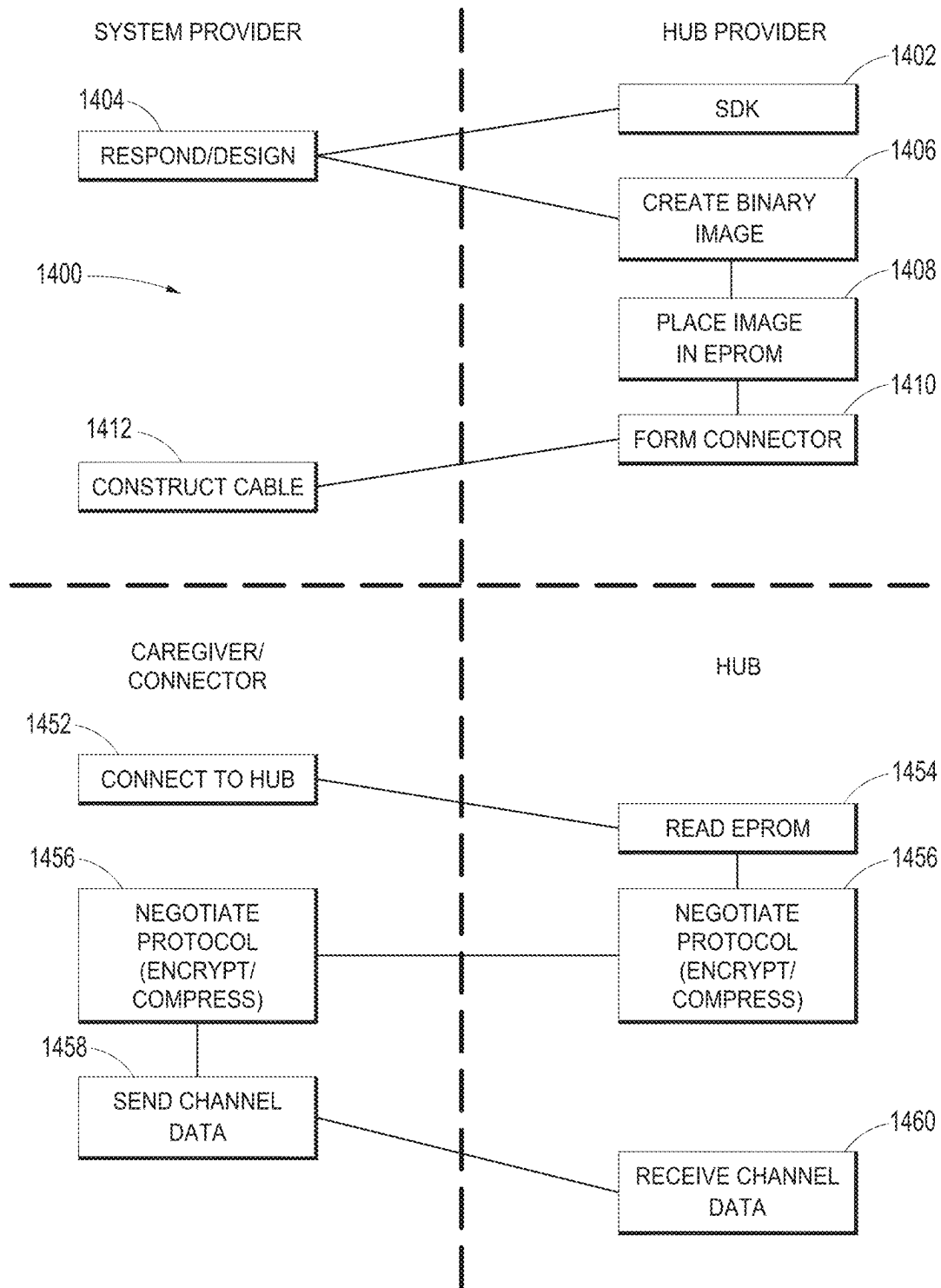
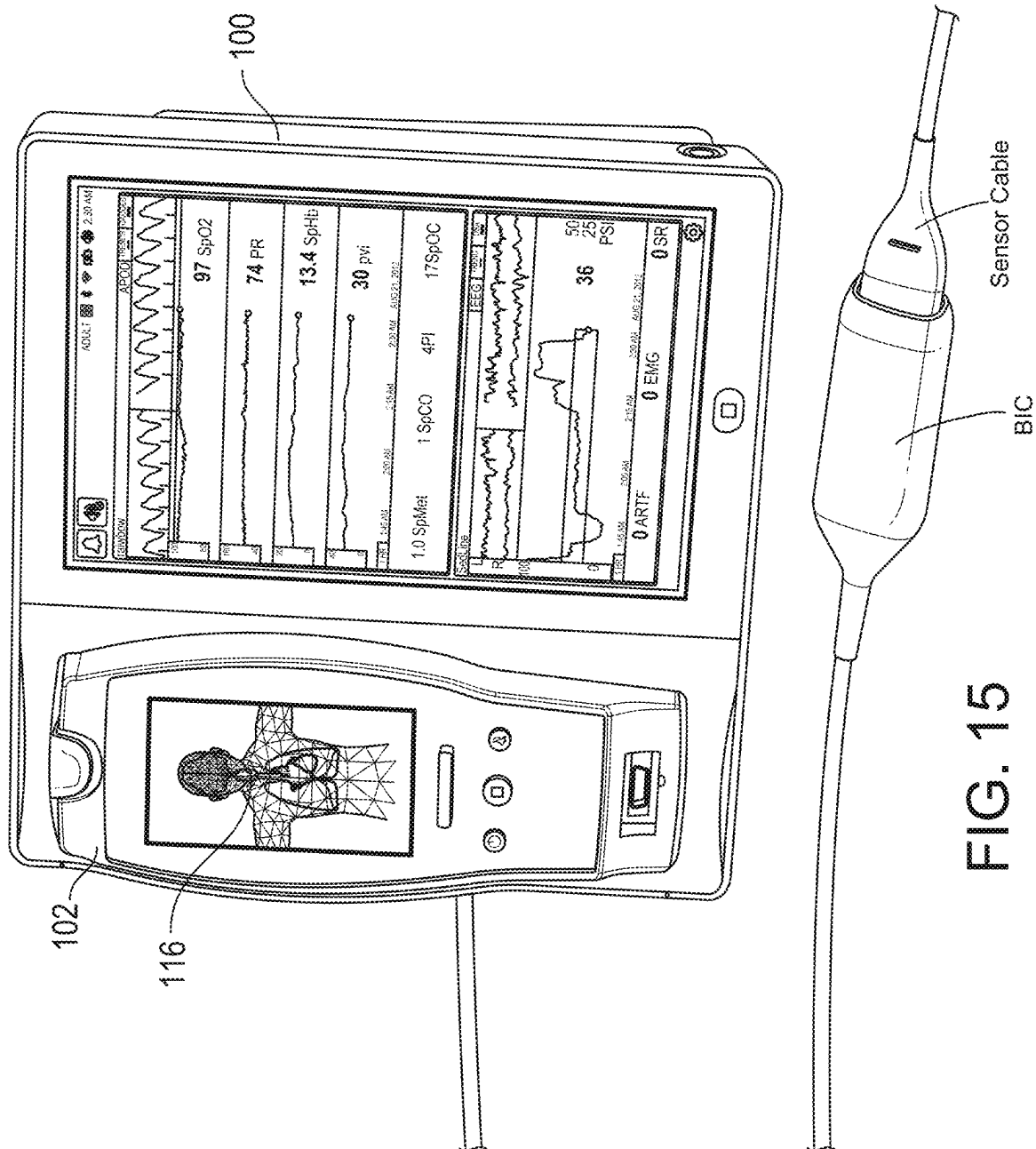


FIG. 14



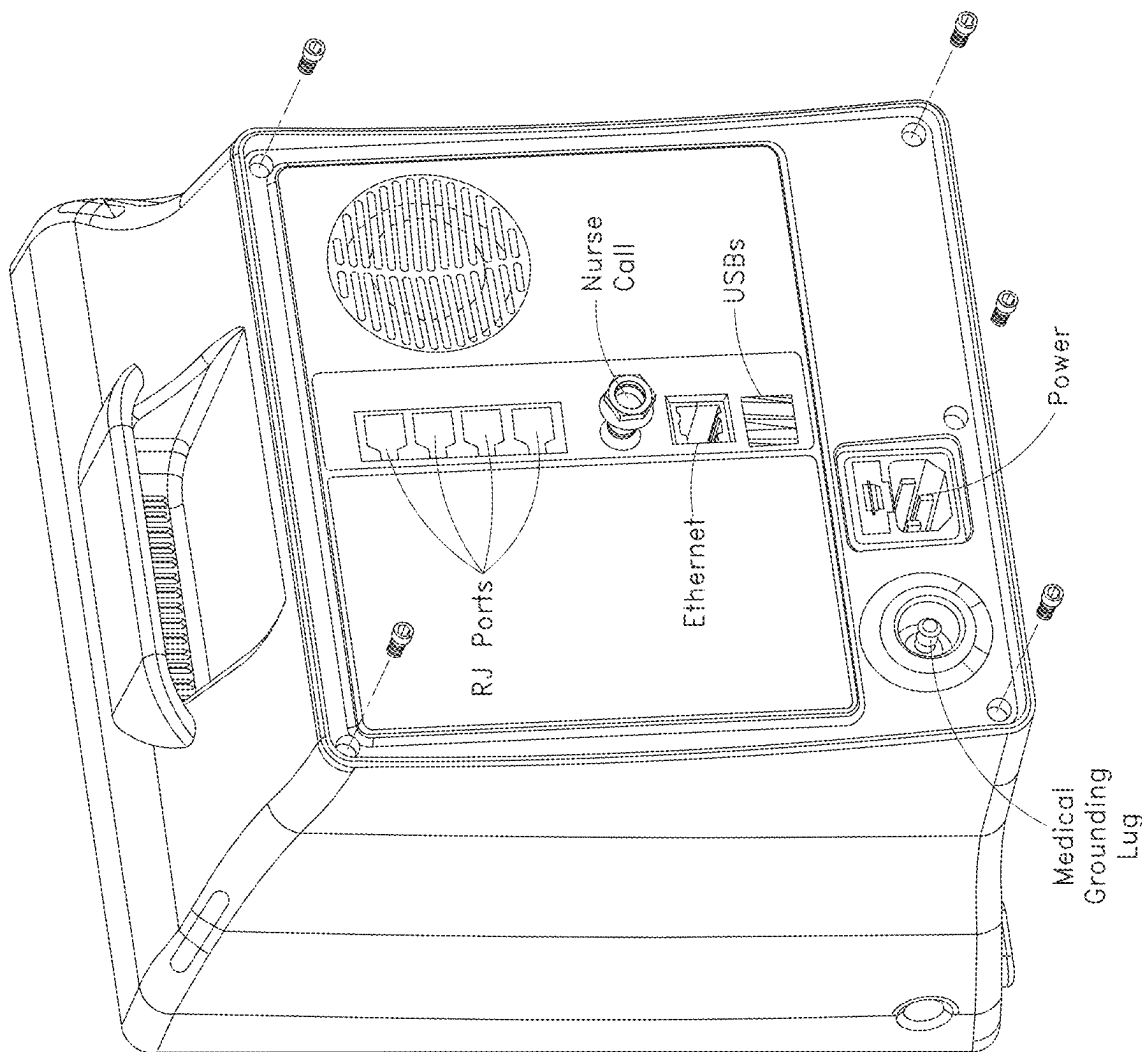
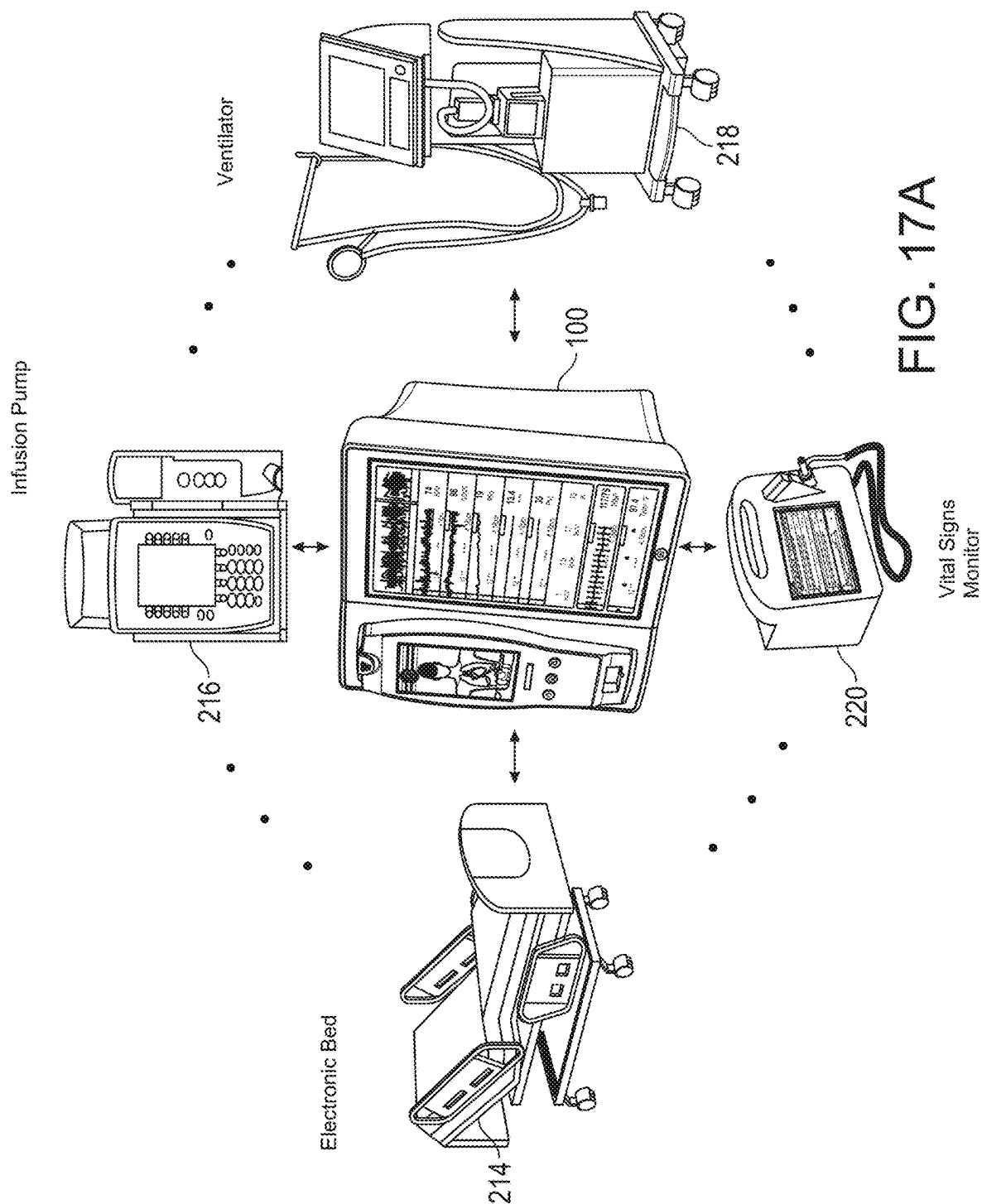


FIG. 16



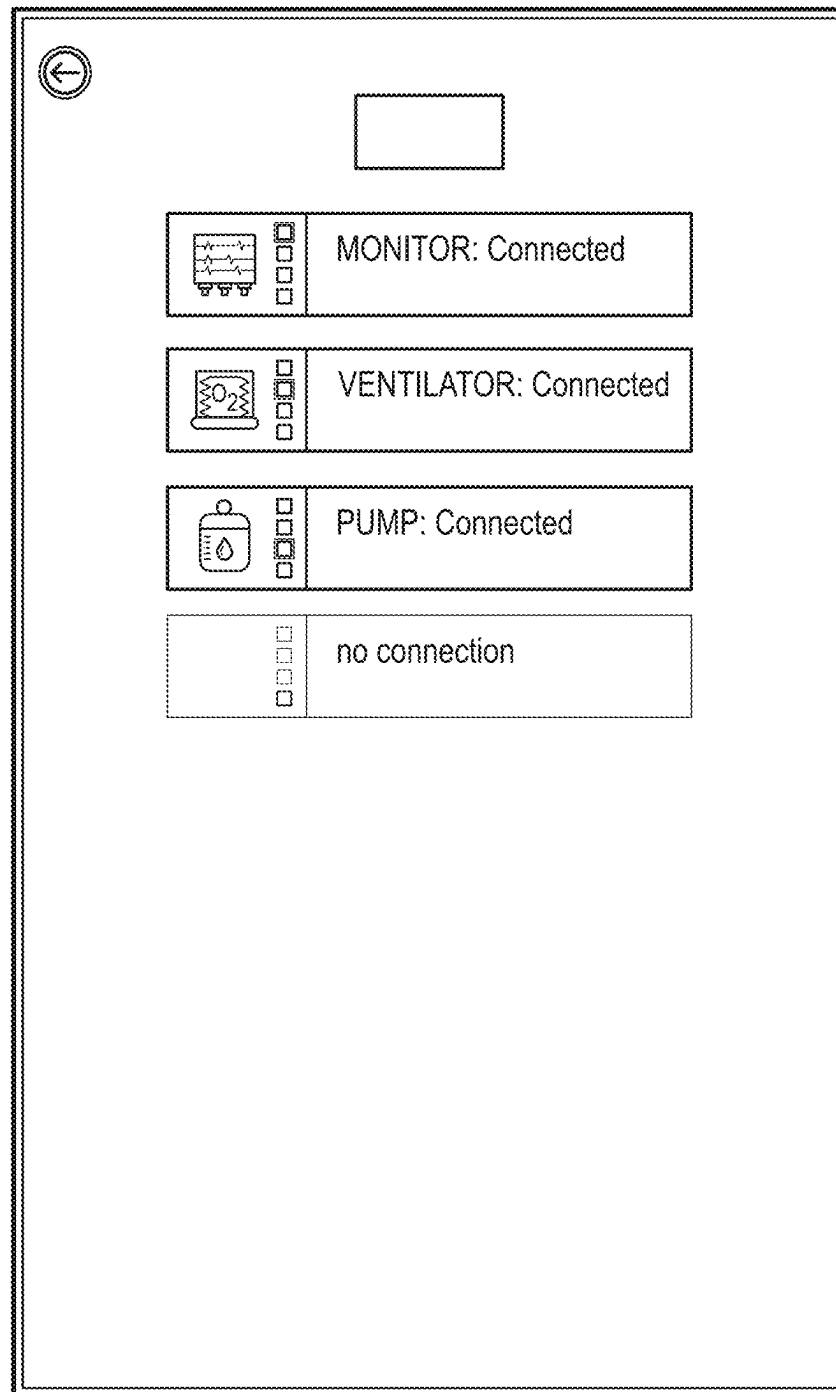


FIG. 17B

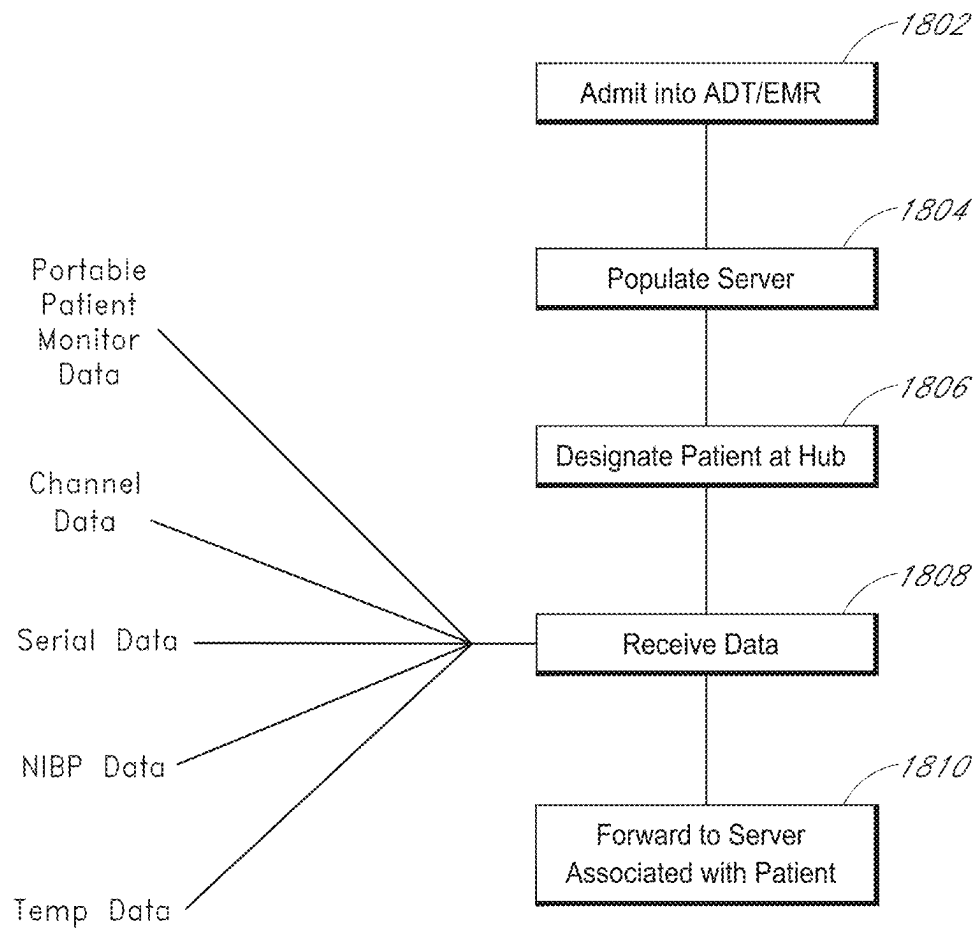


FIG. 18

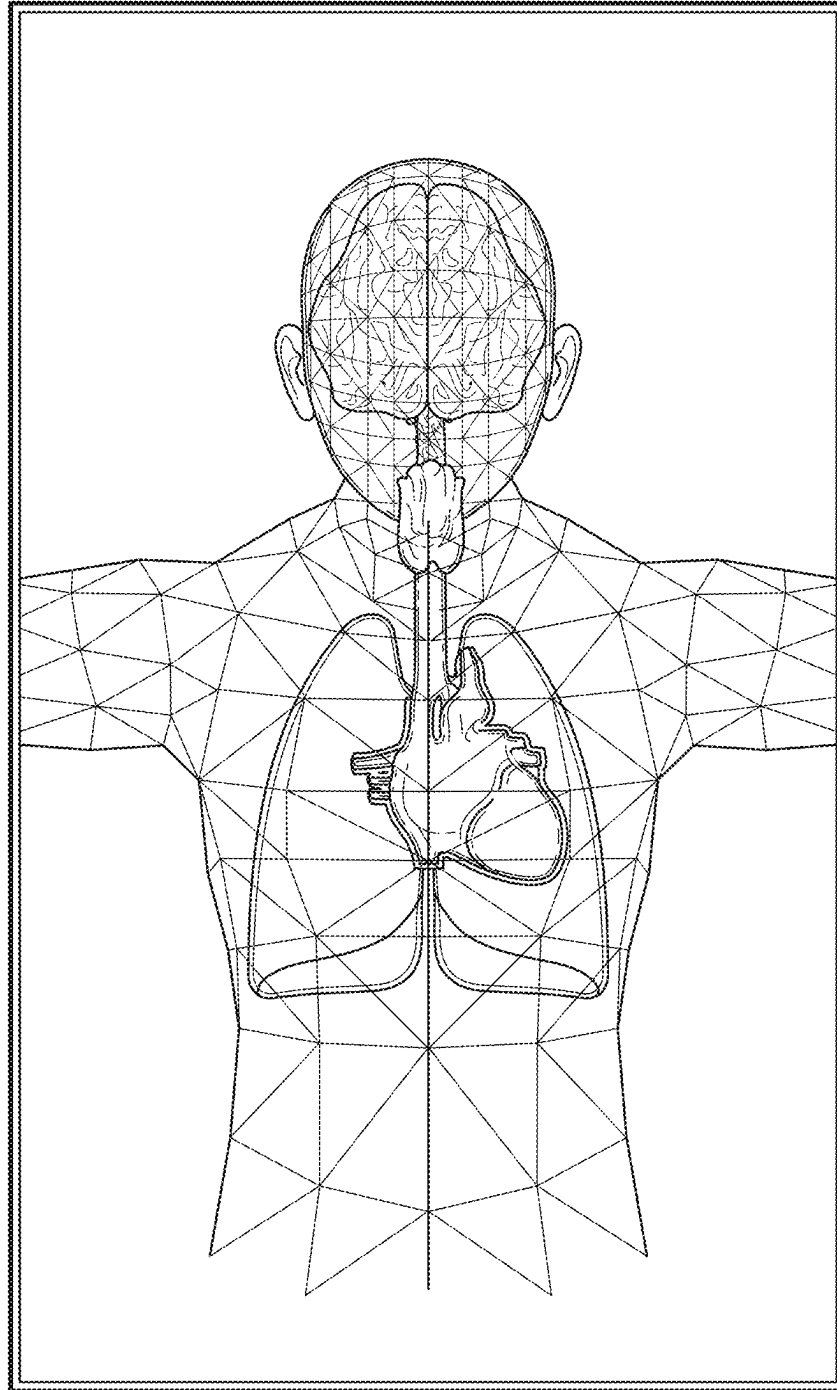


FIG. 19A

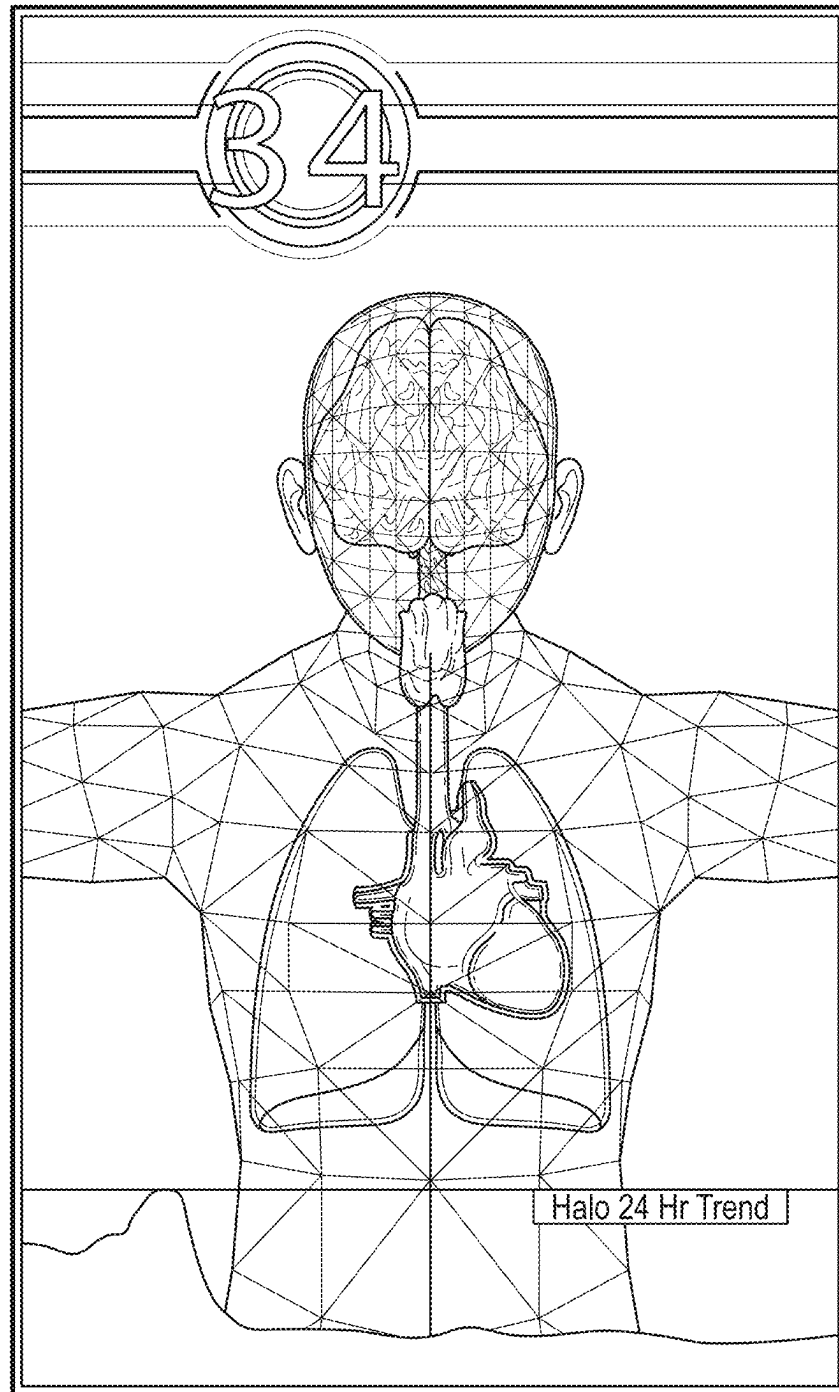


FIG. 19B

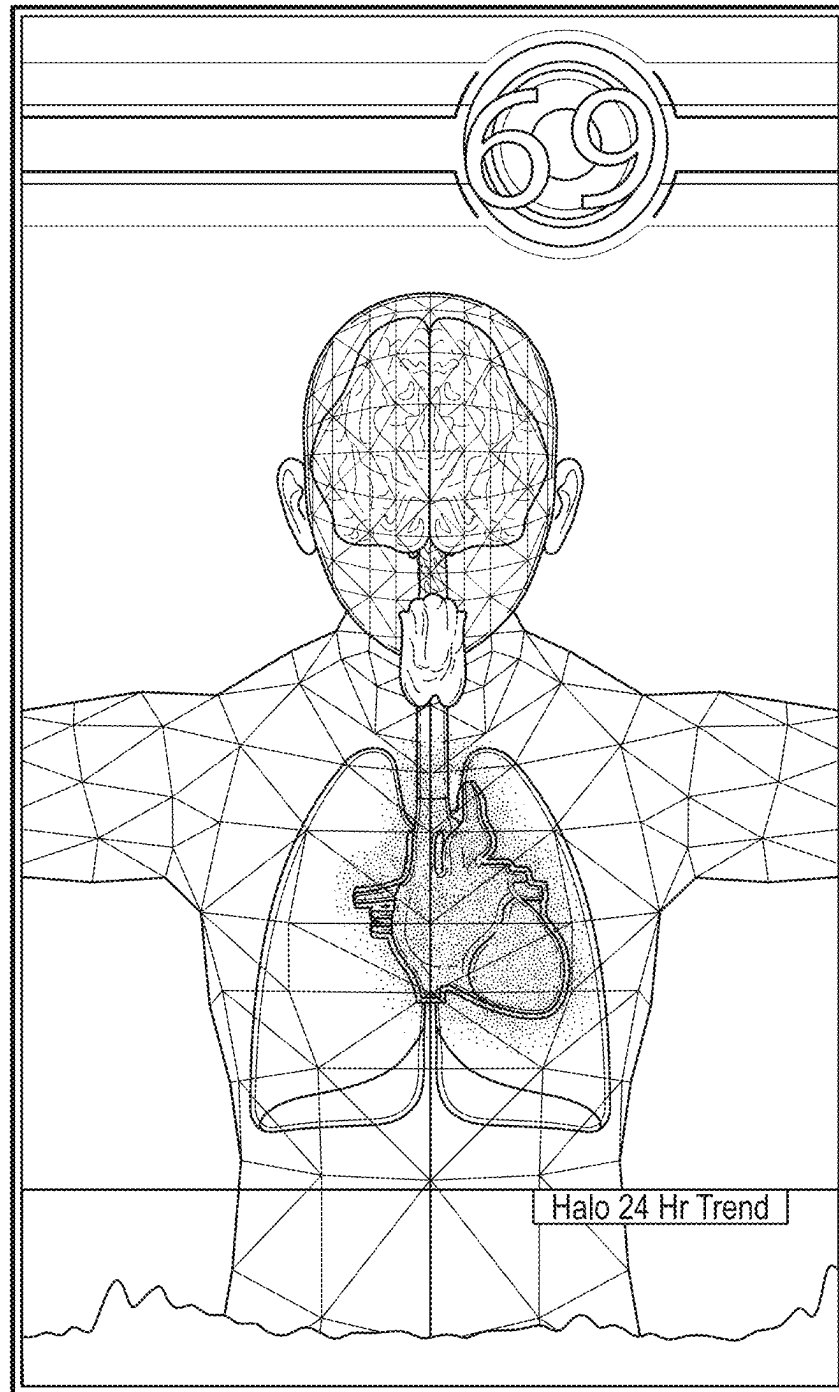


FIG. 19C

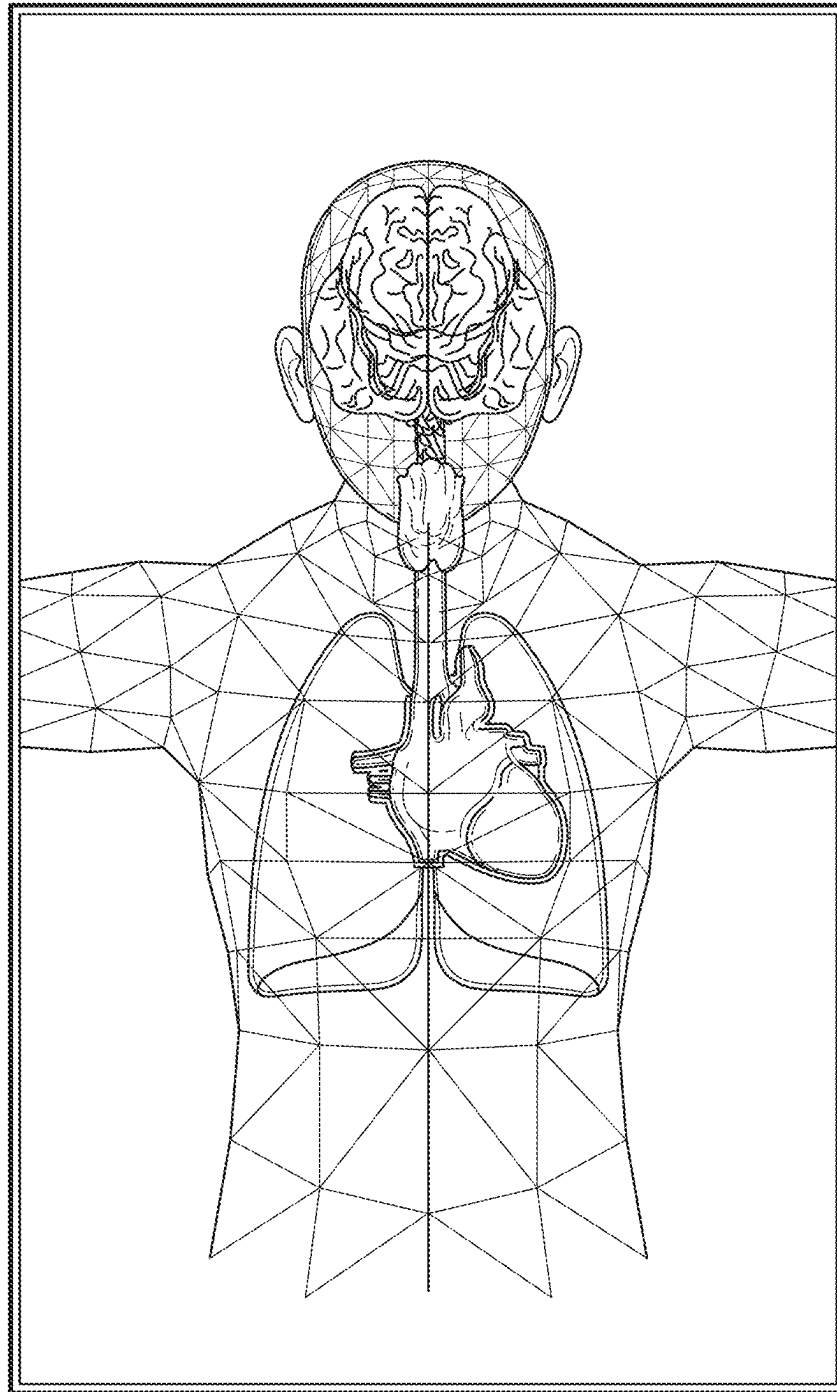


FIG. 19D

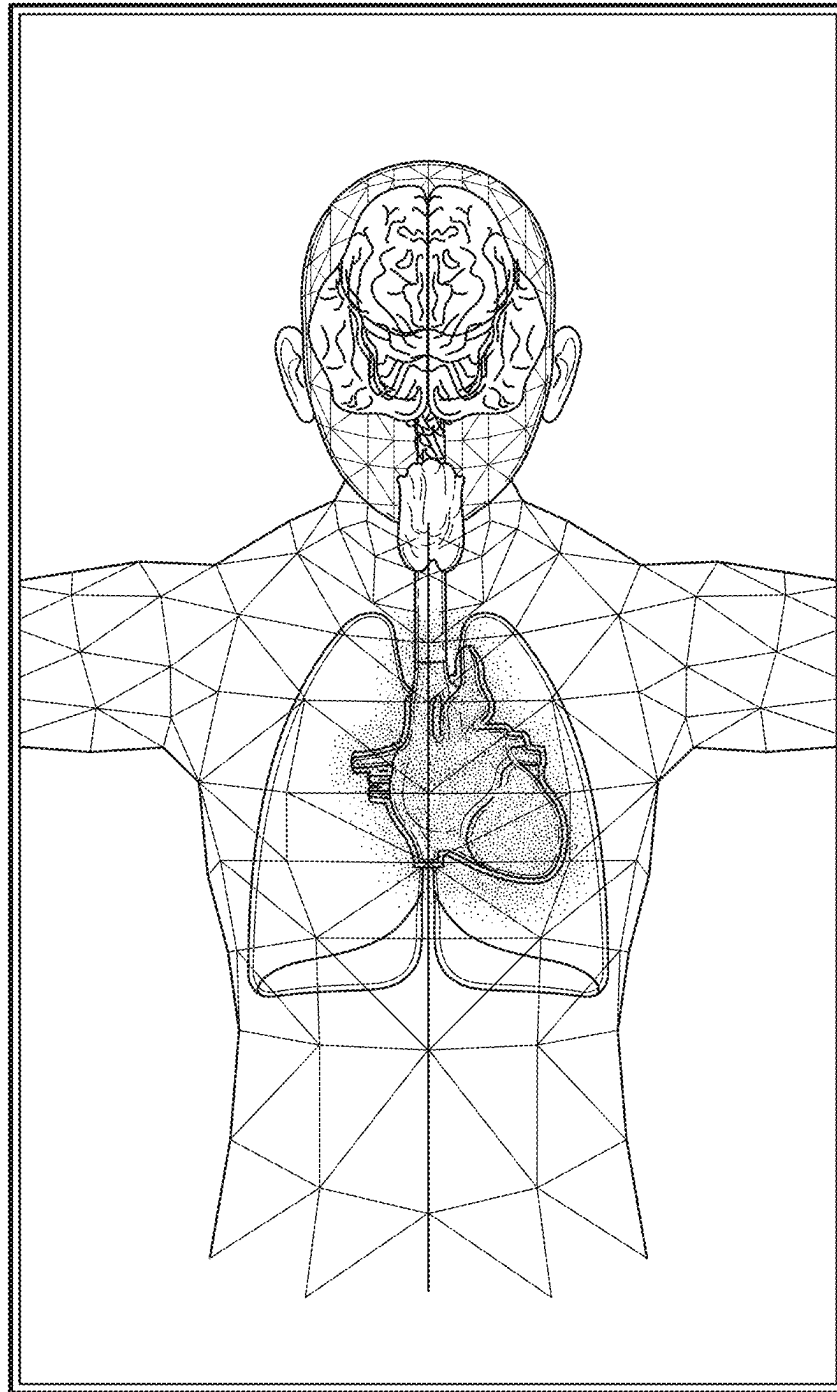


FIG. 19E

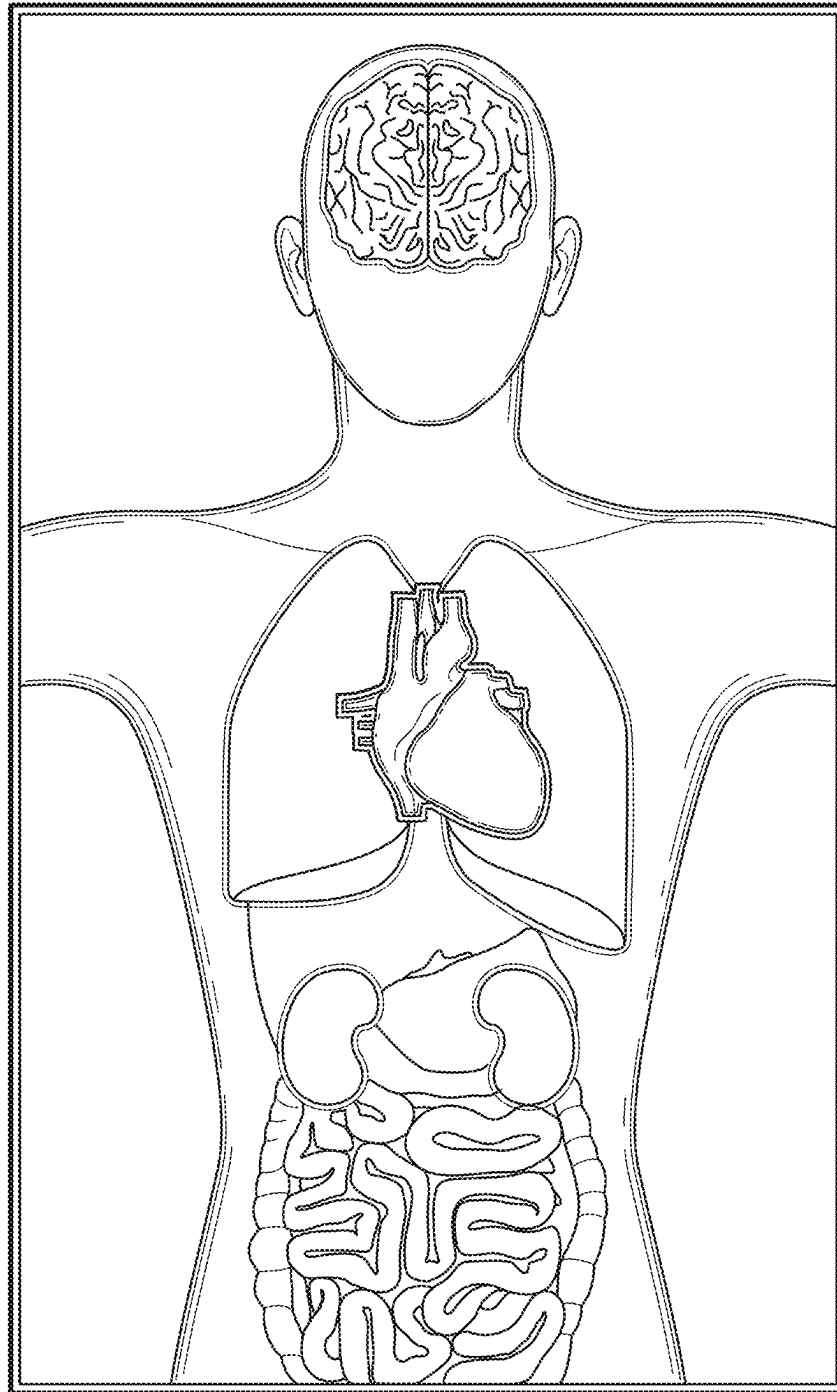


FIG. 19F

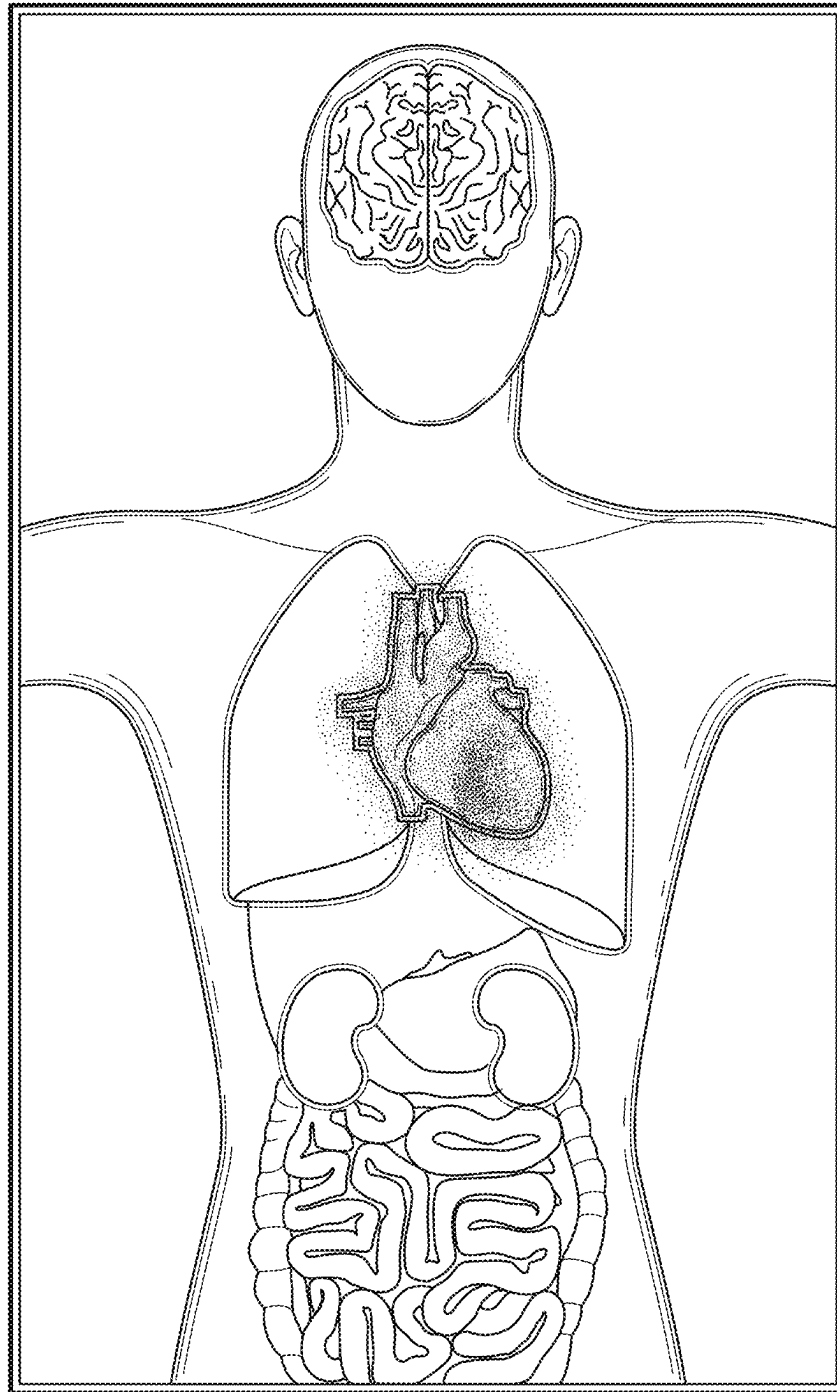


FIG. 19G

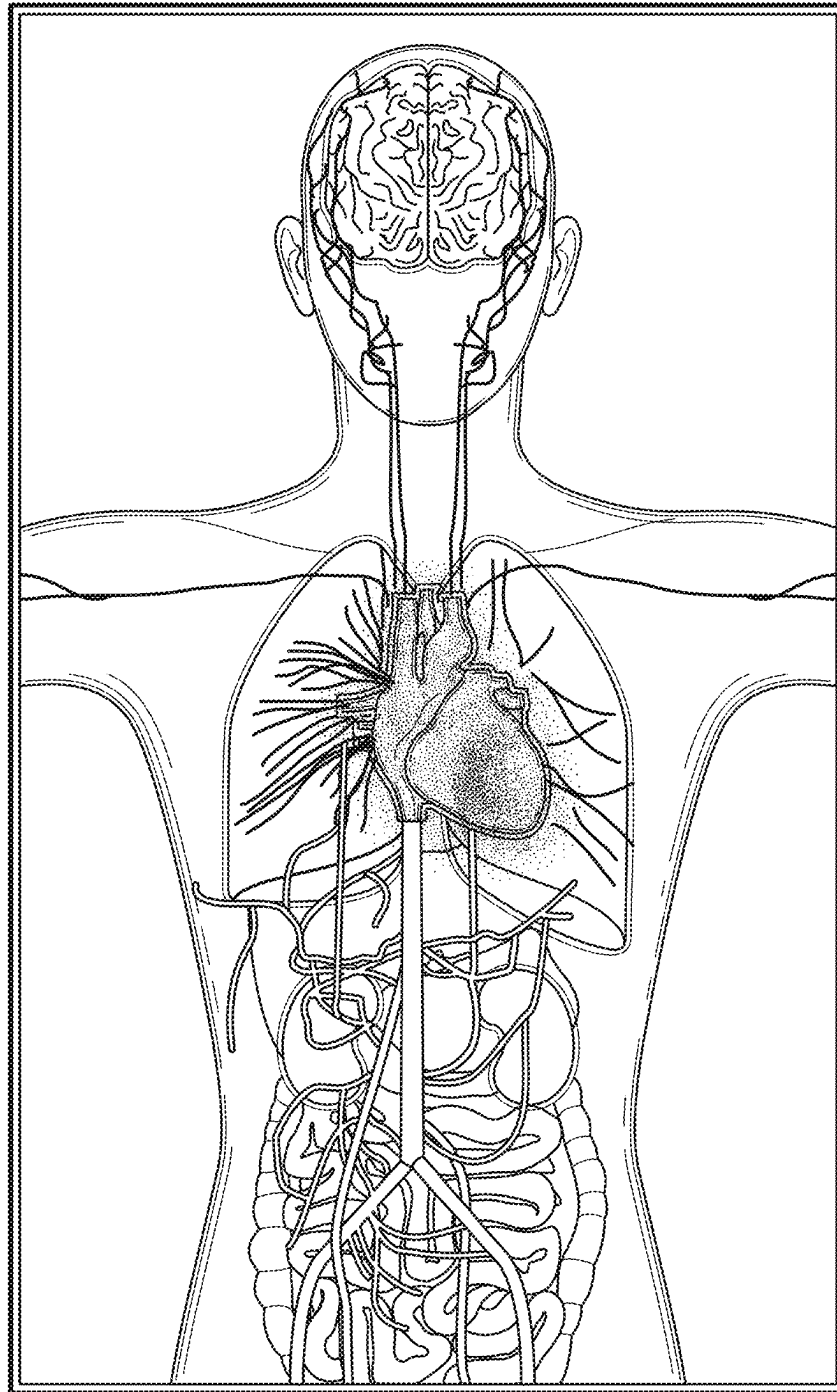


FIG. 19H

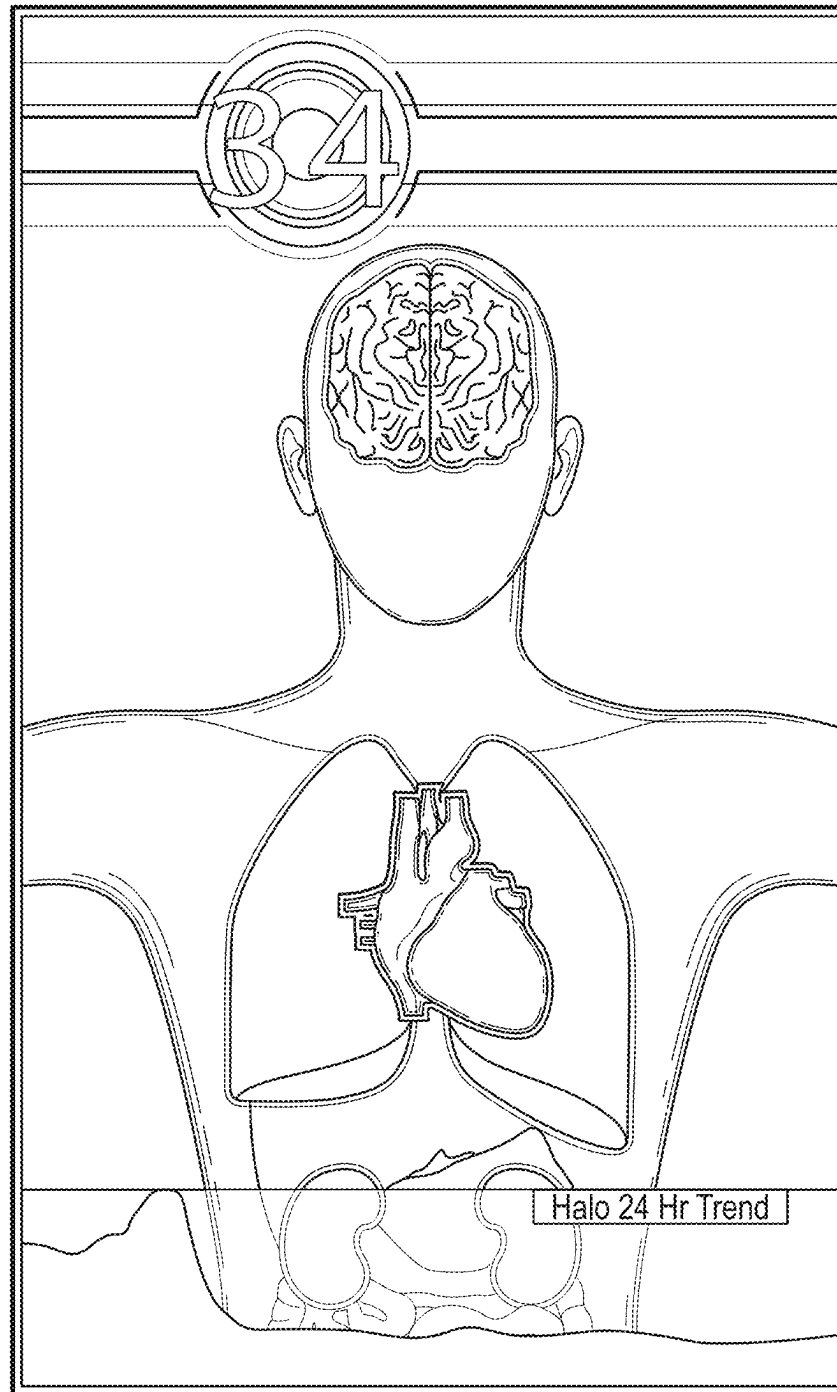


FIG. 19I

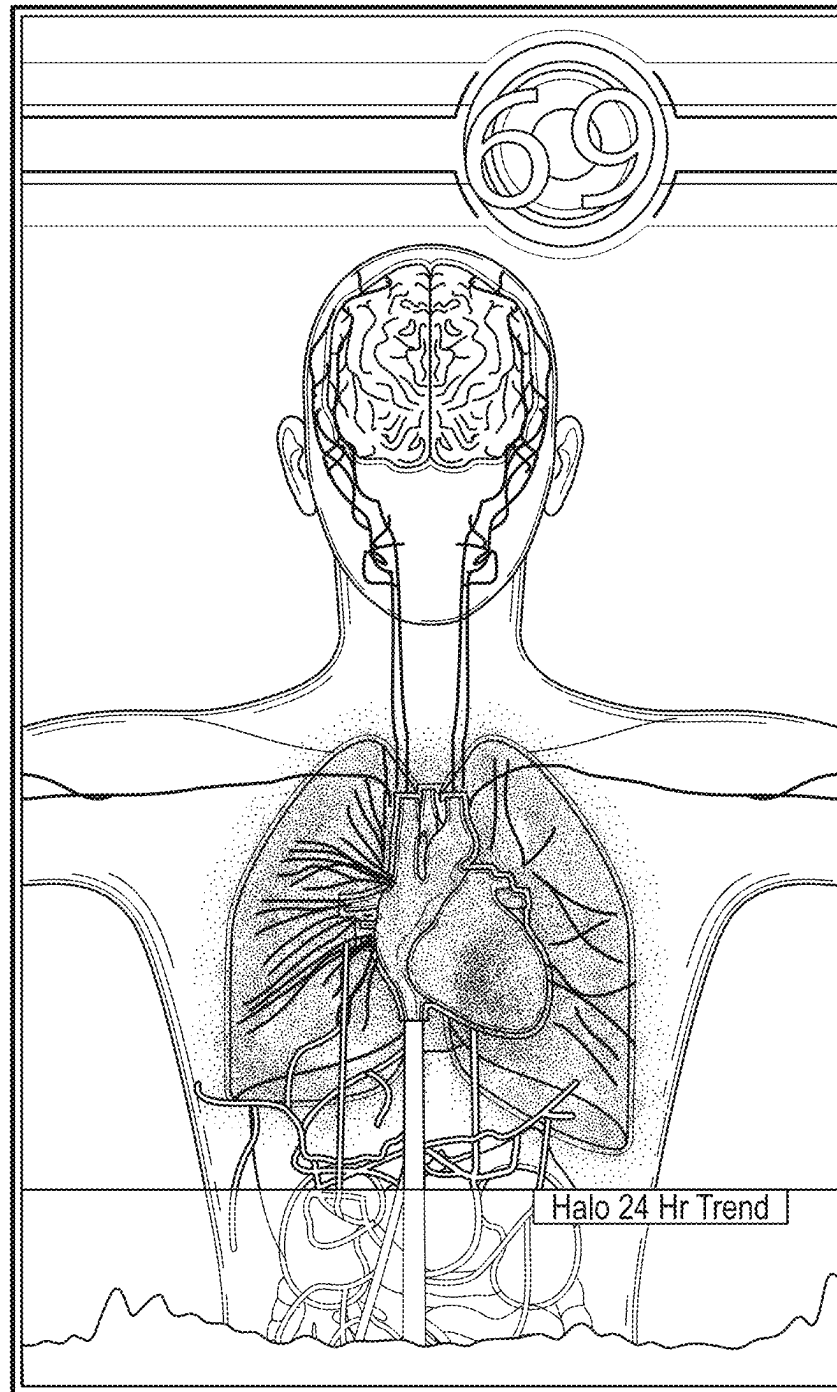


FIG. 19J

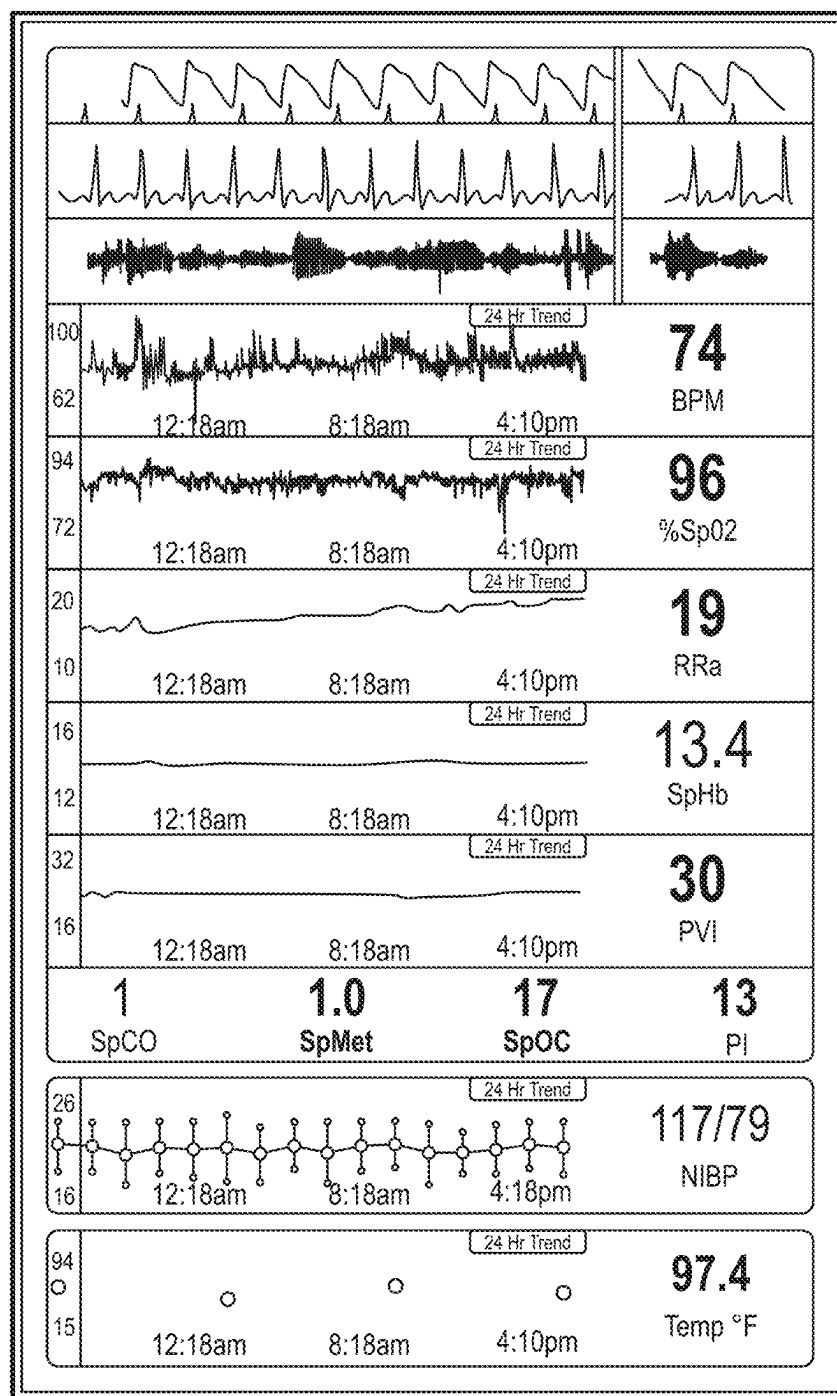


FIG. 20A

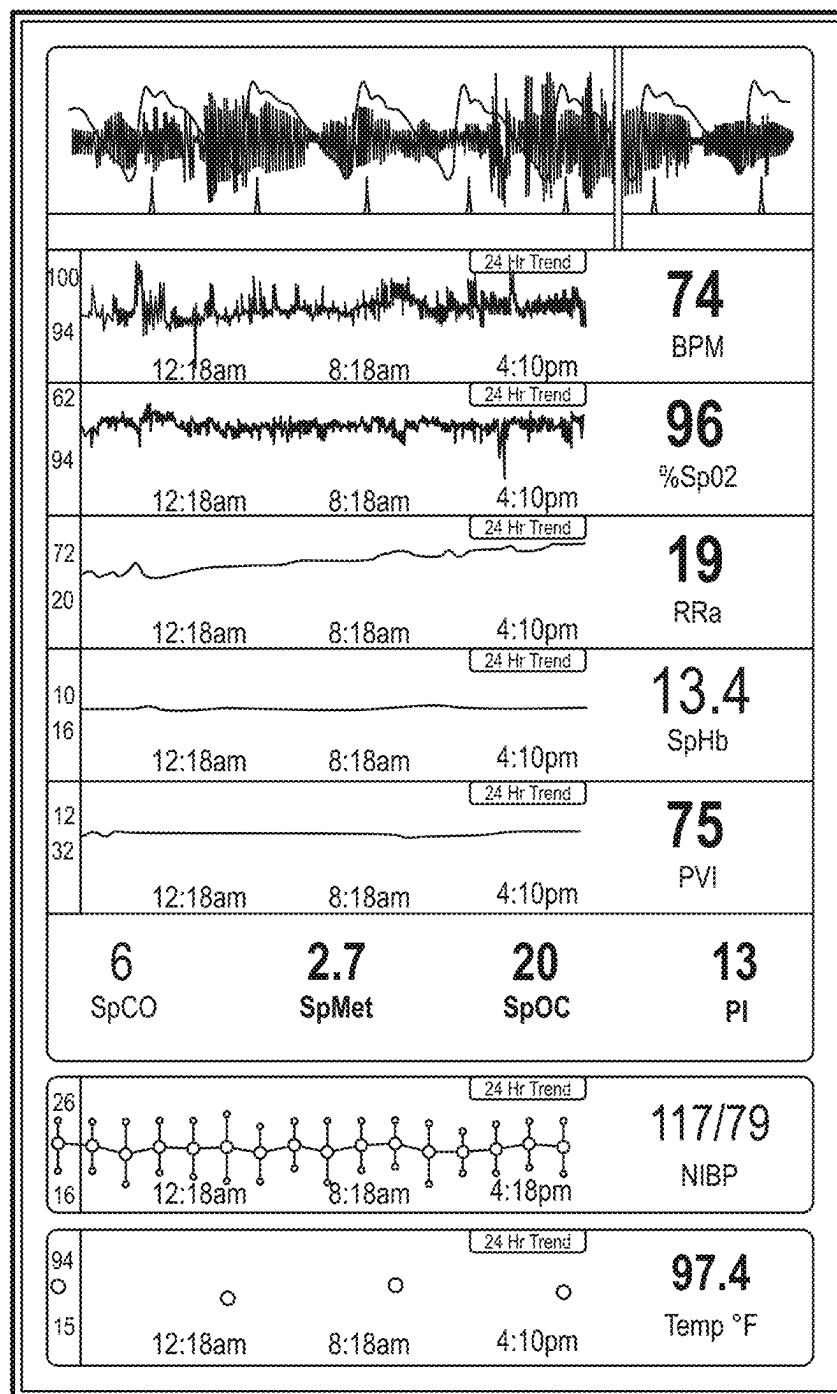


FIG. 20B

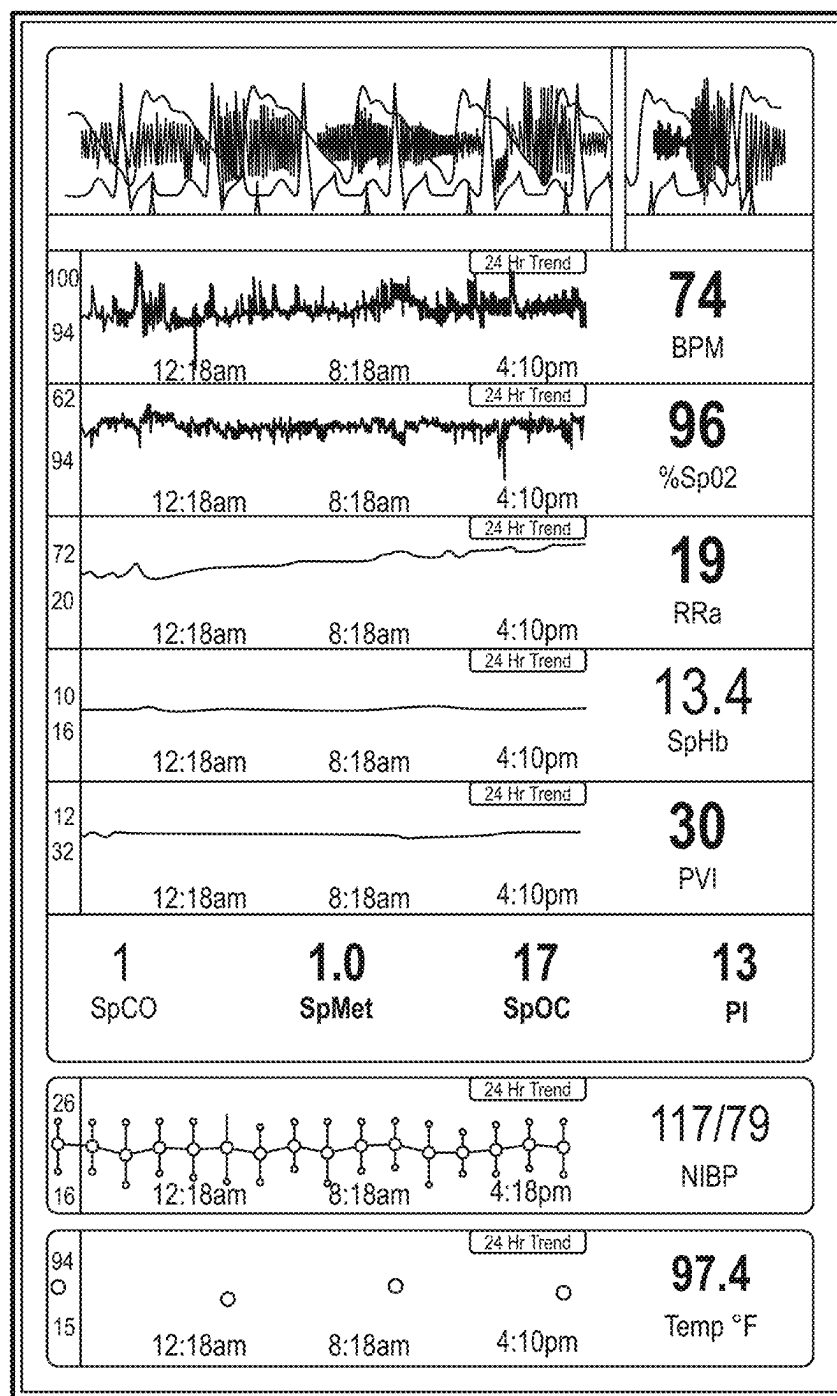


FIG. 20C

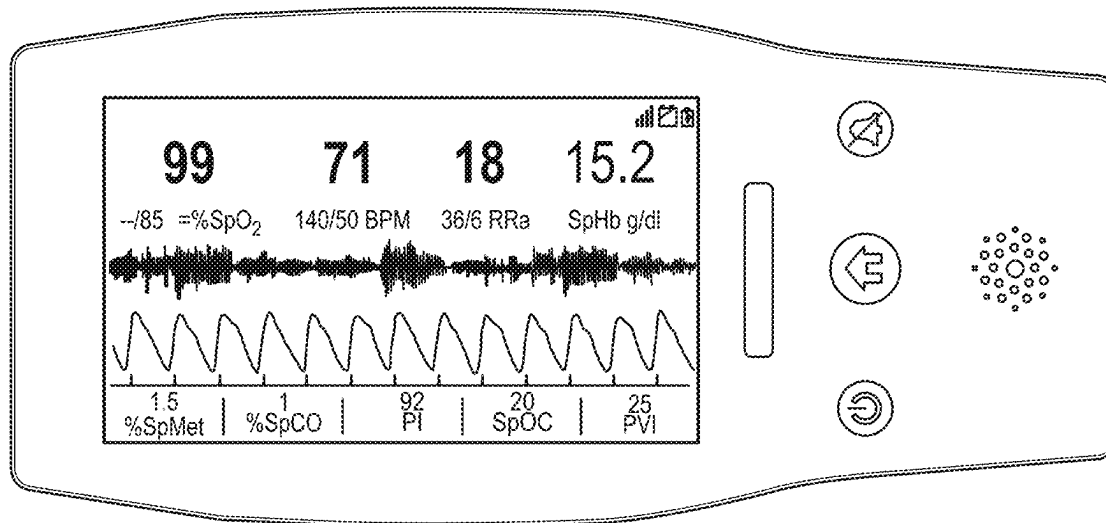


FIG. 21A

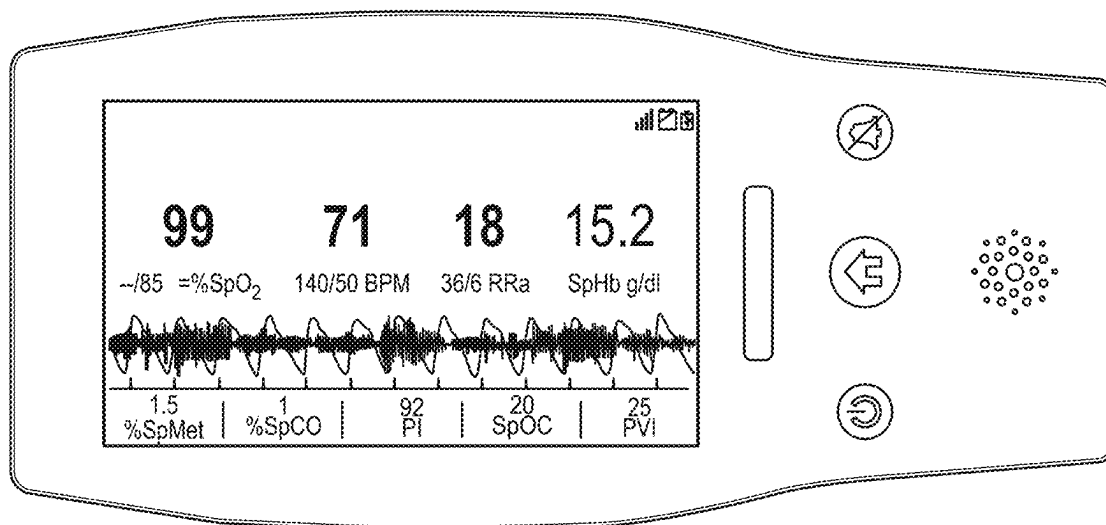


FIG. 21B

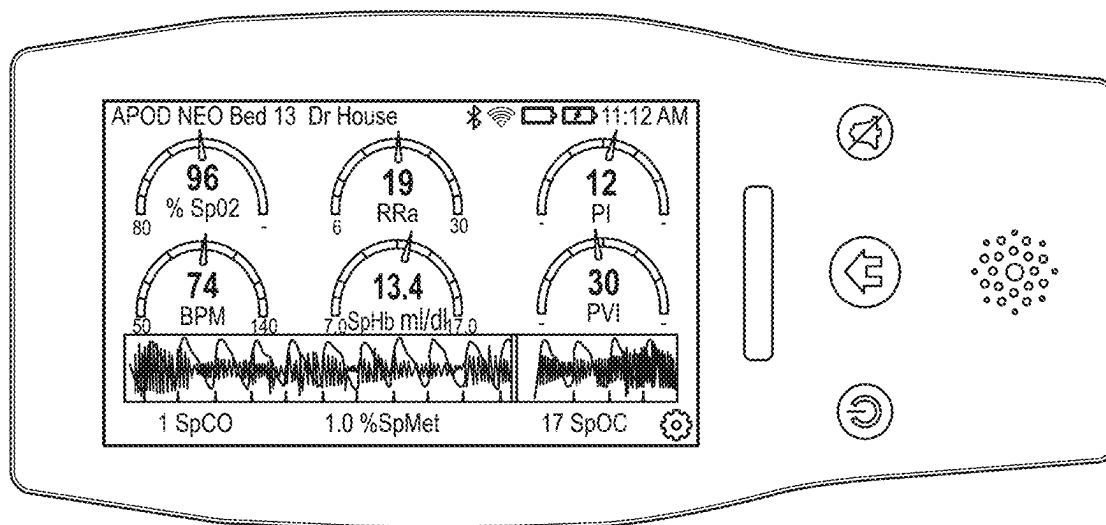


FIG. 22A

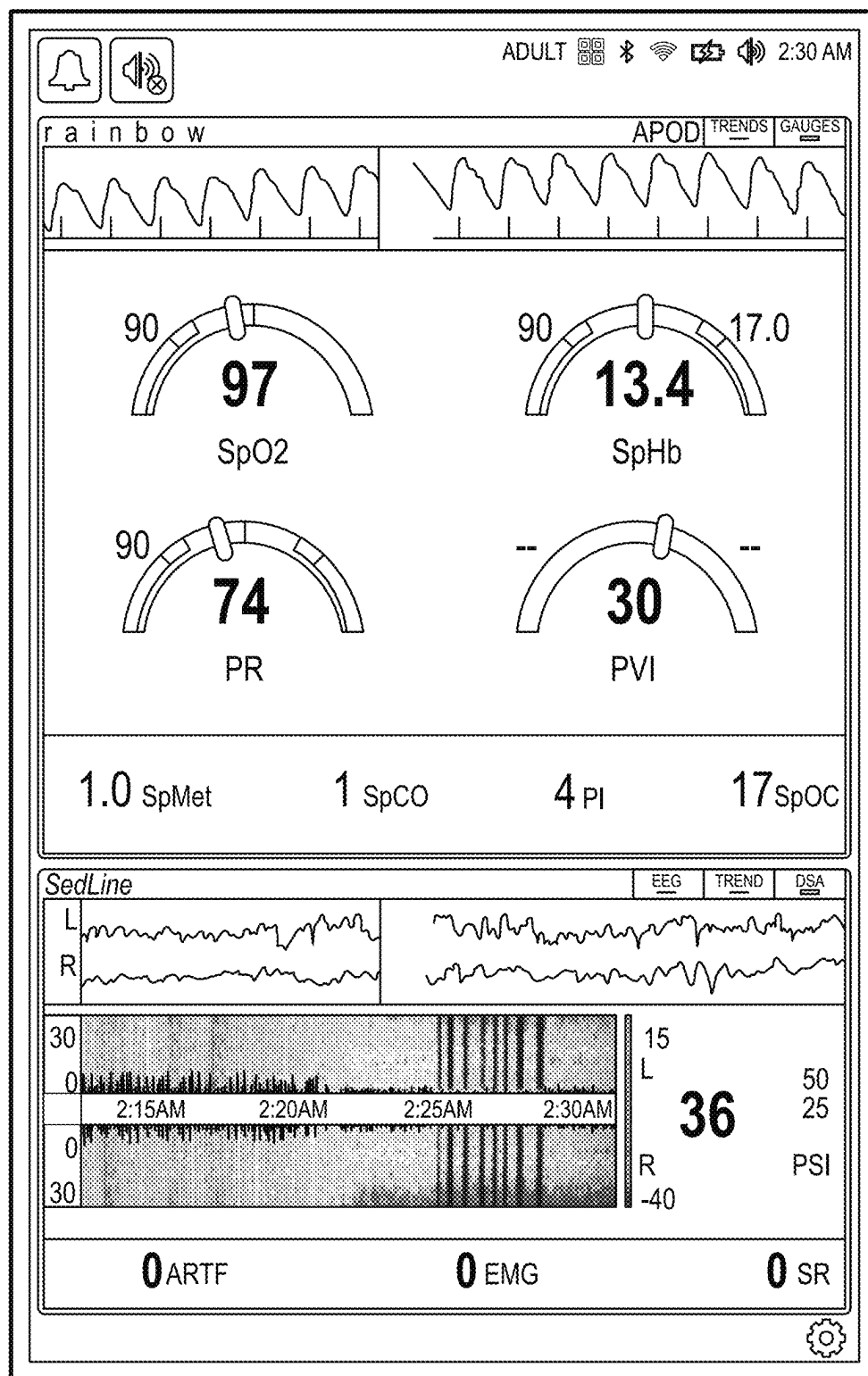


FIG. 22B

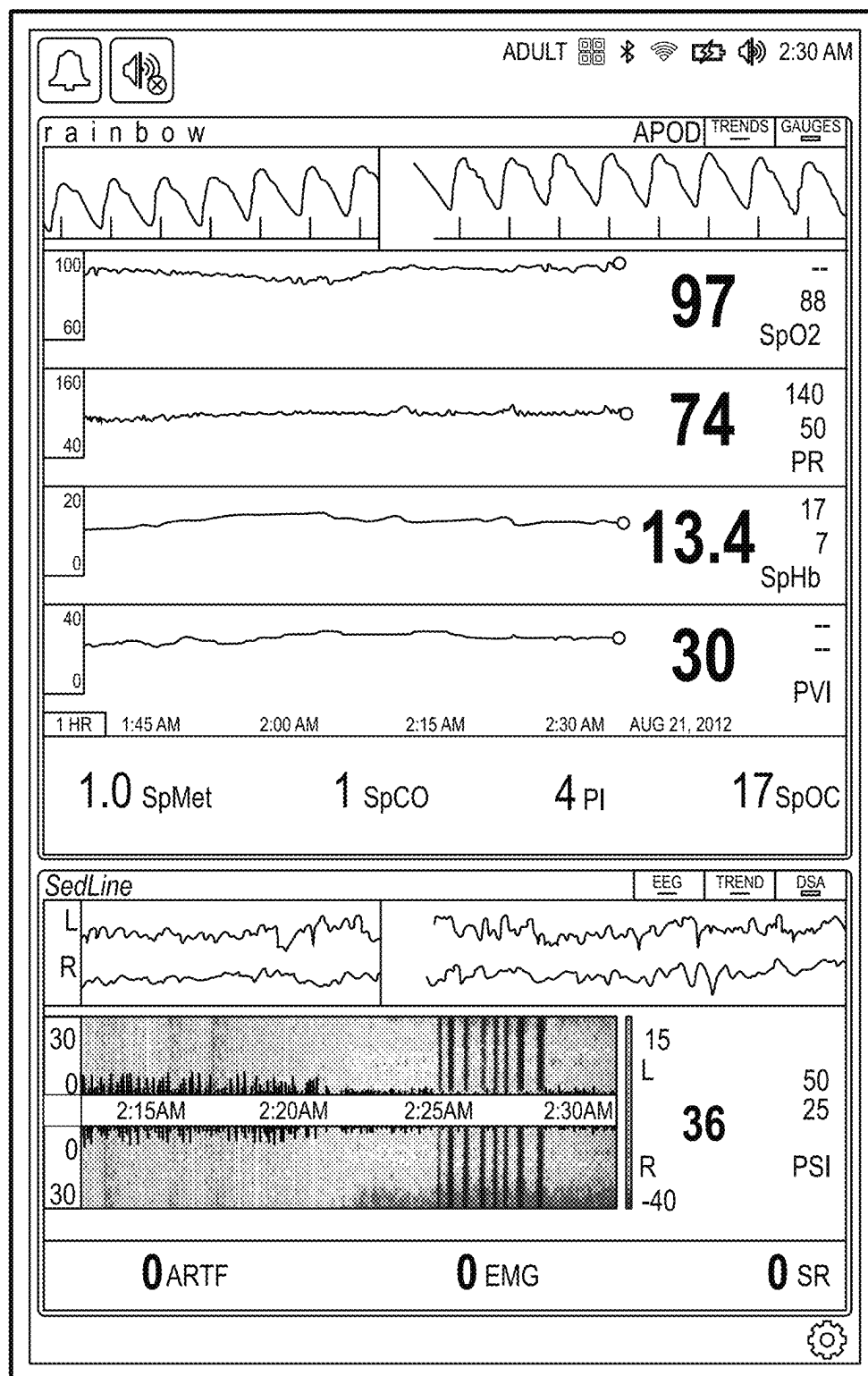


FIG. 23A

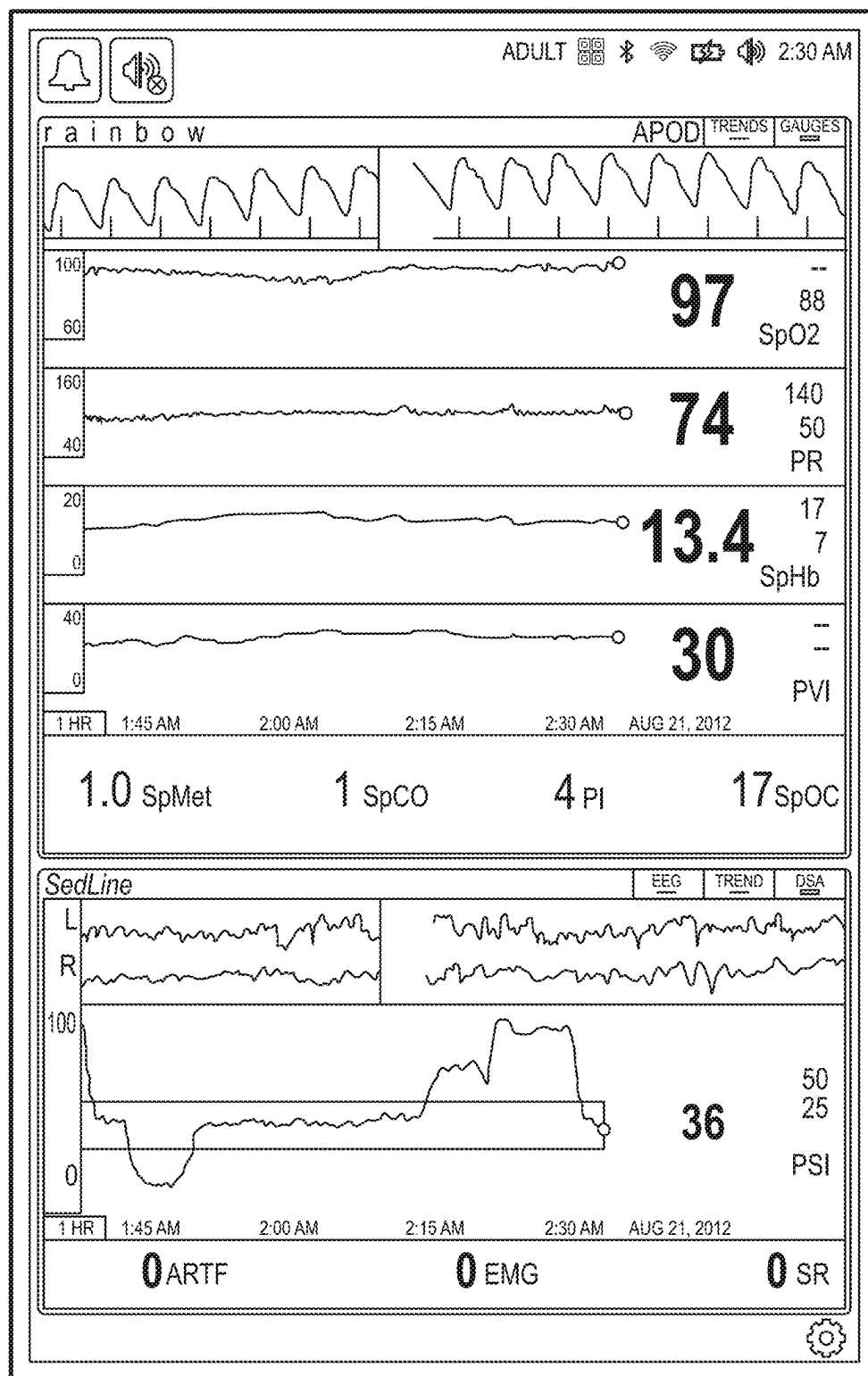


FIG. 23B

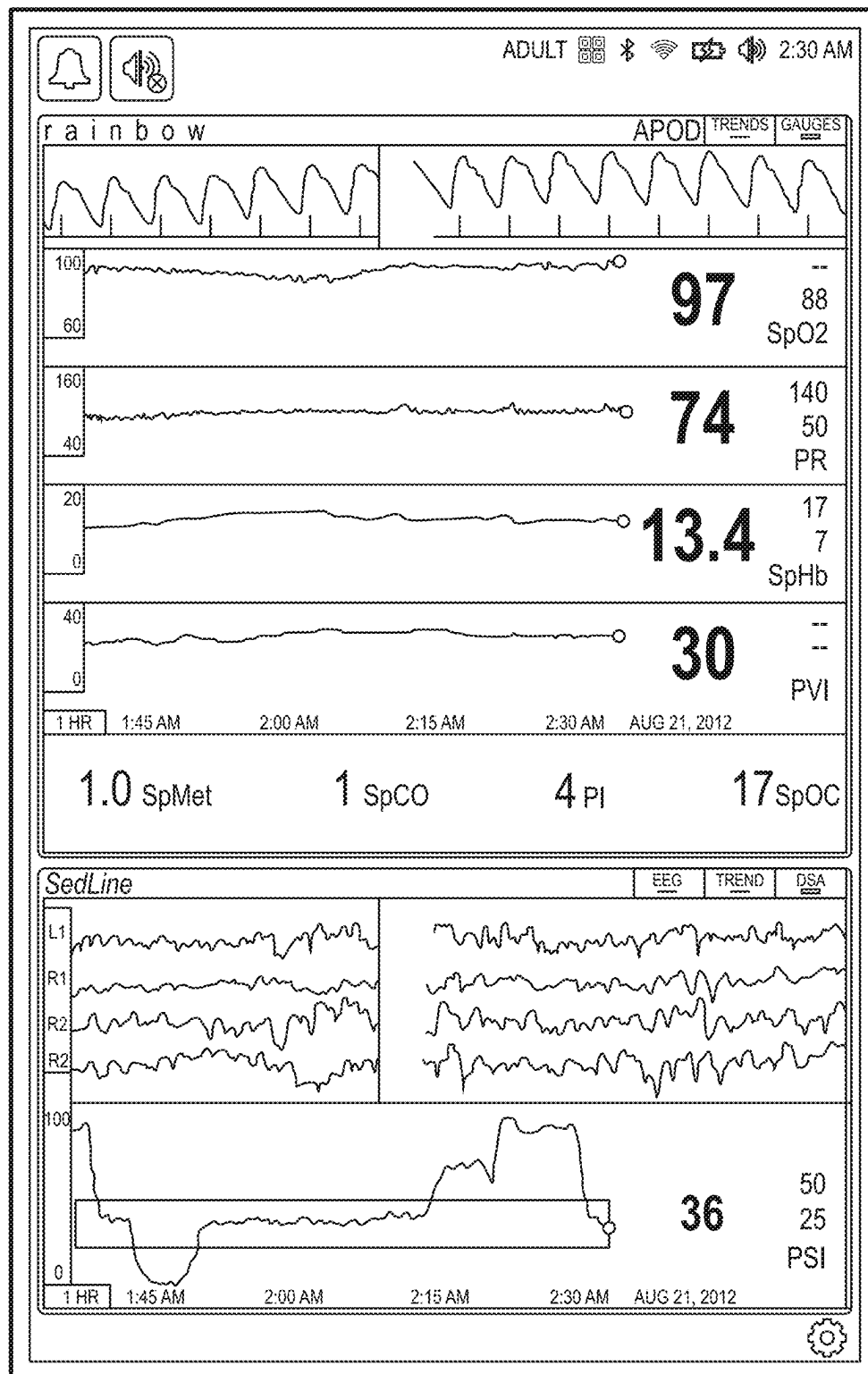


FIG. 23C

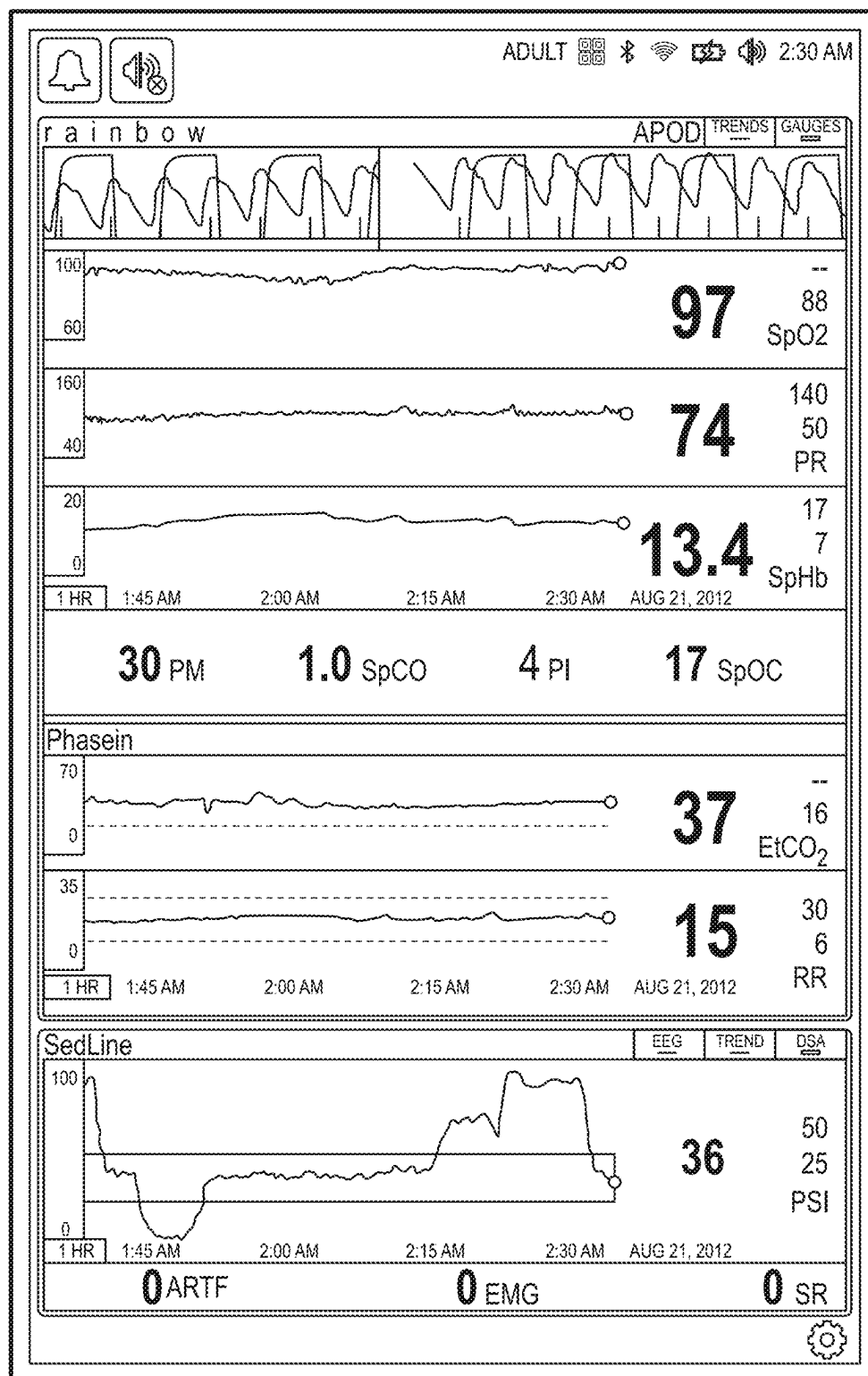


FIG. 23D

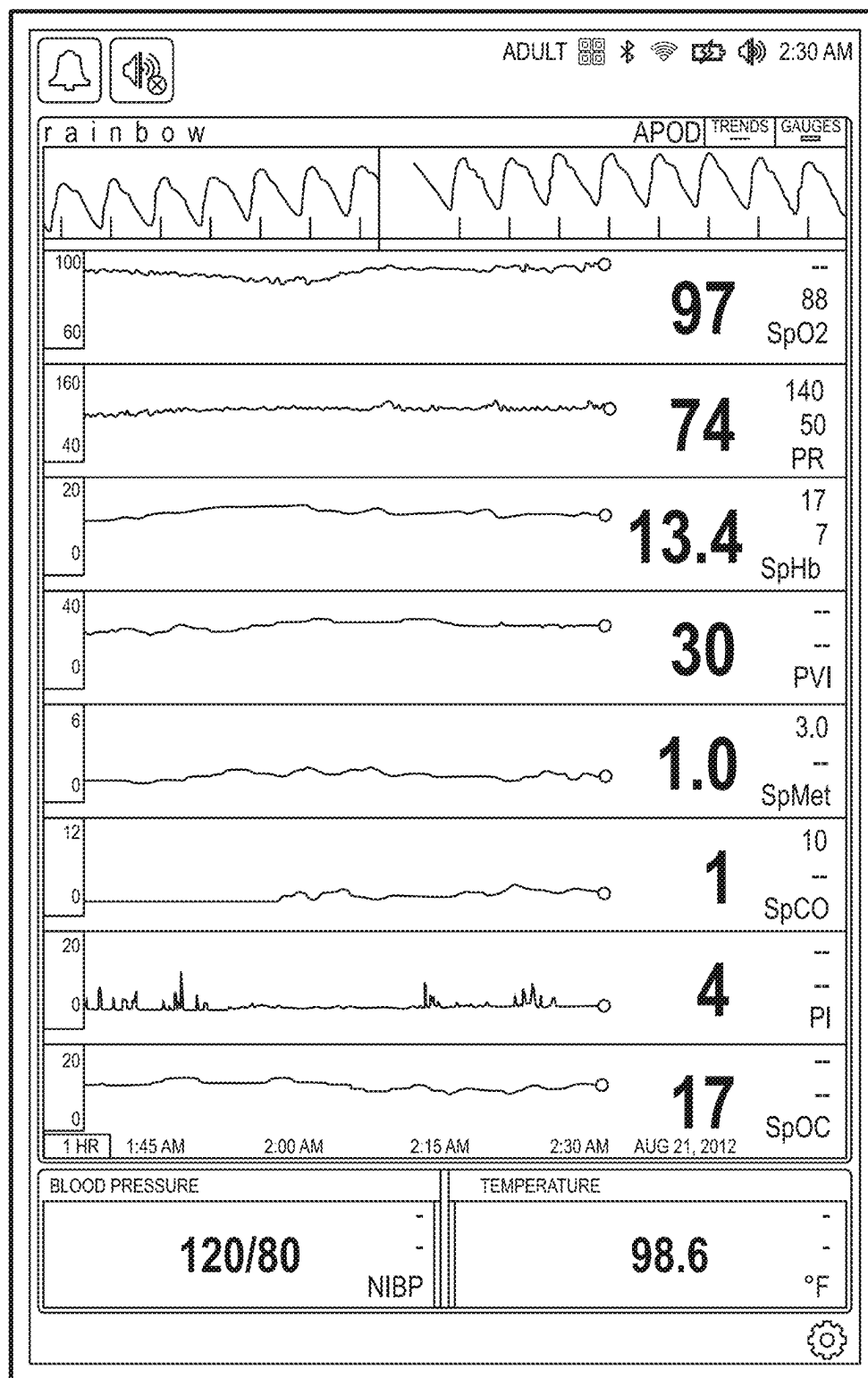


FIG. 23E

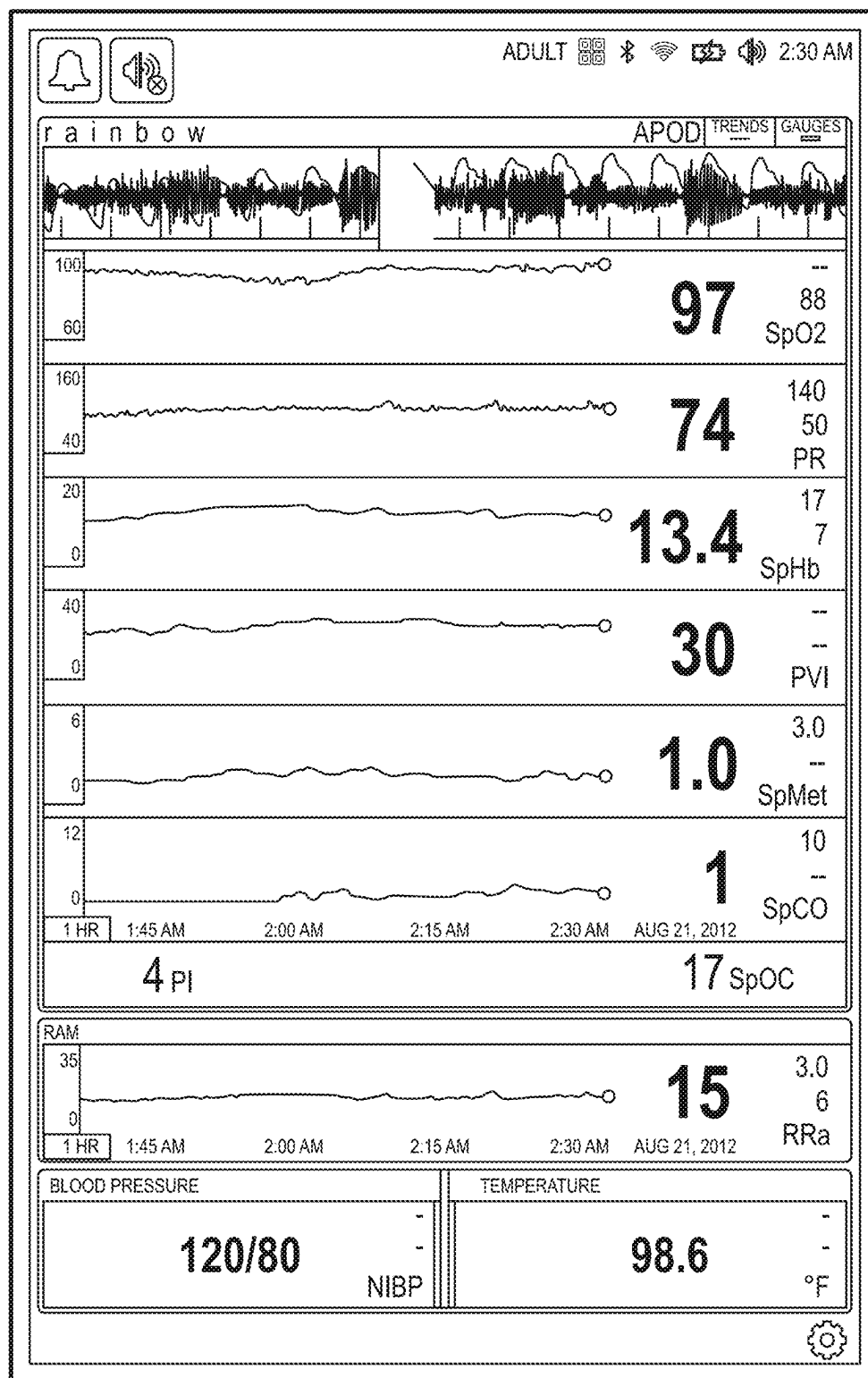


FIG. 23F

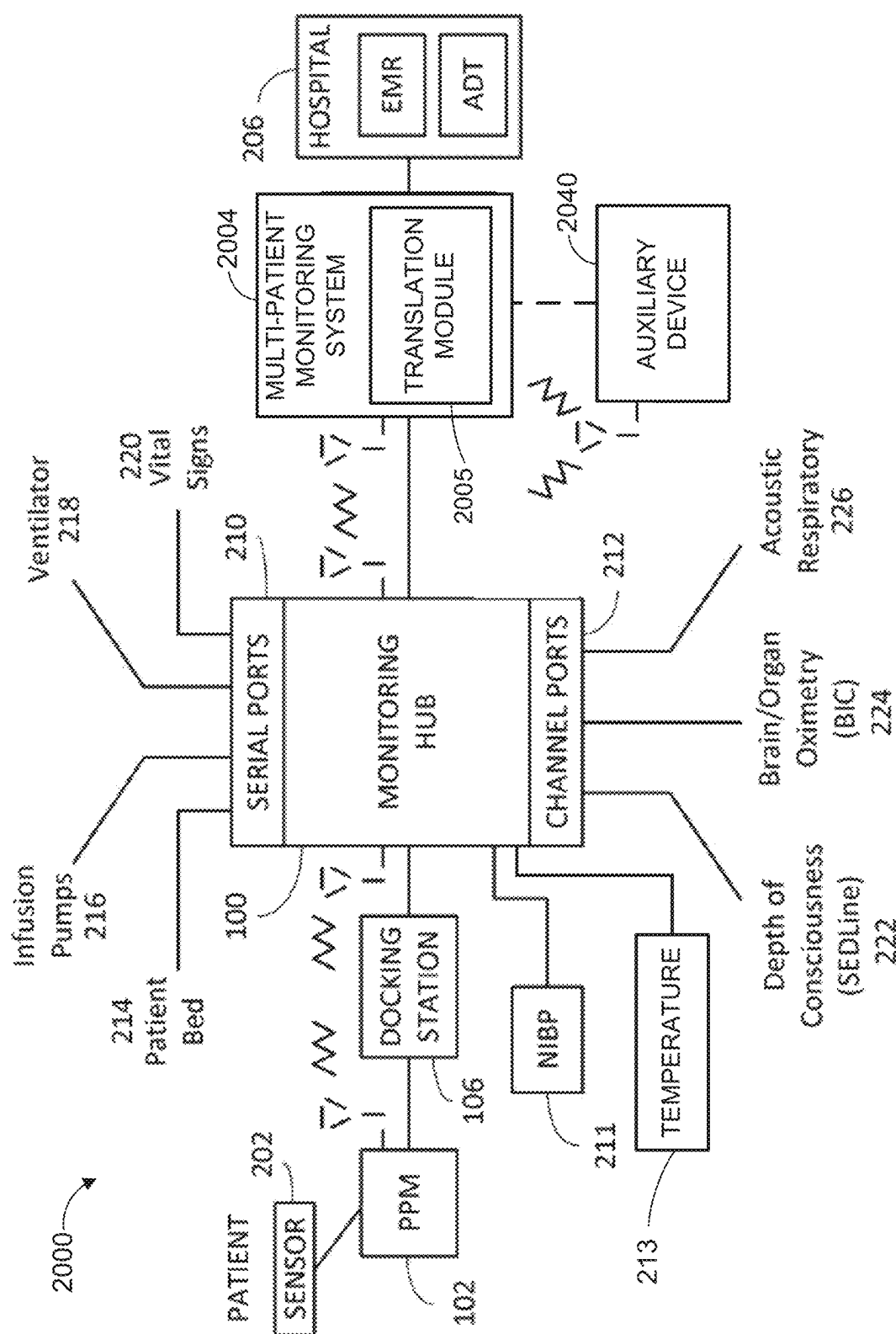
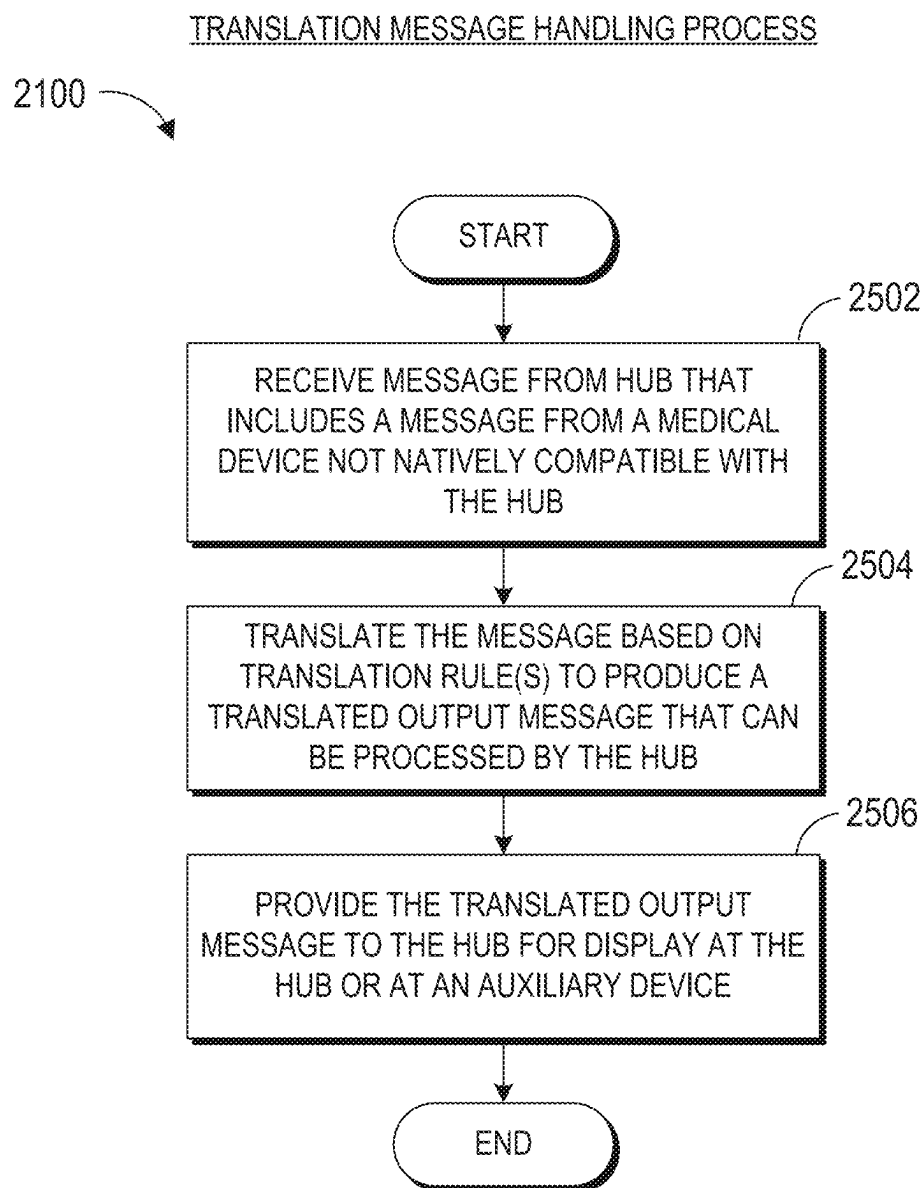


FIG. 24



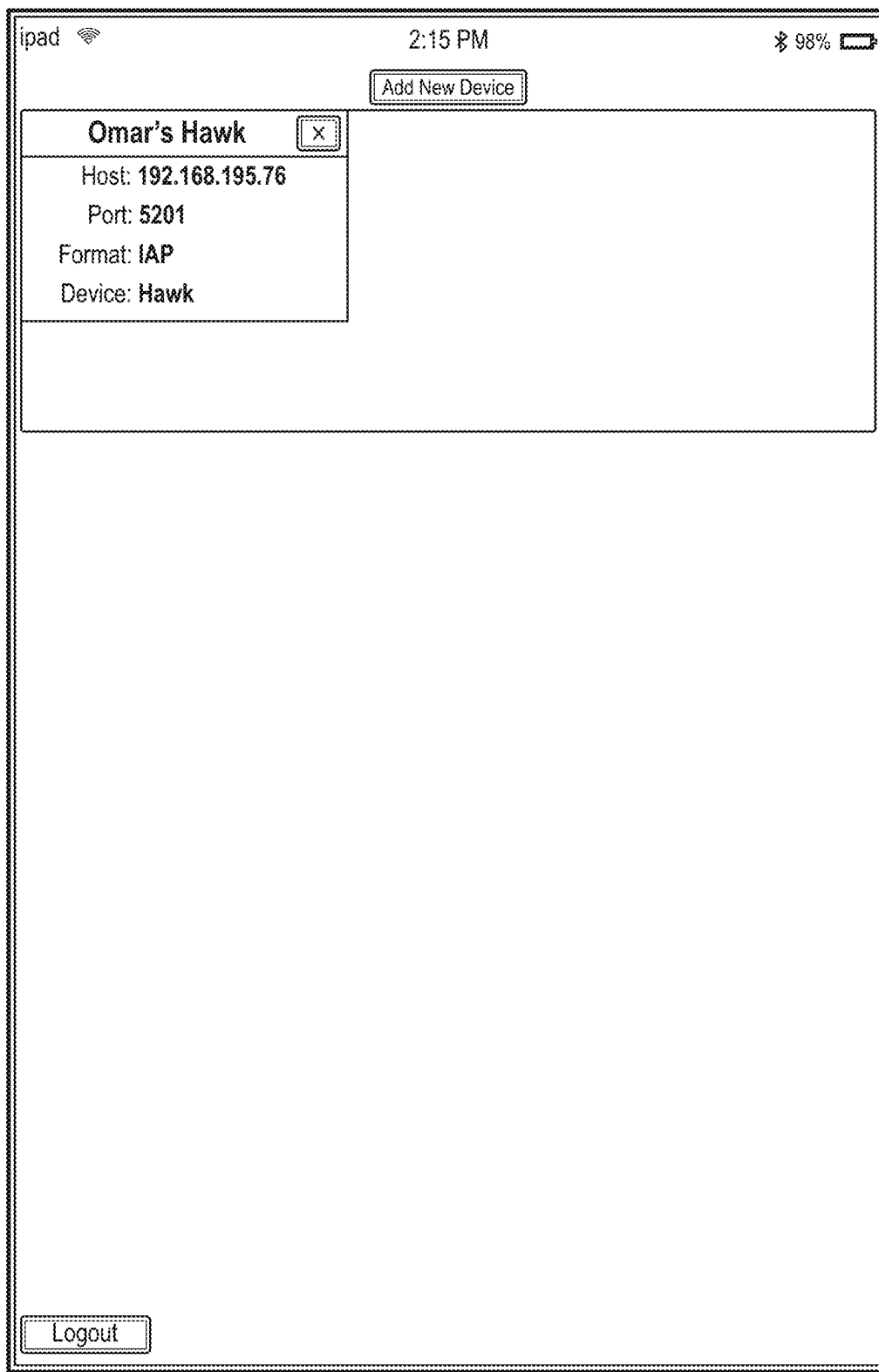


FIG. 26

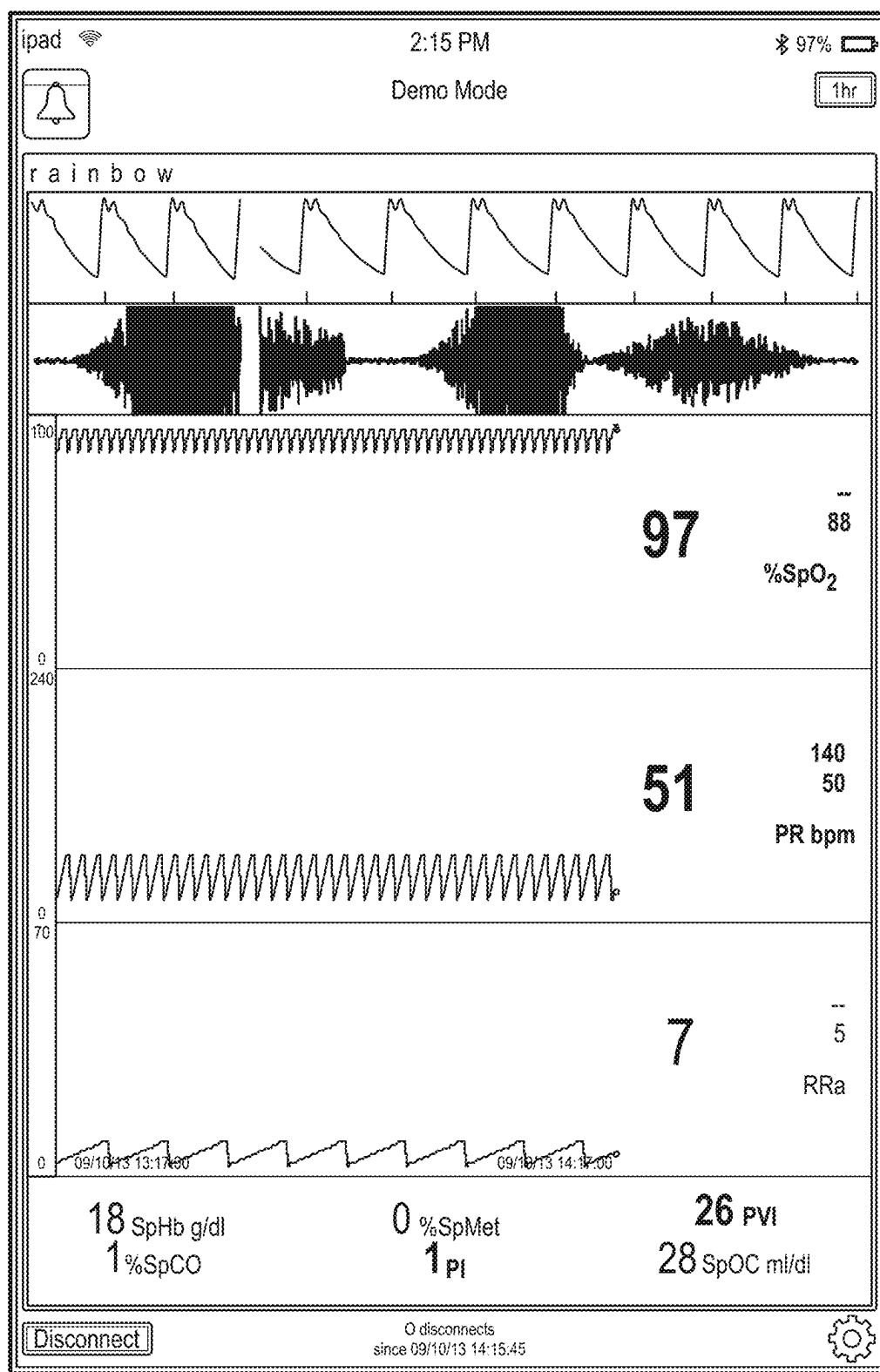


FIG. 27

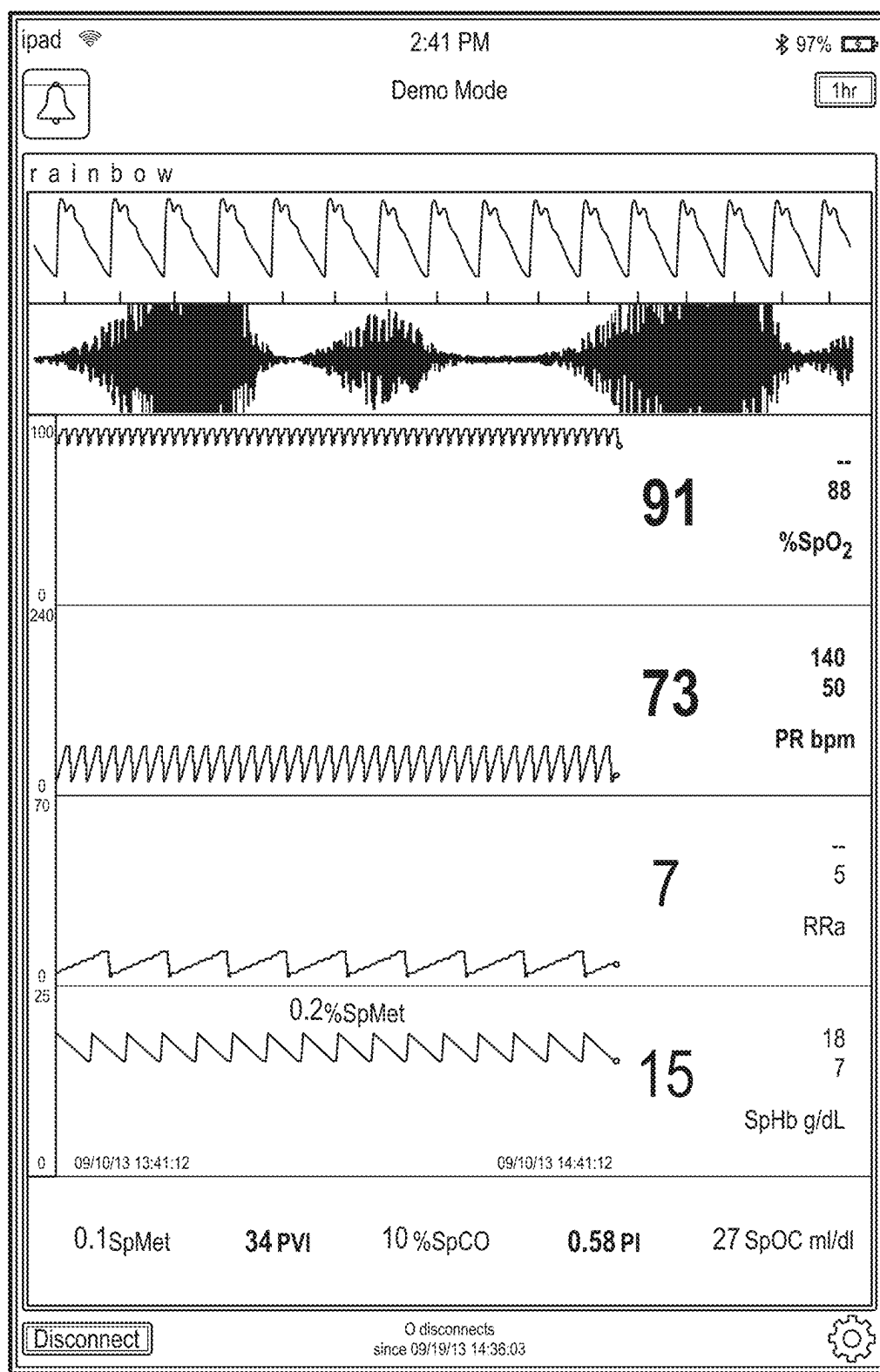


FIG. 28

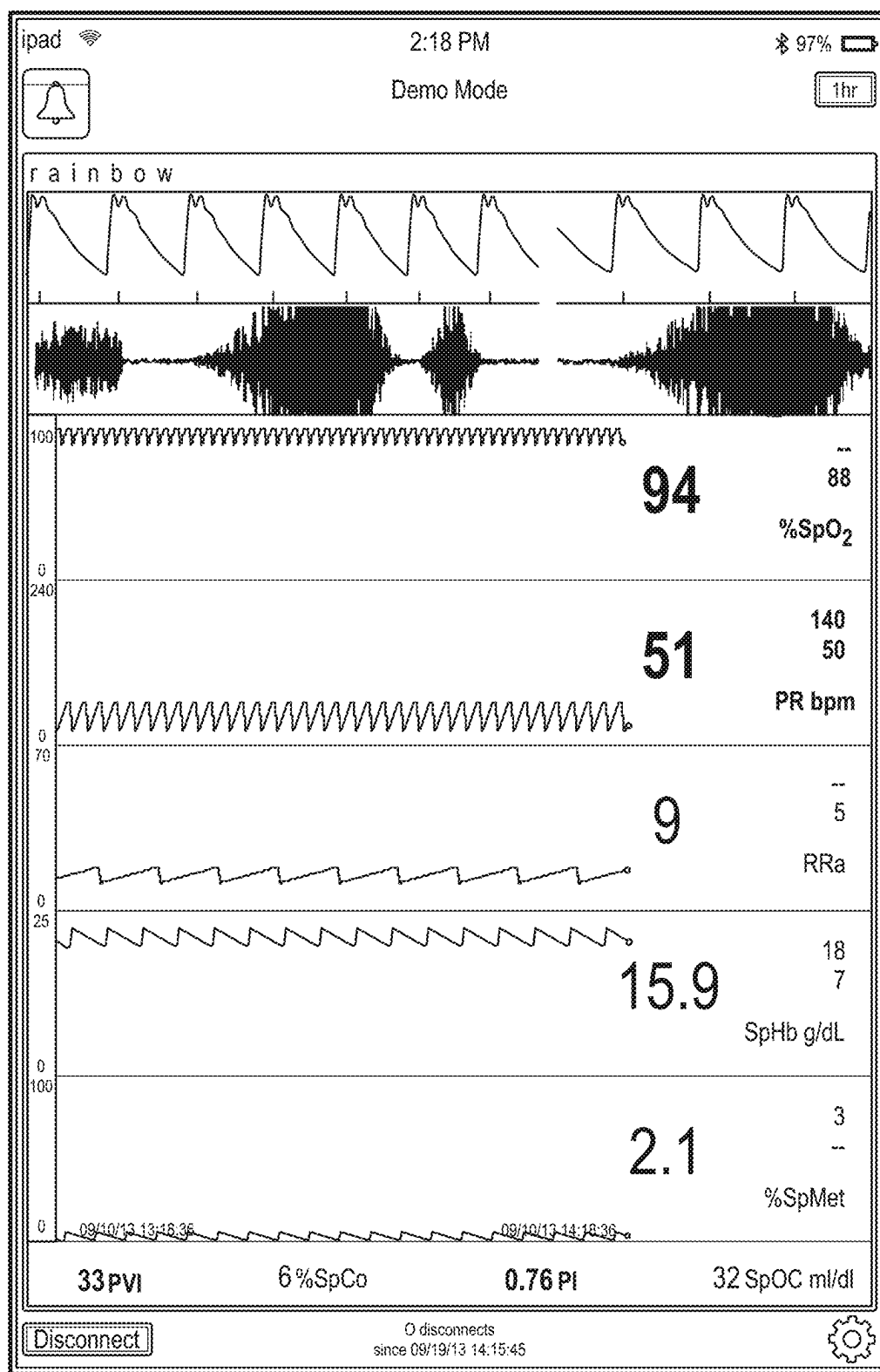


FIG. 29

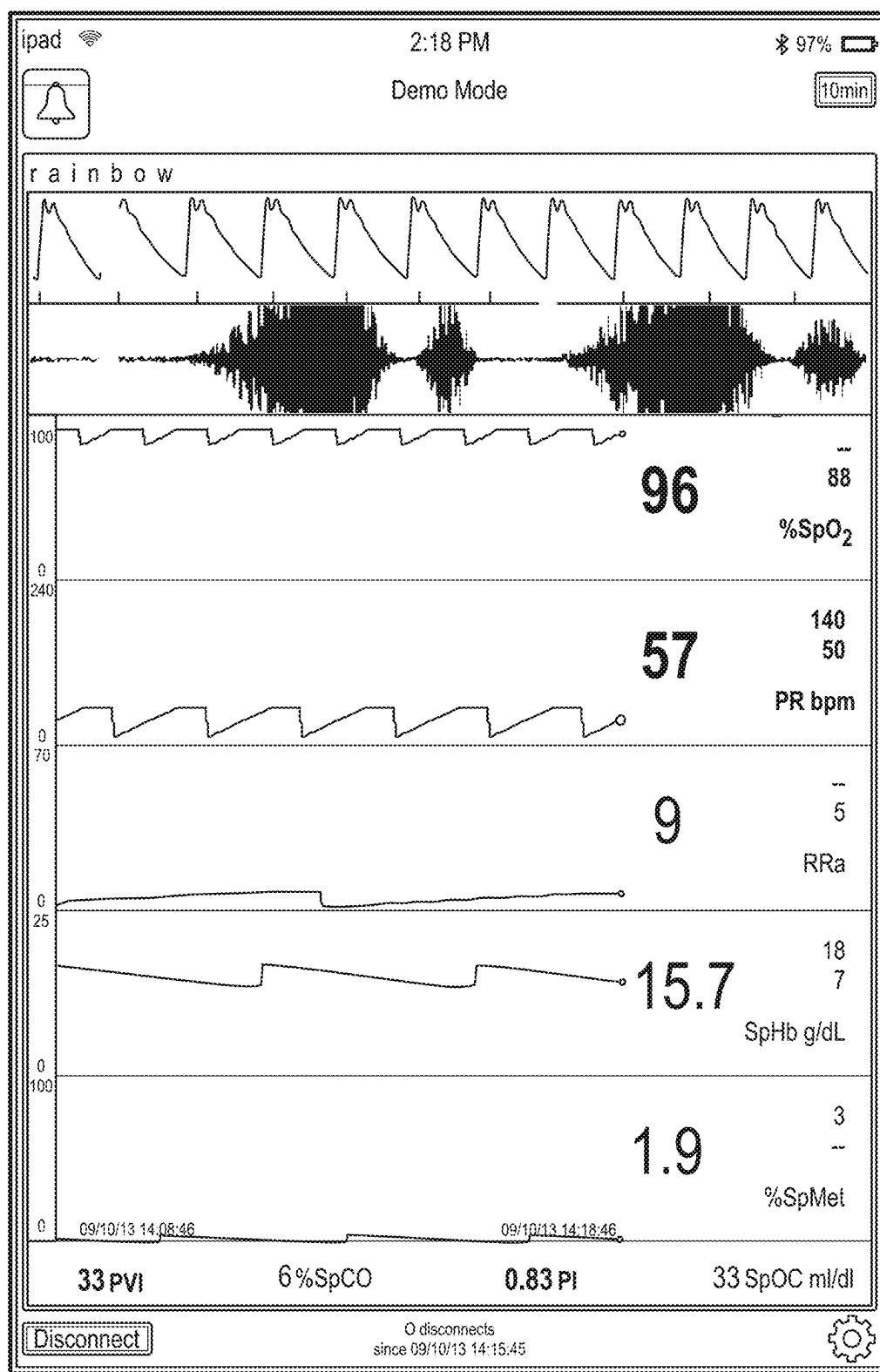


FIG. 30

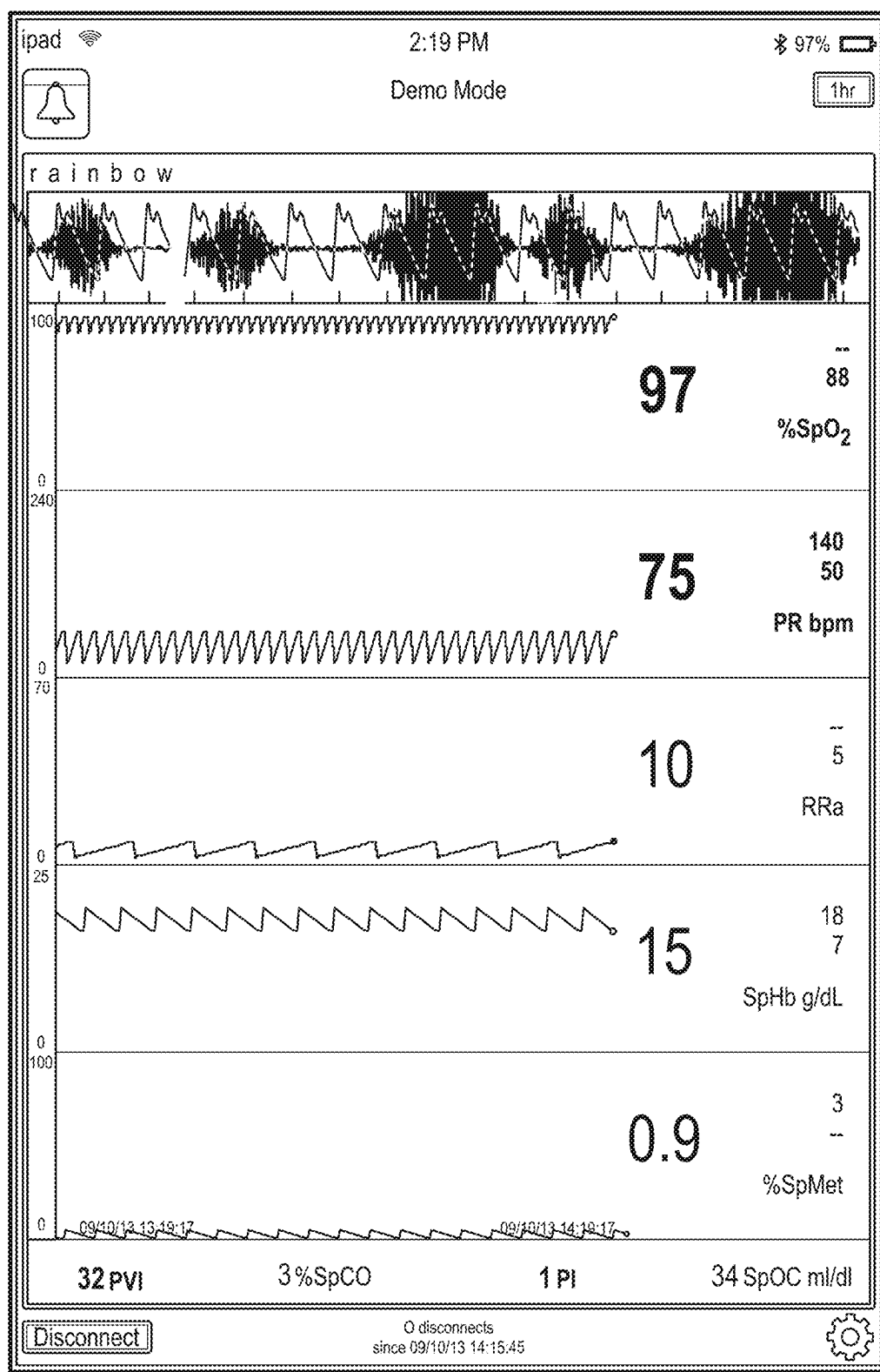


FIG. 31

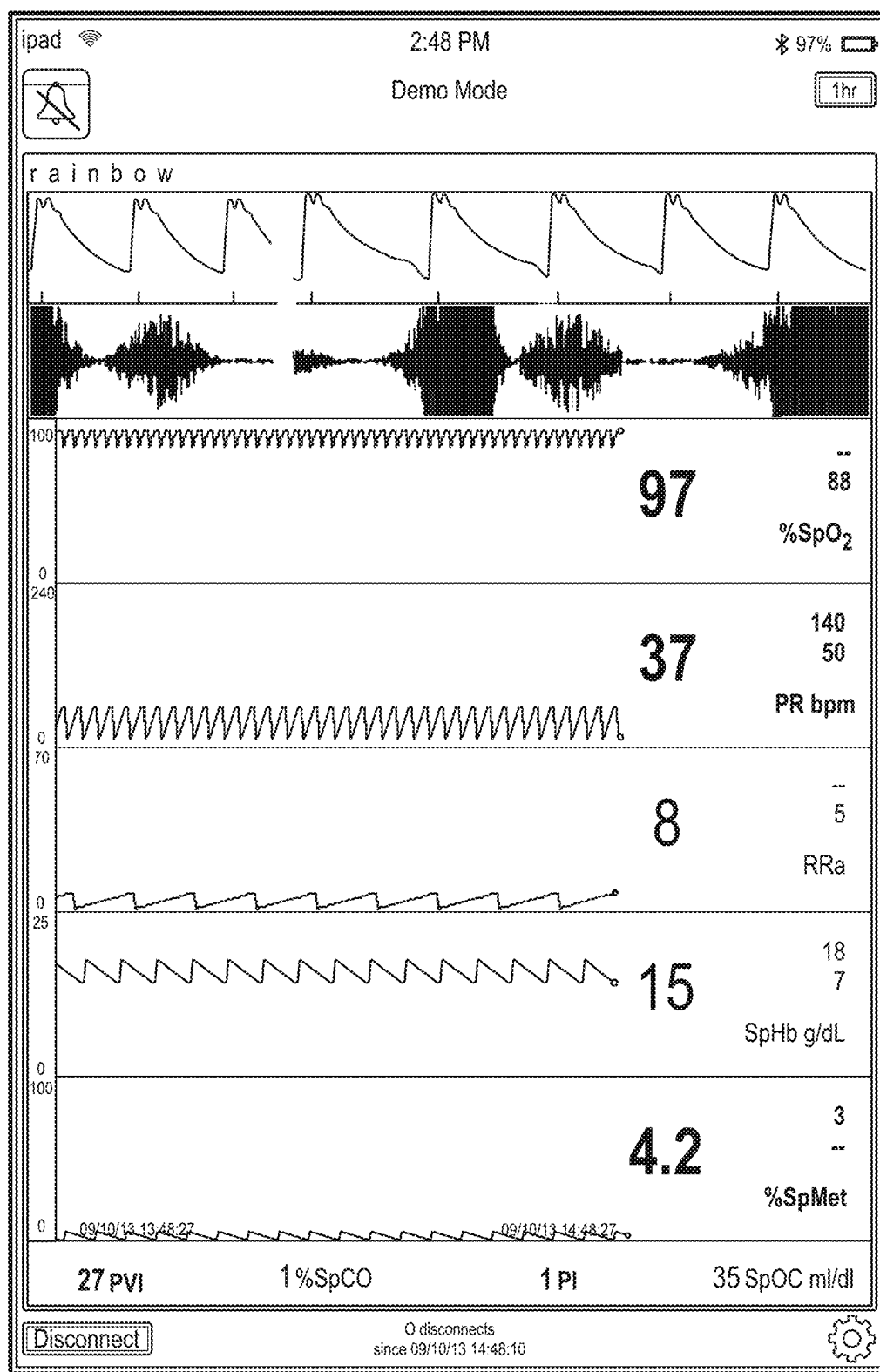


FIG. 32

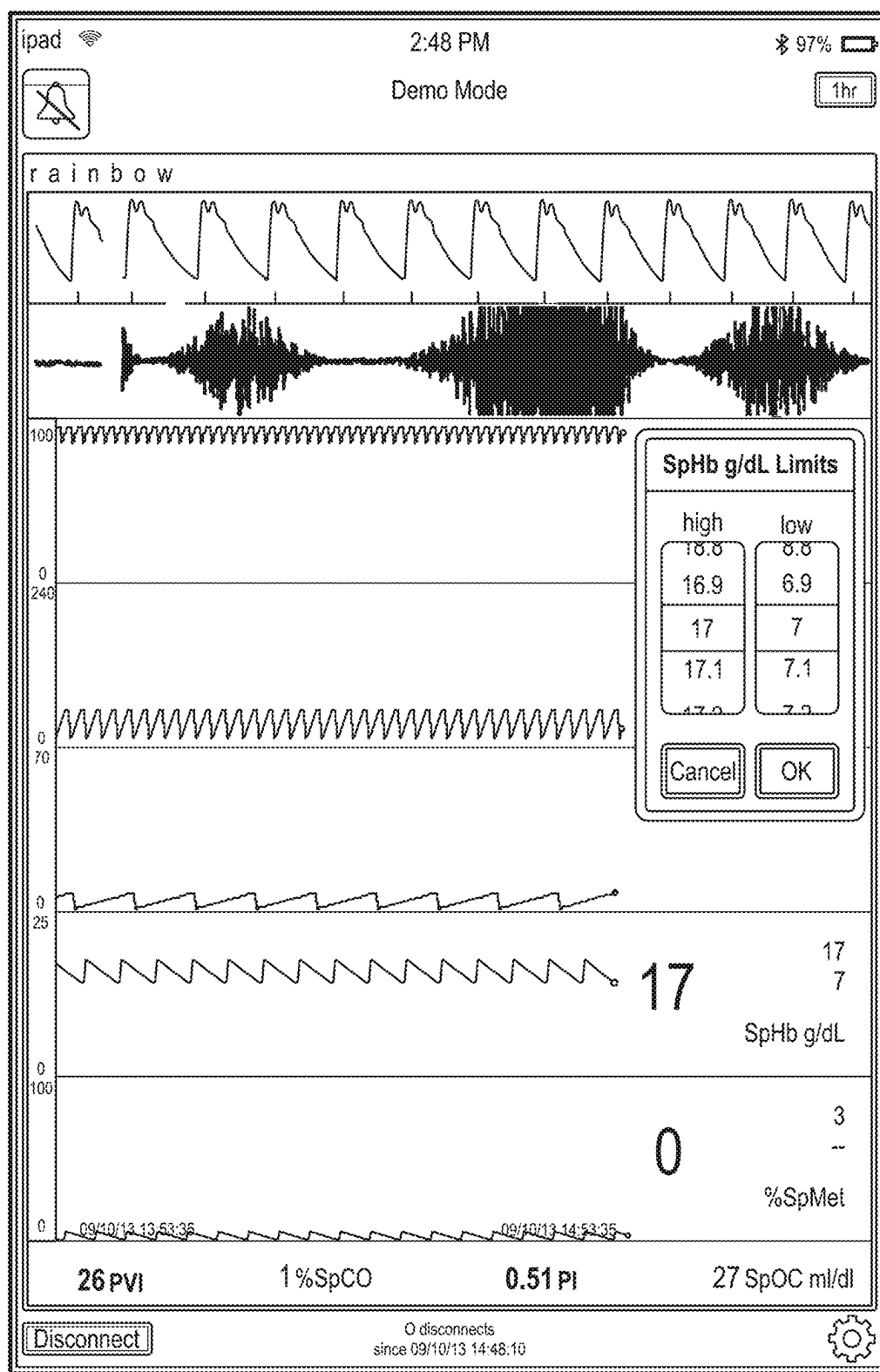


FIG. 33

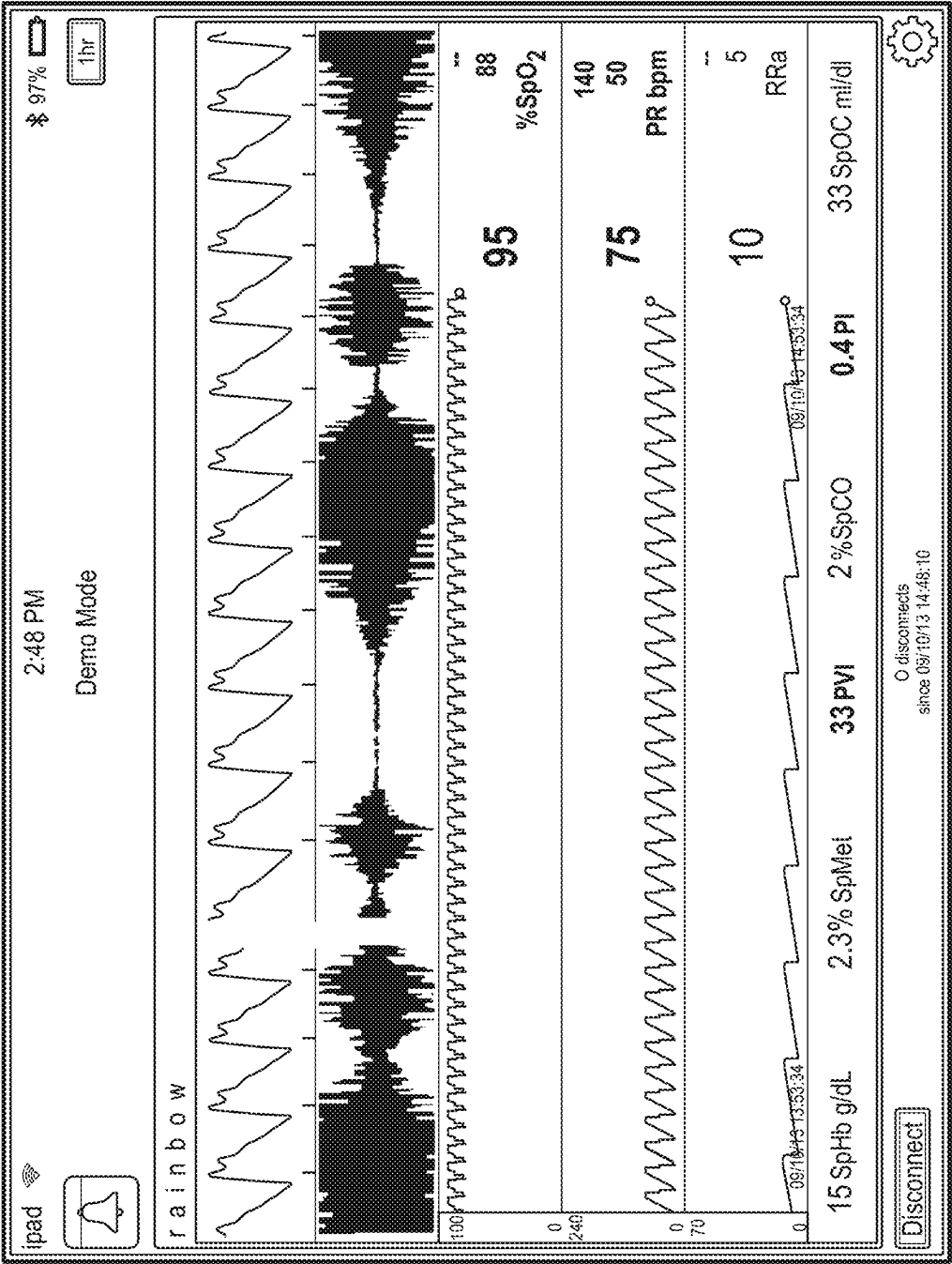


FIG. 34

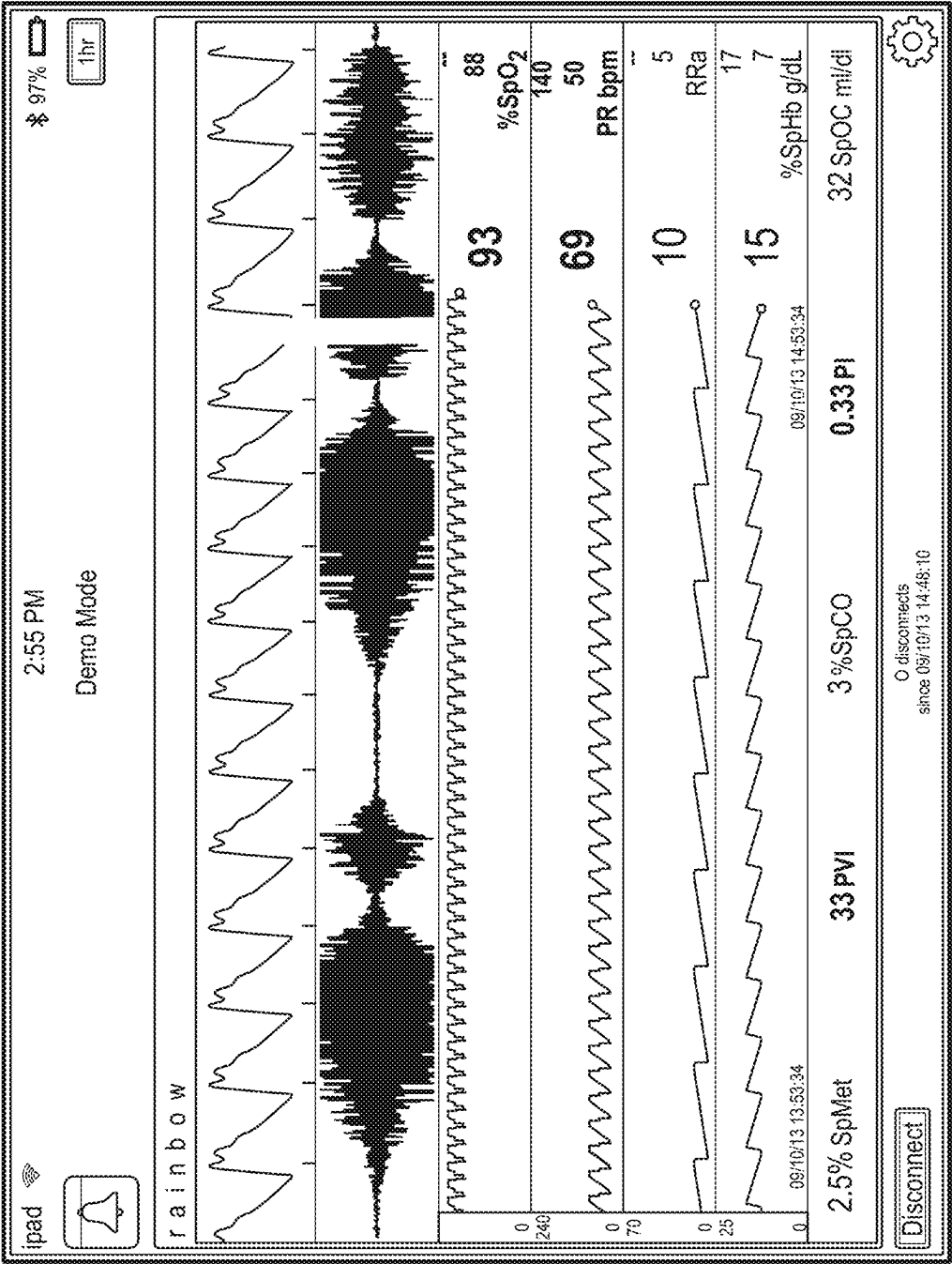


FIG. 35

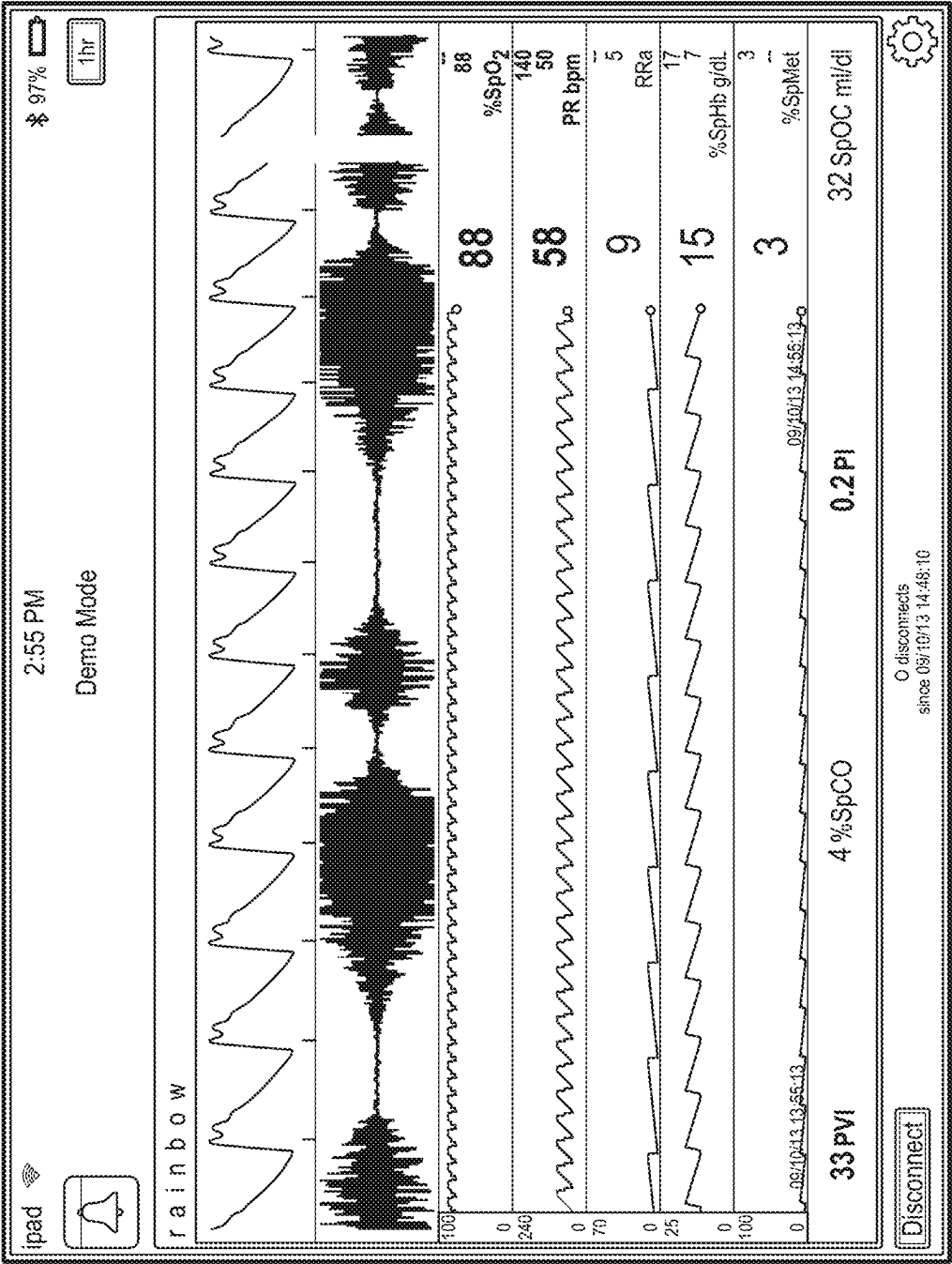


FIG. 36

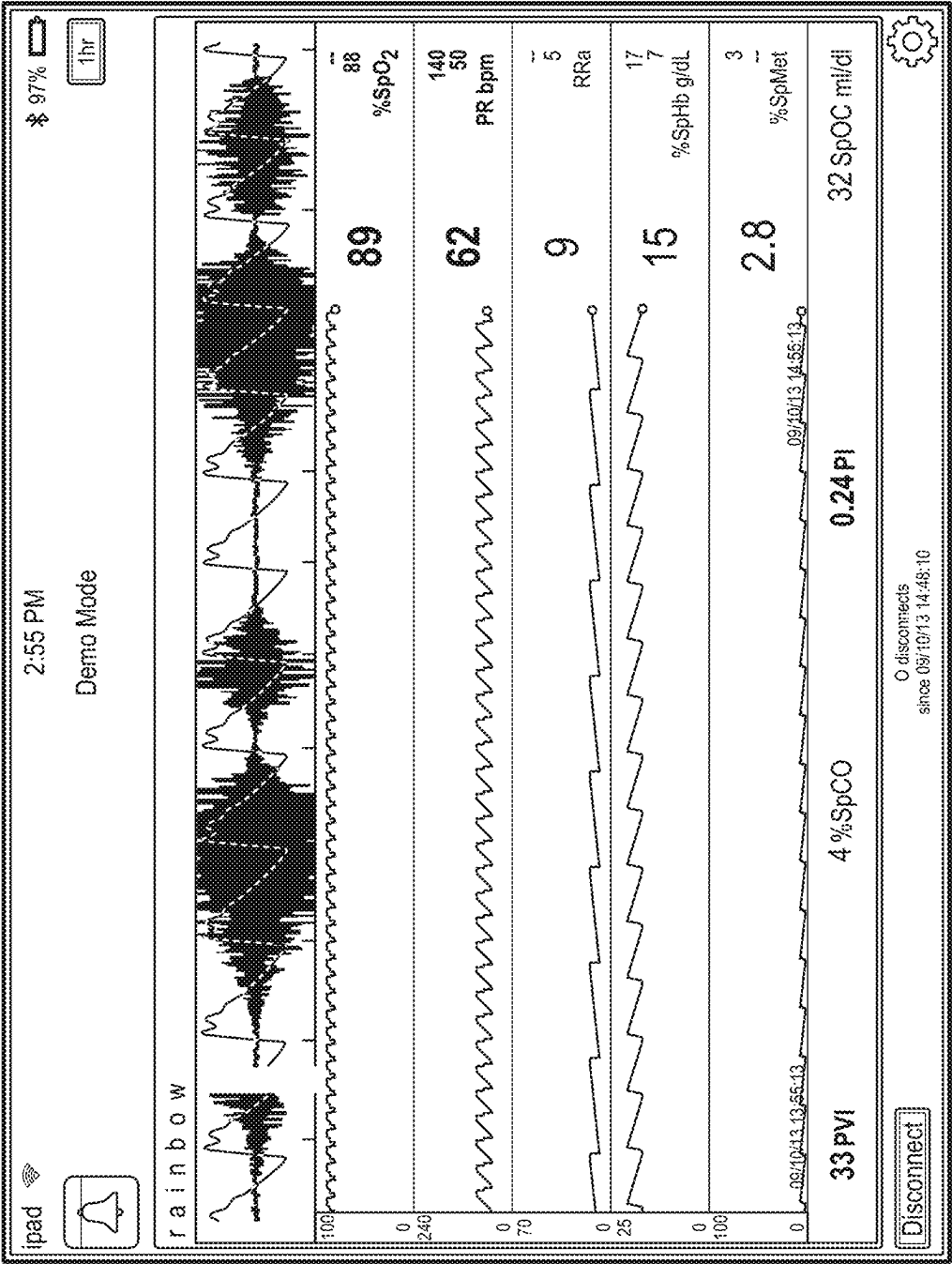
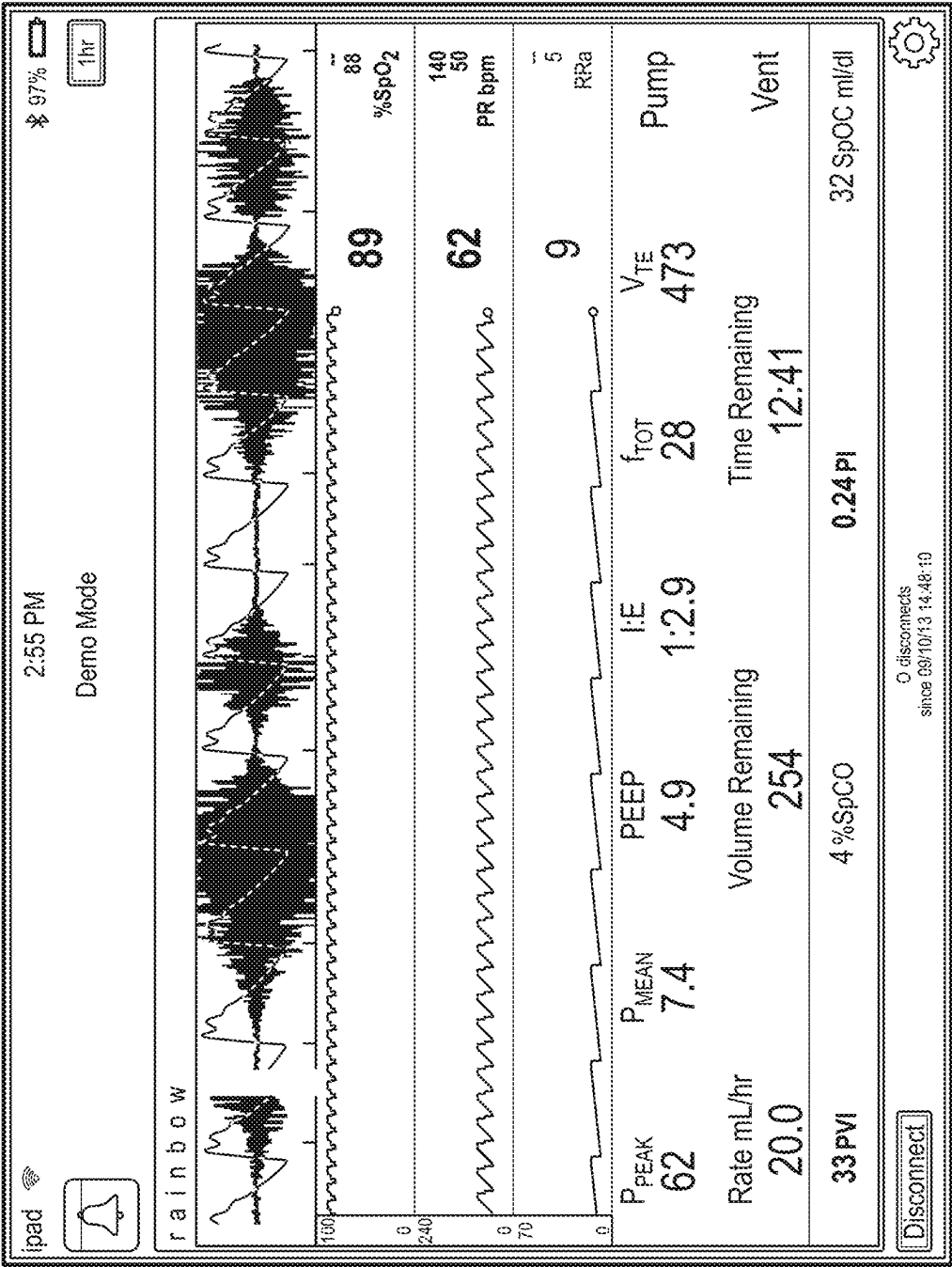


FIG. 37



2210

FIG. 38

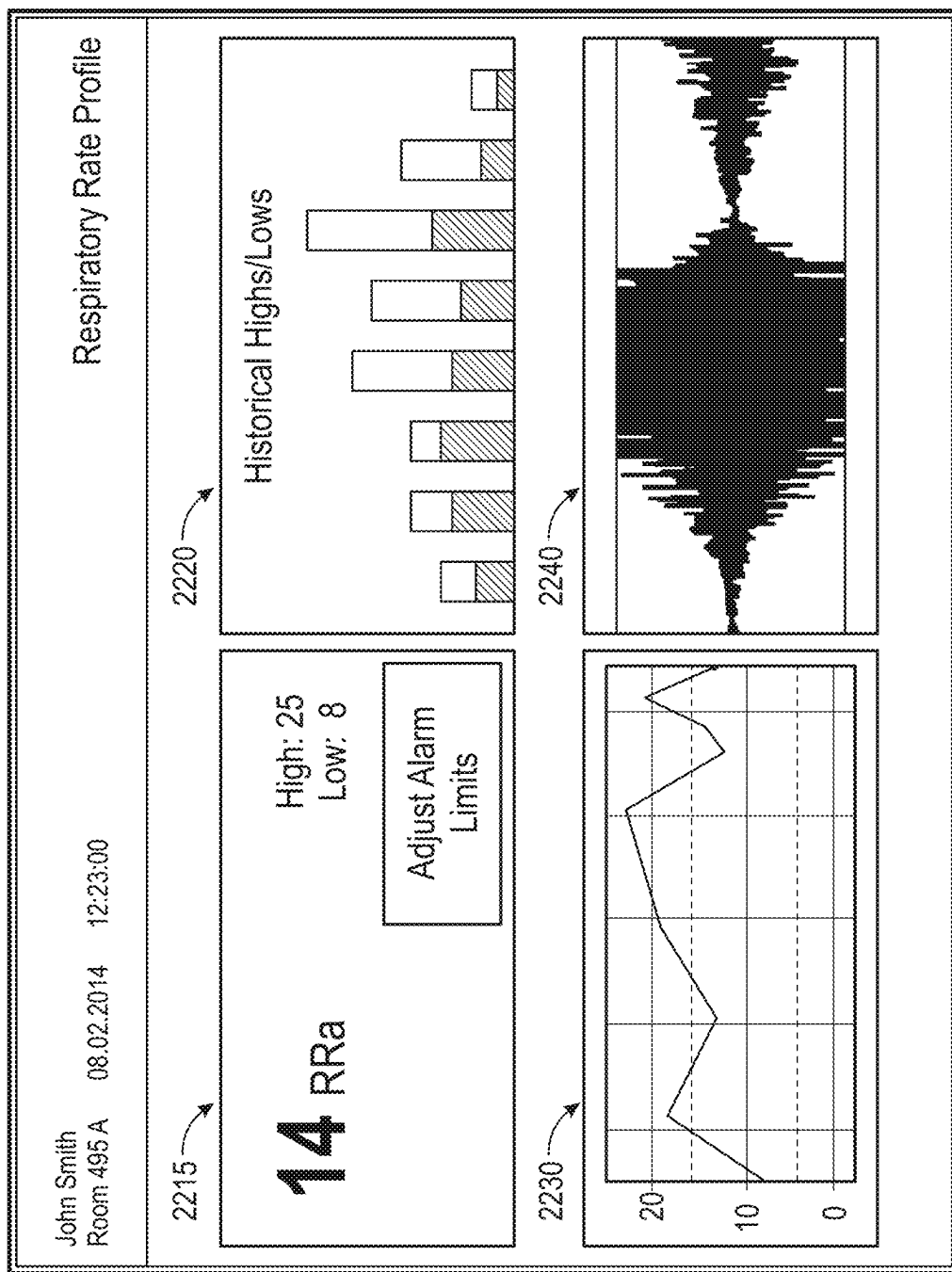


FIG. 39

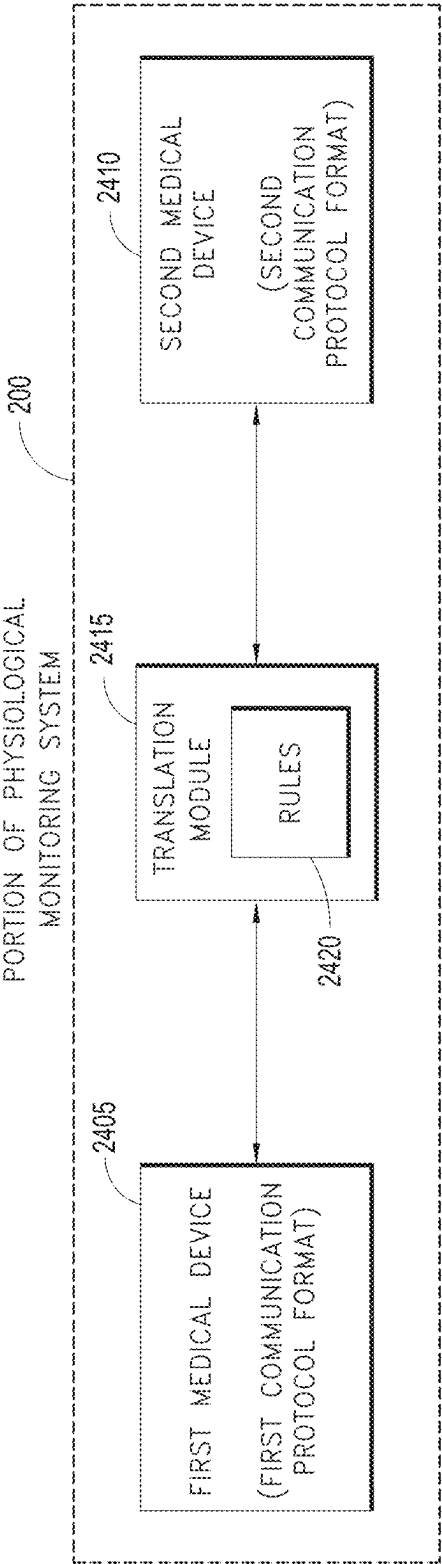


FIG. 40A

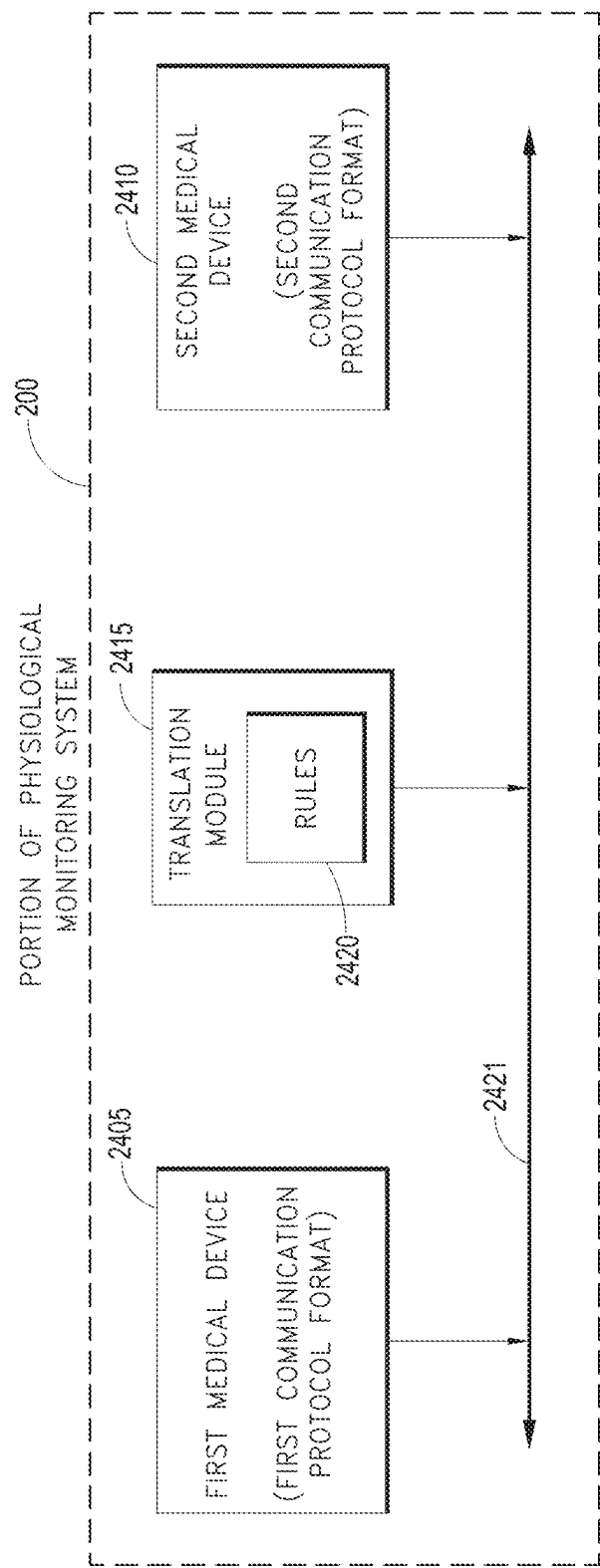


FIG. 40B

[illegible]

```
<?xml version="1.0" encoding="UTF-8" standalone="no" ?>
<ev7>
  <MSH>
    <MSH.1>^</MSH.1>
    <MSH.2>~ | \& ;</MSH.2>
    <MSH.3>VAFC PIMS</MSH.3>
    <MSH.4>50</MSH.4>
    <MSH.5>NPTF-508</MSH.5>
    <MSH.6>200</MSH.6>
    <MSH.7>20091120104609-0600</MSH.7>
    <MSH.9>
      <component n="1">ADT</component>
      <component n="2">ADT</component>
    <MSH.9>
    <MSH.10>58103</MSH.10>
    <MSH.11>P</MSH.11>
    <MSH.17>USA</MSH.17>
  </MSH>
  <EVN>
    <EVN.1>A01</EVN.1>
    <EVN.2>20091120104517-0600</EVN.2>
    <EVN.4>05</EVN.4>
  </EVN>
  ...
</ev7>
```

FIG. 41B

```
<?xml version="1.0" encoding="UTF-8"?>
<ADT_A01>
  <MSH>
    <MSH.1>^</MSH.1>
    <MSH.2>~ | \& ;</MSH.2>
    <MSH.3>
      <HD.1>VAFC PIMS</HD.1>
    </MSH.3>
    <MSH.4>
      <HD.1>50</HD.1>
    </MSH.4>
    <MSH.5>
      <HD.1>NPTF-508</HD.1>
    </MSH.5>
    <MSH.6>
      <HD.1>200</HD.1>
    </MSH.6>
    <MSH.7>
      <TS.1>20091120104609-0600</TS.1>
    </MSH.7>
    <MSH.9>
      <MSG.1>ADT</MSG.1>
      <MSG.2>A01</MSG.2>
    </MSH.9>
    <MSH.10>58103</MSH.10>
    <MSH.11>
      <PT.1>P</PT.1>
    </MSH.11>
    <MSH.17>USA</MSH.17>
  </MSH>
```

FIG. 41C


```
<output-message>
  <ACK>
    <MSH>
      <MSH.1>^</MSH.1>
      <MSH.2>~|\&lt;/MSH.2>
      <MSH.3>
        <HD.1>NPTF-508</HD.1>
      </MSH.3>
      <MSH.4>
        <HD.1>200</HD.1>
      </MSH.4>
      <MSH.5>
        <HD.1>VAFC PIMS</HD.1>
      </MSH.5>
      <MSH.6>
        <HD.1>50</HD.1>
      </MSH.6>
      <MSH.7>
        <TS.1>20091120104609-0600</TS.1>
      </MSH.7>
      <MSH.9>
        <MSG.1>ACK</MSG.1>
        <MSG.3>ACK</MSG.2>
      </MSH.9>
      <MSH.10>856bc9bd-97c8-4aa5-b411-3cd6fe2edd86</MSH.10>
    <MSH>
    <MSA>
      <MSA.1>AA</MSA.1>
      <MSA.2>58103</MSA.2>
    </MSA>
  </ACK>
</output-message>
```

FIG. 41D

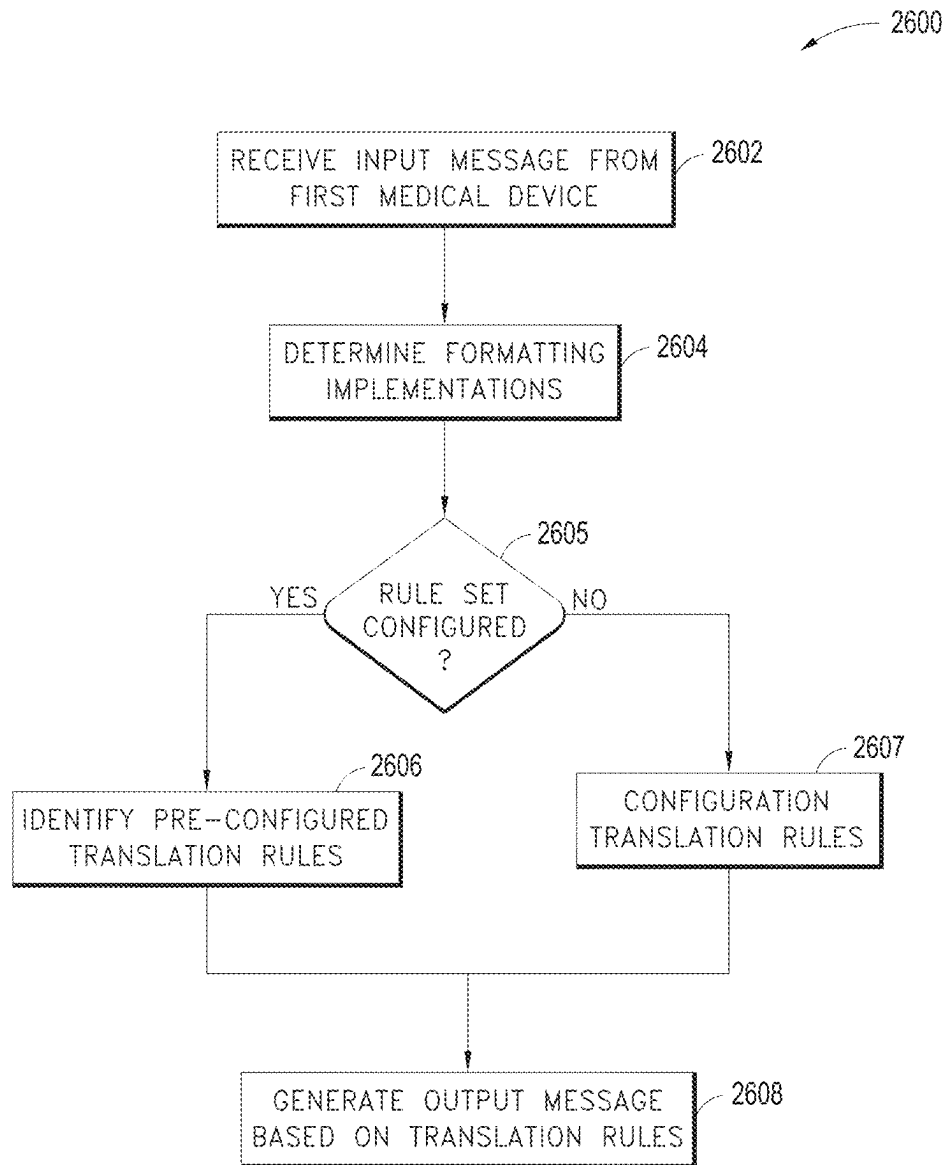


FIG. 42

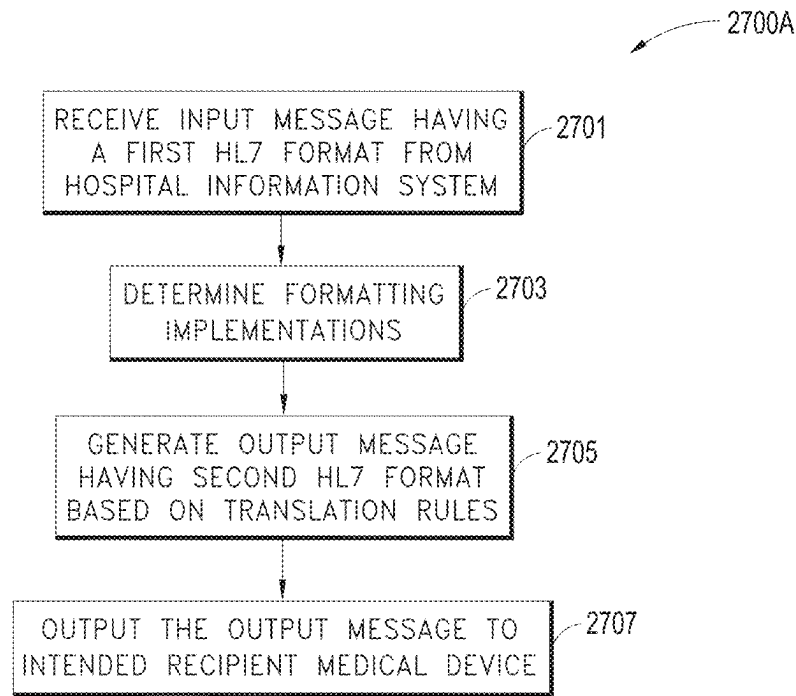


FIG. 43A

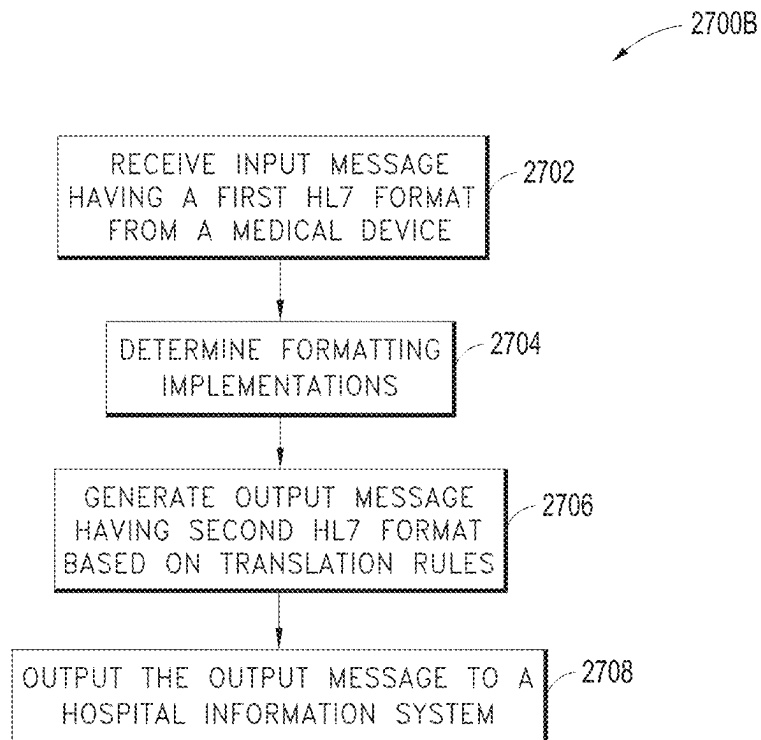


FIG. 43B

File Tools Maint Help

Message Definition Data Sources Display Reports Message Capture

Specification

Organization

HL7 Version

Status

Conformance Type

Msg Type

Event Description

Structure

Spec Version

Implement

Constrain

Extended

Event Type

ADT/ACK - Admit/Visit notification

MSH:{{SFT}}:EVN:PID:{{PID}}:{{ROL}}:{{NK1}}:PV1:{{PV2}}:{{ROL}}:{{DB1}}:{{OBX}}:{{AL1}}:{{DST}}:{{DRC}}:{{C10}}:{{PR}}

Role

Sender

Receiver

Implementation note

Encoding - ER7

Usage Ext

Conformance Profile

Follow HL7 Processing Rules

Query Message

Message Tree

Legend

Discrepancy Log

Set ID--PID

Patient ID

Patient Identifier List

Alternate Patient ID--PID

Patient Name

Mother's Maiden Name

Date/Time of Birth

Administrative Sex

Patient Alias

Race

Patient Address

County Code

Phone Number--Home

Phone Number--Business

Primary Language

Marital Status

Religion

Patient Account Number

SSN Number--Patient

Driver's License Number--Patient

Mother's Identifier

Ethnic Group

Birth Place

Multicode Birth Indicator

Element Parameters

MSH

Message Header

Optional Predicate

Segment

Required

Repetable

Cardinality

Position

In Use

Data Type

Length

Table

Data Link

Example Value

Reference

Implementation Notes

Local Element definition

Fixed Value

Page:

2800

FIG. 44

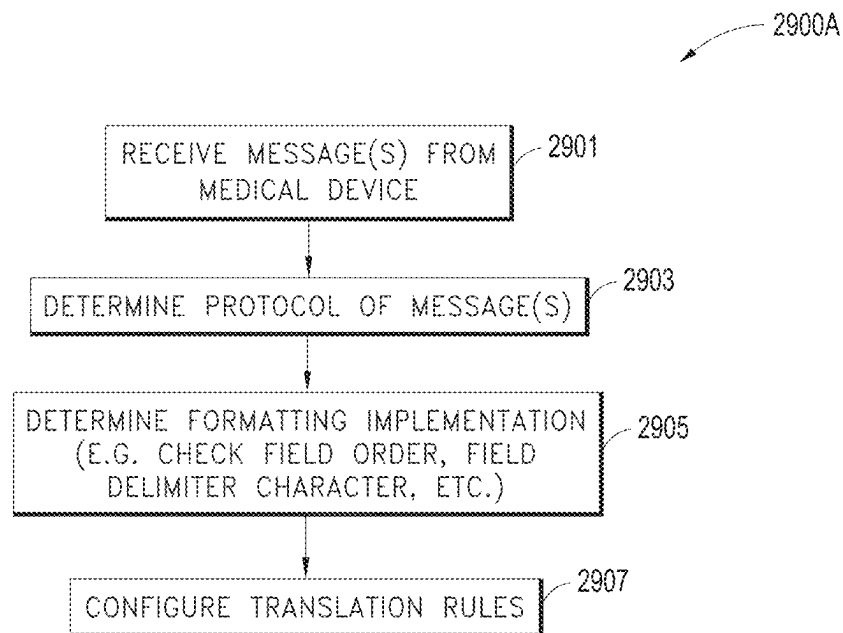


FIG. 45A

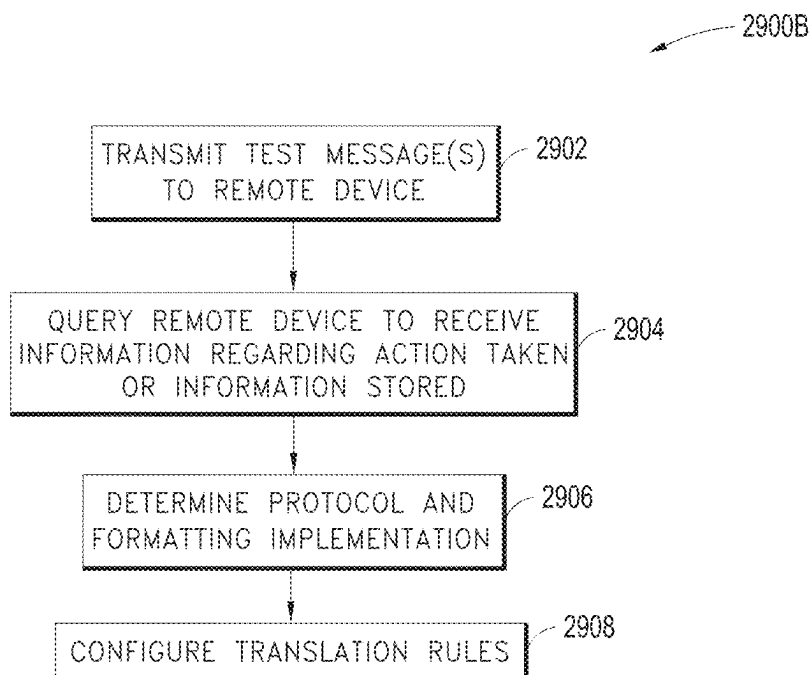


FIG. 45B

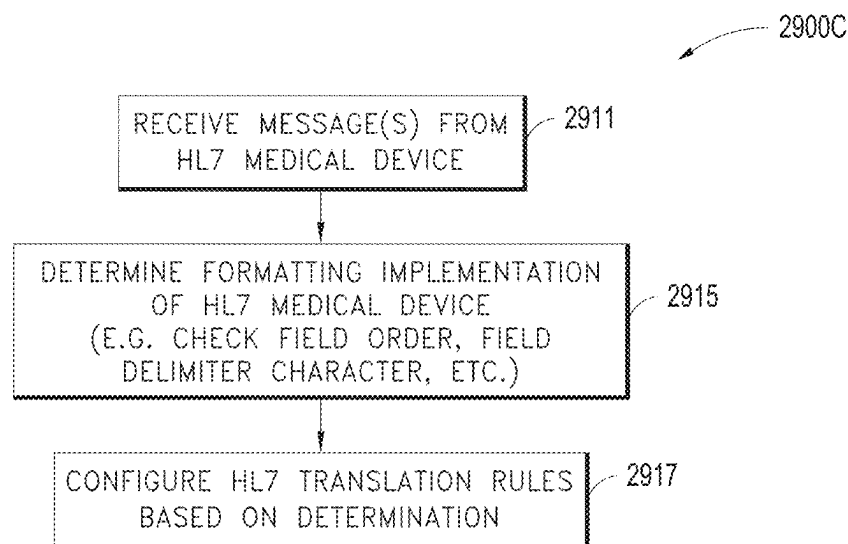


FIG. 45C

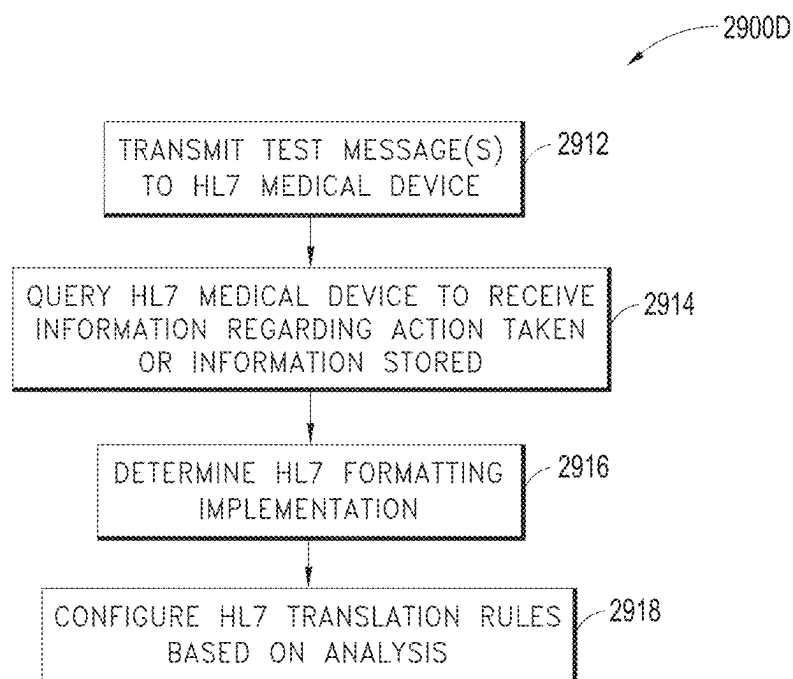


FIG. 45D

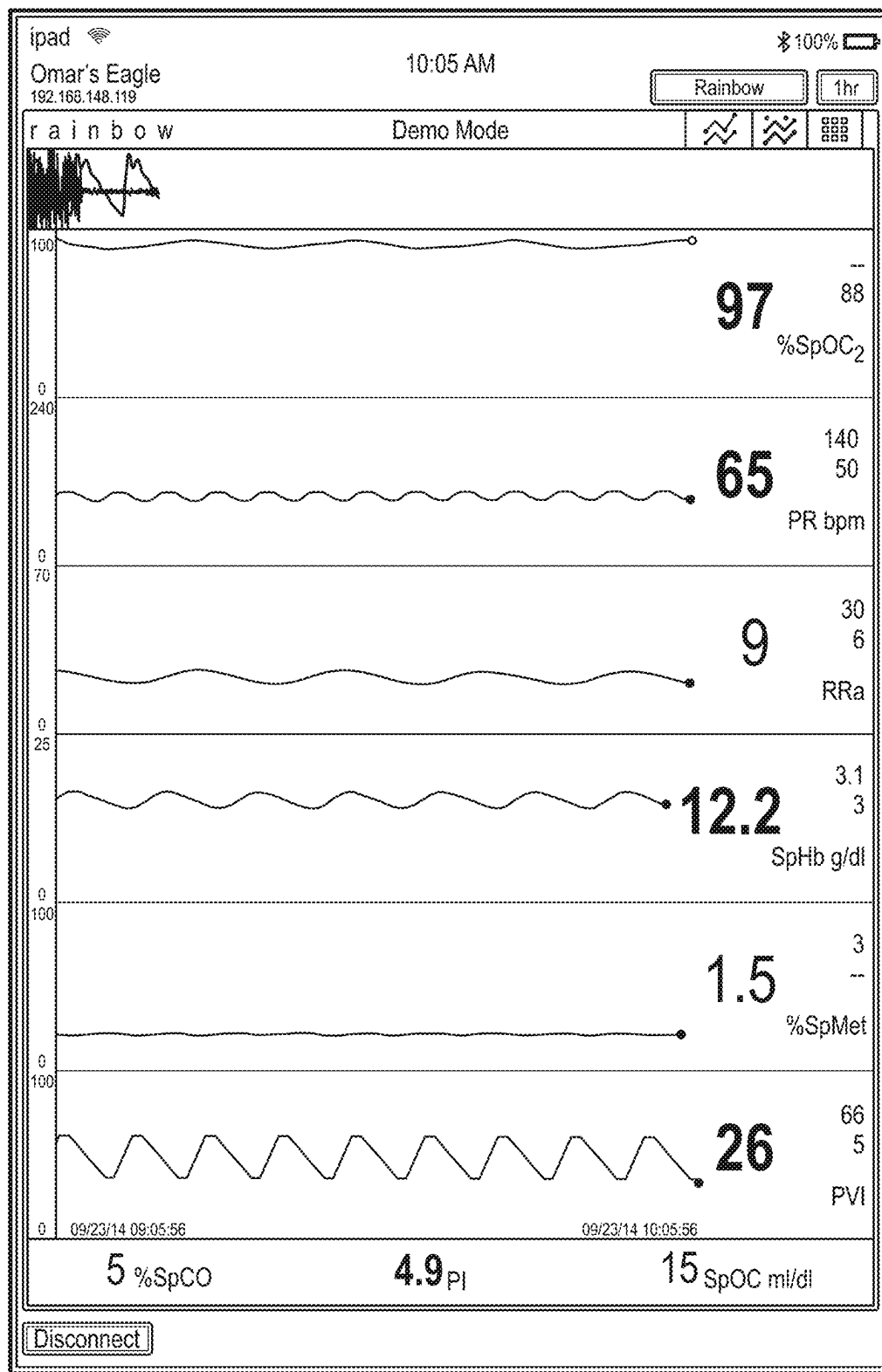


FIG. 46

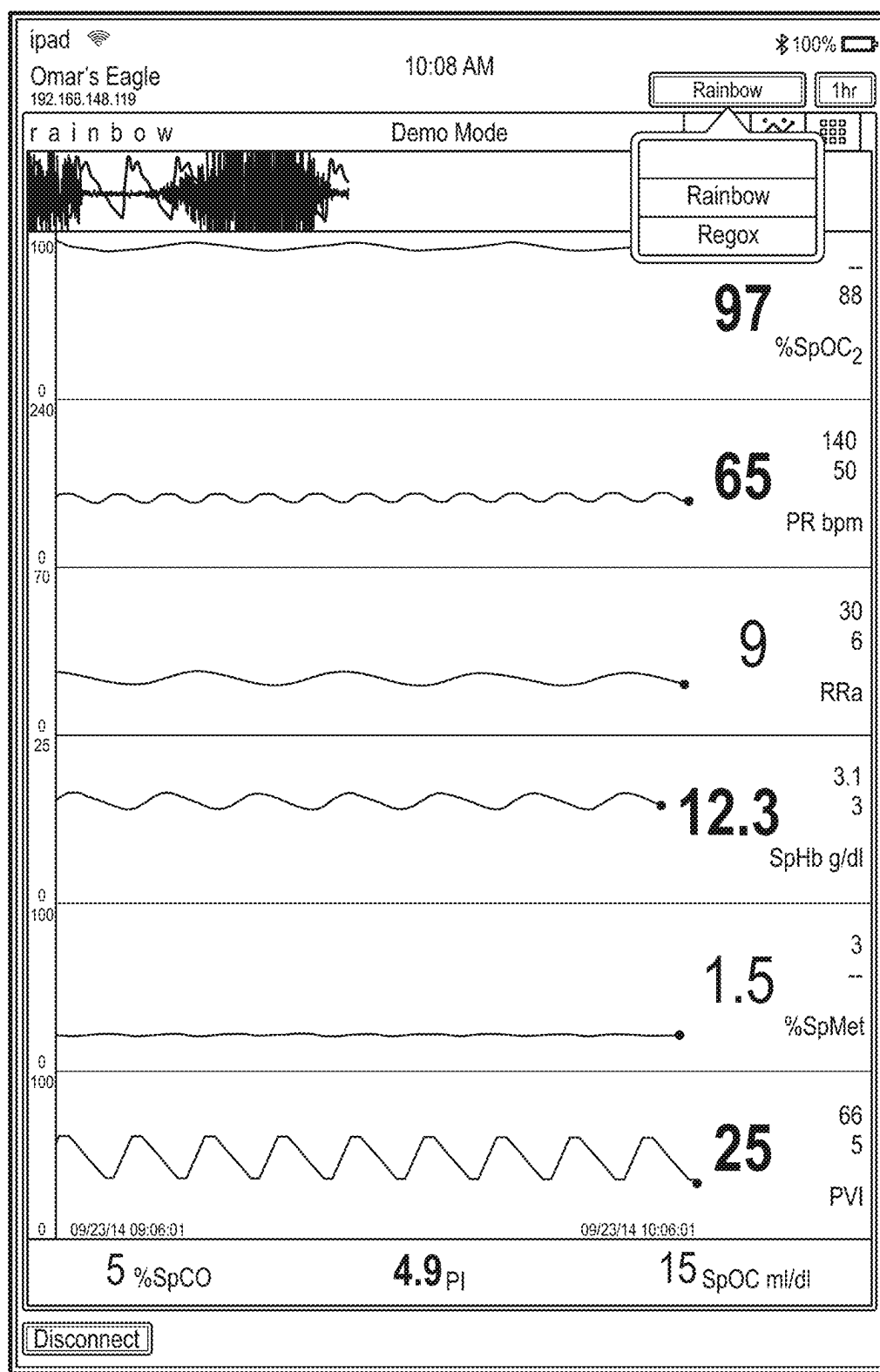


FIG. 47

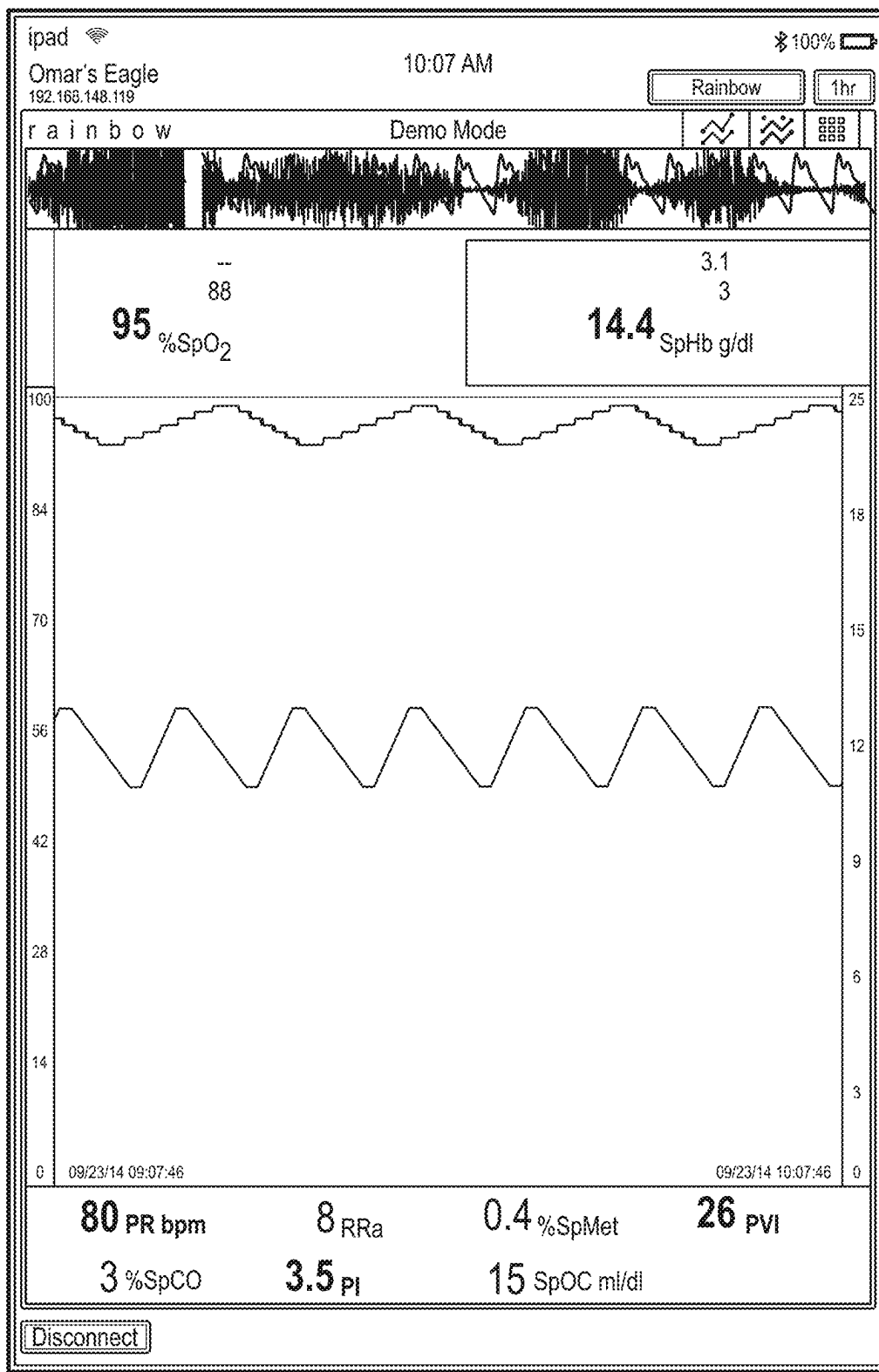


FIG. 48

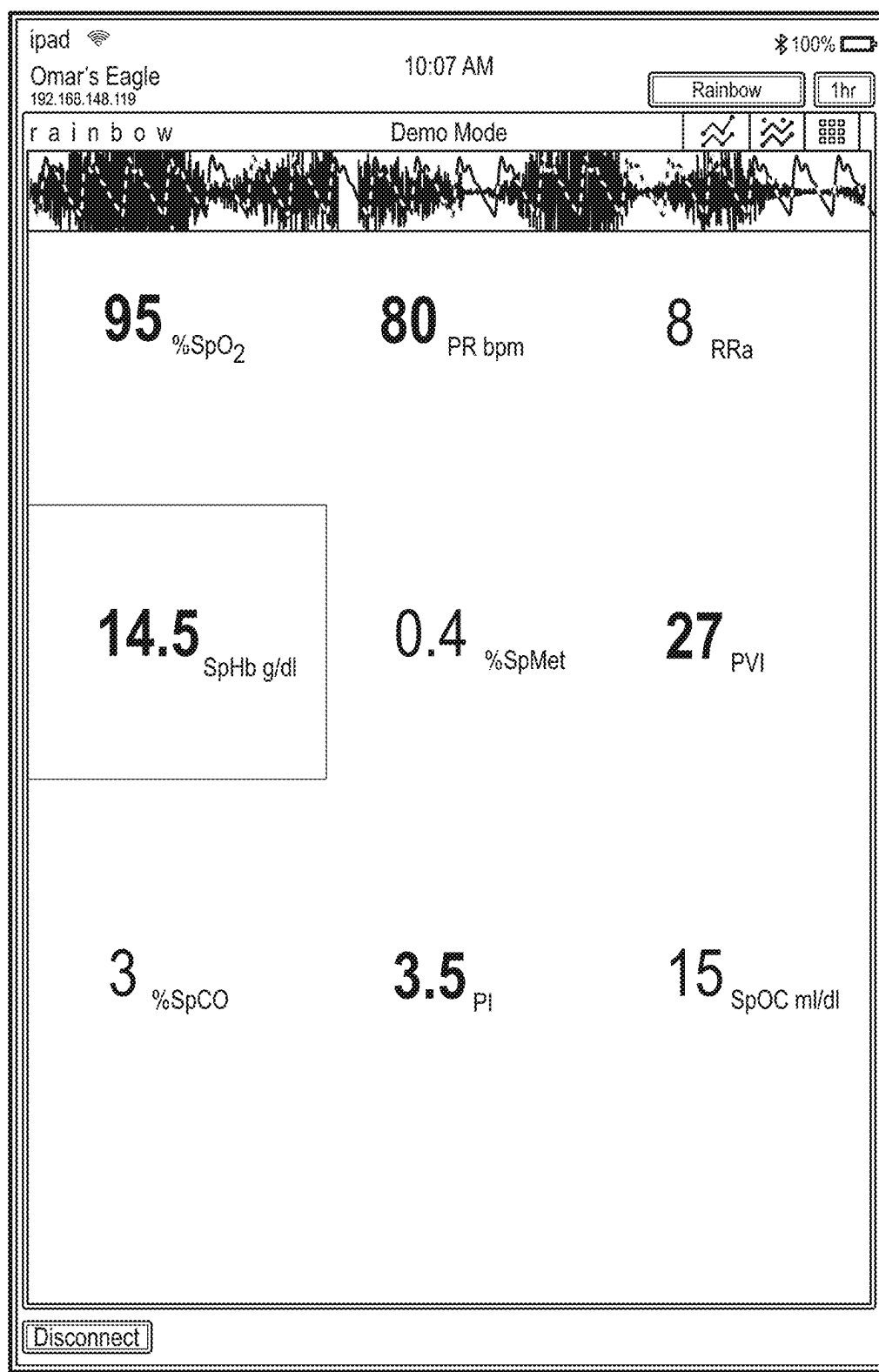


FIG. 49

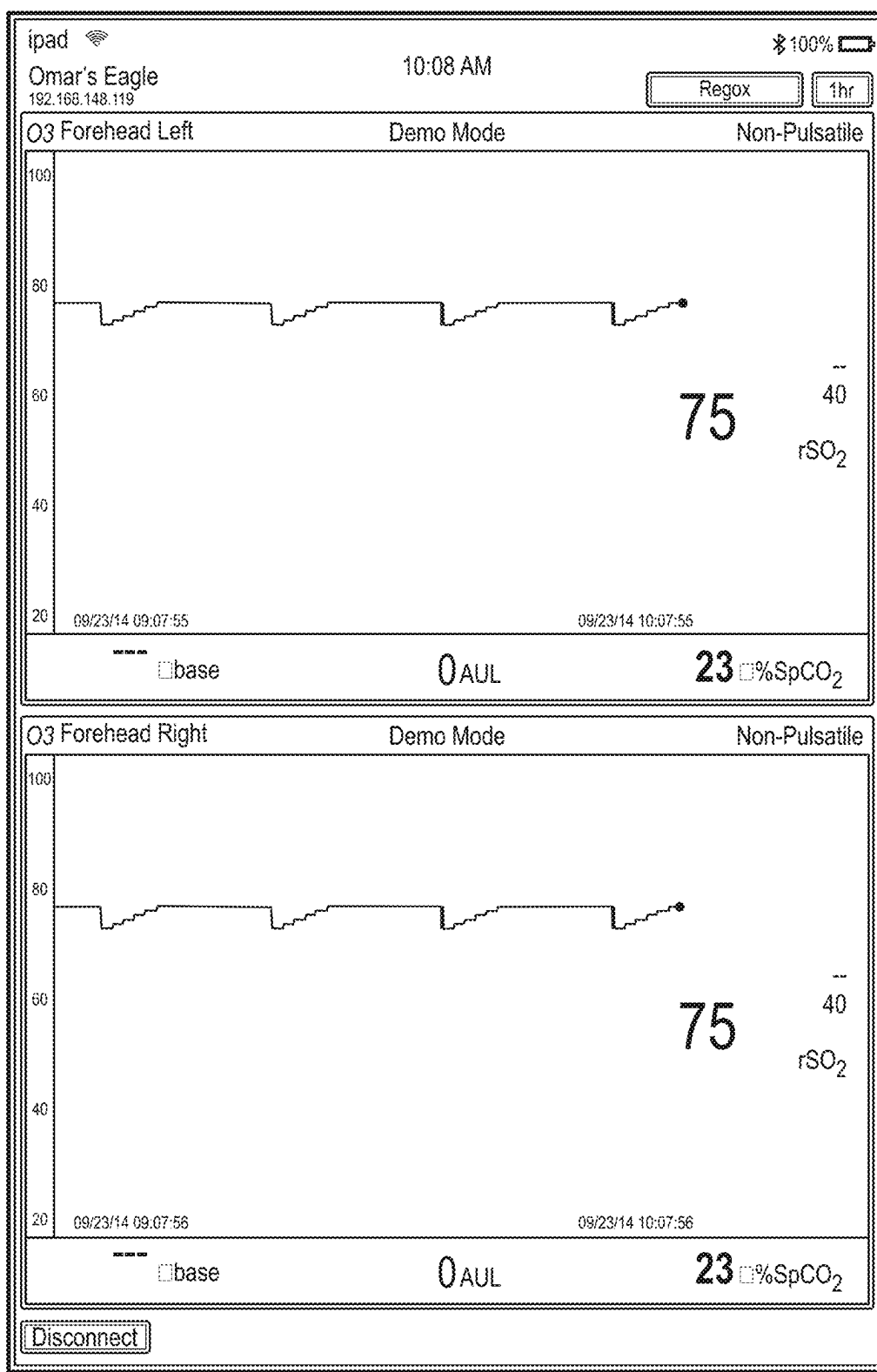


FIG. 50

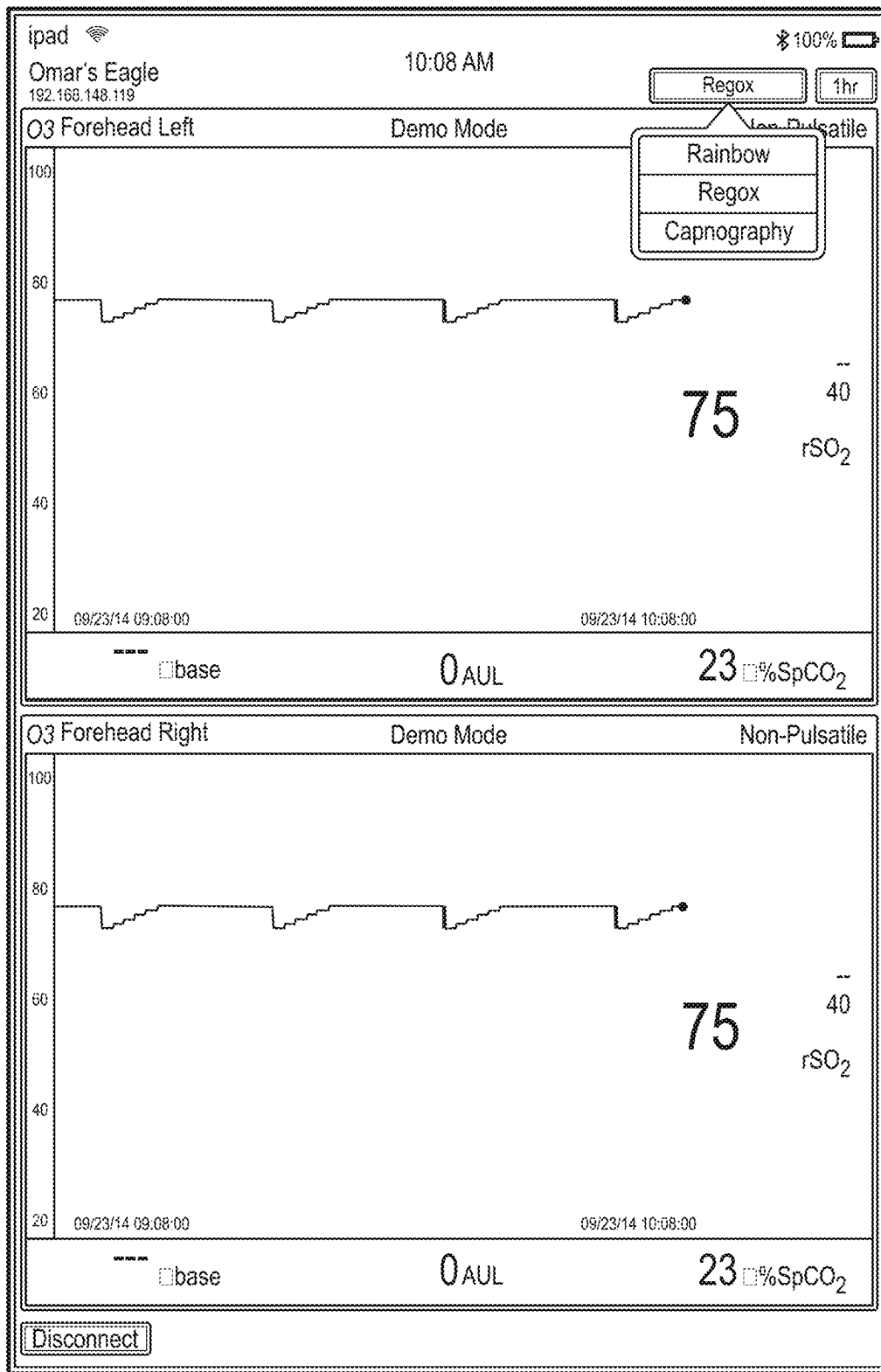


FIG. 51

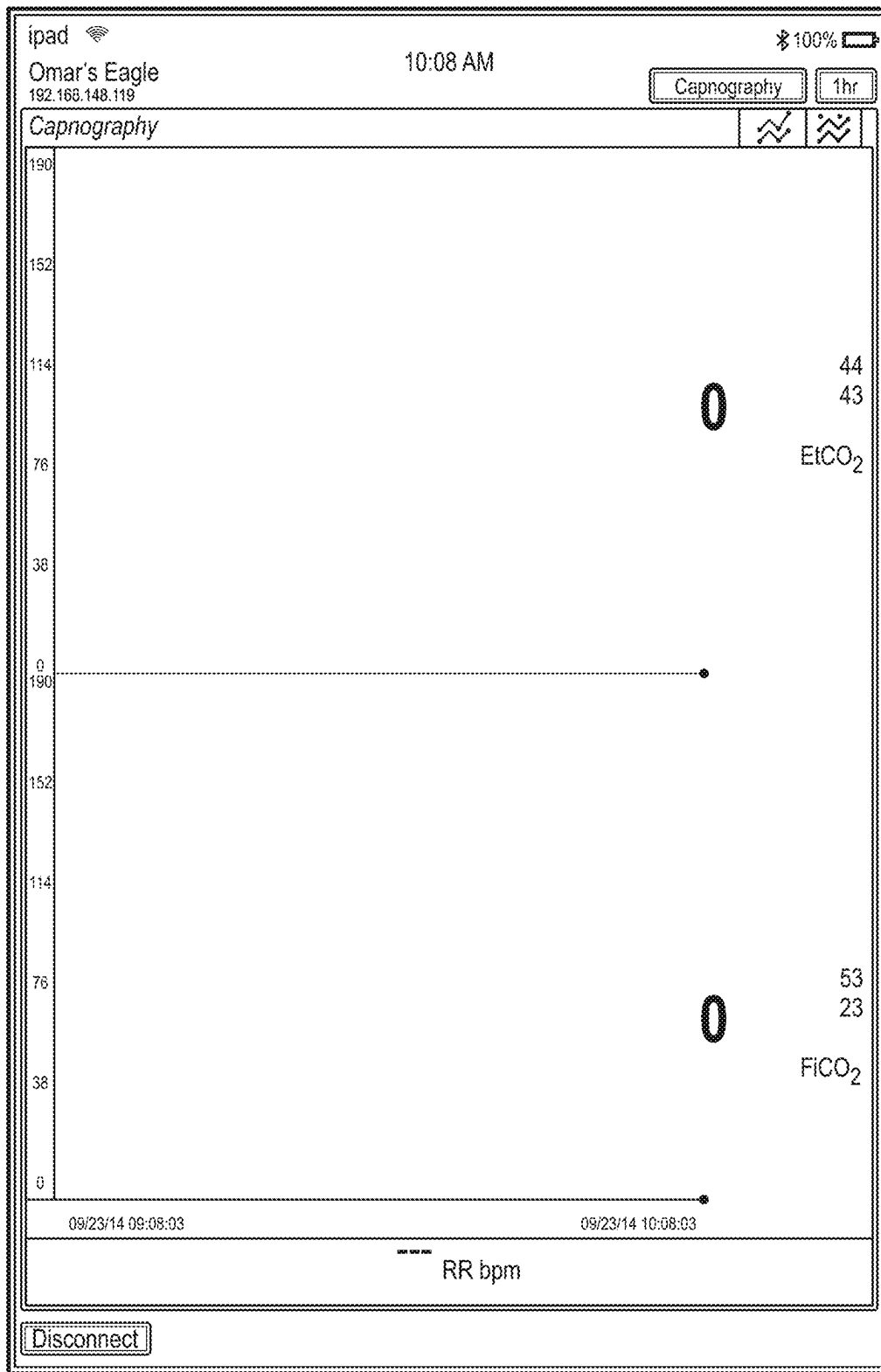


FIG. 52

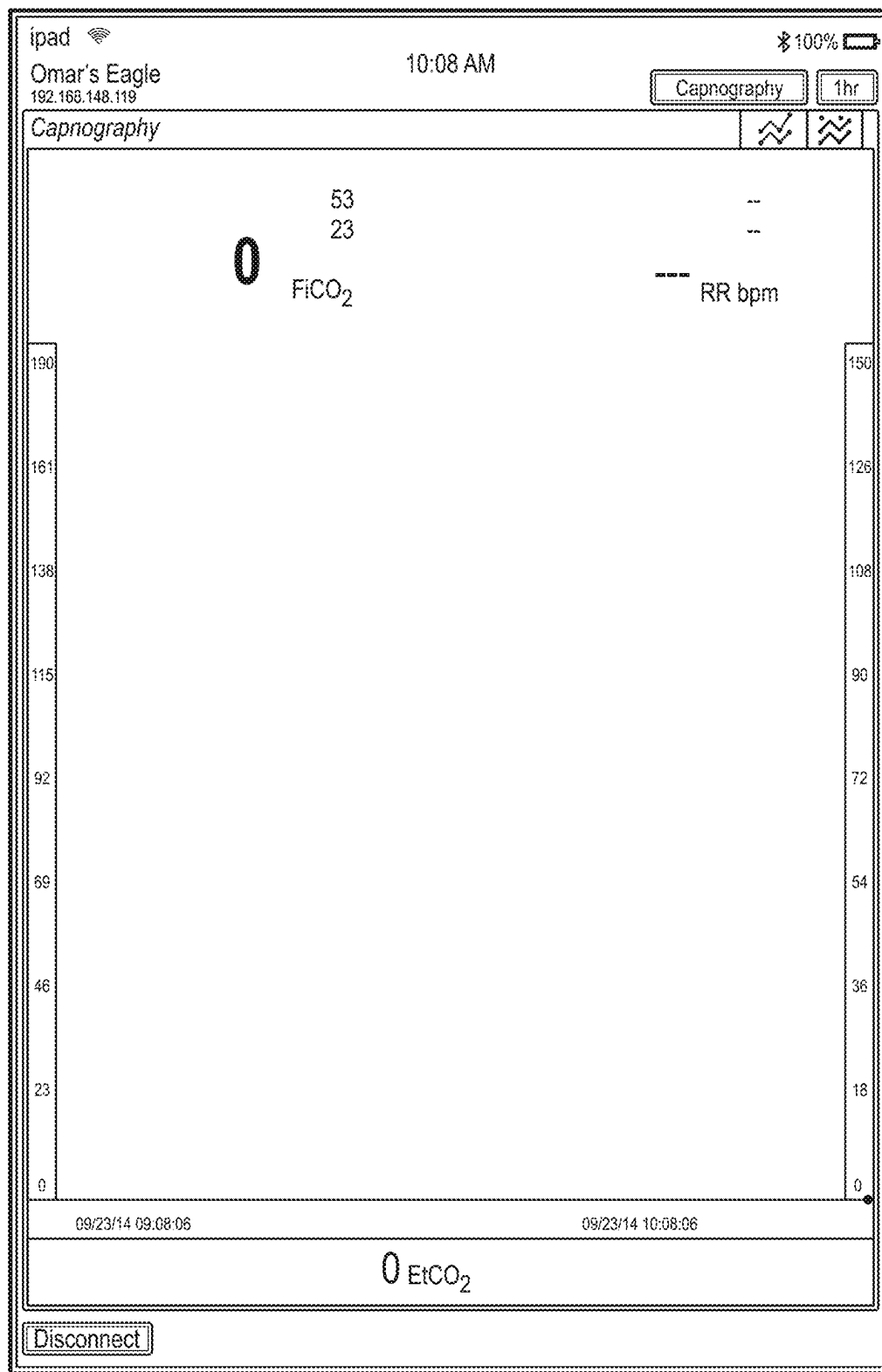


FIG. 53

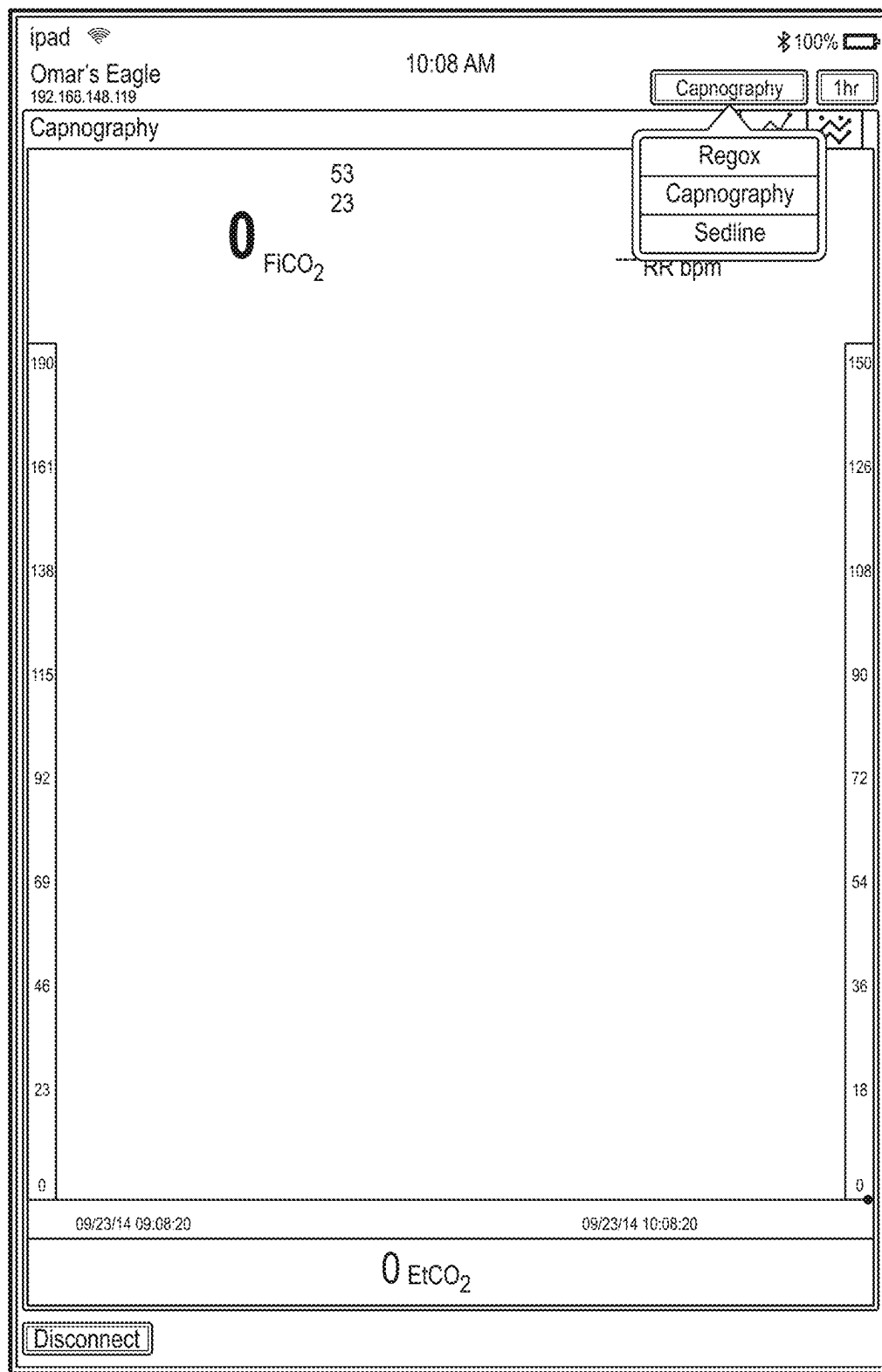


FIG. 54

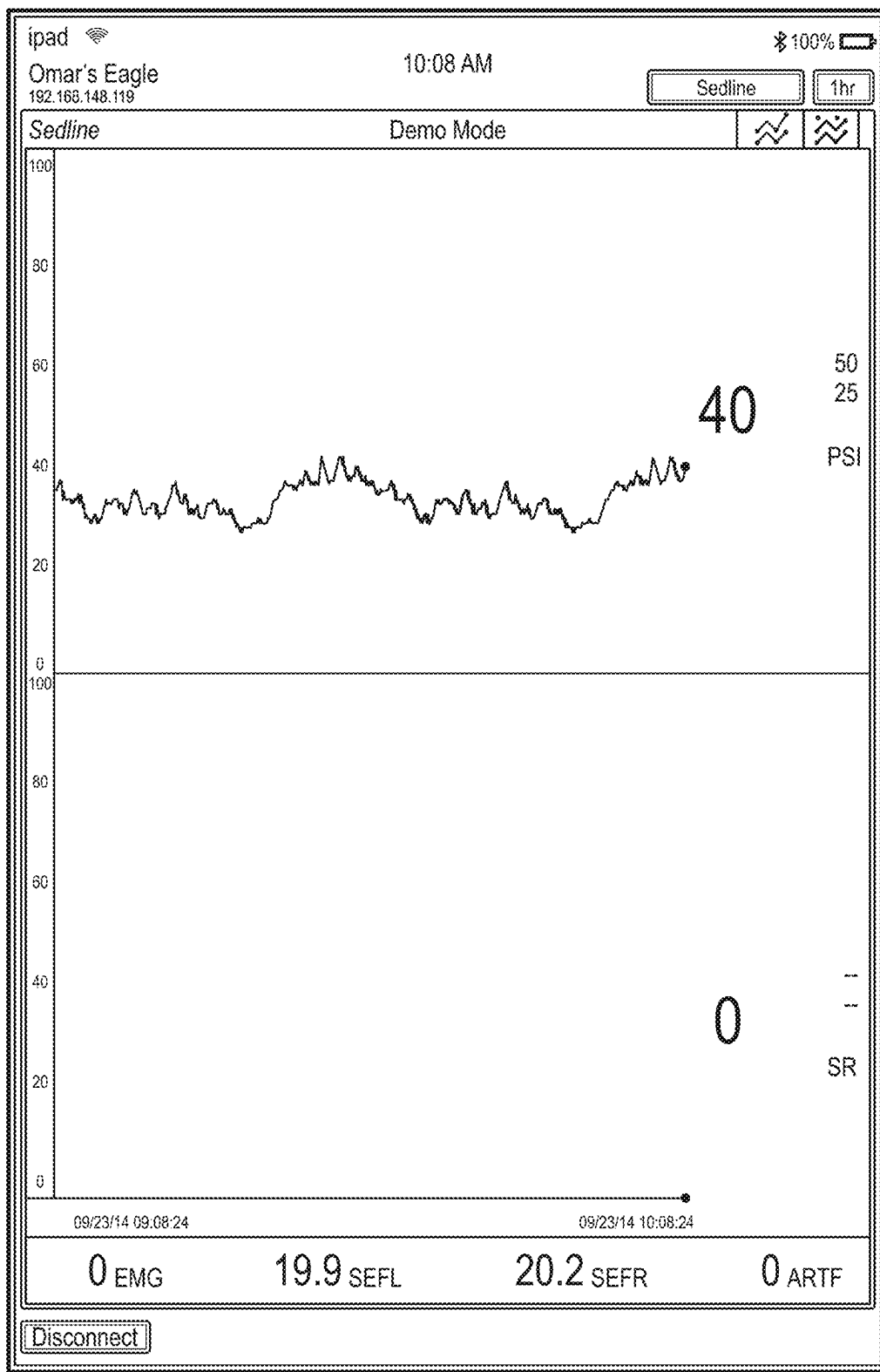


FIG. 55

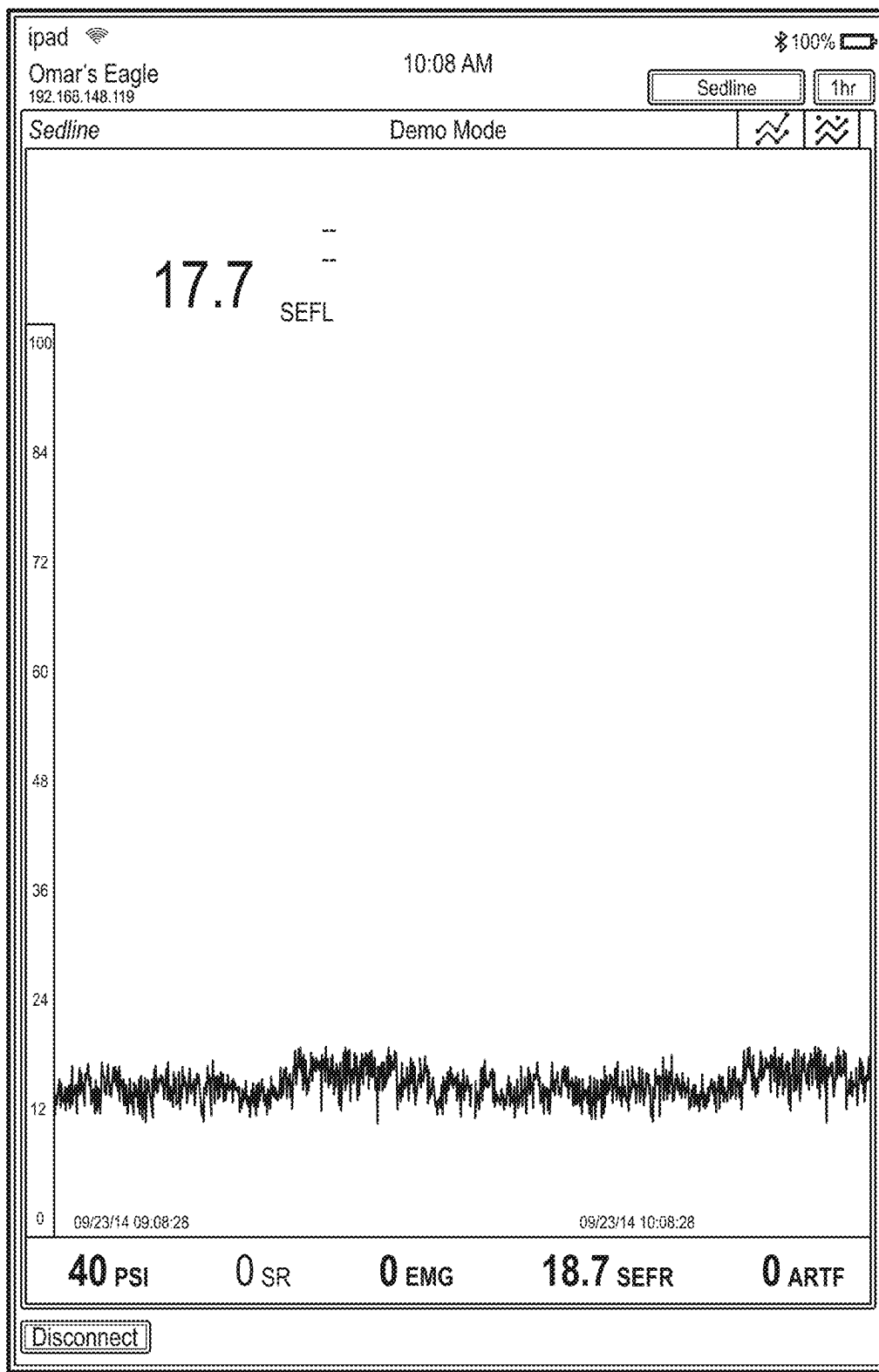


FIG. 56

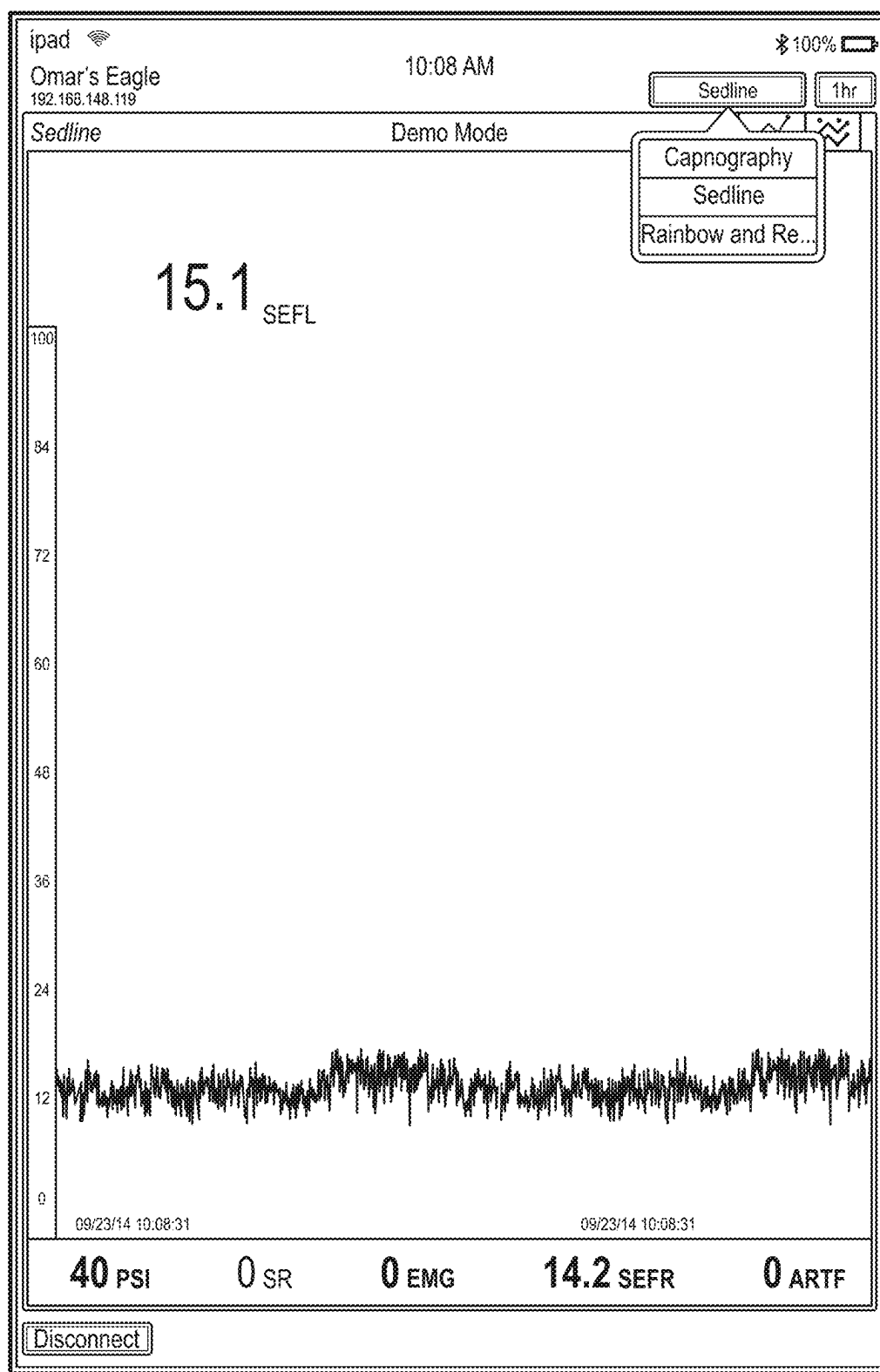


FIG. 57

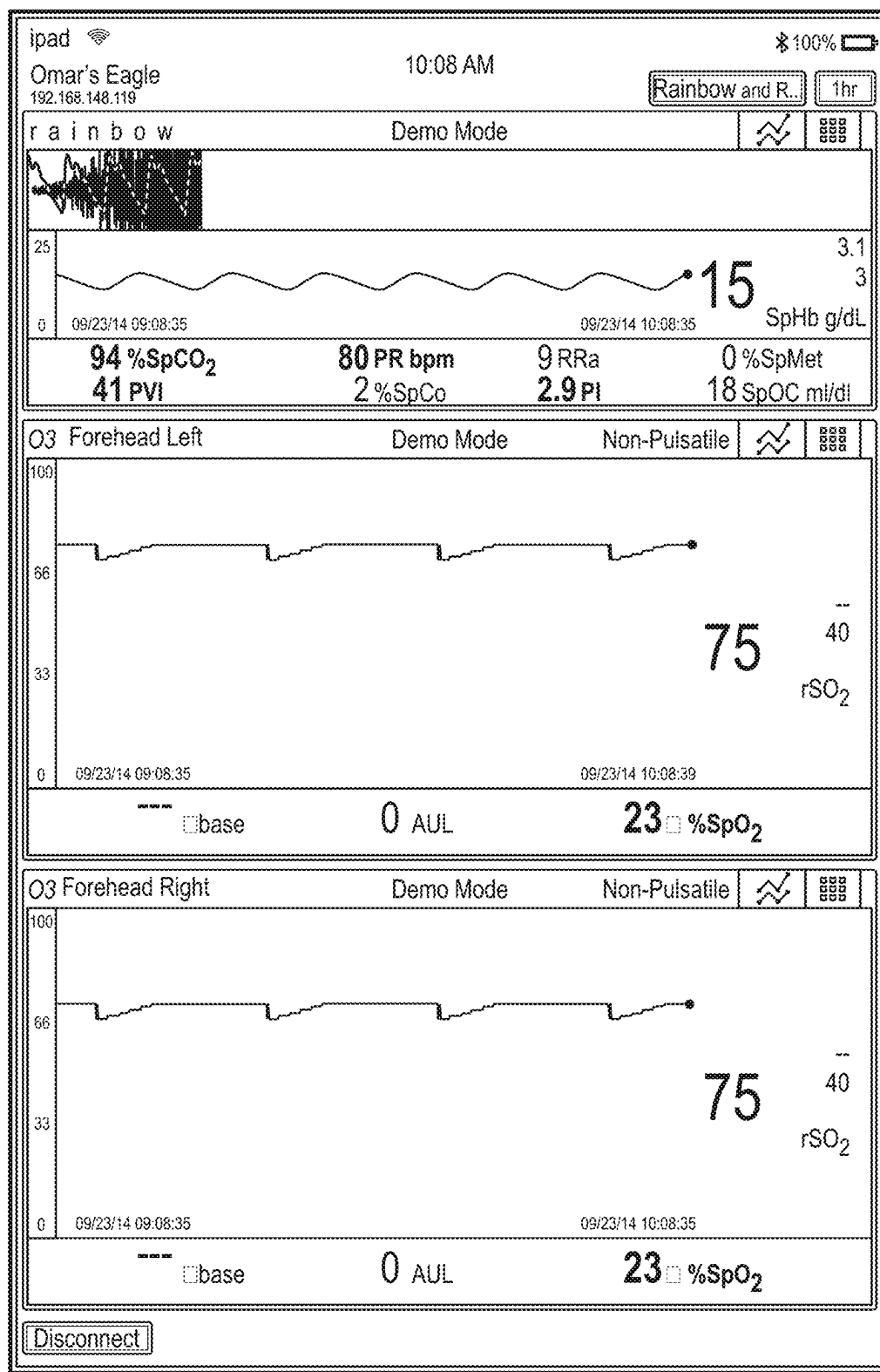


FIG. 58

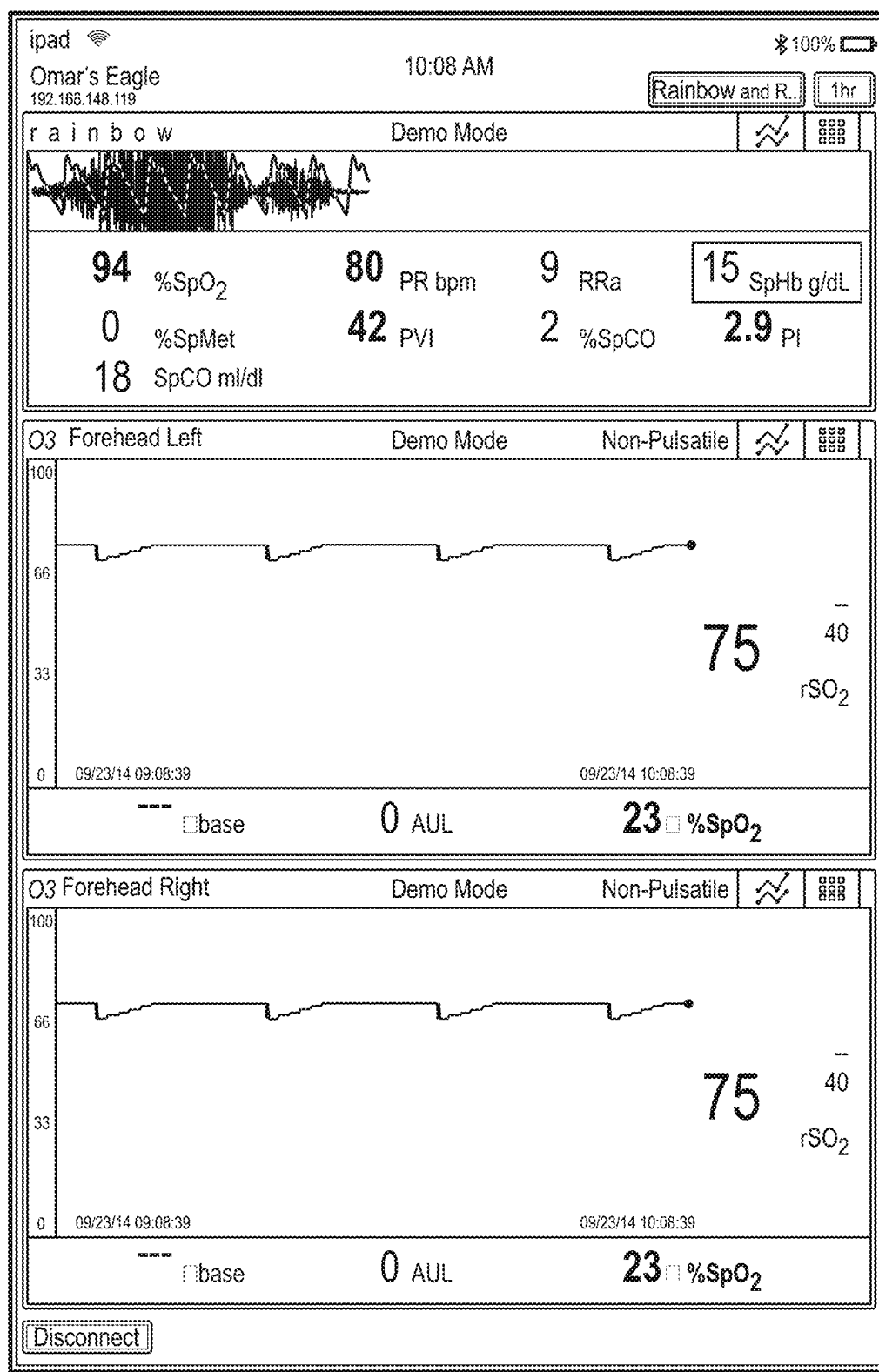


FIG. 59

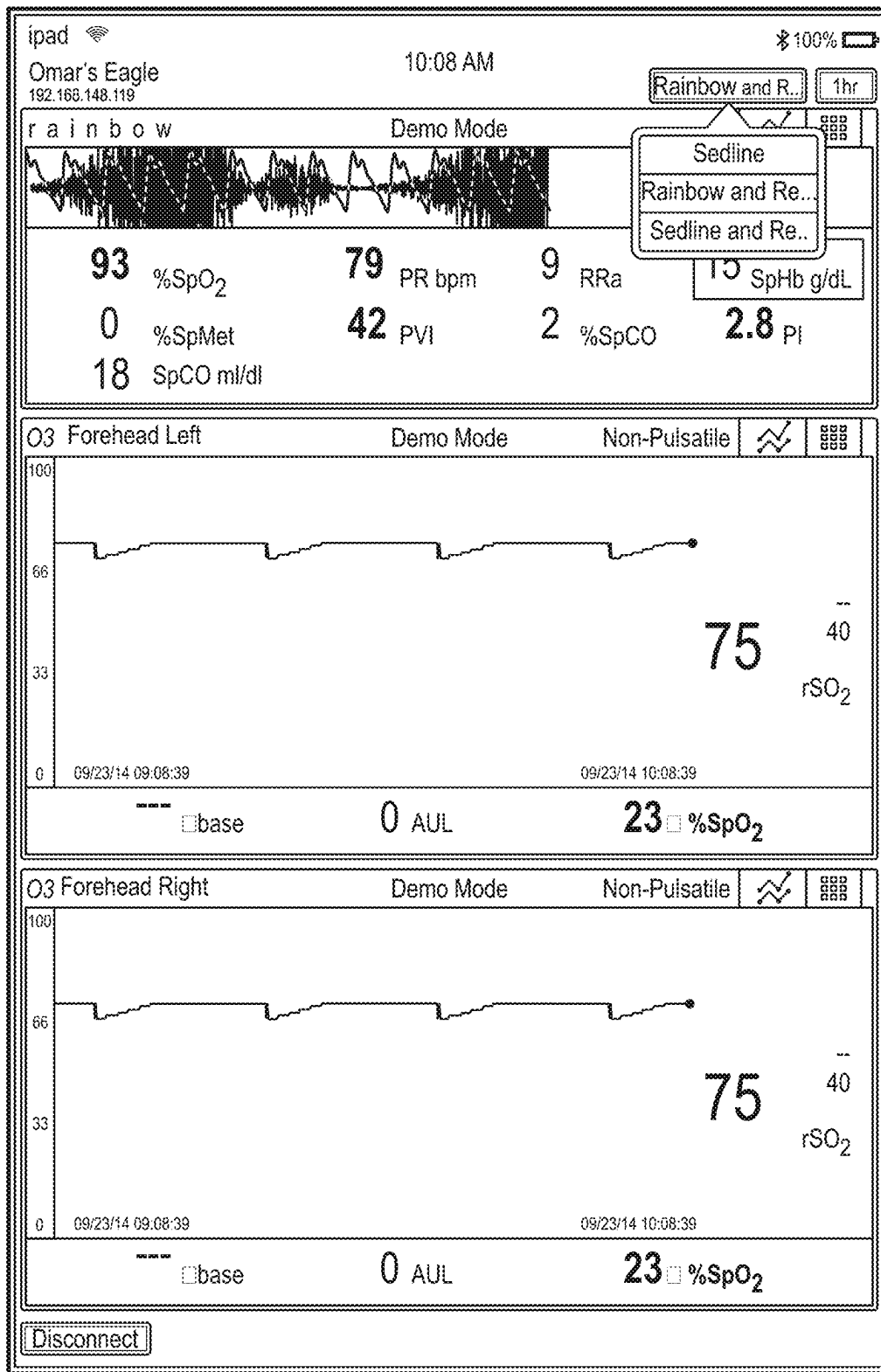


FIG. 60

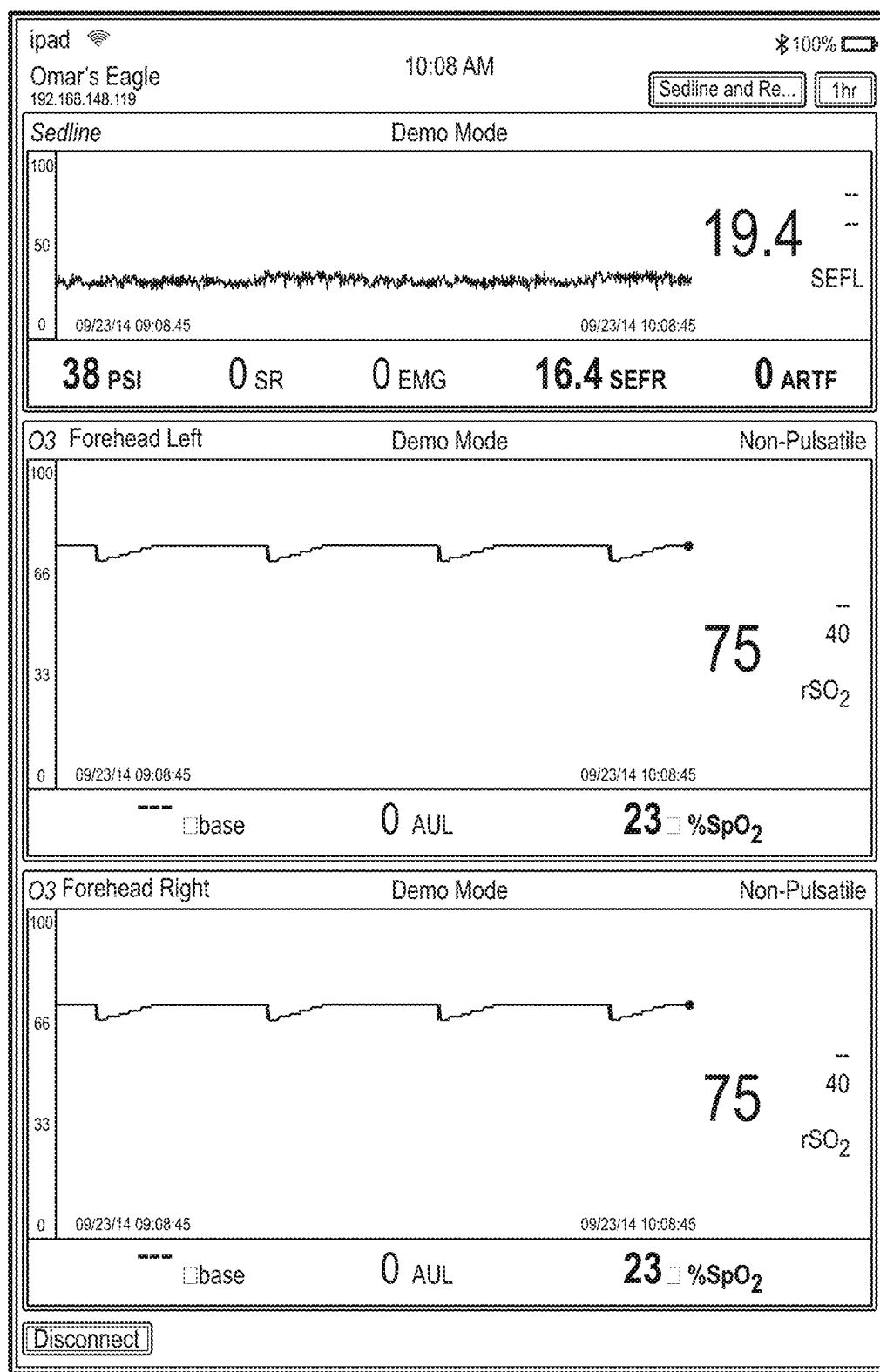


FIG. 61

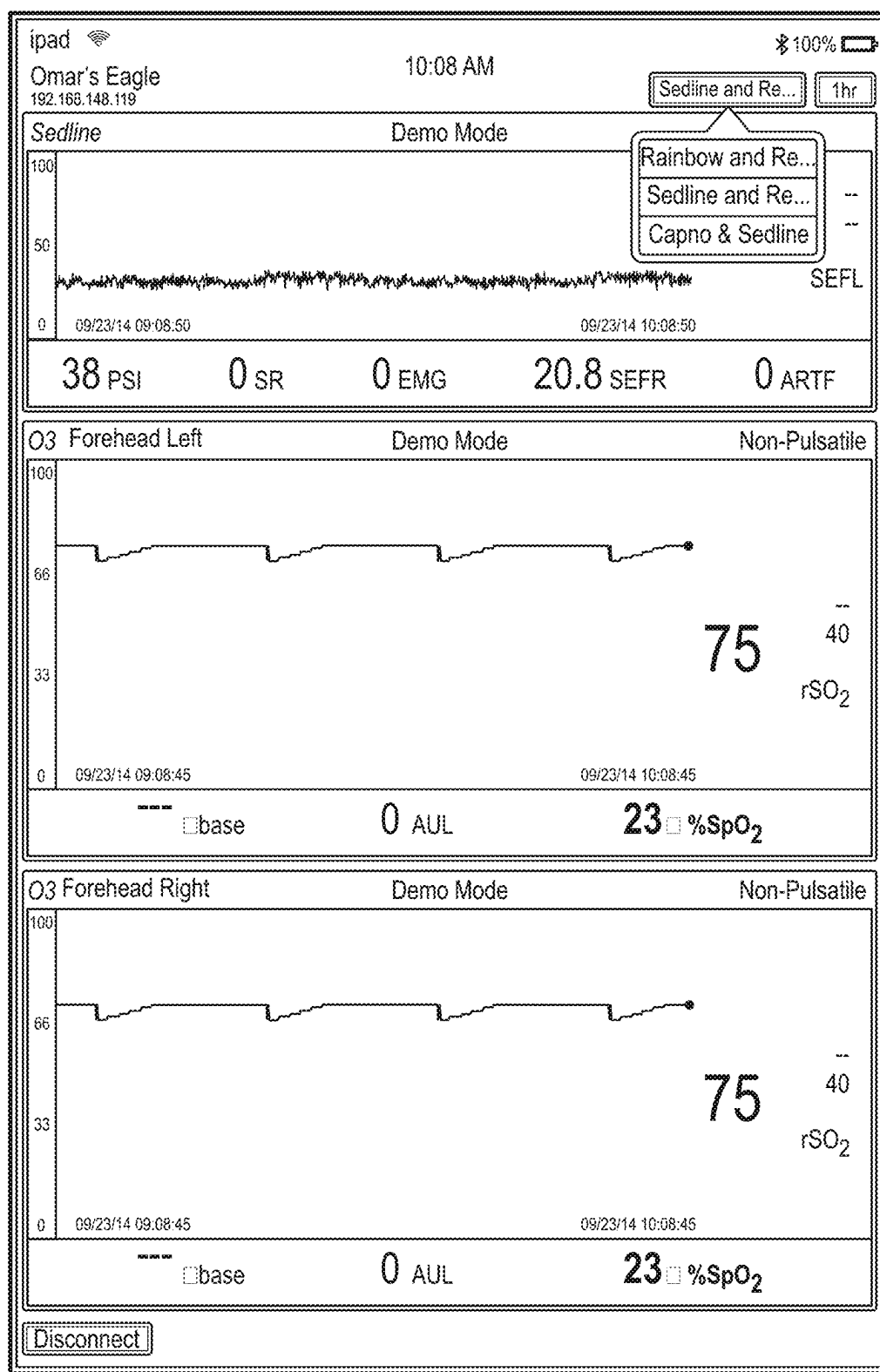


FIG. 62

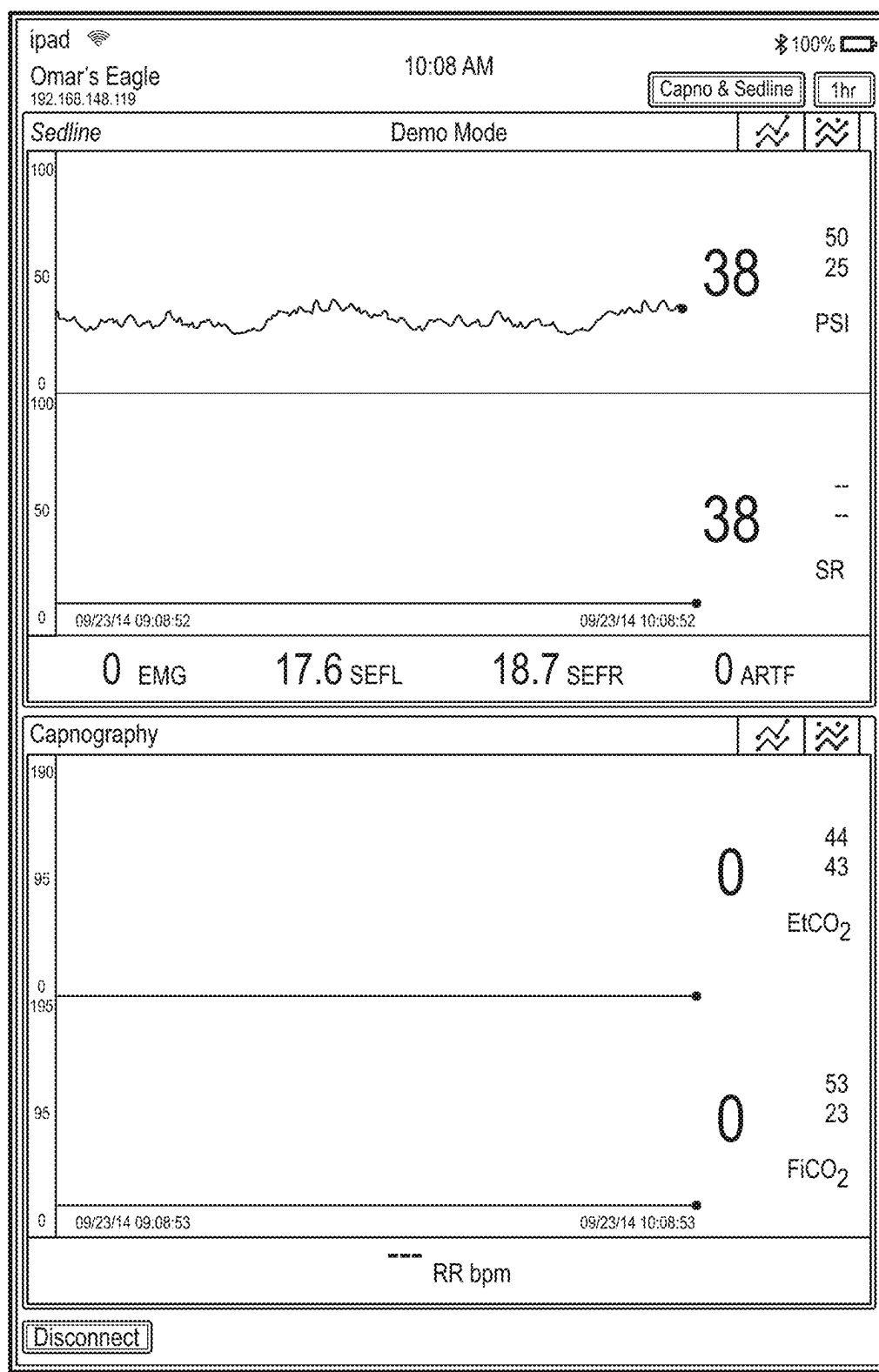


FIG. 63

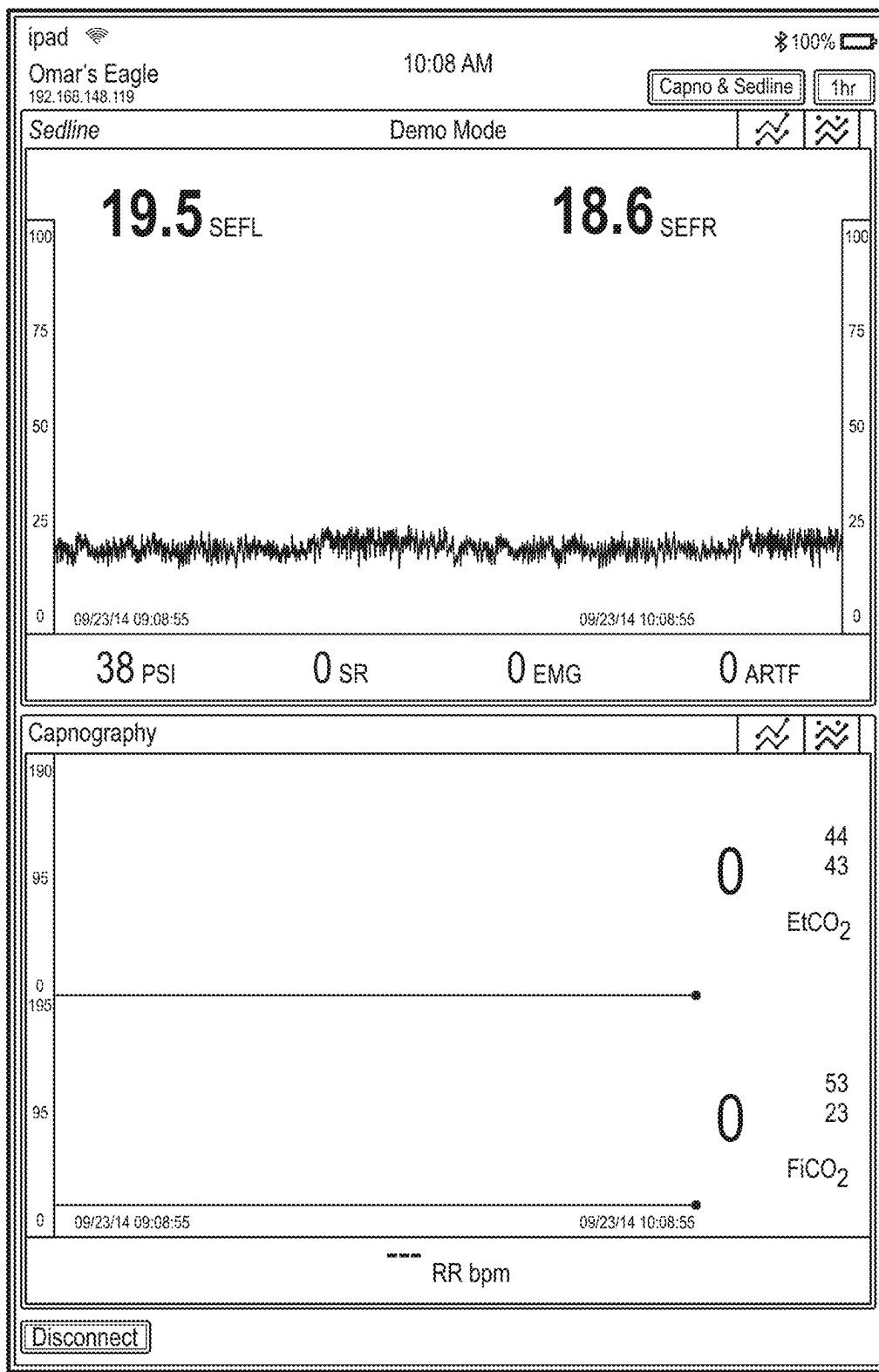


FIG. 64

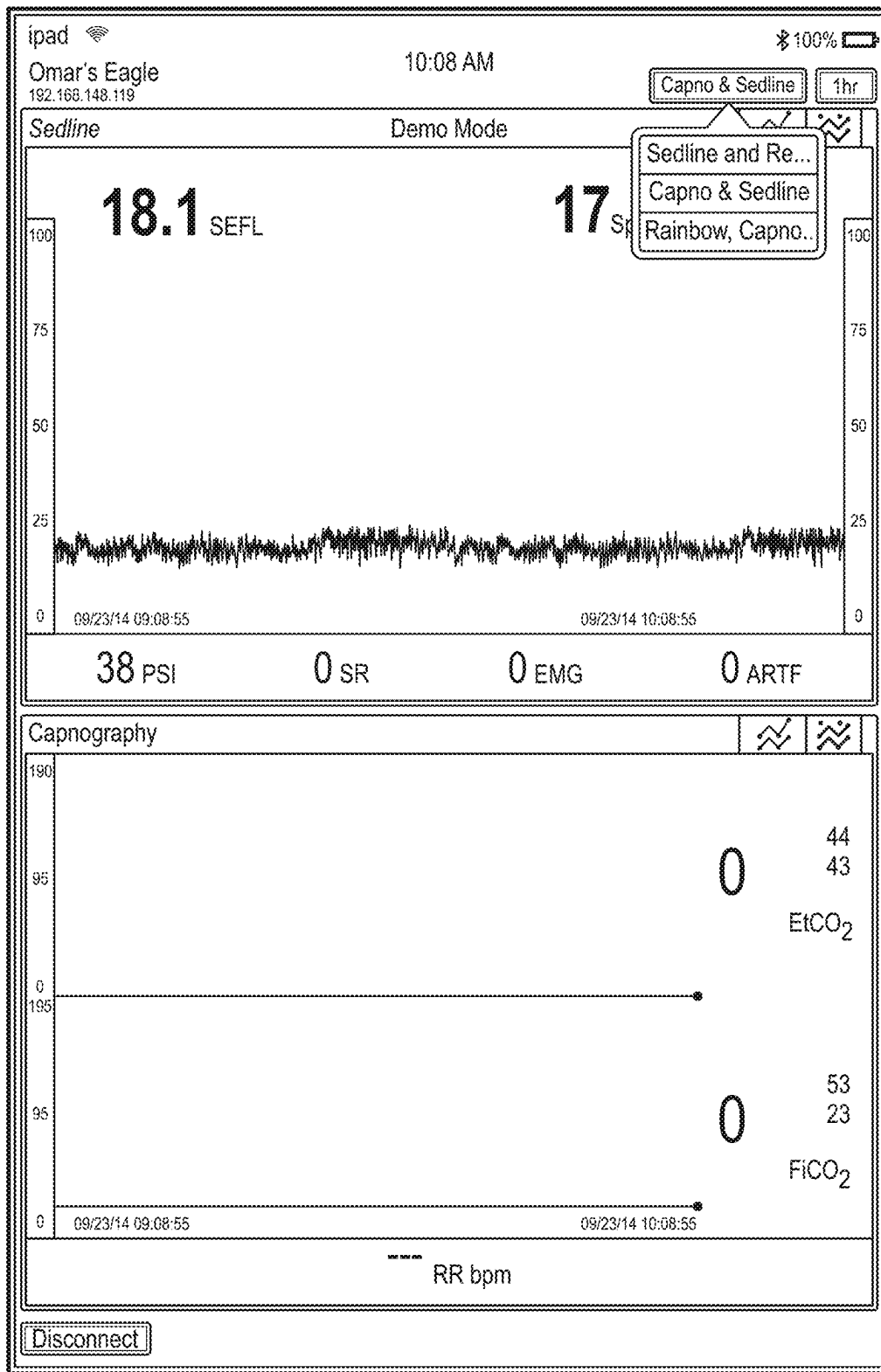


FIG. 65

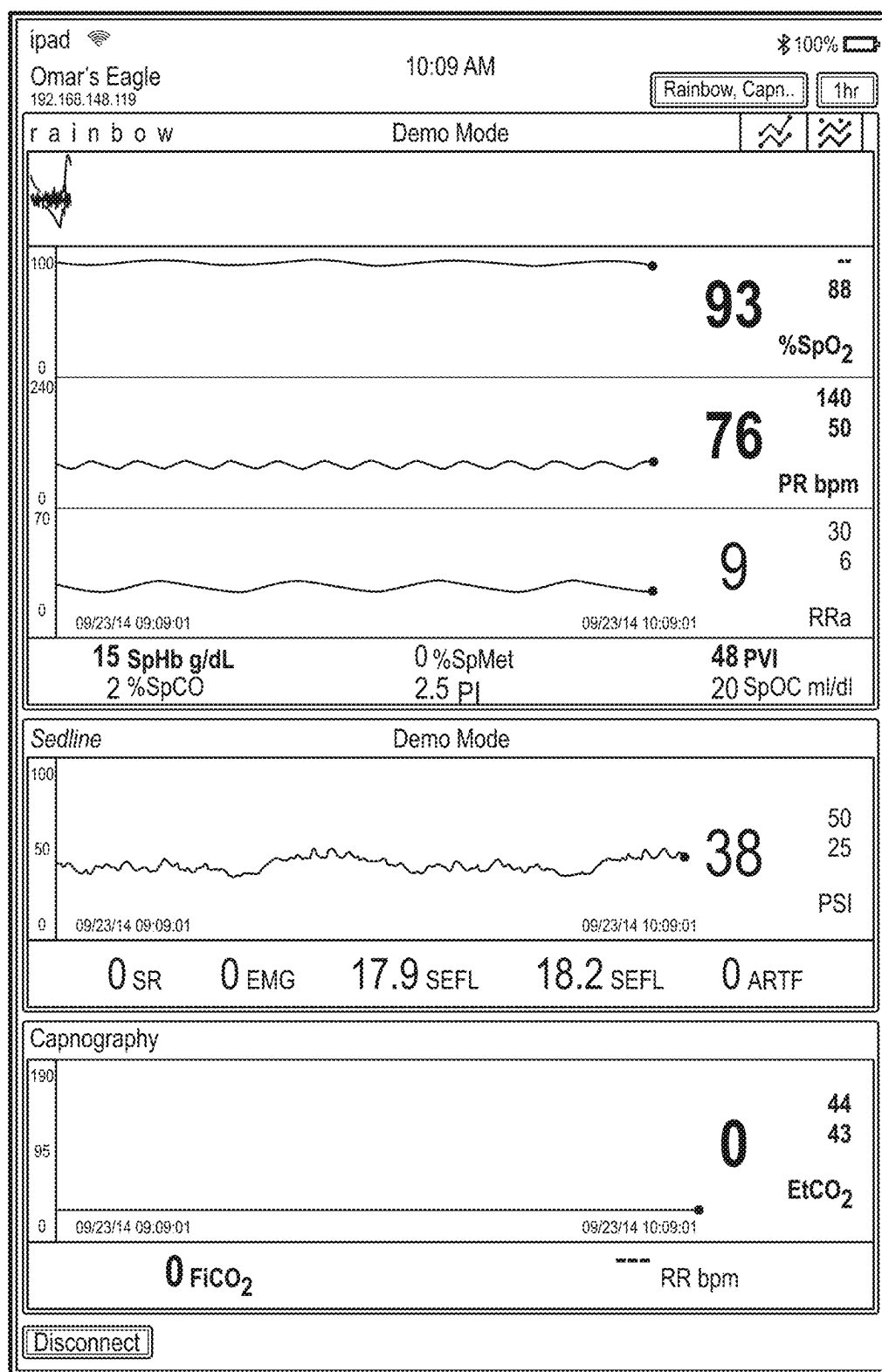


FIG. 66

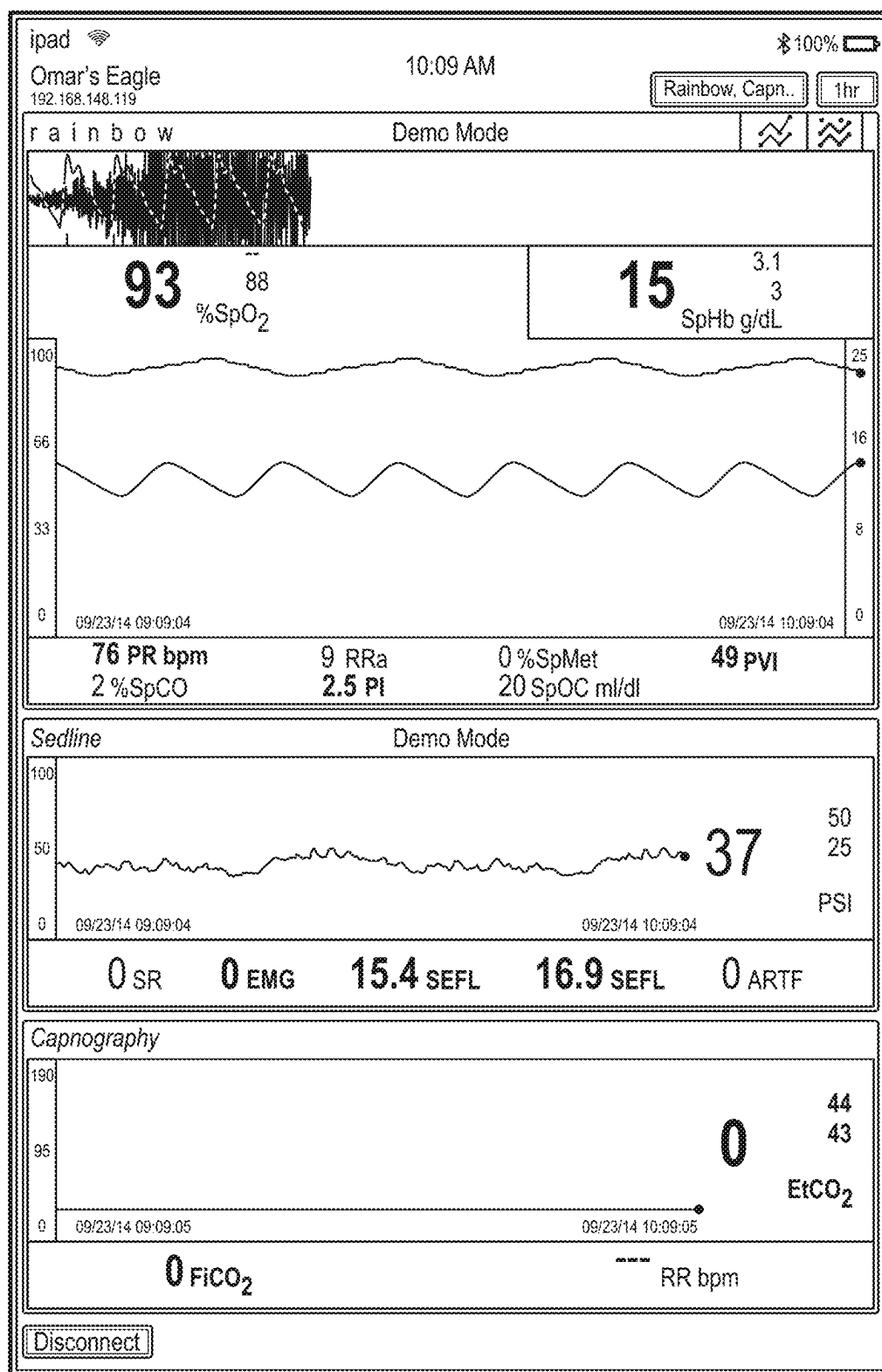


FIG. 67

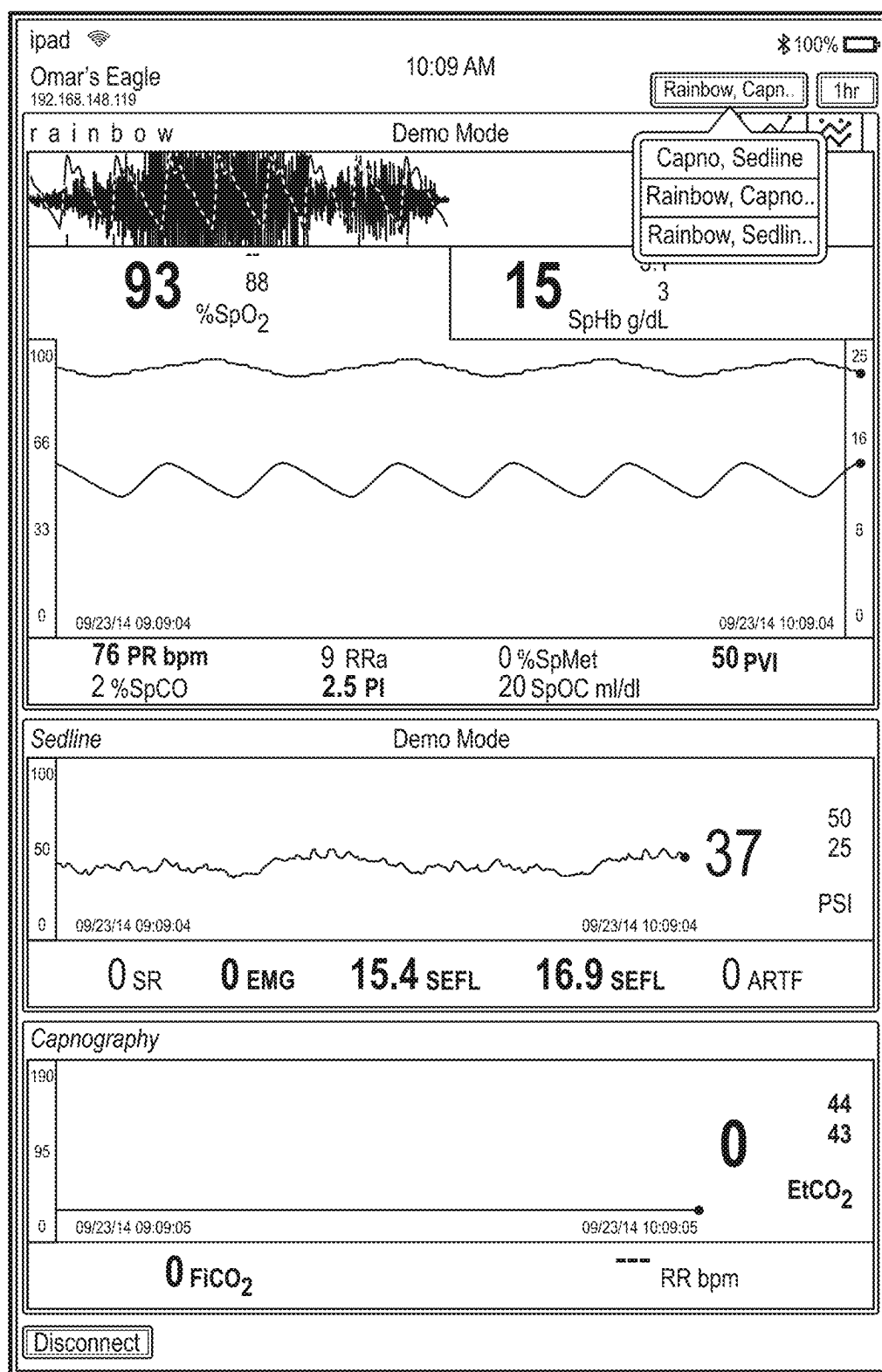


FIG. 68

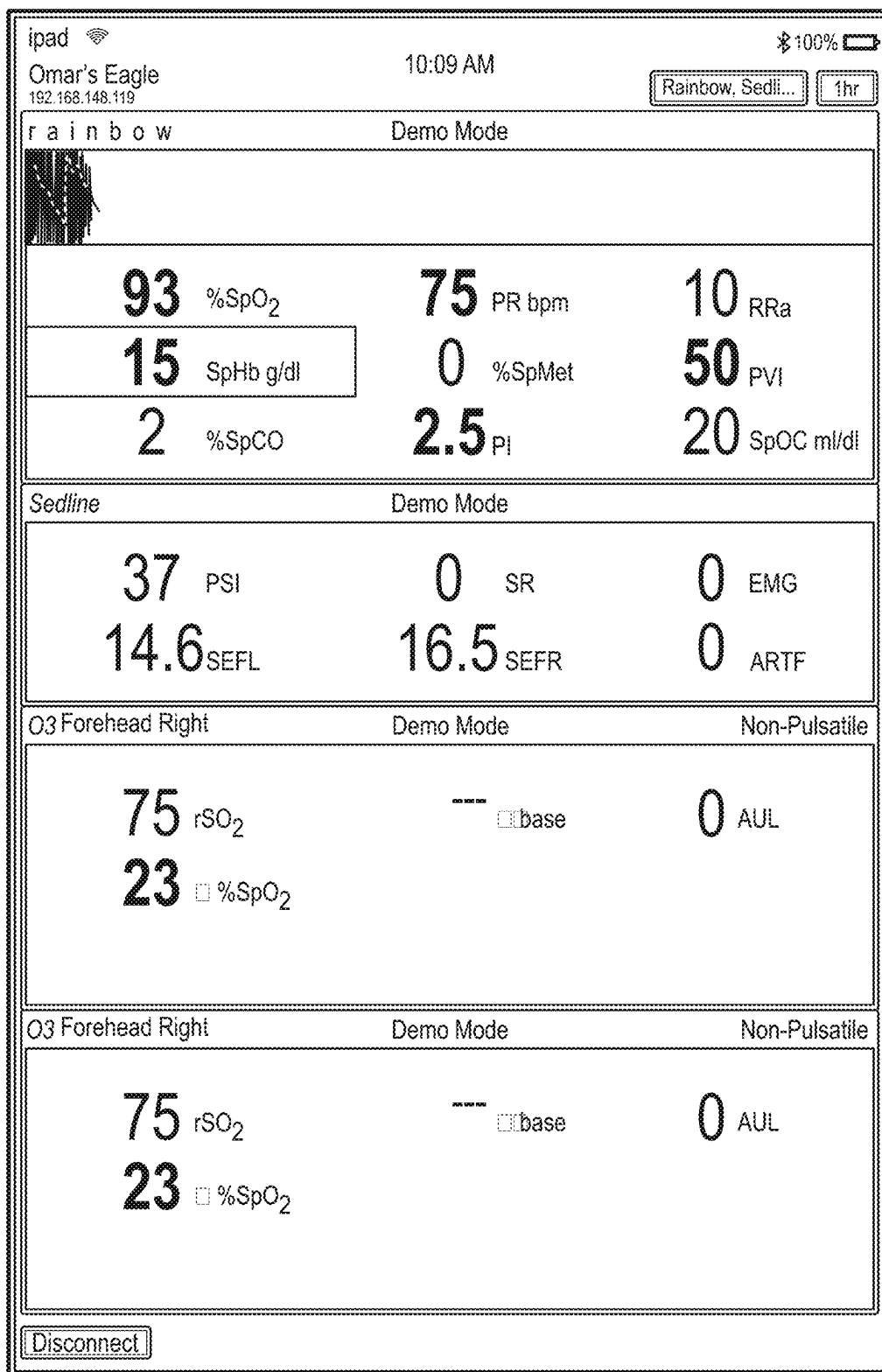


FIG. 69

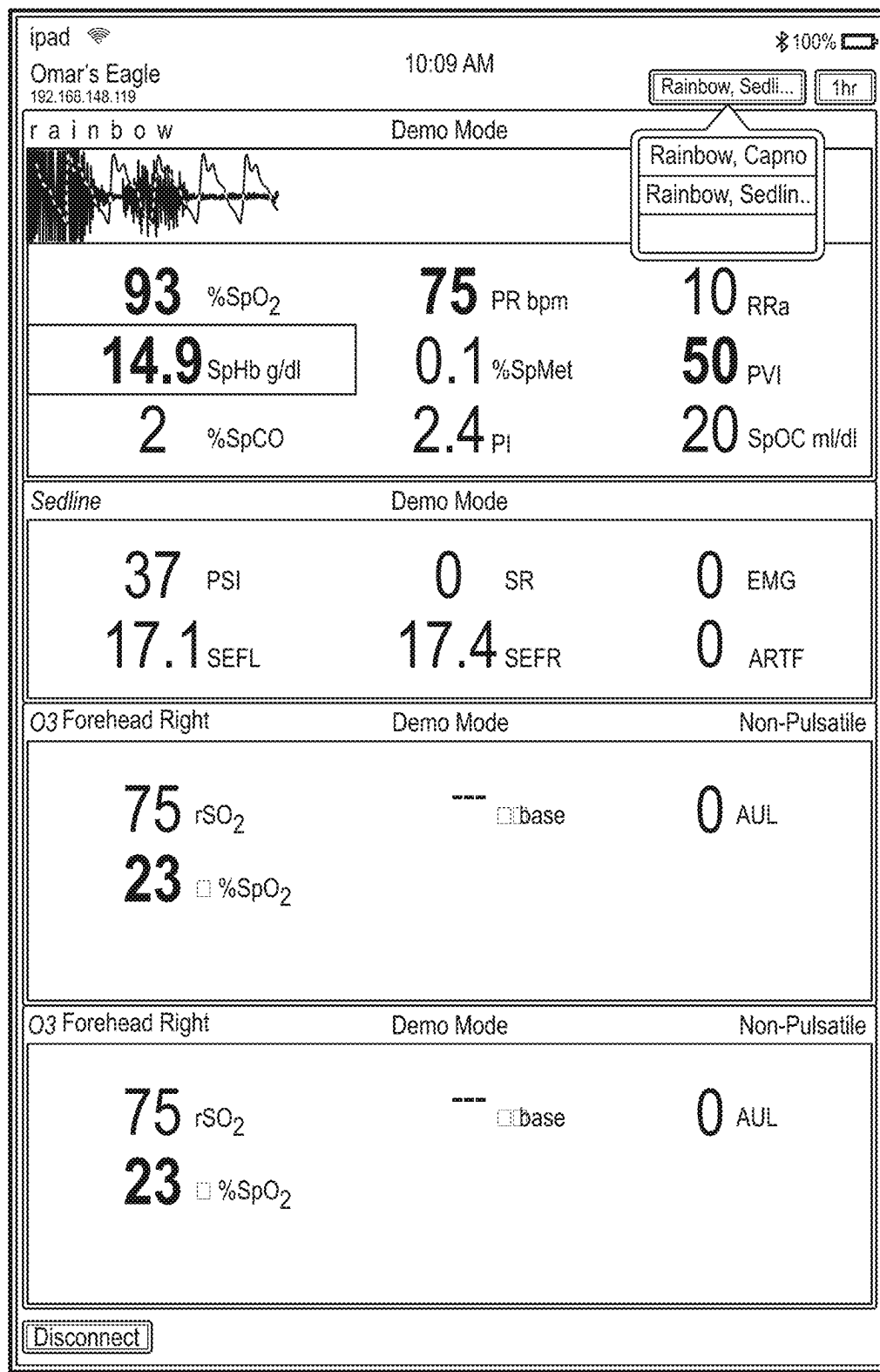


FIG. 70

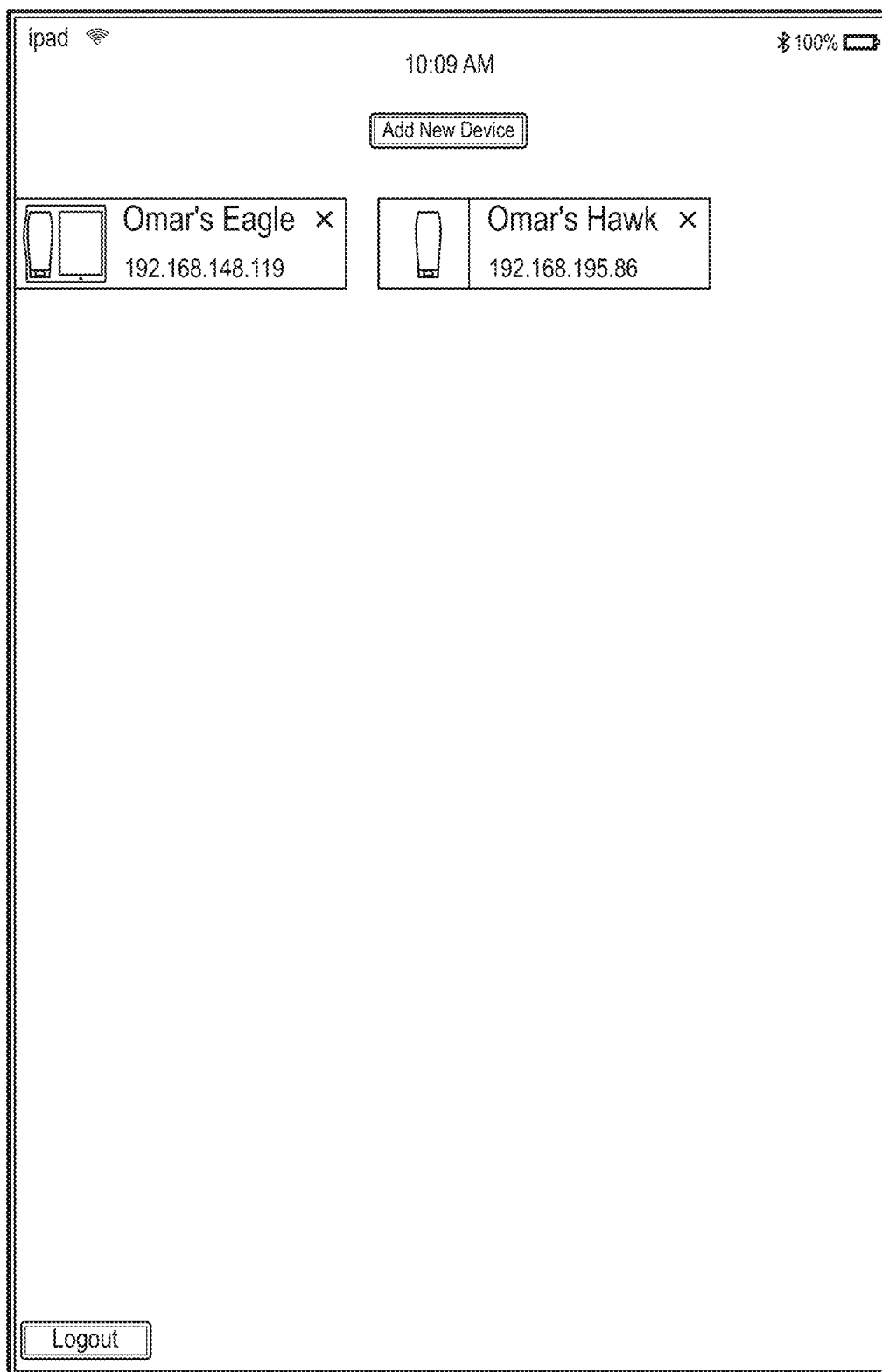


FIG. 71

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SYSTEM FOR DISPLAYING MEDICAL MONITORING DATA

RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/670,051, filed Oct. 31, 2019, titled "System for Displaying Medical Monitoring Data," which is a continuation of U.S. application Ser. No. 15/919,792, filed Mar. 13, 2018, titled "System for Displaying Medical Monitoring Data," which application is a continuation of U.S. application Ser. No. 14/512,237, filed Oct. 10, 2014, titled "System for Displaying Medical Monitoring Data," which application is a non-provisional of U.S. Provisional Application No. 61/889,972, filed Oct. 11, 2013, titled "System for Displaying Medical Monitoring Data," and is also a continuation-in-part of U.S. application Ser. No. 13/651,167, filed Oct. 12, 2012, titled "Medical Monitoring Hub," which is a non-provisional of each of the following U.S. Provisional Patent Applications:

Ser. No.	Date	Title
61/547,017,	Oct. 13, 2011,	Visual Correlation of Physiological Information,
61/547,577,	Oct. 14, 2011,	Visual Correlation of Physiological Information,
61/597,120,	Feb. 9, 2012,	Visual Correlation of Physiological Information,
61/703,773,	Sep. 20, 2012,	Medical Monitoring Hub

All of the above applications are incorporated by reference herein in their entirety.

FIELD OF THE DISCLOSURE

The present disclosure relates generally to patient monitoring devices and specifically to a patient monitor and medical data communication hub.

BACKGROUND OF THE DISCLOSURE

Today's patient monitoring environments are crowded with sophisticated and often electronic medical devices servicing a wide variety of monitoring and treatment endeavors for a given patient. Generally, many if not all of the devices are from differing manufactures, and many may be portable devices. The devices may not communicate with one another and each may include its own control, display, alarms, configurations and the like. Complicating matters, caregivers often desire to associate all types of measurement and use data from these devices to a specific patient. Thus, patient information entry often occurs at each device. Sometimes, the disparity in devices leads to a need to simply print and store paper from each device in a patient's file for caregiver review.

The result of such device disparity is often a caregiver environment scattered with multiple displays and alarms leading to a potentially chaotic experience. Such chaos can be detrimental to the patient in many situations including surgical environments where caregiver distraction is unwanted, and including recovery or monitoring environments where patient distraction or disturbance may be unwanted.

Various manufacturers produce multi-monitor devices or devices that modularly expand to increase the variety of monitoring or treatment endeavors a particular system can

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accomplish. However, as medical device technology expands, such multi-monitor devices begin to be obsolete the moment they are installed.

SUMMARY

This disclosure describes embodiments of a medical monitoring hub as the center of monitoring for a monitored patient. The hub can include configurable medical ports and serial ports for communicating with other medical devices in the patient's proximity. Moreover, the hub can communicate with a portable patient monitor. The monitor, when docked with the hub, may provide display graphics different from when undocked. The display graphics can include anatomical information. The hub can assemble the often vast amount of electronic medical data, associate it with the monitored patient, and in some embodiments, communicate the data to the patient's medical records.

Some aspects of the disclosure describe a first medical device having digital logic circuitry receives a physiological signal associated with a patient from a physiological sensor, obtains a first physiological parameter value based on the physiological signal, and outputs the first physiological parameter value for display. The first medical device can also receive a second physiological parameter value from a second medical device other than the first medical device, where the second physiological parameter value is formatted according to a protocol not used by the first medical device, such that the first medical device is not able to process the second physiological parameter value to produce a displayable output value. The first medical device can pass the physiological parameter data from the first medical device to a separate translation module, receive translated parameter data from the translation module at the first medical device, where the translated parameter data is able to be processed for display by the first medical device, and output a second value from the translated parameter data for display. The first medical device may be, for example, a monitoring hub, a portable physiological monitor, or a multi-patient monitoring system, and the second medical device may be an infusion pump, ventilator, or the like.

For purposes of summarizing the disclosure, certain aspects, advantages and novel features are discussed herein. It is to be understood that not necessarily all such aspects, advantages or features will be embodied in any particular embodiment of the invention and an artisan would recognize from the disclosure herein a myriad of combinations of such aspects, advantages or features.

BRIEF DESCRIPTION OF THE DRAWINGS

The following drawings and the associated descriptions are provided to illustrate embodiments of the present disclosure and do not limit the scope of the claims.

FIGS. 1A-1C illustrate perspective views of an exemplary medical monitoring hub according to an embodiment of the disclosure. For example, FIG. 1A illustrates the hub with an exemplary docked portable patient monitor, FIG. 1B illustrates the hub with a set of medical ports and a noninvasive blood pressure input, and FIG. 1C illustrates the hub with various exemplary temperature sensors attached thereto, all according to various embodiments of the disclosure.

FIG. 2 illustrates a simplified block diagram of an exemplary monitoring environment including the hub of FIG. 1, according to an embodiment of the disclosure.

FIG. 3 illustrates a simplified exemplary hardware block diagram of the hub of FIG. 1, according to an embodiment of the disclosure.

FIG. 4 illustrates a perspective view of an exemplary removable docking station of the hub of FIG. 1, according to an embodiment of the disclosure.

FIG. 5 illustrates a perspective view of exemplary portable patient monitors undocked from the hub of FIG. 1, according to an embodiment of the disclosure. Moreover, FIG. 5 illustrates an exemplary alternative docking station.

FIG. 6 illustrates a simplified block diagram of traditional patient device electrical isolation principles.

FIG. 7A illustrates a simplified block diagram of an exemplary optional patient device isolation system according to an embodiment of the disclosure, while FIG. 7B adds exemplary optional non-isolation power levels for the system of FIG. 7A, also according to an embodiment of the disclosure.

FIG. 8 illustrates a simplified exemplary universal medical connector configuration process, according to an embodiment of the disclosure.

FIGS. 9A-9B, 10, 11A-11F, 11G1-11G2, and 11H-11K illustrate simplified block diagrams of exemplary universal medical connectors having a size and shape smaller in cross section than traditional isolation requirements.

FIG. 10 illustrates a perspective view of a side of the hub of FIG. 1, showing exemplary instrument-side channel inputs for exemplary universal medical connectors, according to an embodiment of the disclosure.

FIGS. 11A-11F, 11G1-11G2, and 11H-11K illustrate various views of exemplary male and mating female universal medical connectors, according to embodiments of the disclosure.

FIG. 12 illustrates a simplified block diagram of a channel system for the hub of FIG. 1, according to an embodiment of the disclosure.

FIG. 13 illustrates an exemplary logical channel configuration, according to an embodiment of the disclosure.

FIG. 14 illustrates a simplified exemplary process for constructing a cable and configuring a channel according to an embodiment of the disclosure.

FIG. 15 illustrates a perspective view of the hub of FIG. 1, including an exemplary attached board-in-cable to form an input channel, according to an embodiment of the disclosure.

FIG. 16 illustrates a perspective view of a back side of the hub of FIG. 1, showing an exemplary instrument-side serial data inputs, according to an embodiment of the disclosure.

FIG. 17A illustrates an exemplary monitoring environment with communication through the serial data connections of FIG. 16, according to embodiments of the disclosure.

FIG. 17B illustrates an exemplary connectivity display of the hub of FIG. 1, according to embodiments of the disclosure.

FIG. 18 illustrates a simplified exemplary patient data flow process, according to an embodiment of the disclosure.

FIGS. 19A-19J illustrate exemplary displays of anatomical graphics for the portable patient monitor of FIG. 1 docked with the hub of FIG. 1, according to embodiments of the disclosure.

FIGS. 20A-20C illustrate exemplary displays of measurement data showing data separation and data overlap on a display of the hub of FIG. 1, respectively, according to embodiments of the disclosure.

FIGS. 21A and 21B illustrate exemplary displays of measurement data showing data separation and data overlap

on a display of the portable patient monitor of FIG. 1, respectively, according to embodiments of the disclosure.

FIGS. 22A and 22B illustrate exemplary analog display indicia according to an embodiment of the disclosure.

FIGS. 23A-23F illustrate exemplary displays of measurement data showing, for example, data presentation in FIGS. 23A-23D when a depth of consciousness monitor is connected to a channel port of the hub of FIG. 1, data presentation in FIG. 23E when temperature and blood pressure sensors communicate with the hub of FIG. 1 and data presentation in FIG. 23F when an acoustic sensor is also communicating with the hub of FIG. 1, according to embodiments of the disclosure.

FIG. 24 illustrates another embodiment of a monitoring environment including the hub of FIG. 1.

FIG. 25 illustrates an embodiment of a translation message handling process.

FIGS. 26-39 illustrate additional example hub displays, including displays of measurement data.

FIG. 40A illustrates an example first medical device and an example second medical device that communicate with one another via a translation module.

FIG. 40B illustrates an example first medical device and an example second medical device that communicate with one another via a translation module and a communication bus.

FIG. 41A illustrates an example input message received by the translation module.

FIG. 41B illustrates an example message header segment of an input message that has been parsed into fields.

FIG. 41C illustrates an example encoded version of the parsed message header segment of FIG. 41B.

FIG. 41D illustrates an example output message of the translation module based on the input message of FIG. 41A.

FIG. 42 illustrates an example translation process for generating an output message based on an input message and a comparison with translation rules associated with the translation module.

FIG. 43A illustrates an example translation process in which the translation module facilitates communication of an HL7 message from a Hospital Information System ("HIS") having a first HL7 format to an intended recipient medical device having a second HL7 format.

FIG. 43B illustrates an example translation process in which the translation module facilitates communication of an HL7 message from a medical device having a first HL7 format to a HIS having a second HL7 format.

FIG. 44 illustrates an example screenshot from a messaging implementation tool for manually configuring translation rules to be used by the translation module.

FIGS. 45A and 45B illustrate example automatic rule configuration processes that can be performed by the translation module.

FIGS. 45C and 45D illustrate example automatic rule configuration processes that can be performed by the translation module for messages utilizing the HL7 protocol.

FIGS. 46-71 illustrate additional example hub displays, including displays of measurement data.

While the foregoing "Brief Description of the Drawings" references generally various embodiments of the disclosure, an artisan will recognize from the disclosure herein that such embodiments are not mutually exclusive. Rather, the artisan would recognize a myriad of combinations of some or all of such embodiments.

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DETAILED DESCRIPTION

I. Introduction

Based on at least the foregoing, a solution is needed that coordinates the various medical devices treating or monitoring a patient. Embodiments of such a solution should provide patient identification seamlessly across the device space and embodiments of such a solution should expand for future technologies without necessarily requiring repeated software upgrades. In addition, embodiments of such a solution may include patient electrical isolation where desired.

Therefore, the present disclosure relates to a patient monitoring hub that is the center of patient monitoring and treatment activities for a given patient. Embodiments of the patient monitoring hub interface with legacy devices without necessitating legacy reprogramming, provide flexibility for interfacing with future devices without necessitating software upgrades, and offer optional patient electrical isolation. In an embodiment, the hub includes a large display dynamically providing information to a caregiver about a wide variety of measurement or otherwise determined parameters. Additionally, in an embodiment, the hub includes a docking station for a portable patient monitor. The portable patient monitor may communicate with the hub through the docking station or through various wireless paradigms known to an artisan from the disclosure herein, including WiFi, Bluetooth, Zigbee, or the like.

In still other embodiments, the portable patient monitor modifies its screen when docked. The undocked display indicia is in part or in whole transferred to a large dynamic display of the hub and the docked display presents one or more anatomical graphics of monitored body parts. For example, the display may present a heart, lungs, a brain, kidneys, intestines, a stomach, other organs, digits, gastrointestinal systems or other body parts when it is docked. In an embodiment, the anatomical graphics may advantageously be animated. In an embodiment, the animation may generally follow the behavior of measured parameters, such as, for example, the lungs may inflate in approximate correlation to the measured respiration rate and/or the determined inspiration portion of a respiration cycle, and likewise deflate according to the expiration portion of the same. The heart may beat according to the pulse rate, may beat generally along understood actual heart contraction patterns, and the like. Moreover, in an embodiment, when the measured parameters indicate a need to alert a caregiver, a changing severity in color may be associated with one or more displayed graphics, such as the heart, lungs, brain, or the like. In still other embodiments, the body portions may include animations on where, when or how to attach measurement devices to measurement sites on the patient. For example, the monitor may provide animated directions for CCHD screening procedures or glucose strip reading protocols, the application of a forehead sensor, a finger or toe sensor, one or more electrodes, an acoustic sensor, and ear sensor, a cannula sensor or the like.

The present disclosure relates to a medical monitoring hub configured to be the center of monitoring activity for a given patient. In an embodiment, the hub comprises a large easily readable display, such as an about ten (10) inch display dominating the majority of real estate on a front face of the hub. The display could be much larger or much smaller depending upon design constraints. However, for portability and current design goals, the preferred display is roughly sized proportional to the vertical footprint of one of

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the dockable portable patient monitors. Other considerations are recognizable from the disclosure herein by those in the art.

The display provides measurement data for a wide variety of monitored parameters for the patient under observation in numerical or graphic form, and in various embodiments, is automatically configured based on the type of data and information being received at the hub. In an embodiment, the hub is moveable, portable, and mountable so that it can be positioned to convenient areas within a caregiver environment. For example, the hub is collected within a singular housing.

In an embodiment, the hub may advantageously receive data from a portable patient monitor while docked or undocked from the hub. Typical portable patient monitors, such as oximeters or co-oximeters can provide measurement data for a large number of physiological parameters derived from signals output from optical and/or acoustic sensors, electrodes, or the like. The physiological parameters include, but not limited to oxygen saturation, carboxy hemoglobin, methemoglobin, total hemoglobin, glucose, pH, bilirubin, fractional saturation, pulse rate, respiration rate, components of a respiration cycle, indications of perfusion including perfusion index, signal quality and/or confidences, plethysmograph data, indications of wellness or wellness indexes or other combinations of measurement data, audio information responsive to respiration, ailment identification or diagnosis, blood pressure, patient and/or measurement site temperature, depth of sedation, organ or brain oxygenation, hydration, measurements responsive to metabolism, combinations of the same or the like, to name a few. In other embodiments, the hub may output data sufficient to accomplish closed-loop drug administration in combination with infusion pumps or the like.

In an embodiment, the hub communicates with other devices in a monitoring environment that are interacting with the patient in a number of ways. For example, the hub advantageously receives serial data from other devices without necessitating their reprogramming or that of the hub. Such other devices include pumps, ventilators, all manner of monitors monitoring any combination of the foregoing parameters, ECG/EEG/EKG devices, electronic patient beds, and the like. Moreover, the hub advantageously receives channel data from other medical devices without necessitating their reprogramming or that of the hub. When a device communicates through channel data, the hub may advantageously alter the large display to include measurement information from that device. Additionally, the hub accesses nurse call systems to ensure that nurse call situations from the device are passed to the appropriate nurse call system.

The hub also communicates with hospital systems to advantageously associate incoming patient measurement and treatment data with the patient being monitored. For example, the hub may communicate wirelessly or otherwise to a multi-patient monitoring system, such as a server or collection of servers, which in turn may communicate with a caregiver's data management systems, such as, for example, an Admit, Discharge, Transfer ("ADT") system and/or an Electronic Medical Records ("EMR") system. The hub advantageously associates the data flowing through it with the patient being monitored thereby providing the electronic measurement and treatment information to be passed to the caregiver's data management systems without the caregiver associating each device in the environment with the patient.

In an embodiment, the hub advantageously includes a reconfigurable and removable docking station. The docking station may dock additional layered docking stations to adapt to different patient monitoring devices. Additionally, the docking station itself is modularized so that it may be removed if the primary dockable portable patient monitor changes its form factor. Thus, the hub is flexible in how its docking station is configured.

In an embodiment, the hub includes a large memory for storing some or all of the data it receives, processes, and/or associates with the patient, and/or communications it has with other devices and systems. Some or all of the memory may advantageously comprise removable SD memory.

The hub communicates with other devices through at least (1) the docking station to acquire data from a portable monitor, (2) innovative universal medical connectors to acquire channel data, (3) serial data connectors, such as RJ ports to acquire output data, (4) Ethernet, USB, and nurse call ports, (5) Wireless devices to acquire data from a portable monitor, (6) other wired or wireless communication mechanisms known to an artisan. The universal medical connectors advantageously provide optional electrically isolated power and communications, are designed to be smaller in cross section than isolation requirements. The connectors and the hub communicate to advantageously translate or configure data from other devices to be usable and displayable for the hub. In an embodiment, a software developers kit ("SDK") is provided to a device manufacturer to establish or define the behavior and meaning of the data output from their device. When the output is defined, the definition is programmed into a memory residing in the cable side of the universal medical connector and supplied as an original equipment manufacture ("OEM") to the device provider. When the cable is connected between the device and the hub, the hub understands the data and can use it for display and processing purposes without necessitating software upgrades to the device or the hub. In an embodiment, the hub can negotiate the schema and even add additional compression and/or encryption. Through the use of the universal medical connectors, the hub organizes the measurement and treatment data into a single display and alarm system effectively and efficiently bringing order to the monitoring environment.

As the hub receives and tracks data from other devices according to a channel paradigm, the hub may advantageously provide processing to create virtual channels of patient measurement or treatment data. In an embodiment, a virtual channel may comprise a non-measured parameter that is, for example, the result of processing data from various measured or other parameters. An example of such a parameter includes a wellness indicator derived from various measured parameters that give an overall indication of the wellbeing of the monitored patient. An example of a wellness parameter is disclosed in U.S. patent application Ser. Nos. 13/269,296, 13/371,767 and 12/904,925, by the assignee of the present disclosure and incorporated by reference herein. By organizing data into channels and virtual channels, the hub may advantageously time-wise synchronize incoming data and virtual channel data.

The hub also receives serial data through serial communication ports, such as RJ connectors. The serial data is associated with the monitored patient and passed on to the multi-patient server systems and/or caregiver backend systems discussed above. Through receiving the serial data, the caregiver advantageously associates devices in the caregiver environment, often from varied manufactures, with a particular patient, avoiding a need to have each individual

device associated with the patient and possible communicating with hospital systems. Such association is vital as it reduces caregiver time spent entering biographic and demographic information into each device about the patient. Moreover, in an embodiment, through the SDK the device manufacturer may advantageously provide information associated with any measurement delay of their device, thereby further allowing the hub to advantageously time-wise synchronize serial incoming data and other data associated with the patient.

In an embodiment, when a portable patient monitor is docked, and it includes its own display, the hub effectively increases its display real estate. For example, in an embodiment, the portable patient monitor may simply continue to display its measurement and/or treatment data, which may be now duplicated on the hub display, or the docked display may alter its display to provide additional information. In an embodiment, the docked display, when docked, presents anatomical graphical data of, for example, the heart, lungs, organs, the brain, or other body parts being measured and/or treated. The graphical data may advantageously animate similar to and in concert with the measurement data. For example, lungs may inflate in approximate correlation to the measured respiration rate and/or the determined inspiration/expiration portions of a respiration cycle, the heart may beat according to the pulse rate, may beat generally along understood actual heart contraction patterns, the brain may change color or activity based on varying depths of sedation, or the like. In an embodiment, when the measured parameters indicate a need to alert a caregiver, a changing severity in color may be associated with one or more displayed graphics, such as the heart, lungs, brain, organs, circulatory system or portions thereof, respiratory system or portions thereof, other body parts or the like. In still other embodiments, the body portions may include animations on where, when or how to attach measurement devices.

The hub may also advantageously overlap parameter displays to provide additional visual information to the caregiver. Such overlapping may be user definable and configurable. The display may also incorporate analog-appearing icons or graphical indicia.

In the interest of clarity, not all features of an actual implementation are described in this specification. An artisan will of course be appreciate that in the development of any such actual implementation (as in any development project), numerous implementation-specific decisions must be made to achieve a developers' specific goals and sub-goals, such as compliance with system- and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of device engineering for those of ordinary skill having the benefit of this disclosure.

To facilitate a complete understanding of the disclosure, the remainder of the detailed description describes the disclosure with reference to the drawings, wherein like reference numbers are referenced with like numerals throughout.

II. Example Hub Embodiments

FIG. 1A illustrates a monitoring environment including a perspective view of an exemplary medical monitoring hub 100 with an exemplary docked portable patient monitor 102 according to an embodiment of the disclosure. The hub 100 includes a display 104, and a docking station 106, which in

an embodiment is configured to mechanically and electrically mate with the portable patient monitor **102**, each housed in a movable, mountable and portable housing **108**. The housing **108** includes a generally upright inclined shape configured to rest on a horizontal flat surface, although the housing **108** can be affixed in a wide variety of positions and mountings and comprise a wide variety of shapes and sizes.

In an embodiment, the display **104** may present a wide variety of measurement and/or treatment data in numerical, graphical, waveform, or other display indicia **110**. In an embodiment, the display **104** occupies much of a front face of the housing **108**, although an artisan will appreciate the display **104** may comprise a tablet or tabletop horizontal configuration, a laptop-like configuration or the like. Other embodiments may include communicating display information and data to a table computer, smartphone, television, or any display system recognizable to an artisan. The upright inclined configuration of FIG. 1A presents display information to a caregiver in an easily viewable manner.

FIG. 1B shows a perspective side view of an embodiment of the hub **100** including the housing **108**, the display **104**, and the docking station **106** without a portable monitor docked. Also shown is a connector for noninvasive blood pressure.

In an embodiment, the housing **108** may also include pockets or indentations to hold additional medical devices, such as, for example, a blood pressure monitor or temperature sensor **112**, such as that shown in FIG. 1C.

The portable patient monitor **102** of FIG. 1A may advantageously comprise an oximeter, co-oximeter, respiratory monitor, depth of sedation monitor, noninvasive blood pressure monitor, vital signs monitor or the like, such as those commercially available from Masimo Corporation of Irvine, CA, and/or disclosed in U.S. Pat. Nos. 2002/0140675, 2010/0274099, 2011/0213273, 2012/0226117, 2010/0030040; U.S. Pat. App. Ser. Nos. 61/242,792, 61/387,457, 61/645,570, 13/554,908 and U.S. Pat. Nos. 6,157,850, 6,334,065, and the like. The monitor **102** may communicate with a variety of noninvasive and/or minimally invasive devices such as optical sensors with light emission and detection circuitry, acoustic sensors, devices that measure blood parameters from a finger prick, cuffs, ventilators, and the like. The monitor **102** may include its own display **114** presenting its own display indicia **116**, discussed below with reference to FIGS. 19A-19J. The display indicia may advantageously change based on a docking state of the monitor **102**. When undocked, the display indicia may include parameter information and may alter orientation based on, for example, a gravity sensor or accelerometer.

In an embodiment, the docking station **106** of the hub **100** includes a mechanical latch **118**, or mechanically releasable catch to ensure that movement of the hub **100** doesn't mechanically detach the monitor **102** in a manner that could damage the same.

Although disclosed with reference to particular portable patient monitors **102**, an artisan will recognize from the disclosure herein a large number and wide variety of medical devices that may advantageously dock with the hub **100**. Moreover, the docking station **106** may advantageously electrically and not mechanically connect with the monitor **102**, and/or wirelessly communicate with the same.

FIG. 2 illustrates a simplified block diagram of an exemplary monitoring environment **200** including the hub **100** of FIG. 1, according to an embodiment of the disclosure. As shown in FIG. 2, the environment may include the portable patient monitor **102** communicating with one or more patient sensors **202**, such as, for example, oximetry optical

sensors, acoustic sensors, blood pressure sensors, respiration sensors or the like. In an embodiment, additional sensors, such as, for example, a NIBP sensor or system **211** and a temperature sensor or sensor system **213** may communicate directly with the hub **100**. The sensors **202**, **211** and **213** when in use are typically in proximity to the patient being monitored if not actually attached to the patient at a measurement site.

As disclosed, the portable patient monitor **102** communicates with the hub **100**, in an embodiment, through the docking station **106** when docked and, in an embodiment, wirelessly when undocked, however, such undocked communication is not required. The hub **100** communicates with one or more multi-patient monitoring servers **204** or server systems, such as, for example, those disclosed with in U.S. Pat. Pub. Nos. 2011/0105854, 2011/0169644, and 2007/0180140. In general, the server **204** communicates with caregiver backend systems **206** such as EMR and/or ADT systems. The server **204** may advantageously obtain through push, pull or combination technologies patient information entered at patient admission, such as demographical information, billing information, and the like. The hub **100** accesses this information to seamlessly associate the monitored patient with the caregiver backend systems **206**. Communication between the server **204** and the monitoring hub **100** may be any recognizable to an artisan from the disclosure herein, including wireless, wired, over mobile or other computing networks, or the like.

FIG. 2 also shows the hub **100** communicating through its serial data ports **210** and channel data ports **212**. As disclosed in the foregoing, the serial data ports **210** may provide data from a wide variety of patient medical devices, including electronic patient bed systems **214**, infusion pump systems **216** including closed loop control systems, ventilator systems **218**, blood pressure or other vital sign measurement systems **220**, or the like. Similarly, the channel data ports **212** may provide data from a wide variety of patient medical devices, including any of the foregoing, and other medical devices. For example, the channel data ports **212** may receive data from depth of consciousness monitors **222**, such as those commercially available from SEDLine, brain or other organ oximeter devices **224**, noninvasive blood pressure or acoustic devices **226**, or the like. In an embodiment, channel device may include board-in-cable ("BIC") solutions where the processing algorithms and the signal processing devices that accomplish those algorithms are mounted to a board housed in a cable or cable connector, which in some embodiments has no additional display technologies. The BIC solution outputs its measured parameter data to the channel port **212** to be displayed on the display **104** of hub **100**. In an embodiment, the hub **100** may advantageously be entirely or partially formed as a BIC solution that communicates with other systems, such as, for example, tablets, smartphones, or other computing systems.

Although disclosed with reference to a single docking station **106**, the environment **200** may include stacked docking stations where a subsequent docking station mechanically and electrically docks to a first docking station to change the form factor for a different portable patient monitor as discussed with reference to FIG. 5. Such stacking may include more than 2 docking stations, may reduce or increase the form factor for mechanical compliance with mating mechanical structures on a portable device.

FIG. 3 illustrates a simplified exemplary hardware block diagram of the hub **100** of FIG. 1, according to an embodiment of the disclosure. As shown in FIG. 3, the housing **108** of the hub **100** positions and/or encompasses an instrument

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board 302, the display 104, memory 304, and the various communication connections, including the serial ports 210, the channel ports 212, Ethernet ports 305, nurse call port 306, other communication ports 308 including standard USB or the like, and the docking station interface 310. The instrument board 302 comprises one or more substrates including communication interconnects, wiring, ports and the like to enable the communications and functions described herein, including inter-board communications. A core board 312 includes the main parameter, signal, and other processor(s) and memory, a portable monitor board ("RIB") 314 includes patient electrical isolation for the monitor 102 and one or more processors, a channel board ("MID") 316 controls the communication with the channel ports 212 including optional patient electrical isolation and power supply 318, and a radio board 320 includes components configured for wireless communications. Additionally, the instrument board 302 may advantageously include one or more processors and controllers, busses, all manner of communication connectivity and electronics, memory, memory readers including EPROM readers, and other electronics recognizable to an artisan from the disclosure herein. Each board comprises substrates for positioning and support, interconnect for communications, electronic components including controllers, logic devices, hardware/software combinations and the like to accomplish the tasks designated above and others.

An artisan will recognize from the disclosure herein that the instrument board 302 may comprise a large number of electronic components organized in a large number of ways. Using different boards such as those disclosed above advantageously provides organization and compartmentalization to the complex system.

FIG. 4 illustrates a perspective view of an exemplary removable docking station 400 of the hub 100 of FIG. 1, according to an embodiment of the disclosure. As shown in FIG. 4, the docking station 400 provides a mechanical mating to portable patient monitor 102 to provide secure mechanical support when the monitor 102 is docked. The docking station 400 includes a cavity 402 shaped similar to the periphery of a housing of the portable monitor 102. The station 400 also includes one or more electrical connectors 404 providing communication to the hub 100. Although shown as mounted with bolts, the docking station 400 may snap fit, may use movable tabs or catches, may magnetically attach, or may employ a wide variety or combination of attachment mechanisms known to an artisan from the disclosure herein. In an embodiment, the attachment of the docking station 400 should be sufficiently secure that when docked, the monitor 102 and docking station cannot be accidentally detached in a manner that could damage the instruments, such as, for example, if the hub 100 was accidentally bumped or the like, the monitor 102 and docking station 400 should remain intact.

The housing 108 of the hub 100 also includes cavity 406 housing the docking station 400. To the extent a change to the form factor for the portable patient monitor 102 occurs, the docking station 400 is advantageously removable and replaceable. Similar to the docking station 400, the hub 100 includes within the cavity 406 of the housing 108 electrical connectors 408 providing electrical communication to the docking station 400. In an embodiment, the docking station 400 includes its own microcontroller and processing capabilities, such as those disclosed in U.S. Pat. Pub. No. 2002/0140675. In other embodiments, the docking station 400 passes communications through to the electrical connector 408.

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FIG. 4 also shows the housing 108 including openings for channel ports 212 as universal medical connectors discussed in detail below.

FIG. 5 illustrates a perspective view of exemplary portable patient monitors 502 and 504 undocked from the hub 100 of FIG. 1, according to an embodiment of the disclosure. As shown in FIG. 5, the monitor 502 may be removed and other monitors, like monitor 504 may be provided. The docking station 106 includes an additional docking station 506 that mechanically mates with the original docking station 106 and presents a form factor mechanically mateable with monitor 504. In an embodiment, the monitor 504 mechanically and electrically mates with the stacked docking stations 506 and 106 of hub 100. As can be readily appreciated by an artisan from the disclosure herein, the stackable function of the docking stations provides the hub 100 with an extremely flexible mechanism for charging, communicating, and interfacing with a wide variety of patient monitoring devices. As noted above, the docking stations may be stacked, or in other embodiments, removed and replaced.

FIG. 6 illustrates a simplified block diagram of traditional patient electrical isolation principles. As shown in FIG. 6, a host device 602 is generally associated with a patient device 604 through communication and power. As the patient device 604 often comprises electronics proximate or connected to a patient, such as sensors or the like, certain safety requirements dictate that electrical surges of energy from, for example, the power grid connected to the host device, should not find an electrical path to the patient. This is generally referred to a "patient isolation" which is a term known in the art and includes herein the removing of direct uninterrupted electrical paths between the host device 602 and the patient device 604. Such isolation is accomplished through, for example, isolation devices 606 on power conductors 608 and communication conductors 610. Isolation devices 606 can include transformers, optical devices that emit and detect optical energy, and the like. Use of isolation devices, especially on power conductors, can be expensive component wise, expensive size wise, and drain power. Traditionally, the isolation devices were incorporated into the patient device 604, however, the patient devices 604 are trending smaller and smaller and not all devices incorporate isolation.

FIG. 7A illustrates a simplified block diagram of an exemplary optional patient isolation system according to an embodiment of the disclosure. As shown in FIG. 7A, the host device 602 communicates with an isolated patient device 604 through isolation devices 606. However, a memory 702 associated with a particular patient device informs the host 602 whether that device needs isolated power. If a patient device 708 does not need isolated power, such as some types of cuffs, infusion pumps, ventilators, or the like, then the host 602 can provide non-isolated power through signal path 710. This power may be much higher than what can cost-effectively be provided through the isolated power conductor 608. In an embodiment, the non-isolated patient devices 708 receive isolated communication as such communication is typically at lower voltages and is not cost prohibitive. An artisan will recognize from the disclosure herein that communication could also be non-isolated. Thus, FIG. 7A shows a patient isolation system 700 that provides optional patient isolation between a host 602 and a wide variety of potential patient devices 604, 708. In an embodiment, the hub 100 includes the channel ports 212 incorporating similar optional patient isolation principles.

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FIG. 7B adds an exemplary optional non-isolation power levels for the system of FIG. 7A according to an embodiment of the disclosure. As shown in FIG. 7B, once the host 602 understands that the patient device 604 comprises a self-isolated patient device 708, and thus does not need isolated power, the host 602 provides power through a separate conductor 710. Because the power is not isolated, the memory 702 may also provide power requirements to the host 602, which may select from two or more voltage or power levels. In FIG. 7B, the host 602 provides either high power, such as about 12 volts, but could have a wide range of voltages or very high power such as about 24 volts or more, but could have a wide range of voltages, to the patient device 708. An artisan will recognize that supply voltages can advantageously be altered to meet the specific needs of virtually any device 708 and/or the memory could supply information to the host 602 which provided a wide range of non-isolated power to the patient device 708.

Moreover, using the memory 702, the host 602 may determine to simply not enable any unused power supplies, whether that be the isolated power or one or more of the higher voltage non-isolated power supplies, thereby increasing the efficiency of the host.

FIG. 8 illustrates a simplified exemplary universal medical connector configuration process 800, according to an embodiment of the disclosure. As shown in FIG. 8, the process includes step 802, where a cable is attached to a universal medical connector incorporating optional patient isolation as disclosed in the foregoing. In step 804, the host device 602 or the hub 100, more specifically, the channel data board 316 or EPROM reader of the instrument board, reads the data stored in the memory 702 and in step 806, determines whether the connecting device requires isolated power. In step 808, when the isolated power is required, the hub 100 may advantageously enable isolated power and in step 810, enable isolated communications. In step 806, when isolated power is not needed, the hub 100 may simply in optional step 812 enable non-isolated power and in embodiments where communications remain isolated, step 810 enable isolated communications. In other optional embodiments, in step 806, when isolated power is not needed, the hub 100 in step 814 may use information from memory 702 to determine the amount of power needed for the patient device 708. When sufficient power is not available, because for example, other connected devices are also using connected power, in step 816 a message may be displayed indicating the same and power is not provided. When sufficient power is available, optional step 812 may enable non-isolated power. Alternatively, optional step 818 may determine whether memory 702 indicates higher or lower power is desired. When higher power is desired, the hub 100 may enable higher power in step 820 and when not, may enable lower power in step 822. The hub 100 in step 810 then enables isolated communication. In an embodiment, the hub 100 in step 818 may simply determine how much power is needed and provide at least sufficient power to the self-isolated device 708.

An artisan will recognize from the disclosure herein that hub 100 may not check to see if sufficient power is available or may provide one, two or many levels of non-isolated voltages based on information from the memory 702.

FIGS. 9A and 9B illustrate simplified block diagrams of exemplary universal medical connectors 900 having a size and shape smaller in cross section than tradition isolation requirements. In an embodiment, the connector 900 physically separates non-isolated signals on one side 910 from isolated signals on another side 920, although the sides could

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be reversed. The gap between such separations may be dictated at least in part by safety regulations governing patient isolation. In an embodiment, the distance between the sides 910 and 920 may appear to be too small.

As shown from a different perspective in FIG. 9B, the distance between connectors "x" appears small. However, the gap causes the distance to include a non-direct path between conductors. For example, any short would have to travel path 904, and the distance of such path is within or beyond such safety regulations, in that the distance is greater than "x." It is noteworthy that the non-straight line path 904 occurs throughout the connector, such as, for example, on the board connector side where solder connects various pins to a PCB board.

FIG. 10 illustrates a perspective view of a side of the hub 100 of FIG. 1, showing exemplary instrument-side channel inputs 1000 as exemplary universal medical connectors. As shown in FIG. 10, the inputs include the non-isolated side 910, the isolated side 920, and the gap. In an embodiment, the memory 710 communicates through pins on the non-isolated side.

FIGS. 11A-11K illustrate various views of exemplary male and mating female universal medical connectors, according to embodiments of the disclosure. For example, FIGS. 11G1 and 11G2 shows various preferred but not required sizing, and FIG. 11H shows incorporation of electronic components, such as the memory 702 into the connectors. FIGS. 11I-11K illustrate wiring diagrams and cabling specifics of the cable itself as it connects to the universal medical connectors.

FIG. 12 illustrates a simplified block diagram of a channel system for the hub of FIG. 1, according to an embodiment of the disclosure. As shown in FIG. 12, a male cable connector, such as those shown in FIG. 11 above, includes a memory such as an EPROM. The memory advantageously stores information describing the type of data the hub 100 can expect to receive, and how to receive the same. A controller of the hub 100 communicates with the EPROM to negotiate how to receive the data, and if possible, how to display the data on display 104, alarm when needed, and the like. For example, a medical device supplier may contact the hub provider and receive a software developers' kit ("SDK") that guides the supplier through how to describe the type of data output from their device. After working with the SDK, a map, image, or other translation file may advantageously be loaded into the EPROM, as well as the power requirements and isolation requirements discussed above. When the channel cable is connected to the hub 100 through the channel port 212, the hub 100 reads the EPROM and the controller of the hub 100 negotiates how to handle incoming data.

FIG. 13 illustrates an exemplary logical channel configuration that may be stored in the EPROM of FIG. 12. As shown in FIG. 13, each incoming channel describes one or more parameters. Each parameter describes whatever the hub 100 should know about the incoming data. For example, the hub 100 may want to know whether the data is streaming data, waveform data, already determined parameter measurement data, ranges on the data, speed of data delivery, units of the data, steps of the units, colors for display, alarm parameters and thresholds, including complex algorithms for alarm computations, other events that are parameter value driven, combinations of the same or the like. Additionally, the parameter information may include device delay times to assist in data synchronization or approximations of data synchronization across parameters or other data received by the hub 100. In an embodiment, the SDK

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presents a schema to the device supplier which self-describes the type and order of incoming data. In an embodiment, the information advantageously negotiates with the hub 100 to determine whether to apply compression and/or encryption to the incoming data stream.

Such open architecture advantageously provides device manufacturers the ability to port the output of their device into the hub 100 for display, processing, and data management as disclosed in the foregoing. By implementation through the cable connector, the device manufacturer avoids any reprogramming of their original device; rather, they simply let the hub 100 know through the cable connector how the already existing output is formatted. Moreover, by describing the data in a language already understood by the hub 100, the hub 100 also avoids software upgrades to accommodate data from “new-to-the-hub” medical devices.

FIG. 14 illustrates a simplified exemplary process for configuring a channel according to an embodiment of the disclosure. As shown in FIG. 14, the hub provider provides a device manufacturer with an SDK in step 1402, who in turn uses the SDK to self-describe the output data channel from their device in step 1404. In an embodiment, the SDK is a series of questions that guide the development, in other embodiments, the SDK provides a language and schema to describe the behavior of the data.

Once the device provider describes the data, the hub provider creates a binary image or other file to store in a memory within a cable connector in step 1405; however, the SDK may create the image and simply communicated it to the hub provider. The cable connector is provided as an OEM part to the provider in step 1410, who constructs and manufactures the cable to mechanically and electrically mate with output ports on their devices in step 1412.

Once a caregiver has the appropriately manufactured cable, with one end matching the device provider’s system and the other OEM’ed to match the hub 100 at its channel ports 212, in step 1452 the caregiver can connect the hub between the devices. In step 1454, the hub 100 reads the memory, provides isolated or non-isolated power, and the cable controller and the hub 100 negotiate a protocol or schema for data delivery. In an embodiment, a controller on the cable may negotiated the protocol, in an alternative embodiment, the controller of the hub 100 negotiates with other processors on the hub the particular protocol. Once the protocol is set, the hub 100 can use, display and otherwise process the incoming data stream in an intelligent manner.

Through the use of the universal medical connectors described herein, connection of a myriad of devices to the hub 100 is accomplished through straightforward programming of a cable connector as opposed to necessitating software upgrades to each device.

FIG. 15 illustrates a perspective view of the hub of FIG. 1 including an exemplary attached board-in-cable (“BIC”) to form an input channel according to an embodiment of the disclosure. As shown in FIG. 15, a SEDLine depth of consciousness board communicates data from an appropriate patient sensor to the hub 100 for display and caregiver review. As described, the provider of the board need only use the SDK to describe their data channel, and the hub 100 understands how to present the data to the caregiver.

FIG. 16 illustrates a perspective view of a back side of the hub 100 of FIG. 1, showing an exemplary serial data inputs. In an embodiment, the inputs include such as RJ 45 ports. As is understood in the art, these ports include a data ports similar to those found on computers, network routers, switches and hubs. In an embodiment, a plurality of these ports are used to associate data from various devices with the

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specific patient identified in the hub 100. FIG. 16 also shows a speaker, the nurse call connector, the Ethernet connector, the USBs, a power connector and a medical grounding lug.

FIG. 17A illustrates an exemplary monitoring environment with communication through the serial data connections of the hub 100 of FIG. 1, according to an embodiment of the disclosure. As shown and as discussed in the foregoing, the hub 100 may use the serial data ports 210 to gather data from various devices within the monitoring environment, including an electronic bed, infusion pumps, ventilators, vital sign monitors, and the like. The difference between the data received from these devices and that received through the channel ports 212 is that the hub 100 may not know the format or structure of this data. The hub 100 may not display information from this data or use this data in calculations or processing. However, porting the data through the hub 100 conveniently associates the data with the specifically monitored patient in the entire chain of caregiver systems, including the foregoing server 214 and backend systems 206. In an embodiment, the hub 100 may determine sufficient information about the incoming data to attempt to synchronize it with data from the hub 100.

In FIG. 17B, a control screen may provide information on the type of data being received. In an embodiment, a green light next to the data indicates connection to a device and on which serial input the connection occurs.

FIG. 18 illustrates a simplified exemplary patient data flow process, according to an embodiment of the disclosure. As shown, once a patient is admitted into the caregiver environment at step 1802, data about the patient is populated on the caregiver backend systems 206. The server 214 may advantageously acquire or receive this information in step 1804, and then make it accessible to the hub 100. When the caregiver at step 1806 assigns the hub 100 to the patient, the caregiver simply looks at the presently available patient data and selects the particular patient being currently monitored. The hub 100 at step 1808 then associates the measurement, monitoring and treatment data it receives and determines with that patient. The caregiver need not again associate another device with the patient so long as that device is communicating through the hub 100 by way of (1) the docking station, (2) the universal medical connectors, (3) the serial data connectors, or (4) other communication mechanisms known to an artisan. At step 1810, some or the entirety of the received, processed and/or determined data is passed to the server systems discussed above.

FIGS. 19A-19J illustrate exemplary displays of anatomical graphics for the portable patient monitor docked with the hub 100 of FIG. 1, according to embodiments of the disclosure. As shown in FIG. 19A, the heart, lungs and respiratory system are shown while the brain is not highlighted. Thus, a caregiver can readily determine that depth of consciousness monitoring or brain oximetry systems are not currently communicating with the hub 100 through the portable patient monitor connection or the channel data ports. However, it is likely that acoustic or other respiratory data and cardiac data is being communicated to or measured by the hub 100. Moreover, the caregiver can readily determine that the hub 100 is not receiving alarming data with respect to the emphasized body portions. In an embodiment, the emphasized portion may animate to show currently measured behavior or, alternatively, animate in a predetermined fashion.

FIG. 19B shows the addition of a virtual channel showing an indication of wellness. As shown in FIG. 19B, the indication is positive as it is a “34” on an increasingly severity scale to “100.” The wellness indication may also be

shaded to show problems. In contrast to FIG. 19B, FIG. 19C shows a wellness number that is becoming or has become problematic and an alarming heart graphic. Thus, a caregiver responding to a patient alarm on the hub 100 or otherwise on another device or system monitoring or treating the patient can quickly determine that a review of vital signs and other parameters relating to heart function is needed to diagnose and/or treat the patient.

FIGS. 19D and 19E show the brain included in the emphasized body portions meaning that the hub 100 is receiving data relevant to brain functions, such as, for example, depth of sedation data or brain oximetry data. FIG. 19E additionally shows an alarming heart function similar to FIG. 19C.

In FIG. 19F, additional organs, such as the kidneys are being monitored, but the respiratory system is not. In FIG. 19G, an alarming hear function is shown, and in FIG. 19H, an alarming circulatory system is being shown. FIG. 19I shows the wellness indication along with lungs, heart, brain and kidneys. FIG. 19J shows alarming lungs, heart, and circulatory system as well as the wellness indication. Moreover, FIG. 19J shows a severity contrast, such as, for example, the heart alarming red for urgent while the circulatory system alarms yellow for caution. An artisan will recognize other color schemes that are appropriate from the disclosure herein.

FIGS. 20A-20C illustrate exemplary displays of measurement data showing data separation and data overlap, respectively, according embodiments of the disclosure. FIGS. 21A and 21B illustrate exemplary displays of measurement data also showing data separation and data overlap, respectively, according embodiments of the disclosure.

For example, acoustic data from an acoustic sensor may advantageously provide breath sound data, while the plethysmograph and ECG or other signals can also be presented in separate waveforms (FIG. 20A, top of the screen capture). The monitor may determine any of a variety of respiratory parameters of a patient, including respiratory rate, expiratory flow, tidal volume, minute volume, apnea duration, breath sounds, riles, rhonchi, stridor, and changes in breath sounds such as decreased volume or change in airflow. In addition, in some cases a system monitors other physiological sounds, such as heart rate to help with probe off detection, heart sounds (S1, S2, S3, S4, and murmurs), and change in heart sounds such as normal to murmur or split heart sounds indicating fluid overload.

Providing a visual correlation between multiple physiological signals can provide a number of valuable benefits where the signals have some observable physiological correlation. As one example of such a correlation, changes in morphology (e.g., envelope and/or baseline) of the plethysmographic signal can be indicative of patient blood or other fluid levels. And, these changes can be monitored to detect hypovolemia or other fluid-level related conditions. A pleth variability index may provide an indication of fluid levels, for example. And, changes in the morphology of the plethysmographic signal are correlated to respiration. For example, changes in the envelope and/or baseline of the plethysmographic signal are correlated to breathing. This is at least in part due to aspects of the human anatomical structure, such as the mechanical relationship and interaction between the heart and the lungs during respiration.

Thus, superimposing a plethysmographic signal and a respiratory signal (FIG. 20B) can give operators an indication of the validity of the plethysmographic signal or signals derived therefrom, such as a pleth variability index. For example, if bursts in the respiration signal indicative of

inhalation and exhalation correlate with changes in peaks and valleys of the plethysmographic envelope, this gives monitoring personnel a visual indication that the plethysmographic changes are indeed due to respiration, and not some other extraneous factor. Similarly, if the bursts in the respiration signal line up with the peaks and valleys in the plethysmographic envelope, this provides monitoring personnel an indication that the bursts in the respiration signal are due to patient breathing sounds, and not some other non-targeted sounds (e.g., patient non-breathing sounds or non-patient sounds).

The monitor may also be configured to process the signals and determine whether there is a threshold level of correlation between the two signals, or otherwise assess the correlation. However, by additionally providing a visual indication of the correlation, such as by showing the signals superimposed with one another, the display provides operators a continuous, intuitive and readily observable gauge of the particular physiological correlation. For example, by viewing the superimposed signals, users can observe trends in the correlation over time, which may not be otherwise ascertainable.

The monitor can visually correlate a variety of other types of signals instead of, or in addition to plethysmographic and respiratory signals. For example, FIG. 20C depicts a screen shot of another example monitoring display. As shown in the upper right portion of FIG. 20C, the display superimposes a plethysmographic signal, an ECG signal, and a respiration signal. In other configurations, more than three different types of signals may be overlaid onto one another.

In one embodiment, the hub 100 provides an interface through which the user can move the signals together to overlay on one another. For example, the user may be able to drag the respiration signal down onto the plethysmographic signal using a touch screen interface. Conversely, the user may be able to separate the signals, also using the touch screen interface. In another embodiment, the monitor includes a button the user can press, or some other user interface allowing the user to overlay and separate the signals, as desired. FIGS. 21A and 21B show similar separation and joining of the signals.

In certain configurations, in addition to providing the visual correlation between the plethysmographic signal and the respiratory signal, the monitor is additionally configured to process the respiratory signal and the plethysmographic signal to determine a correlation between the two signals. For example, the monitor may process the signals to determine whether the peaks and valleys in the changes in the envelope and/or baseline of the plethysmographic signal correspond to bursts in the respiratory signal. And, in response to the determining that there is or is not a threshold level of correlation, the monitor may provide some indication to the user. For example, the monitor may provide a graphical indication (e.g., a change in color of pleth variability index indicator), an audible alarm, or some other indication. The monitor may employ one or more envelope detectors or other appropriate signal processing components in making the determination.

In certain embodiments, the system may further provide an audible indication of the patient's breathing sounds instead of, or in addition to the graphical indication. For example, the monitor may include a speaker, or an earpiece (e.g., a wireless earpiece) may be provided to the monitoring personnel providing an audible output of the patient sounds. Examples of sensors and monitors having such capability are described in U.S. Pat. Pub. No. 2011/0172561 and are incorporated by reference herein.

In addition to the above described benefits, providing both the acoustic and plethysmographic signals on the same display in the manner described can allow monitoring personnel to more readily detect respiratory pause events where there is an absence of breathing, high ambient noise that can degrade the acoustic signal, improper sensor placement, etc.

FIGS. 22A-22B illustrate exemplary analog display indicia, according to an embodiment of the disclosure. As shown in FIGS. 22A and 22B, the screen shots displays health indicators of various physiological parameters, in addition to other data. Each health indicator can include an analog indicator and/or a digital indicator. In embodiments where the health indicator includes an analog and a digital indicator, the analog and digital indicators can be positioned in any number of formations, such as side-by-side, above, below, transposed, etc. In the illustrated embodiment, the analog indicators are positioned above and to the sides of the digital indicators. As shown more clearly in FIG. 22B, the analog displays may include colored warning sections, dashes indicating position on the graph, and digital information designating quantitate information from the graph. In FIG. 22B, for example, the pulse rate PR graph shows that from about 50 to about 140 beats per minute, the graph is either neutral or beginning to be cautionary, whereas outside those numbers the graph is colored to indicate a severe condition. Thus, as the dash moves along the arc, a caregiver can readily see where in the range of acceptable, cautionary, and extreme the current measurements fall.

Each analog indicator of the health indicator can include a dial that moves about an arc based on measured levels of monitored physiological parameters. As the measured physiological parameter levels increase the dial can move clockwise, and as the measured physiological parameter levels decrease, the dial can move counter-clockwise, or vice versa. In this way, a user can quickly determine the patient's status by looking at the analog indicator. For example, if the dial is in the center of the arc, the observer can be assured that the current physiological parameter measurements are normal, and if the dial is skewed too far to the left or right, the observer can quickly assess the severity of the physiological parameter levels and take appropriate action. In other embodiments, normal parameter measurements can be indicated when the dial is to the right or left, etc.

In some embodiments, the dial can be implemented as a dot, dash, arrow, or the like, and the arc can be implemented as a circle, spiral, pyramid, or other shape, as desired. Furthermore, the entire arc can be lit up or only portions of the arc can be lit up based on the current physiological parameter measurement level. Furthermore, the arc can turn colors or be highlighted based on the current physiological parameter level. For example, as the dial approaches a threshold level, the arc and/or dial can turn from green, to yellow, to red, shine brighter, flash, be enlarged, move to the center of the display, or the like.

Different physiological parameters can have different thresholds indicating abnormal conditions. For example, some physiological parameters may upper a lower threshold levels, while others only have an upper threshold or a lower threshold. Accordingly, each health indicator can be adjusted based on the physiological parameter being monitored. For example, the SpO2 health indicator can have a lower threshold that when met activates an alarm, while the respiration rate health indicator can have both a lower and upper threshold, and when either is met an alarm is activated. The thresholds for each physiological parameter can be based on typical, expected thresholds and/or user-specified thresholds.

The digital indicator can provide a numerical representation of the current levels of the physiological parameter the digital indicator may indicate an actual level or a normalized level and can also be used to quickly assess the severity of a patient condition. In some embodiments, the display includes multiple health indicators for each monitored physiological parameter. In certain embodiments, the display includes fewer health indicators than the number of monitored physiological parameters. In such embodiments, the health indicators can cycle between different monitored physiological parameters.

FIGS. 23A-23F illustrate exemplary displays of measurement data showing, for example, data presentation in FIGS. 23A-23D when a depth of consciousness monitor is connected to a channel port of the hub of FIG. 1. As shown in FIGS. 23A-23C, the hub 100 advantageously roughly bifurcates its display 104 to show various information from the, for example, SEDLine device, commercially available from Masimo Corp. of Irvine, CA. In FIG. 23D, the hub 100 includes an attached PhaseIn device, commercially available by PHASEIN AB of Sweden, providing, for example, information about the patient's respiration. The hub 100 also includes the SEDLine information, so the hub 100 has divided the display 104 appropriately. In FIG. 23E, temperature and blood pressure sensors communicate with the hub of FIG. 1 and the hub 100 creates display real estate appropriate for the same. In FIG. 23F, an acoustic sensor is also communicating with the hub of FIG. 1, as well as the forgoing blood pressure and temperature sensor. Accordingly, the hub 100 adjust the display real estate to accommodate the data from each attached device.

The term "and/or" herein has its broadest least limiting meaning which is the disclosure includes A alone, B alone, both A and B together, or A or B alternatively, but does not require both A and B or require one of A or one of B. As used herein, the phrase "at least one of" A, B, "and" C should be construed to mean a logical A or B or C, using a non-exclusive logical or.

The term "plethysmograph" includes its ordinary broad meaning known in the art which includes data responsive to changes in volume within an organ or whole body (usually resulting from fluctuations in the amount of blood or air it contains).

III. Additional Monitoring Environment Embodiments

FIG. 24 illustrates another embodiment of a monitoring environment 2000 including the hub 100 of FIG. 1. The monitoring environment 2000 may include all the features of the monitoring environment 200 of FIG. 2, as well as any of the other features described above. In addition, the monitoring environment 2000 depicts another embodiment of the multi-patient monitoring system 204, namely, the multi-patient monitoring system (MMS) 2004. The MMS 2004 includes a translation module 2005 that can receive serial data, translate the serial data into a format recognizable by the monitoring hub 100, and provide the serial data to the monitoring hub 100 (among possibly other devices). Also shown is an auxiliary device 2040 that may communicate with the MMS 2004, the monitoring hub 100, or the PPM 102, wired or wirelessly.

As described above, the hub 100 may receive serial data from a variety of medical equipment, including the patient's bed 214, infusion pumps 216, a ventilator 218, and other vital signs monitors 220. The hub 100 can pass serial data from these sources on to the MMS 2004. As described

above, the MMS **2004** may then store the serial data in a caregiver backend system **206** such as an EMR system or ADT system.

The medical equipment providing this serial data may use a variety of different proprietary protocols, messaging infrastructure, and the like that may not be natively recognizable by the hub **100**. Accordingly, the hub **100** may not have native capability to read parameter values or other data from this medical equipment, and as a result, may not have the capability to display parameter values or other data from these devices. Advantageously, however, the translation module **2005** at the MMS **2004** can receive serial data from these devices, translate the serial data into a format recognizable by the monitoring hub **100**, and provide the serial data to the monitoring hub **100**. The monitoring hub **100** can then read parameter values and other data from the translated information and output these values or data to a display, such as any of the displays described above.

In an embodiment, the translation module **2005** applies one or more translation rules to the serial data to translate or transform the serial data from one format to another format. The serial data may be formatted according to a Health Level Seven (“HL7”) protocol in one embodiment. The HL7 protocol has been developed to provide a messaging framework for the communication of clinical messages between medical computer systems and devices. However, the HL7 standard is quite flexible and merely provides a framework of guidelines. Consequently, medical devices or clinical computer systems that are all HL7-compliant may still be unable to communicate with each other. For example, the medical equipment **214-220** may each implement a version of the HL7 protocol, but these implementations may be different from an HL7 protocol implemented by the monitoring hub **100**. Accordingly, the monitoring hub **100** may not be able to parse or read messages from the medical equipment **214-220**, even though both use the HL7 standard. Further, the translation module **2005** may translate between different implementations of a common standard other than the HL7 protocol implemented by the hub **100** and medical equipment **214-220** in some embodiments.

In addition to translating between different implementations of a common electronic medical communication protocol (e.g., different formatting of HL7 messages), the translation module **2005** can also translate between input and output messages adhering to different communication protocols. In some embodiments, the translation module **2005** is capable of responding to and translating messages from, for example, one medical communication protocol to a separate medical communication protocol. For example, the translation module **2005** can facilitate communication between messages sent according to the HL7 protocol, the ISO 11073 protocol, other open protocols, or proprietary protocols. Accordingly, the translation module **2005** can translate an input message sent according to the HL7 protocol to an output message according to a different protocol, or vice-versa. In certain embodiments, the translation module **2005** can implement any of the translation features described below in greater detail under the section entitled “Translation Module Embodiments,” as well as further in U.S. application Ser. No. 14/032,132, filed Sep. 19, 2013, titled “Medical Monitoring System,” the disclosure of which is hereby incorporated by reference in its entirety.

Advantageously, in certain embodiments, the translation module **2005** can pass translated serial data back to the hub **100** or PPM **102**. Since the translated data is in a format readable by the hub **100** or PPM **102**, the hub **100** or PPM **102** can output the data from the medical equipment **214-**

220 on the display of the hub **100** or PPM **102**. In addition, the translation module **2005** can provide the translated data to devices other than the hub **100**, including clinician devices (such as cell phones, tablets, or pagers) and an auxiliary device **2040** that will be described below. Moreover, since the serial data provided by the medical equipment **214-220** may include alarm notifications, the translation module **2005** can pass these alarm notifications to the hub **100** or PPM **102**. The hub **100** or PPM **102** can therefore generate visual or audible alarms responsive to these alarm notifications. Further, the translation module **2005** can provide the alarm notifications to clinician devices, e.g., over a hospital network or wide area network (such as the Internet). In addition, the translation module **2005** can provide the alarm notifications to the auxiliary device **2040**.

The translation module **2005** is shown as implemented in the MMS **2004** because it may be beneficial to maintain and update the translation rules of the translation module **2005** in a single location. However, in other embodiments, the translation module **2005** may also be (or instead be) implemented in the hub **100** or PPM **102**. Accordingly, the hub **100** or PPM **102** can access an internal translation module **2005** to translate serial data for output to the display of the hub **100** or PPM **102**.

The auxiliary device **2040** can be a computing device having physical computer hardware, a display, and the like. For example, the auxiliary device **2040** may be a handheld computing device used by a clinician, such as a tablet, laptop, cellphone or smartphone, personal digital assistant (PDA), a wearable computer (such as a smart watch or glasses), or the like. The auxiliary device **2040** may also be simply a display device, such as a computer monitor or digital television. In an embodiment, the auxiliary device **2040** provides second screen functionality for the hub **100**, PPM **102**, or MMS **2004**. As such, the auxiliary device **2040** can communicate wirelessly or through a wired connection with the hub **100**, MMS **2004**, or PPM **102**.

As a second screen device, the auxiliary device **2040** can depict a copy of at least a portion of the display of the hub **100** (or the PPM **102**) or a different version of the hub **100** (or the PPM **102**) display. For instance, the auxiliary device **2040** can receive physiological parameter data, trend data, or waveforms from the hub **100**, PPM **102**, or MMS **2004** and display the parameter data, trend data, or waveforms. The auxiliary device **2040** can output any information available to the hub **100**, PPM **102**, or MMS **2004**. One use of the auxiliary device **2040** is as a clinician device usable by a clinician to view data from the hub **100**, PPM **102**, or MMS **2004** while away from a patient’s room (or even while in a patient’s room). A clinician can use the auxiliary device **2040** to view more detailed information about physiological parameters than is displayed on the hub **100** or PPM **102** in an embodiment (see, e.g., FIG. 39). For instance, the auxiliary device **2040** may include zoom functionality or the like that enables a clinician to zoom into trends or waveforms to more closely inspect parameter activity.

One example reason for copying at least a portion of the display of the hub **100** or PPM **102** is to enable different clinicians to have the same view of the data during a surgical procedure. In some surgical procedures, for instance, two anesthesiologists monitor a patient, one anesthesiologist monitoring the brain function and brain oxygenation of the patient, while the other monitors peripheral oxygenation of the patient. A brain sensor, such as has been described above, may be attached to the patient and provide brain monitoring and oxygenation data that is output to the hub **100** or the PPM **102** for presentation to the first anesthesiologist. A

finger or toe/foot optical sensor can also be attached to the patient and output data to the hub **100** or PPM **102**. The hub **100** or PPM **102** can transmit this data to the auxiliary device **2040**, which the second anesthesiologist can monitor to observe oxygenation in the patient's peripheral limbs. The second anesthesiologist may also need to know the oxygenation at the brain to help interpret the seriousness or lack thereof of poor peripheral oxygenation values. However, in many surgical procedures, a curtain or screen is placed over the patient as part of the procedure, blocking the second anesthesiologist's view of the hub **100** or PPM **102**. Accordingly, the hub **100** or PPM **102** can output a copy of at least a portion of its display to the auxiliary device **2040** so that the second anesthesiologist can monitor brain function or oxygenation.

In one embodiment, the auxiliary device has a larger display area than the display of the hub **100**. For instance, the hub **100** may have a relatively smaller display, such as about 10 inches, while the auxiliary device **2040** may be a television monitor or the like that has a 40 inch or larger display (although any size display may be used for the auxiliary device **2040**). In an embodiment, the auxiliary device **2040** as a television can include a hardware module that includes a processor, memory, and a wireless or wired networking interface or the like. The processor can execute programs from the memory, including programs for displaying physiological parameters, trends, and waveforms on the display of the television. Since a television monitor is larger than embodiments of the hub **100**, the television monitor version of the auxiliary device **2040** can display more fine detail of patient waveforms and trends in some embodiments (see, e.g., FIG. **39**).

In another embodiment, the auxiliary device **2040** may display one portion of any of the displays described herein while the hub **100** displays another portion thereof. For instance, the auxiliary device **2040** may display any of the anatomical graphics described above with respect to FIGS. **19A-19J**, while the hub **100** displays any of the parameter displays described above with respect to FIGS. **20A-23F** (or vice versa). Likewise, the auxiliary device **2040** may display the translated data received from the translation module **2005** while the hub **100** displays channel data (or vice versa). In another embodiment, the auxiliary device **2040** can display both translated data and channel data (see, e.g., FIG. **38**).

In still other embodiments, the auxiliary device **2040** can perform at least some processing of physiological parameters, including any of the functionality of the monitoring hub **100**. For instance, the auxiliary device **2040** may include the translation module **2005** and perform the features thereof.

FIG. **25** illustrates an embodiment of a translation message handling process **2100**. The process **2100** can be implemented by the translation module **2005** described above or by any other computing system. In an embodiment, at block **2502**, the translation module **2005** receives a message from the hub **100** (or PPM **102**) that includes a message from a medical device not natively compatible with the hub **100** (or PPM **102**). At block **2504**, the translation module **2005** translates the message based on one or more translation rules to produce a translated output message that can be processed by the hub **100** (or PPM **102**). At block **2506**, the translation module provides the translated output message to the hub **100** for display at the hub **100** (or PPM **102**) or at an auxiliary device **2040**. The hub **100** (or PPM **102**) may route the translated data to the auxiliary device

2040, or the auxiliary device **2040** may receive the translated data directly from the translation module **2005**.

For example, in one embodiment, a first medical device having digital logic circuitry receives a physiological signal associated with a patient from a physiological sensor, obtains a first physiological parameter value based on the physiological signal, and outputs the first physiological parameter value for display. The first medical device can also receive a second physiological parameter value from a second medical device other than the first medical device, where the second physiological parameter value is formatted according to a protocol not used by the first medical device, such that the first medical device is not able to process the second physiological parameter value to produce a displayable output value. The first medical device can pass the physiological parameter data from the first medical device to a separate translation module, receive translated parameter data from the translation module at the first medical device, where the translated parameter data is able to be processed for display by the first medical device, and output a second value from the translated parameter data for display. The first medical device may be, for example, the hub **100**, PPM **102**, or MMS **2004**, and the second medical device may be the infusion pump **216** or ventilator **218** or the like.

FIGS. **26-38** and **46-71** illustrate additional example hub displays, including displays of measurement data. Each of these displays may be implemented by the auxiliary device **2040**, although similar displays may also be output on the hub **100** (or PPM **102**) directly. The example Figures shown are depicted as being implemented for a tablet computer that includes touchscreen functionality. Touchscreen functionality is optional and can be replaced by other suitable input devices, such as keyboards, mice, track wheels, and the like.

Turning to FIG. **26**, the user interface shown depicts a device connected to the auxiliary device **2040**. The device shown is "Omar's Hawk," which can be an embodiment of the monitoring hub **100**. The auxiliary device **2040** is connected wirelessly to the hub **100** in this embodiment so as to receive data from the hub **100**. The auxiliary device could also connect wirelessly to the MMS **2004** or PPM **102** in other embodiments.

FIG. **27** depicts a default parameter view on the auxiliary device **2040**. Parameter values are shown together with waveforms in an upper portion of the display, and other parameters (such as SpHb, SpMet, PVI, etc.) are shown at the bottom of the display without their corresponding waveforms. Any of these parameters at the bottom of the display may be dragged and dropped onto the upper portion of the display to cause their waveforms to be shown. For instance, FIG. **28** depicts a similar display as in FIG. **27** except that the SpHb parameter has been dragged and dropped onto the upper portion of the display, causing the SpHb waveform and additional details on alarm limits (18 and 7) to be shown. Similarly, FIG. **29** shows the same display as FIG. **28** except that the SpMet parameter has been dragged and dropped on the upper portion of the display, causing its waveform and alarm limit (3) to be shown.

In each of the displays of FIGS. **27-29**, a time window button is shown in the upper right corner. This time window button says "1 hr" in FIGS. **27-29** but may be selected by a user to change the time window, which can affect the window of trend or waveform data shown in the display. A user selection of this time window button and change to a 10 minute window is shown in FIG. **30**. As can be seen, the waveforms in FIG. **30** are shown in a smaller window of time than in the previous Figures.

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FIG. 31 shows another version of the display of FIG. 29 with stacked waveforms, including a stacked SpO2 and respiratory waveform, similar to other stacked waveforms described elsewhere herein. FIG. 32 shows a similar display to FIG. 29 with the pulse rate (PR) and SpMet (methemoglobin) parameters highlighted as being in alarm condition. The alarm condition can be represented as a red box around the parameter values and waveforms in an embodiment, or with red transparency coloring at least a portion of the box. The red box or transparency may also flash in an embodiment, and an audible alarm may sound. Other ways to represent an alarm condition are used in other embodiments.

FIG. 33 shows a popup interface that enables a user to adjust alarm limits for a parameter (in this embodiment, SpHb or total hemoglobin). The popup interface includes scroll wheels that allow a user to quickly scroll among and select possible parameter limit values.

FIGS. 34-38 show landscape display views in contrast to the portrait-oriented displays of FIGS. 26-33. These landscape display views may be accessed by rotating the auxiliary device 2040 (such as tablet etc.) to a landscape orientation. FIG. 34 shows a first set of parameters, while FIGS. 35 and 36 add additional drag-and-dropped parameters with their waveforms and additional alarm limit details, similar to those described above with respect to FIGS. 27-29. FIG. 37 depicts stacked parameter waveforms, stacking SpO2 and respiratory waveforms. FIG. 38 depicts both channel parameters (such as SpO2, PR (pulse rate), and RRa (acoustically-measured respiratory rate)) while also showing translated serial data parameters 2210, including parameters from a pump and a vent. These translated serial data parameters 2210 may have been received from the translation module 2005, either through the hub 100 or directly from the MMS 2004.

Referring again to FIG. 24, as described above, the hub 100 or PPM 102 can output a copy of at least a portion of the display to the auxiliary device 2040. In other embodiments, the hub 100 or PPM 102 can output data with respect to a subset of the full parameters shown on the hub 100 or PPM 102 to the auxiliary device 2040. For instance, the hub 100 or PPM 102 may provide functionality for a clinician to select one or more of the parameters displayed thereon to see just that one or more parameters displayed on the auxiliary device 2040. Doing so may allow the auxiliary device 2040 to show more detail about the selected one or more parameters because fewer parameters may be shown on the auxiliary device's 2040 display than on the hub 100 or PPM 102.

FIG. 39 depicts one example display of an auxiliary device 2040 that depicts data with respect to one parameter, respiratory rate. Unlike the main display of the hub 100 or PPM 102, the display shown in FIG. 39 includes more than just the current value 2215, a recent trend 2230, and small waveform of the respiratory rate. In addition, the display depicts a histogram 2220 of historical highs and lows (e.g., for the past several days) of the patient being monitored. In addition, a detailed waveform 2240 is shown, which may be larger than the waveforms shown on the main display of the hub 100 or PPM 102, which may give the user more detailed insight into the patient's respiratory condition. A user may choose to zoom into the waveform 2240 (or other aspects of the display), causing the waveform 2242 to be enlarged to fill the display in place of the other elements of the display, or the like. Other graphs, tables, waveforms, and data may be shown for the respiratory parameter on the auxiliary

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device display 2040. Of course, parameters other than respiratory rate may also be selected for detailed display on the auxiliary device 2040.

IV. Translation Module Embodiments

Any of the following features described with respect to FIGS. 40A through 45D can be implemented by the translation module 2005 of FIG. 24 or together with any of the devices described above with respect to FIG. 24.

Healthcare costs have been increasing and the demand for reasonably-priced, high-quality patient care is also on the rise. Health care costs can be reduced by increasing the effectiveness of hospital information systems. One factor which may affect the efficacy of a health institution is the extent to which the various clinical computer systems employed at the health institution can interact with one another to exchange information.

Hospitals, patient care facilities, and healthcare provider organizations typically include a wide variety of different clinical computer systems for the management of electronic healthcare information. Each of the clinical computer systems of the overall IT or management infrastructure can help fulfill a particular category or aspect of the patient care process. For example, a hospital can include patient monitoring systems, medical documentation and/or imaging systems, patient administration systems, electronic medical record systems, electronic practice management systems, business and financial systems (such as pharmacy and billing), and/or communications systems, etc.

The quality of care in a hospital or other patient care facility could be improved if each of the different clinical computer systems across the IT infrastructure (or even within the same hospital room; see, e.g., FIGS. 1 and 24) were able to effectively communicate with each other. This could allow for the exchange of patient data that is collected by one clinical computer system with another clinical computer system that could benefit from such patient data. For example, this may allow decisions relating to patient care to be made, and actions to be taken, based on a complete analysis of all the available information.

In current practice, individual clinical computer systems can be, and often are, provided by different vendors. As a result, individual clinical computer systems may be implemented using a proprietary network or communication infrastructure, proprietary communication protocols, etc.; the various clinical computer systems used in the hospital cannot always effectively communicate with each other.

Medical device and medical system vendors sometimes develop proprietary systems that cannot communicate effectively with medical devices and systems of other vendors in order to increase their market share and to upsell additional products, systems, and/or upgrades to the healthcare provider. Thus, healthcare providers are forced to make enterprise or system-wide purchase decisions, rather than selecting the best technology available for each type of individual clinical computer system in use.

One example where this occurs is in the area of life-saving technology available for patient monitoring. For example, many different bedside devices for monitoring various physiological parameters are available from different vendors or providers. One such provider may offer a best-in-class device for monitoring a particular physiological parameter, while another such provider may offer the best-in-class device for another physiological parameter. Accordingly, it may be desirable in some circumstances for a hospital to have the freedom to use monitoring devices from

more than one manufacturer, but this may not be possible if devices from different manufacturers are incapable of interfacing and exchanging patient information. Accordingly, the ability to provide reasonably-priced, high-quality patient care can be compromised. In addition, since each hospital or patient care facility may also implement its own proprietary communication protocols for its clinical computer network environment, the exchange of information can be further hindered.

As described above, the Health Level Seven (“HL7”) protocol has been developed to provide a messaging framework for the communication of clinical messages between medical computer systems and devices. The HL7 communication protocol specifies a number of standards, guidelines, and methodologies which various HL7-compliant clinical computer systems can use to communicate with each other.

The HL7 communication protocol has been adopted by many medical device manufacturers. However, the HL7 standard is quite flexible, and merely provides a framework of guidelines (e.g., the high-level logical structure of the messages); consequently, each medical device or medical system manufacturer or vendor may implement the HL7 protocol somewhat differently while still remaining HL7-compliant. For example, the format of the HL7 messages can be different from implementation to implementation, as described more fully herein. In some cases, the HL7 messages of one implementation can also include information content that is not included in messages according to another HL7 implementation. Accordingly, medical devices or clinical computer systems that are all HL7-compliant still may be unable to communicate with each other.

Consequently, a translation module can be provided that can improve the communication of medical messages between medical devices or systems that use different allowed implementations of an established communication protocol (e.g., HL7), thereby increasing the quality of patient care through the integration of multiple clinical computer systems.

FIG. 40A illustrates a first medical device **2405** and a second medical device **2410** that communicate with one another via a translation module **2415**. The first medical device **2405** is configured to transmit and receive messages according to a first allowed format or implementation of an accepted electronic medical communication protocol, while the second medical device **2410** is configured to transmit and receive messages according to a second allowed format or implementation of the electronic medical communication protocol. In some embodiments, the first and second protocol formats are different implementations of the HL7 communication protocol. Other electronic medical communication protocols besides HL7 can also be used.

The translation module **2415** receives input messages having the first protocol format from the first medical device **2405** and generates output messages to the second medical device **2410** having the second protocol format. The translation module **2415** also receives input messages having the second protocol format from the second medical device **2410** and generates output messages to the first medical device **2405** having the first protocol format. Thus, the translation module **2415** can enable the first and second medical devices **2405**, **2410** to effectively and seamlessly communicate with one another without necessarily requiring modification to the communication equipment or protocol implemented by each device.

In certain embodiments, the translation module **2415** determines the protocol format expected by an intended

recipient of the input message based on, for example, the information in the input message or by referencing a database that stores the protocol format used by various devices, and then generates the output message based on the protocol format used by the intended recipient device or system. The output message can be generated based upon a comparison with, and application of, a set of translation rules **2420** that are accessible by the translation module **2415**.

The translation rules **2420** can include rules that govern how to handle possible variations between formatting implementations within a common protocol. Examples of variations in formatting implementation of an electronic medical communication protocol include, for example, the delimiter or separator characters that are used to separate data fields, whether a particular field is required or optional, the repeatability of portions of the message (e.g., segments, fields, components, sub-components), the sequence of portions of the message (e.g., the order of fields or components), whether a particular portion of a message is included, the length of the message or portions of the message, and the data type used for the various portions of the message.

In certain embodiments, the translation rules **2420** define additions, deletions, swappings, and/or modifications that should be performed in order to “translate” an input message that adheres to a first HL7 implementation into an output message that adheres to a second HL7 implementation. The output message can have, for example, different formatting than the input message, while maintaining all, or a portion of, the substance or content of the input message.

In addition to translating between different implementations of a common electronic medical communication protocol (e.g., different formatting of HL7 messages), the translation module **2415** can also translate between input and output messages adhering to different communication protocols. In some embodiments, the translation module **2415** is capable of responding to and translating messages from, for example, one medical communication protocol to a separate medical communication protocol. For example, the translation module **2415** can facilitate communication between messages sent according to the HL7 protocol, the ISO 11073 protocol, other open protocols, and/or proprietary protocols. Accordingly, an input message sent according to the HL7 protocol can be translated to an output message according to a different protocol, or vice-versa.

The operation of the translation module **2415** and the translation rules **2420** will be described in more detail below. Various embodiments of system architectures including the translation module **2415** will now be described.

In certain embodiments, the first medical device **2405**, the second medical device **2410**, and the translation module **2415** are communicatively coupled via connection to a common communications network or directly (via cables or wirelessly), for example, through the hub **100**, PPM **102**, and/or MMS **2004**. In some embodiments, the translation module **2415** can be communicatively coupled between the first medical device **2405** and the second medical device **2410** (with or without a communications network) such that all messages between the first and second medical devices **2405**, **2410** are routed through the translation module **2415**. Other architectures are also possible.

The first and second medical devices **2405**, **2410** and the translation module **2415** can be included in, for example, a portion of the monitoring environments of FIG. 1 or 24 described above. The first medical device **2405** may be, for example, the infusion pump(s) **216** or ventilator **218**, while the second medical device **2410** may be, for example, the monitoring hub **100**, PPM **102**, MMS **2004**, or auxiliary

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device **2040**. The translation module **2415** is an example implementation of the translation module **2005**.

In certain embodiments, the translation module **2415** can facilitate communication across multiple networks within a hospital environment. In other embodiments, the translation module **2415** can facilitate communication of messages across one or more networks extending outside of the hospital or clinical network environment. For example, the translation module **2415** can provide a communications interface with banking institutions, insurance providers, government institutions, outside pharmacies, other hospitals, nursing homes, or patient care facilities, doctors' offices, and the like.

In some embodiments, the translation module **2415** of FIG. **40** can be a component of, for example, the environment **2000** described above with respect to FIG. **24**. For example, the translation module **2415** can be communicatively coupled with a hospital network or other networks or monitoring environments described above. In such embodiments, the translation module **2415** can facilitate the exchange of patient monitoring information, including, for example, physiological parameter measurements, physiological parameter trend information, and physiological parameter alarm conditions between bedside medical monitor devices, nurses' monitoring stations, a Hospital or Clinical Information System (which may store Electronic Medical Records), and/or many other medical devices and systems. The translation module **2415** can enable seamless communication between different medical devices and systems, each of which may use a different implementation of an electronic medical communication protocol such as, for example, the HL7 communication protocol, within a clinical or hospital network environment.

In certain embodiments, the translation module **2415** can also facilitate communication between a first medical device that is part of the patient monitoring sub-system and a second medical device that is not part of, or is external to, the patient monitoring system **200**. As such, the translation module **2415** can be capable of responding to externally-generated medical messages (such as patient information update messages, status query messages, and the like from an HIS or CIS) and generating external reporting messages (such as event reporting messages, alarm notification messages, and the like from patient monitors or nurses' monitoring stations).

In another embodiment, first and second medical devices **2405**, **2410** communicate with each other over a communication bus **2421**. Communication bus **2421** can include any one or more of the communication networks, systems, and methods described above, including the Internet, a hospital WLAN, a LAN, a personal area network, etc. For example, any of the networks describe above can be used to facilitate communication between a plurality of medical devices, including first and second medical devices **2405**, **2410**, discussed above. One such embodiment is illustrated in FIG. **40B**.

In FIG. **40B**, first medical device **2405** provides a message to the communication bus **2421**. The message is intended for receipt by the second medical device **2410**; however, because first and second medical devices **2405**, **2410** communicate according to different communication protocol format, second medical device **2410** is unable to process the message.

Translation module **2415** monitors the communication bus **2421** for such messages. Translation module receives the message and determines that first medical device **2405** is attempting to communicate with second medical device

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2410. Translation module **2415** determines that message translation would facilitate communication between first and second medical devices **2405**, **2410**. Translation module **2415** therefore utilizes an appropriate translation rule stored in a translation module **2420**. Translation module **2420** can include a memory, EPROM, RAM, ROM, etc.

The translation module **2415** translates the message from the first medical device **2405** according to any of the methods described herein. Once translated, the translation module **2415** delivers the translated message to the communication bus **2421**. The second medical device **2410** receives the translated message and responds appropriately. For example, the second medical device may perform a function and/or attempt to communication with the first medical device **2405**. The translation module **2415** facilitates communication from the second medical device **2410** to the first medical device **2405** in a similar manner.

The first medical device **2405** and the second medical device **2410** can be, for example, any of the medical devices or systems communicatively coupled to a hospital network or hub **100**, PPM **102**, and/or MMS **2004**. These medical devices or systems can include, for example, point-of-care devices (such as bedside patient monitors), data storage units or patient record databases, hospital or clinical information systems, central monitoring stations (such as a nurses' monitoring station), and/or clinician devices (such as pagers, cell phones, smart phones, personal digital assistants (PDAs), laptops, tablet PCs, personal computers, pods, and the like).

In some embodiments, the first medical device **2405** is a patient monitor for communicatively coupling to a patient for tracking a physiological parameter (e.g., oxygen saturation, pulse rate, blood pressure, etc.), and the second medical device **2410** is a hospital information system ("HIS") or clinical information system ("CIS"). In some embodiments, the patient monitor can communicate physiological parameter measurements, physiological parameter alarms, or other physiological parameter measurement information generated during the monitoring of a patient to the HIS or CIS for inclusion with the patient's electronic medical records maintained by the HIS or CIS.

In some embodiments, the first medical device **2405** is an HIS or CIS and the second medical device **2410** is a nurses' monitoring station, as described herein. However, the translation module **2415** can facilitate communication between a wide variety of medical devices and systems that are used in hospitals or other patient care facilities. For example, the translation module **2415** can facilitate communication between patient physiological parameter monitoring devices, between a monitoring device and a nurses' monitoring station, etc.

Using the translation module **2415**, a patient monitoring sub-system, such as those described herein (e.g., physiological monitoring system **200**), can push data to the HIS or pull data from the HIS even if the HIS uses a different implementation of the HL7 protocol, or some other electronic medical communication protocol.

In certain embodiments, the patient monitoring sub-system can be configured to push/pull data at predetermined intervals. For example, a patient monitor or clinician monitoring station can download patient data automatically from the HIS at periodic intervals so that the patient data is already available when a patient is connected to a patient monitor. The patient data sent from the HIS can include admit/discharge/transfer ("ADT") information received upon registration of the patient. ADT messages can be initiated by a hospital information system to inform ancillary

systems that, for example, a patient has been admitted, discharged, transferred or registered, that patient information has been updated or merged, or that a transfer or discharge has been canceled.

In other embodiments, the patient monitoring sub-system can be configured to push/pull data to/from the HIS only when the HIS is solicited by a query. For example, a clinician may make a request for information stored in a patient's electronic medical records on the HIS.

In still other embodiments, the patient monitoring sub-system can be configured to push/pull data to/from the HIS in response to an unsolicited event. For example, a physiological parameter of a patient being monitored can enter an alarm condition, which can automatically be transmitted to the HIS for storing in the patient's electronic medical records. In yet other embodiments, any combination of the above methods or alternative methods for determining when to communicate messages to and from the HIS can be employed.

Example system architectures and example triggers for the communication of messages involving the translation module **2415** have been described. Turning now to the operation of the translation module, FIGS. **25A-25D** illustrate an example medical message at different phases or steps of a translation process. The translation process will be described in more detail below in connection with FIGS. **26**, **27A** and **27B**.

FIG. **41A** illustrates an example ADT input message **2505** received by the translation module **2415** from an HIS. The ADT input message **2505** is implemented according to the HL7 communication protocol and contains information related to the admission of a patient to a hospital. The ADT message **2505** includes multiple segments, including a message header segment **2506**, an event segment, a patient identification segment, a patient visit segment, role segments, a diagnosis segment, and multiple custom segments.

In some embodiments, the message header ("MSH") segment **2506** defines how the message is being sent, the field delimiters and encoding characters, the message type, the sender and receiver, etc. The first symbol or character after the MSH string can define the field delimiter or separator (in this message, a "caret" symbol). The next four symbols or characters can define the encoding characters. The first symbol defines the component delimiter ("~"), the second symbol defines the repeatable delimiter ("|"), the third symbol defines the escape delimiter ("\"), and the fourth symbol defines the sub-component delimiter ("&"). All of these delimiters can vary between HL7 implementations.

In some embodiments, the example header segment **2506** further includes the sending application ("VAFC PIMS"), the receiving application ("NPTF-508"), the date/time of the message ("20091120104609-0600"), the message type ("ADT~A01"), the message control ID ("58103"), the processing ID ("P"), and the country code ("USA"). As represented by the consecutive caret symbols, the header segment also contains multiple empty fields.

FIG. **41B** illustrates the message header segment **2506** after it has been parsed into fields or elements based on an identified field delimiter (the caret symbol). In certain embodiments, the parsed input message comprises an XML message that is configured to be transformed according to extensible stylesheet language transformation (XSLT) rules.

In certain embodiment, the parsed input message can be encoded. FIG. **41C** illustrates the parsed message header

segment of the input message after being encoded (e.g., using a Unicode Transformation Format-8 ("UTF-8") encoding scheme).

The encoded message header segment shows some of the various data types that can be used in the message. For example, the sending application ("VAFC PIMS") of the third parsed field and the receiving application ("NPTF-508") of the fifth parsed field are represented using a hierarchic designator ("HD") name data type. The date/time field (the seventh parsed field) is represented using the time stamp ("TS") data type. The processing ID field (the eleventh parsed field) is represented using the processing type ("PT") data type. The fields that do not include a data type identifier are represented using the string ("ST") data type. Other possible data types include, for example, coded element, structured numeric, timing quantity, text data, date, entry identifier, coded value, numeric, and sequence identification. The data types used for the various fields or attributes of the segments can vary between formatting implementations.

FIG. **41D** illustrates an example output message **2510** from the translation module **2415** based on the example input message **2505** of FIG. **41A**. The output message **2510** includes a message acknowledgement segment **2512**.

Turning to the operation of the translation module, the translation module **2415** can, for example, create, generate, or produce an output message that is reflective of the input message based on an application of the set of translation rules **2420**. In some embodiments, the translation module **2415** can, for example, translate, transform, convert, reformat, configure, change, rearrange, modify, adapt, alter, or adjust the input message based on a comparison with, and application of, the set of translation rules **2420** to form the output message. In some embodiments, the translation module **2415** can, for example, replace or substitute the input message with an output message that retains the content of the input message but has a new formatting implementation based upon a comparison with, and application of, the set of translation rules **2420**.

FIG. **42** illustrates a translation process **2600** for generating an output message based on an input message and a comparison with the set of translation rules **2420** associated with the translation module **2415**. The translation process **2600** starts at block **2602** where the translation module **2415** receives an input message from a first medical device.

At block **2604**, the translation module **2415** determines the formatting implementation of the input message and the formatting implementation to be used for the output message. In certain embodiments, the input message can include one or more identifiers indicative of the formatting implementation. In some embodiments, the determination of the formatting implementation can be made, for example, by analyzing the message itself by identifying the delimiter or encoding characters used, the field order, the repeatability of segments, fields, or components, the data type of the fields, or other implementation variations. In certain embodiments, the translation module **2415** can separate or parse out the formatting from the content of the message (as shown in FIG. **41B**) to aid in the determination of the formatting implementation. In some embodiments, the translation module **2415** determines the formatting implementation of the input message by referencing a database that stores the implementation used by each device with which the translation module **2415** has been configured to interface.

In certain embodiments, the determination of the formatting implementation used by the output message can also be determined from the input message. For example, the input

message can include a field that identifies the intended recipient application, facility, system, device, and/or destination. The input message can alternatively include a field that identifies the type of message being sent (e.g., ADT message) and the translation module **2415** can determine the appropriate recipient from the type of message being sent and/or the sending application, device, or system. The translation module **2415** can then determine the formatting implementation required by the intended recipient of the input message.

At decision block **2605**, the translation module **2415** determines whether a rule set has been configured for the translation from the identified formatting implementation of the input message to the identified formatting implementation to be used for the output message. The rule set may have been manually configured prior to installation of the translation module software or may have been automatically configured prior to receipt of the input message. If a rule set has already been configured, then the translation process **2600** continues to block **2606**. If a rule set has not been configured, then a rule set is configured at block **2607**. The configuration of the rule set can be performed as described below in connection with FIGS. **44** and **45A-2459D**. The translation process **2600** then continues to block **2608**.

At block **2606**, the translation module **2415** identifies the pre-configured rules from the set of translation rules **2420** that govern translation between the determined formatting implementation of the input message and the formatting implementation of the output message. In some embodiments, the identification of the pre-configured rules can be made manually.

At block **2608**, the translation module **2415** generates an output message based on the configured rule set(s) of the translation rules **2420**. In certain embodiments, the output message retains all, or at least a portion of, the content of the input message but has the format expected and supported by the intended recipient of the input message.

The translation rules **2420** can include, for example, unidirectional rules and/or bidirectional rules. A unidirectional rule can be one, for example, that may be applied in the case of a message from a first medical device (e.g., **2405**) to a second medical device (e.g., **2410**) but is not applied in the case of a message from the second medical device to the first medical device. For example, a unidirectional rule could handle a difference in the delimiters used between fields for two different formatting implementations of, for example, the HL7 communication protocol. The translation module **2415** can apply a field delimiter rule to determine if the field delimiter is supported by the intended recipient of the input message. If the field delimiter of the input message is not supported by the intended recipient, the field delimiter rule can replace the field delimiter of the input message with a field delimiter supported by the intended recipient.

For example, an input message from an input medical device can include a formatting implementation that uses a "caret" symbol ("^") as the field delimiter or separator. However, the formatting implementation recognized by the intended recipient medical device may use a "pipe" symbol ("|") as the field delimiter. The translation module **2415** can identify the field delimiter symbol used in the formatting implementation recognized by the intended recipient medical device from the set of translation rules **2420** and generate an output message based on the input message that uses the pipe field delimiter symbol instead of the caret field delimiter symbol used in the input message. The rule to substitute a pipe symbol for a caret symbol would, in this case, only apply to messages that are sent to a recipient device that

recognizes the pipe symbol as a field delimiter. This rule could be accompanied by a complementary rule that indicates that a caret symbol should be substituted for a pipe symbol in the case of a message that is intended for a recipient device that is known to recognize the caret symbol as the field delimiter.

Another unidirectional rule can handle the presence or absence of certain fields between different formatting implementations. For example, an input message from an input medical device can include fields that would not be recognized by the intended recipient medical device. The translation module **2415** can generate an output message that does not include the unrecognized or unsupported fields. In situations where an input message does not include fields expected by the intended recipient medical device, the set of translation rules **2420** can include a rule to insert null entries or empty " " strings in the fields expected by the intended recipient medical device and/or to alert the recipient device of the absence of the expected field. The sender device may also be notified by the translation module **2415** that the recipient device does not support certain portions of the message.

Other unidirectional rules can facilitate, for example, the conversion of one data type to another (for example, string ("ST") to text data ("TX") or structured numeric ("SN") to numeric ("NM")), and the increase or decrease in the length of various portions of the message. Unidirectional rules can also be used to handle variations in repeatability of portions of the message. For example, the translation module **2415** can apply a field repeatability rule to repeated instances of a segment, field, component, or sub-component of the message to determine how many such repeated instances are supported by the recipient device, if any, and deleting or adding any repeated instances if necessary. For example, a phone number field of a patient identification segment can be a repeatable field to allow for entry of home, work, and cell phone numbers.

Bidirectional rules can also be used. Such rules may apply equally to messages between first and second medical devices (e.g., **2405**, **2410**) regardless of which device is the sender and which is the recipient. A bidirectional rule can be used to handle changes in sequence, for example. In certain implementations, an input message from an input medical device can include a patient name field, or fields, in which a first name component appears before a last name component. However, the intended recipient medical device may be expecting an implementation where the last name component appears before the first name component. Accordingly, the set of translation rules **2420** can include a bidirectional rule to swap the order of the first and last name components when communicating between the two medical devices, or between the two formatting implementations. In general, field order rules can be applied to determine whether the fields, components, or sub-components are in the correct order for the intended recipient and rearranging them if necessary. Other bidirectional rules can be included to handle, for example, other sequential variations between formatting implementations or other types of variations.

The translation rules **2420** can also include compound rules. For example, a compound rule can include an if-then sequence of rules, wherein a rule can depend on the outcome of another rule. Some translation rules **2420** may employ computations and logic (e.g., Boolean logic or fuzzy logic), etc.

As discussed above, the messages communicated over the hospital-based communication network can employ the HL7 protocol. FIGS. **43A** and **43B** illustrate translation processes

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2700A, 2700B in which HL7 messages are communicated between a HIS and a medical device over a hospital-based communications network or a clinical network. The translation processes 2700A, 2700B will be described with the assumption that the rules governing “translation” between the first and second HL7 formats have already been configured.

FIG. 43A illustrates a translation process 2700A in which the translation module 2415 facilitates communication of an HL7 message, such as the ADT message of FIG. 41A, from an HIS having a first HL7 format to an intended recipient medical device, such as a patient monitor or a clinician monitoring station, having a second HL7 format.

The translation process 2700A starts at block 2701, where the translation module 2415 receives an input message having a first HL7 format from the HIS. In certain embodiments, the input message includes information regarding, for example, the admission of a patient and/or patient identification and patient medical history information from an electronic medical records database.

At block 2703, the translation module 2415 determines the formatting implementation of the input message and the formatting implementation to be used for the output message. These determinations can be made in a similar manner to the determinations discussed above in connection with block 2604 of FIG. 42.

At block 2705, the translation module 2415 identifies the rules that govern translation between the determined HL7 format of the input message and the HL7 format of the output message and generates an output message having the second HL7 format based on the identified rules. In certain embodiments, the output message retains the content of the input message sent by the HIS but has the format expected and supported by the intended recipient of the input message.

At block 2707, the translation module 2415 can output the output message to the intended recipient over the hospital-based communications network. In certain embodiments, the intended recipient can transmit an acknowledgement message back to the hospital information system acknowledging successful receipt or reporting that an error occurred.

FIG. 43B illustrates a translation process 2700B in which the translation module 2415 facilitates communication of an HL7 message from a medical device, such as a patient monitor, having a first HL7 format to an HIS having a second HL7 format. For example, the patient monitor can transmit reporting event data in such as patient alarm data, to the HIS to store in the patient’s electronic medical records.

The translation process 2700B starts at block 2702, where the translation module 2415 receives an input message having a first HL7 format from the medical device. In certain embodiments, the input message includes patient monitoring data or alarm data regarding one or more physiological parameters of the patient being monitored for storage in an electronic medical records database associated with the HIS.

At block 2704, the translation module 2415 determines the formatting implementation of the input message and the formatting implementation to be used for the output message. These determinations can be made in a similar manner to the determinations discussed above in connection with block 2604 of FIG. 42.

At block 2706, the translation module 2415 identifies the rules that govern translation between the determined HL7 format of the input message and the HL7 format of the output message and generates an output message having the second HL7 format based on the identified rules. In certain

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embodiments, the output message retains the content of the input message sent by the medical device but has the format expected and supported by the HIS.

At block 2708, the translation module 2415 can output the output message to the hospital information system over the hospital-based communications network. In certain embodiments, the HIS can transmit an acknowledgement message back to the medical device acknowledging successful receipt or reporting that an error occurred.

FIGS. 42, 43A and 43B described the operation of the translator module 2415. FIGS. 44 and 45A-45D will be used to illustrate the description of the configuration of the translation rules 2420.

The translation rules 2420 can be implemented as one or more stylesheets, hierarchical relationship data structures, tables, lists, other data structures, combinations of the same, and/or the like. In certain embodiments, the translation rules 2420 can be stored in local memory within the translation module 2415. In other embodiments, the translation rules 2420 can be stored in external memory or on a data storage device communicatively coupled to the translation module 2415.

The translation module 2415 can include a single rule set or multiple rule sets. For example, the translation module 2415 can include a separate rule set for each medical device/system and/or for each possible communication pair of medical devices/systems coupled to the network or capable of being coupled to the network. In some embodiments, the translation module 2415 can include a separate rule set for each possible pair of formatting implementations that are allowed under a medical communication protocol such as, for example, the HL7 protocol.

In certain embodiments, the translation rules 2420 can be manually inputted using, for example, the messaging implementation software tool 2800 illustrated in FIG. 44. For example, the software developer for a particular hospital network can determine the protocol message formats used by the devices and/or systems that are or can be coupled to the hospital network and then manually input rules to facilitate “translation” between the various protocol message formats supported or recognized by the devices and/or systems.

FIG. 44 illustrates an example screenshot from a messaging implementation software tool 2800 for manually configuring translation rules 2420 to be used by the translation module 2415. The screenshot from the messaging implementation software tool 2800 illustrates various parameters that may differ between formatting implementations of an electronic medical communication protocol, such as HL7. The screenshot also includes areas where a user can input information that defines, or is used to define, translation rules for converting between different HL7 implementations. In some embodiments, the messaging implementation software tool 2800 stores a variety of pre-configured rule sets based, for example, on known communication protocol implementations of various medical devices. In such embodiments, a user may configure one or more translation rules 2420 to be used in communications involving such devices by entering identification information, such as the device manufacturer, model number, etc. Based on this identification information, the messaging implementation tool 2800 can identify a pre-configured set of translation rules for communication with that device.

In other embodiments, the translation rules 2420 can be automatically generated. For example, the automatic generation of a new set, or multiple sets, of rules can be triggered by the detection of a newly recognized “commu-

nicating" medical device or system on a network. In certain embodiments, the automatic generation of a new set or multiple sets of rules can occur at the time a first message is received from or sent to a new "communicating" medical device or system coupled to the network. In still other embodiments, the automatic generation of rule sets includes updating or dynamically modifying a pre-existing set of rules.

The automatic generation of translation rule sets can be carried out in a variety of ways. For example, in some embodiments, the translation module **2415** can automatically initiate usage of a pre-configured set of translation rules **2420** based upon, for example, the make and model of a new device that is recognized on the network. In certain embodiments, the translation module **2415** can request one or more messages from the new device or system and then analyze the messages to determine the type of formatting being implemented, as illustrated by the automatic rule configuration process **2900A** of FIG. **45A**. The automatic rule configuration process **2900A** starts at block **2901**, where the translation module **2415** receives one or more messages from a detected medical device or system on the network. The messages can be received upon transmission to an intended recipient medical device or system or in response to a query sent by the translation module **2415** or another medical device or system coupled to the network.

At block **2903**, the translation module **2415** determines the protocol of the one or more received messages by, for example, analyzing the message or by consulting a database that indicates what communication protocol/format is implemented by each medical device or system on the network. In certain embodiments, the translation module **2415** is configured to handle medical messages implemented using a single common protocol, such as HL7. Accordingly, if a determination is made that the received messages are implemented using a non-supported or non-recognized protocol, the translation module can ignore the messages received from the detected medical device or system, output an alert or warning, or allow the messages to be sent without being translated.

At block **2905**, the translation module **2415** determines the formatting implementation of the received message(s). In certain embodiments, the received messages can include one or more identifiers indicative of the formatting implementation. In other embodiments, the determination of the formatting implementation can be made, for example, by analyzing the message itself by checking field order, the delimiter or encoding characters used, or other implementation variations. In certain embodiments, the translation module **2415** can separate or parse out the formatting from the content of the message to aid in the determination of the formatting implementation.

At block **2907**, the translation module **2415** configures one or more rules or rule sets to handle messages received from and/or sent to the detected medical device or system. In certain embodiments, the configuration of the rules involves the creation or generation of new rules. In other embodiments, the configuration of the rules involves the alteration or updating of existing rules. The configured rules or rule sets can be included with the translation rules **2420**. If a set of rules already exists for the formatting implementation used by the new device or system, then the configuration of new translation rules may not be required. Instead, existing translation rules can be associated with the new device or system for use in communication involving that device or system. In other embodiments, the translation module **2415** can create a new set of rules geared specifically

for the new device or system or can modify an existing set of rules based on subtle formatting variations identified.

In other embodiments, the translation module **2415** can generate test message(s) that may be useful in identifying the communication protocol and implementation used by a device or system. For example, the translation module can generate test messages to cause the newly detected device or system to take a particular action (e.g., store information) and then query information regarding the action taken by the newly detected device to determine whether or how the test message was understood. This is illustrated by the automatic rule configuration process **2900B** of FIG. **45B**.

The automatic rule configuration process **2900B** starts at block **2902**, where the translation module **2415** transmits one or more test, or initialization, messages to a remote device or system detected on a network. The test messages can be configured, for example, to instruct the remote device or system to take a particular action (e.g., store patient information). In certain embodiments, the test messages can be configured to generate a response indicative of the type of formatting recognized or supported by the remote device or system. In other embodiments, the test messages can be configured such that only devices or systems supporting a particular formatting implementation will understand and properly act on the test messages.

At block **2904**, the translation module **2415** queries the remote device or system to receive information regarding the action taken based on the test message sent to the remote device or system to determine whether the test message was understood. For example, if the test message instructed the remote device or system to store patient information in a particular location, the translation module **2415** can query the information from the location to determine whether the test message was understood. If the test message was not understood, the translation module **2415** can, for example, continue sending test messages of known formatting implementations until a determination is made that the test message has been understood.

At block **2906**, the translation module **2415** determines the protocol and formatting implementation based on the information received. As an example, in certain embodiments, the test message can include an instruction to store patient name information. The test message can include a patient name field having a first name component followed by a surname component. The translation module **2415** can then query the remote device or system to return the patient surname. Depending on whether the patient surname or the first name is returned, this query can be useful in determining information about the order of fields in the formatting implementation being used by the remote device or system. As another example, the test messages can instruct the detected device or system to store repeated instances of a component. The translation module **2415** can then query the device or system to return the repeated instances to see which, if any, were stored. This repeatability information can also be useful in determining whether certain fields are allowed to be repeated in the formatting implementation being used by the remote device for system, and, if so, how many repeated instances are permitted.

At block **2908**, the translation module **2415** configures one or more rules to handle messages received from and/or sent to the detected medical device or system. For example, the rules can convert messages from the message format used by a first medical device to that used by a second medical device, as described herein. In certain embodiments, the configuration of the rules involves the creation or generation of new rules. In other embodiments, the configu-

ration of the rules involves the alteration or updating of existing rules. If a set of rules already exists for the formatting implementation used by the new device or system, then the configuration of new translation rules may not be required. Instead, existing translation rules can be associated

FIGS. 29C and 29D illustrate automatic rule configuration processes performed by the translation module 2415 for messages utilizing the HL7 protocol. The HL7 protocol can be used, for example, to communicate electronic messages to support administrative, logistical, financial, and clinical processes. For example, HL7 messages can include patient administration messages, such as ADT messages, used to exchange patient demographic and visit information across various healthcare systems.

The automatic rule configuration process 2900C illustrated in FIG. 45C is similar to the process 2900A illustrated in FIG. 45A. At block 2911, the translation module 2415 receives one or more messages from an HL7 medical device. At block 2915, the translation module 2415 determines the formatting implementation of the HL7 medical device from the one or more messages received. As discussed above, the determination of the formatting implementation can be made, for example, by checking field order or sequence, field delimiter characters, repeatability, cardinality, and other HL7 implementation variations.

At block 2917, the translation module 2415 configures one or more rules to handle messages received from and/or sent to the HL7 medical device. In certain embodiments, the configuration of the rules involves the creation or generation of new rules for the detected formatting implementation. In other embodiments, the configuration of the rules involves the dynamic alteration or updating of existing rules. If a set of rules already exists for the formatting implementation used by the new HL7 medical device, then the configuration of new translation rules may not be required. Instead, existing translation rules can be associated with the new HL7 medical device for use in communication involving that device.

The automatic rule configuration process 2900D illustrated in FIG. 45D is similar to the process 2900B illustrated in FIG. 45B. At block 2912, the translation module 2415 transmits one or more test, dummy, or initialization messages to an HL7 medical device. In other embodiments, the translation module 2415 can cause one or more test messages to be transmitted to the new HL7 medical device from another HL7 medical device. As described above, the test messages can include messages having known HL7 formats configured to determine whether the HL7 device understands the test messages. The test messages can include test ADT messages, for example.

At block 2914, the translation module 2415 queries the HL7 medical device to receive information regarding an action taken or information stored in response to the test message. At block 2916, the translation module 2415 determines the formatting implementation of the HL7 device based on the information received. In certain embodiments, the translation module 2415 can analyze the information received to determine whether the test message or messages were properly understood. If none of the test messages were properly understood, the translation module 2415 can send additional test messages having other known HL7 formats and repeat blocks 2914 and 2916.

At block 2918, the translation module 2415 configures one or more translation rules to handle messages received from and/or sent to the detected HL7 medical device. In

certain embodiments, the configuration of the translation rules involves the creation or generation of new translation rules. In other embodiments, the configuration of the rules involves the alteration or updating of existing rules. If a set of translation rules already exists for the formatting implementation used by the new HL7 medical device, then the configuration of new translation rules may not be required. Instead, existing translation rules can be associated with the new HL7 medical device for use in communication involving that HL7 medical device.

The automatic rule configuration processes described above can be triggered by the detection of a network device or system by the translation module 2415. The medical devices referred to in FIGS. 45A-45D can include any of the devices or systems illustrated in FIG. 1 or 24.

In some embodiments, the automatic generation of translation rules can advantageously occur post-installation and post-compilation of the messaging sub-system software, which includes the translation module 2415. In certain embodiments, the automatic generation or dynamic modification of the translation rules 2420 can occur without having to recompile or rebuild the translation module software. This feature can be advantageous in terms of efficiently complying with U.S. Food and Drug Administration ("FDA") requirements regarding validation of software used in healthcare environments.

Take, for example, a situation where a medical device manufacturer plans to use the translation module 2415 to facilitate communication between a particular medical device or system that is to be installed in a hospital (e.g., a patient monitoring system, as described herein), or other patient care facility, and other devices or systems that are already installed at the hospital (e.g., the HIS or CIS). Any software required for the operation of the new medical device to be installed may be at least partially validated for FDA compliance prior to installation at the hospital despite the fact that, for example, the HL7 implementations of other existing devices or systems at the hospital may still be unknown. For example, any aspects of the software for the new medical device that are dependent upon receiving messages from other hospital devices can be validated pre-installation as being capable of fully and correctly operating when the expected message format is received. Then, once the medical device is installed at the hospital, the validation of the software can be completed by showing that the translation module 2415 is able to provide messages of the expected format to the newly installed device. In this way, FDA validation tasks can be apportioned to a greater extent to the pre-installation timeframe where they can be more easily carried out in a controlled manner rather than in the field.

In addition, the translation module 2415 can further help streamline FDA validation, for example, when a medical device or system is expected to be installed at different hospitals whose existing devices use, for example, different implementations of the HL7 protocol. Normally, this type of situation could impose the requirement that the entire functionality of the software for the new medical device be completely validated at each hospital. However, if the translation module 2415 is used to interface between the new medical device and the hospital's existing devices, then much of the software functionality could possibly be validated a single time prior to installation, as just described. Then, once installed at each hospital, the software validation for the medical device can be completed by validating that correct message formats are received from the translation module (the translation rules for which are field-customiz-

able). This may result in making on-site validation procedures significantly more efficient, which will advantageously enable more efficient FDA compliance in order to bring life-saving medical technology to patients more quickly by the use of field-customizable translation rules.

V. Example Embodiments

In certain embodiments, a system for providing medical data translation for output on a medical monitoring hub can include a portable physiological monitor comprising a processor that can: receive a physiological signal associated with a patient from a physiological sensor; calculate a physiological parameter based on the physiological signal, and provide a first value of the physiological parameter to a monitoring hub for display. The monitoring hub can include a docking station that can receive the portable physiological monitor. The monitoring hub can: receive the first value of the physiological parameter from the portable physiological monitor; output the first value of the physiological parameter for display; receive physiological parameter data from a medical device other than the portable physiological monitor, the physiological parameter data formatted according to a protocol other than a protocol natively readable or displayable by the monitoring hub; pass the physiological parameter data to a translation module; receive translated parameter data from the translation module, where the translated parameter data can be readable and displayable by the monitoring hub; and output a second value from the translated parameter data for display.

The system of the preceding paragraph can be combined with any subcombination of the following features: the monitoring hub is further configured to output the first value of the physiological parameter and the second value from the translated parameter data on separate displays; the monitoring hub is further configured to output the second value from the translated parameter data to an auxiliary device having a separate display from a display of the monitoring hub; the auxiliary device is selected from the group consisting of a television, a tablet, a phone, a wearable computer, and a laptop; the physiological parameter data comprises data from an infusion pump; the physiological parameter data comprises data from a ventilator; and the translation module is configured to translate the physiological parameter data from a first Health Level 7 (HL7) format to a second HL7 format.

In certain embodiments, a method of providing medical data translation for output on a medical monitoring hub can include: under the control of a first medical device comprising digital logic circuitry, receiving a physiological signal associated with a patient from a physiological sensor; obtaining a first physiological parameter value based on the physiological signal; outputting the first physiological parameter value for display; receiving a second physiological parameter value from a second medical device other than the first medical device, where the second physiological parameter value is formatted according to a protocol not used by the first medical device, such that the first medical device is not able to process the second physiological parameter value to produce a displayable output value; passing the physiological parameter data from the first medical device to a separate translation module; receiving translated parameter data from the translation module at the first medical device, the translated parameter data able to be processed for display by the first medical device; and outputting a second value from the translated parameter data for display.

The method of the preceding paragraph can be combined with any subcombination of the following features: further including translating the message by at least translating the message from a first Health Level 7 (HL7) format to a second HL7 format; the message can include data from a physiological monitor; the message can include data from an infusion pump or a ventilator; and the message can include data from a hospital bed.

In certain embodiments, a system for providing medical data translation for output on a medical monitoring hub can include a first medical device including electronic hardware that can: obtain a first physiological parameter value associated with a patient; output the first physiological parameter value for display; receive a second physiological parameter value from a second medical device other than the first medical device, the second physiological parameter value formatted according to a protocol not used by the first medical device, such that the first medical device is not able to process the second physiological parameter value to produce a displayable output value; pass the physiological parameter data from the first medical device to a translation module; receive translated parameter data from the translation module at the first medical device, the translated parameter data able to be processed for display by the first medical device; and output a second value from the translated parameter data for display.

The system of the preceding paragraph can be combined with any subcombination of the following features: the first medical device can also output the first value of the physiological parameter and the second value from the translated parameter data on the same display; the first medical device can also output the first value of the physiological parameter and the second value from the translated parameter data on separate displays; the first medical device can also output the second value from the translated parameter data to an auxiliary device; the auxiliary device can be a television monitor; the auxiliary device can be selected from the group consisting of a tablet, a phone, a wearable computer, and a laptop; the first medical device can include the translation module; the first medical device can also pass the physiological parameter data to the translation module over a network; and the physiological parameter data can include data from an infusion pump or a ventilator.

VI. Terminology

Many other variations than those described herein will be apparent from this disclosure. For example, depending on the embodiment, certain acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the algorithms). Moreover, in certain embodiments, acts or events can be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially. In addition, different tasks or processes can be performed by different machines and/or computing systems that can function together.

It is to be understood that not necessarily all such advantages can be achieved in accordance with any particular embodiment of the embodiments disclosed herein. Thus, the embodiments disclosed herein can be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

The various illustrative logical blocks, modules, and algorithm steps described in connection with the embodiments disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

The various illustrative logical blocks and modules described in connection with the embodiments disclosed herein can be implemented or performed by a machine, such as a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor can be a microprocessor, but in the alternative, the processor can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor can include electrical circuitry or digital logic circuitry configured to process computer-executable instructions. In another embodiment, a processor includes an FPGA or other programmable device that performs logic operations without processing computer-executable instructions. A processor can also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. A computing environment can include any type of computer system, including, but not limited to, a computer system based on a microprocessor, a mainframe computer, a digital signal processor, a portable computing device, a device controller, or a computational engine within an appliance, to name a few.

The steps of a method, process, or algorithm described in connection with the embodiments disclosed herein can be embodied directly in hardware, in a software module stored in one or more memory devices and executed by one or more processors, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of non-transitory computer-readable storage medium, media, or physical computer storage known in the art. An example storage medium can be coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor. The storage medium can be volatile or nonvolatile. The processor and the storage medium can reside in an ASIC.

Conditional language used herein, such as, among others, “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without

author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Further, the term “each,” as used herein, in addition to having its ordinary meaning, can mean any subset of a set of elements to which the term “each” is applied.

Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, certain embodiments of the inventions described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others.

Additionally, all publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

What is claimed is:

1. A method of displaying medical data, the method comprising:
 - by a first medical device comprising digital logic circuitry, receiving a first physiological signal from a physiological sensor;
 - determining, based on the first physiological signal, a first physiological parameter value of a first physiological parameter associated with a patient;
 - outputting, in a first section of a display, the first physiological parameter value for display together with a corresponding waveform of the first physiological parameter;
 - receiving a second physiological parameter value of a second physiological parameter associated with the patient;
 - outputting, in a second section of the display, the second physiological parameter value for display;
 - outputting, in the first section of the display, indications of alarm limits associated with the first physiological parameter, wherein the indications of the alarm lim-

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its are displayed adjacent to the displayed first physiological parameter value, and wherein the second physiological parameter value is displayed in the second section of the display without indications of corresponding alarm limits;

in response to receiving a first user input dragging the second physiological parameter value from the second section of the display to the first section of the display, causing the second physiological parameter value together with a corresponding waveform of the second physiological parameter to be displayed in the first section of the display adjacent to the first physiological parameter value together with the corresponding waveform of the first physiological parameter; and

in response to receiving a second user input dragging the corresponding waveform of the second physiological parameter onto the corresponding waveform of the first physiological parameter;

causing the corresponding waveforms of the first and second physiological parameters to overlay on one another in a same portion of the display such that both of the corresponding waveforms are visible, continuous, and aligned in time, wherein the corresponding waveforms comprise a plethysmograph and respiration;

determining a level of correlation between the first physiological signal and a second physiological signal associated with the second physiological parameter, wherein the first physiological signal comprises a plethysmographic signal, wherein the second physiological signal comprises a respiratory signal, and wherein determining the level of correlation comprises determining a correlation between (1) peaks or valleys in the plethysmographic signal, and (2) bursts in the respiratory signal; and

in response to the determined level of correlation satisfying a correlation threshold, outputting at least one of: a graphical indication, or an audible alarm.

2. The method of claim 1, wherein:

the second physiological parameter value is outputted for display in the second section of the display without the corresponding waveform of the second physiological parameter.

3. The method of claim 2 further comprising:

by the first medical device comprising digital logic circuitry:

in response to determining an alarm condition associated with the second physiological parameter, causing highlighting of a portion of the first section of the display surrounding the displayed second physiological parameter value and the corresponding waveform of the second physiological parameter.

4. The method of claim 3, wherein the highlighting comprises a colored box surrounding the displayed second physiological parameter value and the corresponding waveform of the second physiological parameter.

5. The method of claim 2 further comprising:

by the first medical device comprising digital logic circuitry:

further in response to receiving the first user input dragging and dropping the second physiological parameter value from the second section of the display to the first section of the display, causing indications of alarm limits associated with the sec-

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ond physiological parameter to be displayed in the first section of the display adjacent to the second physiological parameter value.

6. The method of claim 5, wherein the alarm limits comprise adjustable alarm limits, and wherein the method further comprises, by the first medical device comprising digital logic circuitry:

in response to receiving a third user input, causing display of a popup user interface adjacent to the waveform of the second physiological parameter in the first section of the display and configured to receive adjustments to the alarm limits for the second physiological parameter.

7. The method of claim 1 further comprising:

by the first medical device comprising digital logic circuitry:

obtaining from a second medical device, a third physiological parameter value formatted according to a protocol not used by the first medical device, such that the first medical device is not able to process the third physiological parameter value to produce a displayable output value;

passing the third physiological parameter value from the first medical device to a translation module;

receiving translated parameter data from the translation module at the first medical device, the translated parameter data able to be processed for display by the first medical device; and

outputting the third physiological parameter value from the translated parameter data for display.

8. The method of claim 7 further comprising:

by the first medical device comprising digital logic circuitry:

determining the translated parameter data by translating the third physiological parameter value from a first Health Level 7 (HL7) format to a second HL7 format.

9. The method of claim 8, wherein the third physiological parameter value comprises data from at least one of: a physiological monitor, an infusion pump, a ventilator, or a hospital bed.

10. The method of claim 7 further comprising:

by the first medical device comprising digital logic circuitry:

outputting the first physiological parameter value and the third physiological parameter value from the translated parameter data for display on the same display.

11. The method of claim 7 further comprising:

by the first medical device comprising digital logic circuitry:

outputting the third physiological parameter value from the translated parameter data to an auxiliary device having a separate display from the display of the first medical device.

12. The method of claim 11, wherein the auxiliary device is configured to:

display, in the separate display, at least one of: a histogram of highs and lows for the past several days of the patient, or a detailed waveform that is larger than the waveforms shown on the display of the first medical device; and

in response to receiving a third user input, zooming the detailed waveform to cause the detailed waveform to be enlarged to fill the separate display in place of the other elements of the separate display.

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13. The method of claim 1 further comprising:

by the first medical device comprising digital logic circuitry:

displaying, in the display, a time window button selectable by a user to change a time window of displayed waveforms.

14. A system for displaying medical data, the system comprising:

a first medical device comprising electric hardware configured to:

receive a first physiological signal from a physiological sensor;

determine, based on the first physiological signal, a first physiological parameter value of a first physiological parameter associated with a patient;

output, in a first section of a display, the first physiological parameter value for display together with a corresponding waveform of the first physiological parameter;

receive a second physiological parameter value of a second physiological parameter associated with the patient;

output, in a second section of the display, the second physiological parameter value for display;

output, in the first section of the display, indications of alarm limits associated with the first physiological parameter, wherein the indications of the alarm limits are displayed adjacent to the displayed first physiological parameter value, and wherein the second physiological parameter value is displayed in the second section of the display without indications of corresponding alarm limits;

in response to receiving a first user input dragging the second physiological parameter value from the second section of the display to the first section of the display, cause the second physiological parameter value together with a corresponding waveform of the second physiological parameter to be displayed in the first section of the display adjacent to the first physiological parameter value together with the corresponding waveform of the first physiological parameter; and

in response to receiving a second user input dragging the corresponding waveform of the second physiological parameter onto the corresponding waveform of the first physiological parameter:

cause the corresponding waveforms of the first and second physiological parameters to overlay on one another in a same portion of the display such that both of the corresponding waveforms are visible, continuous, and aligned in time, wherein the corresponding waveforms comprise a plethysmograph and respiration;

determine a level of correlation between the first physiological signal and a second physiological signal associated with the second physiological parameter, wherein the first physiological signal comprises a plethysmographic signal, wherein the second physiological signal comprises a respiratory signal, and wherein determining the level of correlation comprises determining a correlation between (1) peaks or valleys in the plethysmographic signal, and (2) bursts in the respiratory signal; and

in response to the determined level of correlation satisfying a correlation threshold, output at least one of: a graphical indication, or an audible alarm.

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15. The system of claim 14, wherein:

the second physiological parameter value is outputted for display in the second section of the display without the corresponding waveform of the second physiological parameter.

16. The system of claim 15, wherein the first medical device is further configured to:

in response to determining an alarm condition associated with the second physiological parameter, cause highlighting of a portion of the first section of the display surrounding the displayed second physiological parameter value and the corresponding waveform of the second physiological parameter.

17. The system of claim 16, wherein the highlighting comprises a colored box surrounding the displayed second physiological parameter value and the corresponding waveform of the second physiological parameter.

18. The system of claim 15, wherein the first medical device is further configured to:

further in response to receiving the first user input dragging and dropping the second physiological parameter value from the second section of the display to the first section of the display, cause indications of alarm limits associated with the second physiological parameter to be displayed in the first section of the display adjacent to the second physiological parameter value.

19. The system of claim 18, wherein the alarm limits comprise adjustable alarm limits, and wherein the first medical device is further configured to:

in response to receiving a third user input, cause display of a popup user interface adjacent to the waveform of the second physiological parameter in the first section of the display and configured to receive adjustments to the alarm limits for the second physiological parameter.

20. The system of claim 14, wherein the first medical device is further configured to:

obtain from a second medical device, a third physiological parameter value formatted according to a protocol not used by the first medical device, such that the first medical device is not able to process the third physiological parameter value to produce a displayable output value;

pass the third physiological parameter value from the first medical device to a translation module;

receive translated parameter data from the translation module at the first medical device, the translated parameter data able to be processed for display by the first medical device; and

output the third physiological parameter value from the translated parameter data for display.

21. The system of claim 20, wherein the first medical device is further configured to:

determine the translated parameter data by translating the third physiological parameter value from a first Health Level 7 (HL7) format to a second HL7 format.

22. The system of claim 21, wherein the third physiological parameter value comprises data from at least one of: a physiological monitor, an infusion pump, a ventilator, or a hospital bed.

23. The system of claim 20, wherein the first medical device is further configured to:

output the first physiological parameter value and the third physiological parameter value from the translated parameter data for display on the same display.

24. The system of claim **20**, wherein the first medical device is further configured to:

output the third physiological parameter value from the translated parameter data to an auxiliary device having a separate display from the display of the first medical device. 5

25. The system of claim **24**, wherein the auxiliary device is configured to:

display, in the separate display, at least one of: a histogram of highs and lows for the past several days of the patient, or a detailed waveform that is larger than the waveforms shown on the display of the first medical device; and 10

in response to receiving a third user input, zooming the detailed waveform to cause the detailed waveform to be enlarged to fill the separate display in place of the other elements of the separate display. 15

26. The system of claim **14**, wherein the first medical device is further configured to:

display, in the display, a time window button selectable by a user to change a time window of displayed waveforms. 20

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